



Draft Report #2

Title: Repeat Intravenous Dose Range Finding Toxicity
Study of LT3114 in Sprague Dawley Rats with
Toxicokinetics (non-GLP)

**Calvert
Study No.:** 0437RL35.002

**Testing
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Date: 19 January 2015

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II. List of Abbreviations

ASCP	American Society of Clinical Pathologists
CFR	Code of Federal Regulations
DACVP	Diplomate of the American College of Veterinary Pathologists
DVM	Doctor of Veterinary Medicine
EDTA	Ethylenediamine tetraacetic acid
FDA	Food and Drug Administration
GLP	Good Laboratory Practice
IACUC	Institutional Animal Care and Use Committee
ILAR	Institute for Laboratory Animal Resources
MSDS	Material Safety Data Sheet
SOP	Standard Operating Procedure
USDA	United States Department of Agriculture

III. Regulatory Compliance and Signatures

Calvert Study No.: 0437RL35.002

Title: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

A. Regulatory Compliance

1. Calvert Laboratories-Conducted Study Elements

This was a non-regulated study. However, it was run according to the Standard Operating Procedures (SOPs) of Calvert. There was no formal involvement of the Quality Assurance Unit. The toxicokinetic subreport and main study report were in the process of being finalized at the time that this draft was issued.

2. Sponsor-Conducted Study Elements

The following portions of the study were conducted by the Sponsor or a Sponsor-enlisted subcontractor:

- Analysis of test article composition and purity

B. Study Director Signature

I, the undersigned, hereby declare that this report is a true and accurate record of the results obtained. No circumstances occurred during the study that were considered to have significantly affected the overall quality or integrity of the data obtained. All protocol deviations were documented in the study records and are listed in Appendix VII of this report.

Jay Pennell, M.S.
Study Director
Calvert Laboratories, Inc.

Date

C. Signatures of Other Responsible Personnel

Scientific Oversight and Report Review

Scientific Management
Calvert Laboratories, Inc.

Date

IV. Summary

A. Title

Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

B. Objective

The purpose of this study was to evaluate the toxicity of LT3114 when administered intravenously to Sprague Dawley rats. Animals received three doses, one week apart, and toxicokinetics were assessed following each dose.

C. Methods

The test article, LT3114, was supplied by the Sponsor already formulated at 32.1 mg/ml in sterile containers. Diluent was also provided in separate sterile containers. The test article was then prepared into dosing solutions for intravenous administration. Seventy-six experimentally naïve Sprague Dawley rats (38 males and 38 females), 9 weeks old and weighing 202-297 grams for males and females at the outset of the study were assigned to treatment groups as shown in the tables below.

Toxicology Groups

Group	Dose Level (mg/kg/dose) *	Concentration (mg/ml)	Dose Volume (ml/kg)	Number of Animals**	
				Male	Female
1. Control (vehicle)	0	0	5	5	5
2. Test Article Low-dose	10	2	5	5	5
3. Test Article Mid-dose	50	10	5	5	5
4. Test Article High-dose	150	30	5	5	5

*Dosing occurs on Day 1, Day 8, and Day 15

**Toxicology animals were euthanized and necropsied on Day 16

Toxicokinetic Groups

Group	Dose Level (mg/kg dose)*	Concentration (mg/ml)	Dose Volume (ml/kg)	Number of Animals**	
				Male	Female
5. Test Article Low-dose	10	2	5	6	6
6. Test Article Mid-dose	50	10	5	6	6
7. Test Article High-dose	150	30	5	6	6

*Dosing occurs on Day 1, Day 8, and Day 15

**Toxicokinetic animals were euthanized after the last toxicokinetic sample was taken

Animals were dosed on Days 1, 8, and 15. Mortality/morbidity observations were performed twice daily on Days 1 to 15 and once on Day 16. On Days 1, 8, and 15, animals in Groups 1-4 were observed prior to dose administration, immediately post-dose, and 1-2 hours post-dose. Animals were observed once daily on non-dosing days. Body weights were recorded on Days 1, 8, and 15. A fasted body weight was recorded for animals in Groups 1-4 on Day 16. Food consumption was recorded weekly for animals in Groups 1-4. Body temperatures were recorded once daily on Day -1 through Day 15 for animals in Groups 1-4. Temperatures were recorded at approximately the same time each day. Blood for evaluation of hematology, coagulation and clinical chemistry and urine for urinalysis was collected on Day 16. Blood for antibody sampling was collected from animals in Group 1-4 on Day 16. Blood for toxicokinetic evaluation was collected from animals in Groups 5-7 on Day 1 predose and at 30 minutes and 6 hours post-dose, Day 2 (~24 hours post Day 1 dose), Day 3 (~48 hours post Day 1 dose), Day 5 (~96 hours post Day 1 dose), Day 8 (~168 hours post Day 1 dose and prior to Day 8 dose), and Day 15 (predose). All surviving animals in Groups 1-4 were sacrificed on Day 16. Selected tissues were harvested at necropsy, selected organs weighed, and selected tissues from the control and high dose groups evaluated microscopically.

D. Results and Conclusions

Male and female Sprague Dawley rats tolerated LT3114 when given three doses, once per week, at 0, 10, 50 or 150 mg/kg. All animals survived to scheduled sacrifice on Day 16. Test article related toxicity was not observed during the study, as evidenced by clinical observation, body weight, food consumption, clinical pathology, necropsy, organ weight and histopathology evaluation. Slight changes in body temperature were noted during the study but appeared to be

related to normal variation and suggest there were no test article related effects on body temperature.

Animals were exposed to LT3114 following an intravenous dose of LT3114 at 10, 50, and 150 mg/kg/dose. Exposure was LT3114 dose dependent and this exposure increased as the LT3114 dose increased; this exposure appeared to be dose proportional. There did not appear to be any gender-related differences with regard to exposure or elimination. $T_{1/2}$ ranged from 147 to 225 hours. Cl was ~0.2 to ~0.3 ml/hr/kg. V_{ss} ranged from 55 to 80 ml/kg and V_c ranged from 36 to 55 ml/kg.

In this study to evaluate the toxicity of LT3114 when administered intravenously to Sprague-Dawley rats, the plasma toxicokinetics of LT3114 were assessed following each dose level per sex. The predose plasma of all animals in Groups 5, 6, and 7 were devoid of LT3114 on Day 1. Animals were exposed to LT3114 following an intravenous dose of LT3114 at 10, 50, and 150 mg/kg/dose. Exposure was LT3114 dose dependent and this exposure increased as the LT3114 dose increased; this exposure appeared to be dose proportional. There did not appear to be any gender-related differences with regard to exposure or elimination. $T_{1/2}$ ranged from 147 to 225 hours. Cl was ~0.2 to ~0.3 ml/hr/kg. V_{ss} ranged from 55 to 80 ml/kg and V_c ranged from 36 to 55 ml/kg.

V. General Information

A. Key Study Dates

Study Initiation Date:	1 Jul 2014
Animal Receipt Date:	18 Jun 2014
Experimental Start Date:	2 Jul 2014
First Day of Dosing:	7 Jul 2014
Necropsy:	22 Jul 2014
Experimental Completion Date:	22 Jul 2014

B. Responsible Personnel at the Testing Facility

Study Director:	Jay Pennell, M.S.
Study Coordinator:	Renée D. Falzone, B.S.
Project Leader:	Jennifer Jenson, B.S.
Primary Technician:	Michelle Zuerblis, B.S.
Pharmacy Technicians:	Jillian YonKondy, B.S. and Katelynn Henrich, B.S.
Staff Veterinarian:	Jaclyn Tontini, D.V.M.
Medical Technologist:	Barbara Soya, B.S., MT (ASCP) and Amanda O'Brien, B.S., MT (ASCP)

C. Contributing Scientists

Study Monitor:	Marlene W. Modi, Ph.D. MWM Consulting Group, Inc. for Lpath Inc. 14 Joseph Court Monmouth Junction, NJ 08852
Sponsor Representative:	Gary Woodnutt, Ph.D. Sr. Vice President Research Lpath Inc. 4025 Sorrento Valley Blvd San Diego, CA 92121
Principal Investigator – Toxicokinetic Data Evaluation:	Steven Sloneker Calvert Laboratories, Inc. Scott Technology Park

130 Discovery Drive
Scott Township, PA 18447

Principal Investigator – Dosing Formulation
and Toxicokinetic Sample Analysis:

Victor J. Johnson, Ph.D.
Burleson Research Technologies
120 First Flight Lane
Morrisville, NC 27560

Principal Investigator – Histology Slide
Preparation:

Craig Zook
Histo-Scientific Research Laboratories,
Inc. (HSRL)
5930 Main St.
Mt. Jackson, VA 22842

Principal Investigator – Histopathology
Evaluation:

Brett H. Saladino, DVM, DACVP
Director of Pathology
Calvert Laboratories, Inc.
Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447

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D. Objective

The purpose of this study was to evaluate the toxicity of LT3114 when administered intravenously to Sprague Dawley rats. Animals received three doses, one week apart, and toxicokinetics were assessed following each dose.

E. Experimental Design Overview

1. General Description

Seventy-six experimentally naïve Sprague Dawley rats (38 males and 38 females) were assigned to treatment groups as shown in the tables below.

Toxicology Groups

Group	Dose Level (mg/kg/dose)*	Concentration (mg/ml)	Dose Volume (ml/kg)	Number of Animals**	
				Male	Female
1. Control (vehicle)	0	0	5	5	5
2. Test Article Low-dose	10	2	5	5	5
3. Test Article Mid-dose	50	10	5	5	5
4. Test Article High-dose	150	30	5	5	5

*Dosing occurs on Day 1, Day 8, and Day 15

**Toxicology animals were euthanized and necropsied on Day 16

Toxicokinetic Groups

Group	Dose Level (mg/kg dose)*	Concentration (mg/ml)	Dose Volume (ml/kg)	Number of Animals**	
				Male	Female
5. Test Article Low-dose	10	2	5	6	6
6. Test Article Mid-dose	50	10	5	6	6
7. Test Article High-dose	150	30	5	6	6

*Dosing occurs on Day 1, Day 8, and Day 15

**Toxicokinetic animals were euthanized after the last toxicokinetic sample was taken

Animals were dosed on Days 1, 8, and 15. Mortality/morbidity observations were performed twice daily on Days 1 to 15 and once on Day 16. On Days 1, 8, and 15, animals in Groups 1-4 were observed prior to dose administration, immediately post-dose, and 1-2 hours post-dose. Animals were observed once daily on non-dosing days. Body weights were recorded on Days 1, 8, and 15. A fasted body weight was recorded for animals in Groups 1-4 on Day 16. Food consumption was recorded weekly for animals in Groups 1-4. Body temperatures were recorded once daily on Day -1 through Day 15 for animals in Groups 1-4. Temperatures were recorded at approximately the same time each day. Blood for evaluation of hematology, coagulation and clinical chemistry and urine for urinalysis was collected on Day 16. Blood for antibody sampling was collected from animals in Group 1-4 on Day 16. Blood for toxicokinetic evaluation was collected from animals in Groups 5-7 on Day 1 predose and at 30 minutes and 6 hours post-dose, Day 2 (~24 hours post Day 1 dose), Day 3 (~48 hours post Day 1 dose), Day 5 (~96 hours post Day 1 dose), Day 8 (~168 hours post Day 1 dose and prior to Day 8 dose), and Day 15 (predose). All surviving animals in Groups 1-4 were sacrificed on Day 16. Selected tissues were harvested at necropsy, selected organs weighed, and

selected tissues from the control and high dose groups evaluated microscopically.

VI. Materials and Methods

A. Test Article

1. Test Article

Identification: LT3114

Lot No. : 103948

Supplier: Gallus Biopharmaceuticals, LLC

Physical Description: Clear, colorless liquid

Storage Conditions: Stored refrigerated (4.4-6.5°C) and protected from light.

2. Dose Preparation

Before dose preparation, 2 ml of the 10% polysorbate 80 solution was added to each liter of diluent formulation buffer under aseptic conditions and mixed gently to ensure distribution throughout the volume of buffer. The Sponsor provided the polysorbate 80 solution and the diluent formulation. This diluent may have then been used on multiple studies for placebo dosing as well as for diluting test article to achieve the appropriate dosing formulations.

The Sponsor provided the test article already formulated at 32.1 mg/ml in sterile containers. Diluent was also provided in separate sterile containers. Diluent and test article needed to come to room temperature before the formulation was prepared. The formulation was not shaken. Group 1 control animals received the diluent. When necessary depending on the dose level, doses were prepared by dilution using the provided 32.1 mg/ml and diluent solutions under aseptic conditions. Before dosing, visual inspection of the formulation should have been conducted against a black background for any irregularities.

3. Dose Formulation Samples

On Days 1, 8 and 15, duplicate 1-ml samples of each dosing solution, including the vehicle control, were obtained to determine the concentration of the test article in vehicle. These samples were stored at refrigerated temperatures (2.1-8.0°C).

4. Dose Formulation Analysis

Dose formulation samples were returned to, and analytical evaluation was done under the direction of:

Victor J. Johnson, Ph.D.
Burleson Research Technologies
120 First Flight Lane
Morrisville, NC 27560

Targeted Total

Number of Samples: 12 in duplicate (24 total collected)

Sample Storage:

Conditions: Refrigerated temperatures (2-8°C)

Testing Requirements: Concentration and homogeneity

5. Accountability and Disposition

Test material accountability was maintained according to Calvert SOPs. After the end of the study, unused material was retained for use on future studies.

B. Test System (Animals and Animal Care)

1. Description

Species: Rat

Stock: Sprague-Dawley, Hsd:SD

Total Number: 76 (38 males and 38 females)

Gender: Male and female

Age Range: 9 weeks at start of dosing; records of dates of birth for animals used in this study are retained in the Calvert archives.

Body Weight Range: 202 to 297 g at the outset (Day 1) of the study.

Animal Source: Harlan Laboratories, Inc.

Experimental History: Purpose-bred and experimentally naïve at the outset of the study.

Identification: Eartag and cage card.

2. *Rationale for Choice of Species and Number of Animals*

The Sprague Dawley rat was chosen for this study as it is a species that is used for non-clinical toxicity and pharmacokinetic evaluations and satisfies the regulatory requirement for non-clinical safety studies in rodents (1). Previous research has been conducted with this test article in rats. The test article, a human anti-LPA antibody (LT3114) and the murine parental molecule have both been evaluated for efficacy in a variety of rat and mouse animal models of disease. In models using Sprague Dawley rats efficacy has been examined in the STZ induced model of diabetic neuropathy, a chemo-therapeutic-induced model of peripheral neuropathy and in models of acute inflammatory pain (carrageen or Freund's adjuvant induced with doses (intravenous or subcutaneous) ranging from 10 to 40 mg/kg. In addition, single and multiple doses pharmacokinetic studies have been conducted with LT 3114 in murine at dose levels ranging from 10 to 30 mg/kg.

The number of animals used in this study was needed to provide an assessment of the acute intravenous dose toxicity of the test article at three dose levels, account for variability among animals and allow for statistical evaluation of numerical data (1, 2).

3. *Husbandry*

Housing: Animals were group housed by sex upon receipt and individually housed upon assignment to study in compliance with National Research Council "Guide for the Care and Use of Laboratory Animals". The

room in which the animals were kept is documented in the study records. No other species were kept in the same room.

- Lighting: 12 hours light/12 hours dark, except when room lights were turned on during the dark cycle to accommodate blood sampling or other study procedures.
- Room Temperature: 21.3 to 23.4°C
- Relative Humidity: 44.2-62.9%
- Food: All animals had access to Harlan Teklad Rodent Diet (certified) or equivalent *ad libitum*, unless otherwise specified. The lot number(s) and specifications of each lot used are archived at Calvert. No contaminants were known to be present in the certified diet at levels that would be expected to interfere with the results of this study. Analysis of the diet was limited to that performed by the manufacturer, records of which are maintained in the Calvert archives.
- Water: Water was available *ad libitum*, to each animal via an automatic watering device. The water is routinely analyzed for contaminants as per Calvert SOPs. No contaminants were known to be present in the water at levels that would be expected to interfere with the results of this study. Results of the water analysis are maintained in the Calvert archives.
- Acclimation: Study animals were acclimated to their housing for 19 days prior to their first day of dosing.

4. Prestudy Health Screen and Selection Criteria

All animals received for this study were assessed as to their general health by a member of the veterinary staff or other authorized personnel. During the

acclimation period, each animal was observed at least once daily for any abnormalities or for the development of infectious disease. Only animals that were suitable for use were assigned to this study.

5. Assignment to Study Groups

Animals were assigned to study groups using Pristima version 6.3.2 (Xybion Corporation, 201 Littleton Road, Morris Plains, NJ 07950). Animals were randomly assigned to groups using selection designed to achieve similar group mean body weights.

Rats were given the following identification numbers and identified by ear tag:

Toxicology Groups

Group Number	Males	Females
1	2701-2705	2706-2710
2	2711-2715	2716-2720
3	2721-2725	2726-2730
4	2731-2735	2736-2740

Toxicokinetic Groups

Group Number	Males	Females
5	2741-2746	2747-2752
6	2753-2758	2759-2764
7	2765-2770	2771-2776

6. Humane Care of Animals

Treatment of animals was in accordance with the study protocol and also in accordance with Calvert SOPs which adhere to the regulations outlined in the USDA Animal Welfare Act (9 CFR Parts 1, 2 and 3) and the conditions specified in the Guide for the Care and Use of Laboratory Animals (ILAR publication, 1996, National Academy Press). The Calvert IACUC approved the study protocol prior to dose administration.

C. Test Article Administration**1. Group Assignments and Dose Levels****Toxicology Groups**

Group	Dose Level (mg/kg/dose) [*]	Concentration (mg/ml)	Dose Volume (ml/kg)	Number of Animals ^{**}	
				Male	Female
1. Control (vehicle)	0	0	5	5	5
2. Test Article Low-dose	10	2	5	5	5
3. Test Article Mid-dose	50	10	5	5	5
4. Test Article High-dose	150	30	5	5	5

^{*}Dosing occurs on Day 1, Day 8, and Day 15

^{**}Toxicology animals were euthanized and necropsied on Day 16

Toxicokinetic Groups

Group	Dose Level (mg/kg dose) [*]	Concentration (mg/ml)	Dose Volume (ml/kg)	Number of Animals ^{**}	
				Male	Female
5. Test Article Low-dose	10	2	5	6	6
6. Test Article Mid-dose	50	10	5	6	6
7. Test Article High-dose	150	30	5	6	6

^{*}Dosing occurs on Day 1, Day 8, and Day 15

^{**}Toxicokinetic animals were euthanized after the last toxicokinetic sample was taken

2. Dosing

Route: Slow intravenous injection (~1 minute)

Frequency: On Days 1, 8, and 15

Procedure: Doses were administered on Days 1, 8, and 15 by intravenous injection via the lateral tail vein at a dose volume of 5 ml/kg. Each intravenous dose was administered as a slow bolus injection (~1 minute). Each animal received a mg/kg dose based on its most recent body weight.

3. Justification for Route, Dose Levels and Dosing Schedule

The intravenous route was chosen as it is the intended route of administration in humans.

Dose levels were selected by the Sponsor. The anticipated lowest pharmacologically active dose is 30 mg/kg. The high dose selected for this study represents a several-fold increase from the anticipated lowest pharmacologic dose. The study animals were administered the test formulations in order of ascending dosage on every dosing day. Should any adverse signs have been observed during administration of the low- or mid-dose levels, dosing was aborted and the next highest dose(s) were not administered.

The frequency of dosing was chosen by the Sponsor based on the test article half-life in rodents (~100 hr) and supports the safety of the product under its intended clinical use.

D. In-Life Observations and Measurements

1. Mortality/Morbidity:

Frequency: Twice daily (a.m. and p.m.) on Day 1 to Day 15.
Once on Day 16 prior to euthanasia.
Each animal observed for evidence of death or impending death (as per Calvert SOP VET-14).

2. Clinical Observations:

Frequency: On Days 1, 8, and 15, animals in Groups 1-4 were observed prior to dose administration, immediately post-dose, and 1-2 hours post-dose. Animals were observed once daily on non-dosing days.

3. Body Weight:

Frequency: Prior to dose administration on Days 1, 8, and 15.

A fasted body weight was recorded for animals in Groups 1-4 on Day 16 for calculation of organ to body weight ratios.

4. Food Consumption:

Frequency: Full feeder weights and/or feeder weigh backs were recorded weekly for Groups 1-4 for determination of food consumption.

5. Body Temperature:

Frequency: Body temperature was recorded once daily on Day -1 through Day 15 for animals in Groups 1-4. On dosing days, body temperatures were recorded post-dose. Temperatures were recorded at approximately the same time on each day of measurement.

E. Clinical Pathology Evaluation**1. Sample Collection**

Blood samples for evaluation of serum chemistry, hematology and coagulation parameters were collected from all surviving animals in Groups 1-4 at terminal sacrifice on Day 16. Animals were anesthetized by CO₂ inhalation prior to blood collection. Immediately following exsanguination by cardiocentesis for terminal blood collection, rats were returned to the CO₂ chamber to ensure euthanasia. Urine was collected overnight (approximately 17 hours). Animals were fasted overnight (approximately 18-20 hours) prior to blood collection for clinical pathology evaluation.

2. Collection Procedures, Processing and Analysis**a) Hematology**

Method of Collection: Cardiocentesis

Anticoagulant: EDTA

Parameters Analyzed:

Hematology Parameters	
Red Blood Cell Count (RBC) and Morphology	Platelet count (PLT)
White Blood Cell Count (WBC)*	Hematocrit (HCT)
Mean Corpuscular Hemoglobin (MCH)	Hemoglobin (HGB)
Mean Corpuscular Hemoglobin Concentration (MCHC)	Reticulocyte Count (Retic)
Mean Corpuscular Volume (MCV)	

*Total and differential white blood cell counts, including neutrophils, basophils, eosinophils, monocytes, lymphocytes and large unstained cells

b) Coagulation

Method of Collection: Cardiocentesis

Anticoagulant: Sodium Citrate

Parameters Analyzed:

Coagulation Parameters	
Activated Partial Thromboplastin Time (APTT)	Prothrombin Time (PT)

c) Serum Clinical Chemistry

Method of Collection: Cardiocentesis

Anticoagulant: None

Parameters Analyzed:

Clinical Chemistry Parameters	
Alanine Aminotransferase (ALT)	Globulin (calculated)(GLOB)
Albumin (ALB)	Glucose (GLU)
Albumin/Globulin ratio (calculated)(A/G)	Phosphorus (PHOS)
Alkaline Phosphatase (ALP)	Potassium (K)
Aspartate Aminotransferase (AST)	Sodium (NA)
Calcium (CA)	Total Bilirubin (T-BIL)
Chloride (CL)	Total Protein (TP)
Cholesterol (CHOL)	Triglycerides (TRIG)
Creatinine (CREAT)	Urea Nitrogen (BUN)

d) Urinalysis

Method of Collection: Collection in metabolism cages

Parameters Analyzed:

Urinalysis Parameters	
Specific gravity	Appearance/Color
pH	Bilirubin
Protein	Blood
Glucose	Leukocytes
Ketone	Nitrites
Urobilinogen	Microscopic examination of formed elements

F. Antibody Sampling

Whole blood samples (approximately 05 ml/sample) were collected from all animals in Groups 1-4 via cardiocentesis at terminal sacrifice on Day 16 into K₂ EDTA tubes. Animals will be anesthetized by CO₂ inhalation prior to blood collection. Blood samples were spun in a centrifuge (approximately 3000 rpm for approximately 10 minutes) to separate the plasma. Plasma was then collected and stored frozen at approximately -70°C. Plasma samples were shipped to the Sponsor for use in assay development. There are no assay results to report for this study as these plasma samples were used for assay development.

G. Toxicokinetics (Group 5-7 only)

1. Blood Sampling

On Day 1, whole blood samples (approximately 0.5 ml/sample) were collected from 3 animals/sex/group at the following timepoints:

0 hour (pre-dose)	30 minutes post-dose
	6 hours post-dose

On the following days post the Day 1 dosing, whole blood samples (approximately 0.5 ml/sample) were collected from 3 animals/sex/group animals at the following timepoints:

Day 2 (~24 hours post Day 1 dose)

Day 3 (~48 hours post Day 1 dose)

Day 5 (~96 hours post Day 1 dose)

Day 8 (~168 hours post Day 1 dose and prior to next dose)

Day 15 (~pre-dose)

The total volume collected, in consideration with other parameters that required blood collection, did not exceed 1% of body weight for the animals during a two-week period. Animals were not bled more than 4 times throughout the duration of the study. Animals were anesthetized by CO₂ inhalation prior to blood collection. Whole blood (approximately 0.5 ml/sample) was collected by retroorbital puncture into K₂EDTA plasma collection tubes. Blood samples were spun in a centrifuge (approximately 3000 rpm for approximately 10 minutes) to separate the serum. Plasma was then collected and stored frozen at approximately -70°C.

2. Bioanalytical Evaluation

Plasma samples were returned to, and bioanalytical evaluation was done under the direction of:

Victor J. Johnson, Ph.D.
Burleson Research Technologies
120 First Flight Lane
Morrisville, NC 27560

Targeted Number
of Samples: 144 (8 timepoints * 6 animals/group/timepoint * 3 groups)

Sample Storage
Conditions: approximately -70° C or lower

Testing Requirements: Testing for levels of test article

3. *Toxicokinetic Evaluation*

Data from the bioanalytical evaluation was provided to, and toxicokinetic evaluation was done under the direction of:

Steven Sloneker
Calvert Laboratories, Inc.
Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447

Parameters Examined: At least the following areas will be examined,
 AUC_{0-inf} , AUC_{0-last} , C_{max} , Cl , C_{last} , C_0 , T_{last} , T_{max} , $T_{1/2}$,
 V_{ss} , V_c

Note: If the terminal half-life cannot be reliably calculated, the dependent PK parameters of AUC_{0-inf} , Cl , V_{ss} , and V_c will not be reported.

Data Evaluation

Software: Win Nonlin Version 5.3

H. Terminal Procedures and Anatomic Pathology

1. *Termination*

a) Scheduled Sacrifice

All animals were euthanized by CO₂ asphyxiation following terminal blood collection on Day 16.

b) Final Body Weight

A fasted terminal body weight was recorded prior to sacrifice on Day 16. This body weight was used to calculate organ to body weight ratios.

2. Gross Necropsy

A complete gross necropsy was performed on all animals that were sacrificed during the study. The necropsy included examination of:

- the external body surface
- all orifices
- the cranial, thoracic and abdominal cavities and their contents.

All abnormalities were described completely and recorded.

3. Organ Weights

At scheduled sacrifice on Day 16, the following organs (when present) were weighed before fixation, after dissection of excess fat and other excess tissues. Paired organs were weighed together unless gross abnormalities were present, in which case they were weighed separately. The thyroids/parathyroids were weighed fixed at the histology laboratory.

Organs Weighed	
Adrenals	Testes
Brain	Ovaries
Heart	Spleen
Kidneys	Thyroids/parathyroids
Liver	

Organ to body weight ratios were calculated (using the final body weight obtained prior to necropsy), as well as organ to brain weight ratios.

4. Tissue Collection and Preservation

For all animals necropsied, the tissues listed in the table below were preserved in 10% neutral buffered formalin (except for the eyes, which were preserved in Davidson's fixative and the testes that were preserved in modified Davidsons's fixative for optimum fixation).

Tissues Collected	
Cardiovascular	Urogenital
Aorta	Kidneys
Heart	Urinary Bladder
Digestive	Ovaries
Salivary gland(s)	Uterus
Tongue	Cervix
Esophagus	Vagina
Stomach	Testes
Small Intestine	Epididymides
Duodenum	Prostate
Jejunum	Seminal Vesicles
Ileum	Endocrine
Large Intestine	Adrenals
Cecum	Pituitary
Colon	Thyroid/Parathyroid
Rectum	Skin/Musculoskeletal
Pancreas	Skin
Liver	Mammary Gland
Respiratory	Skeletal Muscle (thigh)
Trachea	Femur with articular surface
Larynx	Nervous/Special Sense
Lung with mainstem bronchus	Eye with optic nerve
Lymphoid/Hematopoietic	Sciatic Nerve
Sternum with bone marrow	Brain
Thymus	Spinal Cord – cervical
Spleen	Spinal Cord – midthoracic
Lymph Nodes	Spinal Cord – lumbar
Mandibular	Lacrimal Glands
Mesenteric	Other
	Animal Eartag
	Gross Findings
	Injection Site(s)

5. Histology

Preparation of slides was performed under the direction of:

Craig Zook

Histo-Scientific Research Laboratories, Inc. (HSRL)

5930 Main St.

Mt. Jackson, VA 22842

Tissues for evaluation were processed to paraffin blocks and prepared to slides. Slides were stained with hematoxylin and eosin. Occasionally, other stains may have been required by the Study Pathologist to aid in the diagnosis of lesions; these were documented in the final report.

Slides were prepared for animals in Groups 1 and 4 for the following tissues: adrenals, brain, heart, kidneys, liver, lung, spleen, ovaries and testes.

6. Histopathology Evaluation

Histopathology evaluation was performed by:

Brett H. Saladino, DVM, Diplomate ACVP

Director of Pathology

Calvert Laboratories, Inc.

Scott Technology Park

130 Discovery Drive

Scott Township, PA 18447

Histopathological evaluations were done for animals in Groups 1 and 4 for the following tissues: adrenals, brain, heart, kidneys, liver, lung, spleen, ovaries and testes.

VII. Records and Reports

A. Data Collection and Analysis

In-Life data (clinical observations, body weights, feeder weights, dose administration and/or other related data) was collected using Pristima version 6.3.2 or on paper when necessary. Hematology data was collected using the Advia 120 Hematology analyzer. Coagulation data was collected using the Diagnostica STA

Compact Coagulation Analyzer. Clinical chemistry data was collected using the AU400 Chemistry analyzer. Once collected on their respective instruments, hematology, coagulation and clinical chemistry data was transferred to Pristima version 6.3.2 via a validated interface. In the event of hardware or software maintenance, testing or malfunction or network connection issues, data was collected on paper or on the appropriate clinical pathology instrument and then entered into Pristima version 6.3.2 at a later date. Necropsy data was collected on paper. Anatomic pathology findings were directly entered into Pristima version 6.3.2. Any other data not collected on-line was manually tabulated for inclusion in the report.

In-Life data, clinical pathology data, necropsy and anatomic pathology data were tabulated and/or statistically evaluated using Pristima version 6.3.2. Pristima software was developed by Xybion Corporation (201 Littleton Road, Morris Plains, NJ 07950).

Statistical evaluation was performed on In-Life, Clinical Pathology, and Organ Weight numerical data. For In-Life, Clinical Pathology, and Organ Weight parameters, the software determined statistical significance by following a decision tree. First, the homogeneity of the data was determined by Bartlett's Test. If the data was homogeneous, a one-way analysis of variance was performed to assess statistical significance. If statistically significant differences between the means were found, Dunnett's test was used to determine the degree of significance from the control means ($p < 0.05$, $p < 0.01$ and $p < 0.001$). If the data was non-homogeneous, the Kruskal-Wallis non-parametric analysis was performed to assess statistical significance. If statistically significant differences between the means are found ($p < 0.05$, $p < 0.01$ and $p < 0.001$), the Mann-Whitney U-Test was used to determine the degree of significance from the control means ($p < 0.05$, $p < 0.01$ and $p < 0.001$).

B. Storage of Records

Test article preparation and test article information, test article tracking, in-life data, necropsy data, clinical pathology data, protocol, protocol amendments, slides, pathology evaluations (which may include blocks and wet tissues), and the original final report generated as a result of this study will be archived at Calvert, 105 Edella Road, Suite 100, Clarks Summit, PA 18411. After 2 years, the Sponsor will be contacted to determine final disposition of all study materials.

VIII. Results and Discussion

A. Dose Formulation Analysis

The analytical report is presented in Appendix IV. Dose formulation samples were provided for three dose concentrations, 2 mg/mL, 10 mg/mL, and 30 mg/mL. Dose formulation samples were collected from each dosing solution and the vehicle from each of three independent preparations during the study. Analyses demonstrated the absence of LT3114 in the vehicle dosing solutions. The LT3114 concentration ranges were 2.08 – 2.31 mg/mL, 9.18 – 10.08 mg/mL, and 27.3 – 35.2 mg/mL for the 2, 10, and 30 mg/mL dosing solutions, respectively. The LT3114 concentrations determined for all dose formulation samples met acceptance criteria for accuracy and precision.

B. In-Life Observations

1. Mortality/Morbidity

There were no unscheduled mortalities.

2. Clinical Observations

There were no clinical signs observed throughout the study (Table 1).

3. Body Temperature

Animals in the LT3114 treatment groups had slightly but statistically significantly higher group mean body temperatures than the controls sporadically throughout the study (Table 2). However, the treatment groups had higher body temperatures than the controls prior to treatment initiation, and in two groups (Group 3 and 4 females), the differences at baseline prior to treatment with LT3114 were statistically significant. Individual animal body temperatures recorded during the treatment phase were generally within the range of the baseline temperatures for males (35.1 – 37.2°C) and females (36.1 – 38.4°C). There were some instances in which animals in the LT3114 groups had individual body temperatures slightly above the range of baseline data. However, on the days that this occurred, the control animals also had higher body temperatures than their respective baseline recordings. This trend suggests that the changes in body temperature throughout the dosing phase were not related to LT3114 and may have been due to normal variation or

other factors (animal restraint, dosing procedures, fluctuations in room temperature, etc.). Taken together, the differences in group mean body temperature prior to study initiation, and the similarity between temperature trends in the control and treatment groups during the dosing phase, suggest that there were no test article related effects on body temperature.

4. Body Weight

There were no significant differences in group mean body weights (Table 3) or body weight gains (Table 4).

5. Food Consumption

There were no significant differences in group mean food consumption (Table 5).

C. Clinical Pathology

1. Hematology

There were no findings related to hematology (Table 6) or erythrocyte morphology (Table 7).

2. Coagulation

There were no findings related to coagulation parameters (Table 8).

3. Clinical Chemistry

There were no test article related differences in group mean serum chemistry parameters (Table 9). Males in Group 4 had statistically significantly higher mean globulin than the controls on Day 16, but all globulin values were within the range of historical control data.

4. Urinalysis

There were no findings related to quantitative urinalysis (Table 10) or qualitative urinalysis (Table 11).

D. Postmortem Observations

1. Gross Necropsy Findings

No test article-related macroscopic observations were noted at the terminal sacrifice on Day 16 (Table 12). One male in Group 2 (Animal No. 2711) had bilaterally small testes that correlated microscopically with diffuse, bilateral atrophied/hypoplastic seminiferous tubules, which was likely an incidental congenital anomaly.

2. Organ Weights

There were no differences in absolute organ weights (Table 13), organ to body weight ratios (Table 14) or organ to brain weight ratios (Table 15) that were related to treatment. Group 2 females had statistically significantly lower liver to body weight ratios than controls on Day 16. However, Group 2 absolute liver weights and liver to brain weight ratios were not significantly different than controls, and no liver weight findings were noted in Group 3 or 4 animals.

3. Histopathology

No test article-related microscopic observations were noted at the terminal sacrifice on Day 16. All microscopic findings were considered incidental background findings of no toxicologic significance because they were also noted in control animals or are known background findings in animals of this species, strain and age. The high incidence of findings in the kidneys of both Group 1 and Group 4 animals represent early changes associated with the condition known as chronic progressive nephropathy of rats.

E. Toxicokinetics

The toxicokinetic report is presented in Appendix V. In this study to evaluate the toxicity of LT3114 when administered intravenously to Sprague-Dawley rats, the plasma toxicokinetics of LT3114 were assessed following each dose level per sex.

The predose plasma of all animals in Groups 5, 6, and 7 were devoid of LT3114 on Day 1.

Animals were exposed to LT3114 following an intravenous dose of LT3114 at 10, 50, and 150 mg/kg/dose. Exposure was LT3114 dose dependent and this exposure increased as the LT3114 dose increased; this exposure appeared to be

dose proportional. There did not appear to be any gender-related differences with regard to exposure or elimination. $T_{1/2}$ ranged from 147 to 225 hours. Cl was ~ 0.2 to ~ 0.3 ml/hr/kg. V_{ss} ranged from 55 to 80 ml/kg and V_c ranged from 36 to 55 ml/kg.

IX. Conclusion

Male and female Sprague Dawley rats tolerated LT3114 when given three intravenous doses, one dose per week, at 0, 10, 50 or 150 mg/kg. All animals survived to scheduled sacrifice on Day 16. Test article related toxicity was not observed during the study, as evidenced by clinical observation, body weight, food consumption, clinical pathology, necropsy, organ weight and histopathology evaluation. Slight changes in body temperature were noted during the study but appeared to be related to normal variation and were not test article related effects on body temperature..

X. References

1. Gad, Shayne C. (ed.). (1995). Safety Assessment for Pharmaceuticals. Van Nostrand Reinhold, New York, pp. 111-128.
2. Speid, L.H., Lumley, C.E., and Walker, S.R.. (1990). Harmonization of Guidelines for Toxicity Testing of Pharmaceuticals by 1992. Regulatory Toxicology and Pharmacology, 12: 179-211.

XI. Tables

DRAFT

TABLE 1 – Clinical Signs Summary Report By Session

Group	Sex	Dose Level
1*	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1*	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

* Control

TABLE 1 - Clinical Signs Summary Report By Session

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

		Males							
		Dosing Phase							
Group #	Category, Observations	Day of Phase:	1			8			15
		Session #:	1	2	3	1	2	3	1
		Scheduled:	S	S	S	S	S	S	S
Control	Number Examined:	5	5	5	5	5	5	5	
	Number Normal:	5	5	5	5	5	5	5	
2	Number Examined:	5	5	5	5	5	5	5	
	Number Normal:	5	5	5	5	5	5	5	
3	Number Examined:	5	5	5	5	5	5	5	
	Number Normal:	5	5	5	5	5	5	5	
4	Number Examined:	5	5	5	5	5	5	5	
	Number Normal:	5	5	5	5	5	5	5	

Session 1 = Predose Session 2 = Immediately Post-dose Session 3 = 1-2 Hours Post-dose

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Group #	Category, Observations	Males	
		Dosing Phase	
		Day of Phase: 15	
		Session #: 2	3
		Scheduled: S	S
Control		Number Examined: 5	5
		Number Normal: 5	5
2		Number Examined: 5	5
		Number Normal: 5	5
3		Number Examined: 5	5
		Number Normal: 5	5
4		Number Examined: 5	5
		Number Normal: 5	5

Session 1 = Predose Session 2 = Immediately Post-dose Session 3 = 1-2 Hours Post-dose

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

		Females							
		Dosing Phase							
Group #	Category, Observations	Day of Phase:	1			8		15	
		Session #:	1	2	3	1	2	3	1
		Scheduled:	S	S	S	S	S	S	S
Control	Number Examined:	5	5	5	5	5	5	5	
	Number Normal:	5	5	5	5	5	5	5	
2	Number Examined:	5	5	5	5	5	5	5	
	Number Normal:	5	5	5	5	5	5	5	
3	Number Examined:	5	5	5	5	5	5	5	
	Number Normal:	5	5	5	5	5	5	5	
4	Number Examined:	5	5	5	5	5	5	5	
	Number Normal:	5	5	5	5	5	5	5	

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Group #	Category, Observations	Females	
		Dosing Phase	
		Day of Phase: 15	
		Session #: 2	3
		Scheduled: S	S
Control		Number Examined: 5	5
		Number Normal: 5	5
2		Number Examined: 5	5
		Number Normal: 5	5
3		Number Examined: 5	5
		Number Normal: 5	5
4		Number Examined: 5	5
		Number Normal: 5	5

Session 1 = Predose Session 2 = Immediately Post-dose Session 3 = 1-2 Hours Post-dose

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

		Males							
Group #	Category, Observations	Dosing Phase							
		Day of Phase:	2	3	4	5	6	7	9
		Session #:	1	1	1	1	1	1	1
		Scheduled:	S	S	S	S	S	S	S
Control	Number Examined:	5	5	5	5	5	5	5	
	Number Normal:	5	5	5	5	5	5	5	
2	Number Examined:	5	5	5	5	5	5	5	
	Number Normal:	5	5	5	5	5	5	5	
3	Number Examined:	5	5	5	5	5	5	5	
	Number Normal:	5	5	5	5	5	5	5	
4	Number Examined:	5	5	5	5	5	5	5	
	Number Normal:	5	5	5	5	5	5	5	

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Group #	Category, Observations	Males						
		Dosing Phase						
		Day of Phase:	10	11	12	13	14	16
		Session #:	1	1	1	1	1	1
		Scheduled:	S	S	S	S	S	S
Control	Number Examined:	5	5	5	5	5	5	
	Number Normal:	5	5	5	5	5	5	
2	Number Examined:	5	5	5	5	5	5	
	Number Normal:	5	5	5	5	5	5	
3	Number Examined:	5	5	5	5	5	5	
	Number Normal:	5	5	5	5	5	5	
4	Number Examined:	5	5	5	5	5	5	
	Number Normal:	5	5	5	5	5	5	

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Group #	Category, Observations	Females							
		Dosing Phase							
		Day of Phase:	2	3	4	5	6	7	9
		Session #:	1	1	1	1	1	1	1
		Scheduled:	S	S	S	S	S	S	S
Control	Number Examined:	5	5	5	5	5	5	5	
	Number Normal:	5	5	5	5	5	5	5	
2	Number Examined:	5	5	5	5	5	5	5	
	Number Normal:	5	5	5	5	5	5	5	
3	Number Examined:	5	5	5	5	5	5	5	
	Number Normal:	5	5	5	5	5	5	5	
4	Number Examined:	5	5	5	5	5	5	5	
	Number Normal:	5	5	5	5	5	5	5	

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Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

		Females						
		Dosing Phase						
Group #	Category, Observations	Day of Phase:	10	11	12	13	14	16
		Session #:	1	1	1	1	1	1
		Scheduled:	S	S	S	S	S	S
Control	Number Examined:	5	5	5	5	5	5	
	Number Normal:	5	5	5	5	5	5	
2	Number Examined:	5	5	5	5	5	5	
	Number Normal:	5	5	5	5	5	5	
3	Number Examined:	5	5	5	5	5	5	
	Number Normal:	5	5	5	5	5	5	
4	Number Examined:	5	5	5	5	5	5	
	Number Normal:	5	5	5	5	5	5	

TABLE 2 – Body Temperature Summary Report

Group	Sex	Dose Level
1*	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1*	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

* Control

Statistical evaluation performed using Prisma version 6.3.2

Bartlett's Test for homogeneity: used to determine homogeneity of the data.
Homogeneous data: ANOVA with Dunnett's test was used to determine statistical significance.

Non-homogeneous data: ANOVA with Kruskal Wallis test was used to determine statistical significance followed by Mann-Whitney U-Test to determine which groups were statistically different.

Statistically significant differences were determined from control group

*D = Dunnett Exact Homogeneous Test Significant: 0.05 level

+D = Dunnett Exact Homogeneous Test Significant: 0.01 level

#D = Dunnett Exact Homogeneous Test Significant: 0.001 level

*U = Mann Whitney U Test Significant: 0.05 level

+U = Mann Whitney U Test Significant: 0.01 level

#U = Mann Whitney U Test Significant: 0.001 level

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Pretest (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #		BWTP (°C)
Control	(n)	5
	Means	36.26
	Sdevs	0.493
2	(n)	5
	Means	36.42
	Sdevs	0.782
3	(n)	5
	Means	36.50
	Sdevs	0.860
4	(n)	5
	Means	36.84
	Sdevs	0.351

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Pretest (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #		BWTP (°C)
Control	(n)	5
	Means	36.88
	Sdevs	0.630
2	(n)	5
	Means	37.58
	Sdevs	0.471
3	(n)	5
	Means	37.90+D
	Sdevs	0.436
4	(n)	5
	Means	37.66*D
	Sdevs	0.279

+D = Dunnett LSD Test Significant at the 0.01 level

*D = Dunnett LSD Test Significant at the 0.05 level

TABLE 2 - Body Temperature Summary Report

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Study: 0437RL35.002

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 1 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #		BWTP (°C)
Control	(n)	5
	Means	36.16
	Sdevs	0.577
2	(n)	5
	Means	36.18
	Sdevs	0.630
3	(n)	5
	Means	36.84
	Sdevs	0.230
4	(n)	5
	Means	36.68
	Sdevs	0.239

TABLE 2 - Body Temperature Summary Report

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 1 (Scheduled Animal Room - Session 1)

		Females
		Body Temperature
Group #	BWTP (°C)	
Control	(n)	5
	Means	36.96
	Sdevs	0.207
2	(n)	5
	Means	37.56*D
	Sdevs	0.422
3	(n)	5
	Means	37.68*D
	Sdevs	0.370
4	(n)	5
	Means	37.79+D
	Sdevs	0.311

*D = Dunnett LSD Test Significant at the 0.05 level

+D = Dunnett LSD Test Significant at the 0.01 level

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 2 (Scheduled Animal Room - Session 1)

		Males	
		Body Temperature	
Group #		BWTP (°C)	
Control	(n)	5	
	Means	36.42	
	Sdevs	0.148	
2	(n)	5	
	Means		
	Sdevs	0.515	
3	(n)	5	
	Means	36.58	
	Sdevs	0.449	
4	(n)	5	
	Means	37.19*D	
	Sdevs	0.408	

*D = Dunnett LSD Test Significant at the 0.05 level

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 2 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #		BWTP (°C)
Control	(n)	5
	Means	37.64
	Sdevs	0.568
2	(n)	5
	Means	38.10
	Sdevs	0.245
3	(n)	5
	Means	38.32
	Sdevs	0.327
4	(n)	5
	Means	37.78
	Sdevs	0.540

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 3 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #		BWTP (°C)
Control	(n)	5
	Means	35.70
	Sdevs	1.012
2	(n)	5
	Means	36.48
	Sdevs	0.164
3	(n)	5
	Means	36.82
	Sdevs	0.409
4	(n)	5
	Means	36.82
	Sdevs	0.482

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 3 (Scheduled Animal Room - Session 1)

		Females	
		Body Temperature	
Group #		BWTP (°C)	
Control	(n)	5	
	Means	37.00	
	Sdevs	0.200	
2	(n)	5	
	Means	37.72*D	
	Sdevs	0.192	
3	(n)	5	
	Means	38.24+D	
	Sdevs	0.288	
4	(n)	5	
	Means	37.68*D	
	Sdevs	0.622	

*D = Dunnett LSD Test Significant at the 0.05 level

+D = Dunnett LSD Test Significant at the 0.01 level

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 4 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #		BWTP (°C)
Control	(n)	5
	Means	36.42
	Sdevs	0.415
2	(n)	5
	Means	36.90
	Sdevs	0.524
3	(n)	5
	Means	37.12
	Sdevs	0.130
4	(n)	5
	Means	37.16
	Sdevs	0.737

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 4 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #		BWTP (°C)
Control	(n)	5
	Means	37.94
	Sdevs	0.493
2	(n)	5
	Means	38.26
	Sdevs	0.089
3	(n)	5
	Means	38.56*U
	Sdevs	0.089
4	(n)	5
	Means	38.16
	Sdevs	0.498

*U = Mann-Whitney U Test Significant at 0.05 level.

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 5 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #		BWTP (°C)
Control	(n)	5
	Means	36.48
	Sdevs	0.130
2	(n)	5
	Means	36.82
	Sdevs	0.268
3	(n)	5
	Means	36.48
	Sdevs	0.383
4	(n)	5
	Means	36.94
	Sdevs	0.555

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 5 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #		BWTP (°C)
Control	(n)	5
	Means	37.48
	Sdevs	0.460
2	(n)	5
	Means	38.20*D
	Sdevs	0.235
3	(n)	5
	Means	38.26*D
	Sdevs	0.195
4	(n)	5
	Means	38.00
	Sdevs	0.604

*D = Dunnett LSD Test Significant at the 0.05 level

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 6 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #		BWTP (°C)
Control	(n)	5
	Means	36.44
	Sdevs	0.114
2	(n)	5
	Means	37.10*U
	Sdevs	0.469
3	(n)	5
	Means	37.46+U
	Sdevs	0.279
4	(n)	5
	Means	37.52*U
	Sdevs	0.602

*U = Mann-Whitney U Test Significant at 0.05 level.

+U = Mann-Whitney U Test Significant at 0.01 level.

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 6 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #		BWTP (°C)
Control	(n)	5
	Means	37.78
	Sdevs	0.432
2	(n)	5
	Means	38.32
	Sdevs	0.277
3	(n)	5
	Means	38.38*D
	Sdevs	0.228
4	(n)	5
	Means	38.36*D
	Sdevs	0.397

*D = Dunnett LSD Test Significant at the 0.05 level

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 7 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #		BWTP (°C)
Control	(n)	5
	Means	36.54
	Sdevs	0.182
2	(n)	5
	Means	37.22*D
	Sdevs	0.217
3	(n)	5
	Means	37.36+D
	Sdevs	0.385
4	(n)	5
	Means	37.44+D
	Sdevs	0.550

*D = Dunnett LSD Test Significant at the 0.05 level

+D = Dunnett LSD Test Significant at the 0.01 level

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 7 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #		BWTP (°C)
Control	(n)	5
	Means	37.76
	Sdevs	0.270
2	(n)	5
	Means	38.06
	Sdevs	0.270
3	(n)	5
	Means	38.30
	Sdevs	0.122
4	(n)	5
	Means	38.22
	Sdevs	0.676

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 8 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #		BWTP (°C)
Control	(n)	5
	Means	36.52
	Sdevs	0.277
2	(n)	5
	Means	36.60
	Sdevs	0.424
3	(n)	5
	Means	36.48
	Sdevs	0.327
4	(n)	5
	Means	36.74
	Sdevs	0.503

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 8 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #		BWTP (°C)
Control	(n)	5
	Means	37.48
	Sdevs	0.303
2	(n)	5
	Means	38.12
	Sdevs	0.743
3	(n)	5
	Means	37.82
	Sdevs	0.835
4	(n)	5
	Means	37.66
	Sdevs	0.654

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 9 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #		BWTP (°C)
Control	(n)	5
	Means	36.44
	Sdevs	0.167
2	(n)	5
	Means	37.08
	Sdevs	0.502
3	(n)	5
	Means	37.02
	Sdevs	0.726
4	(n)	5
	Means	37.20
	Sdevs	0.652

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 9 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #		BWTP (°C)
Control	(n)	5
	Means	37.90
	Sdevs	0.122
2	(n)	5
	Means	38.26*U
	Sdevs	0.207
3	(n)	5
	Means	38.52+U
	Sdevs	0.192
4	(n)	5
	Means	38.04
	Sdevs	0.767

*U = Mann-Whitney U Test Significant at 0.05 level.

+U = Mann-Whitney U Test Significant at 0.01 level.

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 10 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #		BWTP (°C)
Control	(n)	5
	Means	36.68
	Sdevs	0.179
2	(n)	5
	Means	37.12
	Sdevs	0.444
3	(n)	5
	Means	37.12
	Sdevs	0.756
4	(n)	5
	Means	37.40
	Sdevs	0.728

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 10 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #		BWTP (°C)
Control	(n)	5
	Means	38.08
	Sdevs	0.342
2	(n)	5
	Means	38.16
	Sdevs	0.167
3	(n)	5
	Means	38.48*D
	Sdevs	0.179
4	(n)	5
	Means	38.56+D
	Sdevs	0.114

*D = Dunnett LSD Test Significant at the 0.05 level

+D = Dunnett LSD Test Significant at the 0.01 level

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 11 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #		BWTP (°C)
Control	(n)	5
	Means	36.50
	Sdevs	0.122
2	(n)	5
	Means	37.32
	Sdevs	0.444
3	(n)	5
	Means	37.14
	Sdevs	0.688
4	(n)	5
	Means	37.34
	Sdevs	0.856

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 11 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #		BWTP (°C)
Control	(n)	5
	Means	38.12
	Sdevs	0.228
2	(n)	5
	Means	38.18
	Sdevs	0.192
3	(n)	5
	Means	38.48
	Sdevs	0.192
4	(n)	5
	Means	38.38
	Sdevs	0.319

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 12 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #		BWTP (°C)
Control	(n)	5
	Means	33.86
	Sdevs	0.518
2	(n)	5
	Means	34.84
	Sdevs	0.611
3	(n)	5
	Means	34.18
	Sdevs	0.698
4	(n)	5
	Means	34.36
	Sdevs	0.305

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 12 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #		BWTP (°C)
Control	(n)	5
	Means	35.46
	Sdevs	0.820
2	(n)	5
	Means	35.96
	Sdevs	0.351
3	(n)	5
	Means	35.96
	Sdevs	0.594
4	(n)	5
	Means	35.84
	Sdevs	0.261

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 13 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #		BWTP (°C)
Control	(n)	5
	Means	34.28
	Sdevs	1.038
2	(n)	5
	Means	34.94
	Sdevs	0.792
3	(n)	5
	Means	35.02
	Sdevs	0.589
4	(n)	5
	Means	35.32
	Sdevs	0.884

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 13 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #		BWTP (°C)
Control	(n)	5
	Means	35.58
	Sdevs	0.249
2	(n)	5
	Means	36.92+D
	Sdevs	0.634
3	(n)	5
	Means	37.64+D
	Sdevs	0.385
4	(n)	5
	Means	38.02+D
	Sdevs	0.319

+D = Dunnett LSD Test Significant at the 0.01 level

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 14 (Scheduled Animal Room - Session 1)

		Males	
		Body Temperature	
Group #		BWTP (°C)	
Control	(n)	5	
	Means	36.68	
	Sdevs	0.415	
2	(n)	5	
	Means	37.40*D	
	Sdevs	0.339	
3	(n)	5	
	Means	37.62+D	
	Sdevs	0.327	
4	(n)	5	
	Means	37.66+D	
	Sdevs	0.559	

*D = Dunnett LSD Test Significant at the 0.05 level

+D = Dunnett LSD Test Significant at the 0.01 level

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 14 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #		BWTP (°C)
Control	(n)	5
	Means	37.98
	Sdevs	0.277
2	(n)	5
	Means	38.36*D
	Sdevs	0.089
3	(n)	5
	Means	38.42+D
	Sdevs	0.217
4	(n)	5
	Means	38.66+D
	Sdevs	0.167

*D = Dunnett LSD Test Significant at the 0.05 level

+D = Dunnett LSD Test Significant at the 0.01 level

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 15 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #		BWTP (°C)
Control	(n)	5
	Means	35.08
	Sdevs	0.383
2	(n)	5
	Means	35.70*U
	Sdevs	0.374
3	(n)	5
	Means	34.88
	Sdevs	0.779
4	(n)	5
	Means	34.74
	Sdevs	0.114

*U = Mann-Whitney U Test Significant at 0.05 level.

TABLE 2 - Body Temperature Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 15 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #		BWTP (°C)
Control	(n)	5
	Means	37.18
	Sdevs	0.487
2	(n)	5
	Means	36.88
	Sdevs	0.572
3	(n)	5
	Means	37.28
	Sdevs	0.277
4	(n)	5
	Means	37.02
	Sdevs	0.798

TABLE 3 – Body Weight Summary Report

Group	Sex	Dose Level
1*	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1*	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

* Control

Statistical evaluation performed using Pristima version 6.3.2

Bartlett's Test for homogeneity: used to determine homogeneity of the data.
Homogeneous data: ANOVA with Dunnett's test was used to determine statistical significance.

Non-homogeneous data: ANOVA with Kruskal Wallis test was used to determine statistical significance followed by Mann-Whitney U-Test to determine which groups were statistically different.

Statistically significant differences were determined from control group

*D = Dunnett Exact Homogeneous Test Significant: 0.05 level

+D = Dunnett Exact Homogeneous Test Significant: 0.01 level

#D = Dunnett Exact Homogeneous Test Significant: 0.001 level

*U = Mann Whitney U Test Significant: 0.05 level

+U = Mann Whitney U Test Significant: 0.01 level

#U = Mann Whitney U Test Significant: 0.001 level

TABLE 3 - Body Weight Summary Report (g)

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

		Males		
Group #		Dosing Phase		
		Day: 1 Session 1	Day: 8 Session 1	Day: 15 Session 1
Control	(n)	5	5	5
	Means	285.6	307.8	326.4
	Sdevs	7.47	10.06	10.33
2	(n)	5	5	5
	Means	285.6	309.6	330.4
	Sdevs	5.94	9.40	12.82
3	(n)	5	5	5
	Means	283.2	311.0	331.4
	Sdevs	6.53	13.77	13.74
4	(n)	5	5	5
	Means	288.0	313.2	328.8
	Sdevs	7.28	11.28	14.18

TABLE 3 - Body Weight Summary Report (g)

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

		Females		
Group #		Dosing Phase		
		Day: 1 Session 1	Day: 8 Session 1	Day: 15 Session 1
Control	(n)	5	5	5
	Means	214.6	229.6	241.2
	Sdevs	4.28	9.84	10.28
2	(n)	5	5	5
	Means	212.8	225.6	239.2
	Sdevs	4.15	1.82	9.58
3	(n)	5	5	5
	Means	214.2	234.0	248.0
	Sdevs	2.17	9.97	13.15
4	(n)	5	5	5
	Means	211.4	227.2	234.8
	Sdevs	6.02	5.36	5.45

TABLE 4 – Body Weight Gain Per Interval Summary Report

Group	Sex	Dose Level
1*	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1*	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

* Control

Statistical evaluation performed using Prisma version 6.3.2

Bartlett's Test for homogeneity: used to determine homogeneity of the data.
Homogeneous data: ANOVA with Dunnett's test was used to determine statistical significance.

Non-homogeneous data: ANOVA with Kruskal Wallis test was used to determine statistical significance followed by Mann-Whitney U-Test to determine which groups were statistically different.

Statistically significant differences were determined from control group

*D = Dunnett Exact Homogeneous Test Significant: 0.05 level

+D = Dunnett Exact Homogeneous Test Significant: 0.01 level

#D = Dunnett Exact Homogeneous Test Significant: 0.001 level

*U = Mann Whitney U Test Significant: 0.05 level

+U = Mann Whitney U Test Significant: 0.01 level

#U = Mann Whitney U Test Significant: 0.001 level

TABLE 4 - Body Weight Gain Per Interval Summary Report (g)

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley		Males		Subchronic/Intravenous Dosing
Group #		Dosing Phase		
		Day: 8 Session 1	Day: 15 Session 1	
Control	(n)	5	5	
	Means	22.2	18.6	
	Sdevs	4.15	3.65	
2	(n)	5	5	
	Means	24.0	20.8	
	Sdevs	4.95	5.76	
3	(n)	5	5	
	Means	27.8	20.4	
	Sdevs	7.56	3.21	
4	(n)	5	5	
	Means	25.2	15.6	
	Sdevs	4.15	3.29	

TABLE 4 - Body Weight Gain Per Interval Summary Report (g)

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

		Females	
Group #		Dosing Phase	
		Day: 8 Session 1	Day: 15 Session 1
Control	(n)	5	5
	Means	15.0	11.6
	Sdevs	6.52	4.83
2	(n)	5	5
	Means	12.8	13.6
	Sdevs	4.49	10.14
3	(n)	5	5
	Means	19.8	14.0
	Sdevs	9.73	4.85
4	(n)	5	5
	Means	15.8	7.6
	Sdevs	3.03	3.29

TABLE 5 –Food Consumption Summary Report

Group	Sex	Dose Level
1*	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1*	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

* Control

Statistical evaluation performed using Pristima version 6.3.2

Bartlett's Test for homogeneity: used to determine homogeneity of the data.
Homogeneous data: ANOVA with Dunnett's test was used to determine statistical significance.

Non-homogeneous data: ANOVA with Kruskal Wallis test was used to determine statistical significance followed by Mann-Whitney U-Test to determine which groups were statistically different.

Statistically significant differences were determined from control group

*D = Dunnett Exact Homogeneous Test Significant: 0.05 level

+D = Dunnett Exact Homogeneous Test Significant: 0.01 level

#D = Dunnett Exact Homogeneous Test Significant: 0.001 level

*U = Mann Whitney U Test Significant: 0.05 level

+U = Mann Whitney U Test Significant: 0.01 level

#U = Mann Whitney U Test Significant: 0.001 level

TABLE 5 - Food Consumption Summary Report (g)

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

		Males	
Group #		Dosing Phase	
		Day: 8	Day: 15
Control	Cages (N)	5	5
	Means	158.4	157.8
	Sdevs	8.99	8.98
2	Cages (N)	5	5
	Means	163.4	161.2
	Sdevs	8.62	8.98
3	Cages (N)	5	5
	Means	161.0	162.6
	Sdevs	11.64	7.83
4	Cages (N)	5	5
	Means	165.0	159.6
	Sdevs	8.97	7.16

TABLE 5 - Food Consumption Summary Report (g)

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

		Females	
Group #		Dosing Phase Day: 8	Day: 15
Control	Cages (N)	5	5
	Means	132.4	138.2
	Sdevs	11.59	12.66
2	Cages (N)	5	5
	Means	122.4	126.6
	Sdevs	5.13	14.54
3	Cages (N)	5	5
	Means	134.4	138.0
	Sdevs	11.10	13.84
4	Cages (N)	5	5
	Means	135.2	133.2
	Sdevs	7.26	9.93

TABLE 6 – Hematology Summary Report

Group	Sex	Dose Level
1*	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1*	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

* Control

Statistical evaluation performed using Prisma version 6.3.2

Bartlett's Test for homogeneity: used to determine homogeneity of the data.
Homogeneous data: ANOVA with Dunnett's test was used to determine statistical significance.

Non-homogeneous data: ANOVA with Kruskal Wallis test was used to determine statistical significance followed by Mann-Whitney U-Test to determine which groups were statistically different.

Statistically significant differences were determined from control group

*D = Dunnett Exact Homogeneous Test Significant: 0.05 level

+D = Dunnett Exact Homogeneous Test Significant: 0.01 level

#D = Dunnett Exact Homogeneous Test Significant: 0.001 level

*U = Mann Whitney U Test Significant: 0.05 level

+U = Mann Whitney U Test Significant: 0.01 level

#U = Mann Whitney U Test Significant: 0.001 level

TABLE 6 – Hematology Summary Report

Abbr	Description	Units
WBC	White Blood Cells	x10e3/ μ L
RBC	Red Blood Cells	x10e6/ μ L
HGB	Hemoglobin	g/dL
HCT	Hematocrit	%
#LYMPH	Lymphocytes	x10e3/ μ L
%LYMPH	Lymphocytes - %	%
MCV	Mean Corpuscular Volume	fL
MCHC	Mean Corpuscular Hemoglobin Concentration	g/dL
MCH	Mean Corpuscular Hemoglobin	pg
#MONO	Absolute Monocytes	x10e3/ μ L
%MONO	Monocytes - %	%
#NEUT	Absolute Neutrophil	x10e3/ μ L
%NEUT	Neutrophil - %	%
PLT	Platelets	x10e3/ μ L
#BASO	Absolute Basophil	x10e3/ μ L
%BASO	Basophil - %	%
#EOS	Absolute Eosinophil	x10e3/ μ L
%EOS	Eosinophil - %	%
%LUC	Large Unstained Cells - %	%
#LUC	Absolute Large Unstained Cells	x10e3/ μ L
%RETIC	Reticulocytes - %	%
#RETIC	Absolute Reticulocyte	x10e9/L

TABLE 6 - Hematology Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

		Males								
Group #		Complete Blood Count								
		WBC (x10 ³ /μL)	RBC (x10 ⁶ /μL)	HGB (g/dL)	HCT (%)	MCV (fL)	MCH (pg)	MCHC (g/dL)	PLT (10 ³ /μL)	%NEUT (%)
Control	(n)	5	5	5	5	5	5	5	5	5
	Means	10.966	8.694	15.56	51.04	58.68	17.88	30.50	900.2	11.92
	Sdevs	3.8474	0.5764	1.195	4.072	1.619	0.377	0.200	78.47	1.918
2	(n)	5	5	5	5	5	5	5	5	5
	Means	10.246	8.336	15.04	49.38	59.26	18.04	30.40	943.2	12.38
	Sdevs	2.2561	0.1290	0.472	1.277	1.288	0.385	0.406	50.33	0.811
3	(n)	5	5	5	5	5	5	5	5	5
	Means	11.172	8.340	15.14	49.90	59.84	18.14	30.36	912.0	13.26
	Sdevs	2.6185	0.1476	0.207	0.857	1.311	0.251	0.358	76.39	2.806
4	(n)	5	5	5	5	5	5	5	5	5
	Means	10.402	8.496	15.22	50.06	58.96	17.96	30.44	948.4	12.74
	Sdevs	1.7791	0.3441	0.303	1.593	0.688	0.391	0.439	135.61	1.957

TABLE 6 - Hematology Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

		Males								
		Complete Blood Count								
Group #		%LYMP (%)	%MONO (%)	%EOS (%)	%BASO (%)	%LUC (%)	#NEUT (x10 ³ /μL)	#LYMPH (x10 ³ /μL)	#MONO (x10 ³ /μL)	#EOS (x10 ³ /μL)
Control	(n)	5	5	5	5	5	5	5	5	5
	Means	83.00	3.10	0.56	0.64	0.78	1.300	9.090	0.350	0.064
	Sdevs	2.223	0.648	0.207	0.182	0.217	0.5260	3.1106	0.1636	0.0434
2	(n)	5	5	5	5	5	5	5	5	5
	Means	82.02	3.88	0.60	0.46	0.70	1.260	8.404	0.404	0.058
	Sdevs	1.059	0.760	0.255	0.134	0.200	0.2651	1.8230	0.1596	0.0286
3	(n)	5	5	5	5	5	5	5	5	5
	Means	81.14	3.84	0.54	0.44	0.80	1.456	9.082	0.432	0.062
	Sdevs	3.331	1.001	0.089	0.089	0.100	0.3585	2.2155	0.1861	0.0205
4	(n)	5	5	5	5	5	5	5	5	5
	Means	81.12	4.30	0.70	0.44	0.70	1.310	8.458	0.444	0.070
	Sdevs	1.445	0.863	0.187	0.089	0.224	0.1957	1.5661	0.1099	0.0274

TABLE 6 - Hematology Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males

		Complete Blood Count			
Group #		#BASO (x10 ³ /μL)	#LUC (x10 ³ /μL)	%RETIC (%)	#RETIC (x10 ⁹ /L)
Control	(n)	5	5	5	5
	Means	0.076	0.090	1.668	145.60
	Sdevs	0.0416	0.0469	0.2446	28.555
2	(n)	5	5	5	5
	Means	0.048	0.072	1.924	160.24
	Sdevs	0.0192	0.0327	0.2169	16.200
3	(n)	5	5	5	5
	Means	0.054	0.086	1.828	152.66
	Sdevs	0.0241	0.0219	0.2651	23.616
4	(n)	5	5	5	5
	Means	0.046	0.076	1.884	160.08
	Sdevs	0.0167	0.0336	0.1537	14.667

TABLE 6 - Hematology Summary Report

Calvert Labs Inc.
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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females

Group #		Complete Blood Count								
		WBC (x10 ³ /μL)	RBC (x10 ⁶ /μL)	HGB (g/dL)	HCT (%)	MCV (fL)	MCH (pg)	MCHC (g/dL)	PLT (10 ³ /μL)	%NEUT (%)
Control	(n)	4	4	4	4	4	4	4	4	4
	Means	7.033	8.065	14.90	47.55	58.88	18.50	31.43	909.5	16.60
	Sdevs	1.9300	0.4615	0.852	3.783	1.648	0.327	0.690	98.02	2.834
2	(n)	5	5	5	5	5	5	5	5	5
	Means	8.668	8.306	15.38	48.70	58.62	18.48	31.56	865.4	15.08
	Sdevs	1.8640	0.5452	0.844	3.341	0.890	0.342	0.666	197.46	3.497
3	(n)	5	5	5	5	5	5	5	5	5
	Means	9.804	8.240	15.28	48.80	59.20	18.56	31.34	983.0	16.38
	Sdevs	3.2810	0.1872	0.438	1.306	0.977	0.297	0.167	127.54	4.092
4	(n)	5	5	5	5	5	5	5	5	5
	Means	9.918	8.080	15.16	48.32	59.80	18.76	31.42	973.6	14.78
	Sdevs	2.8423	0.1488	0.378	1.094	1.298	0.378	0.517	127.69	5.387

TABLE 6 - Hematology Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

		Females								
		Complete Blood Count								
Group #		%LYMP (%)	%MONO (%)	%EOS (%)	%BASO (%)	%LUC (%)	#NEUT (x10 ³ /μL)	#LYMPH (x10 ³ /μL)	#MONO (x10 ³ /μL)	#EOS (x10 ³ /μL)
Control	(n)	4	4	4	4	4	4	4	4	4
	Means	76.93	3.98	1.15	0.53	0.78	1.145	5.435	0.278	0.080
	Sdevs	2.651	0.932	0.191	0.263	0.222	0.2310	1.6192	0.0964	0.0183
2	(n)	5	5	5	5	5	5	5	5	5
	Means	77.86	4.08	1.24	0.52	1.24	1.346	6.718	0.340	0.110
	Sdevs	3.632	1.354	0.378	0.110	0.397	0.5624	1.3178	0.0860	0.0510
3	(n)	5	5	5	5	5	5	5	5	5
	Means	77.46	4.02	0.86	0.48	0.88	1.592	7.616	0.390	0.080
	Sdevs	4.469	0.976	0.288	0.148	0.311	0.6946	2.7186	0.1306	0.0381
4	(n)	5	5	5	5	5	5	5	5	5
	Means	79.60	3.54	1.04	0.46	0.60	1.462	7.894	0.352	0.112
	Sdevs	5.833	1.205	0.371	0.089	0.187	0.6067	2.3620	0.1460	0.0576

TABLE 6 - Hematology Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

		Females			
		Complete Blood Count			
Group #		#BASO (x10 ³ /μL)	#LUC (x10 ³ /μL)	%RETIC (%)	#RETIC (x10 ⁹ /L)
Control	(n)	4	4	4	4
	Means	0.043	0.058	2.108	170.15
	Sdevs	0.0330	0.0206	0.3683	31.926
2	(n)	5	5	5	5
	Means	0.044	0.110	1.960	161.24
	Sdevs	0.0182	0.0524	0.4829	33.092
3	(n)	5	5	5	5
	Means	0.050	0.080	2.272	186.78
	Sdevs	0.0274	0.0235	0.5837	46.382
4	(n)	5	5	5	5
	Means	0.046	0.062	1.872	151.18
	Sdevs	0.0251	0.0286	0.3421	27.614

TABLE 7 – Red Blood Cell Morphology Summary Report By Session

Group	Sex	Dose Level
1*	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1*	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

* Control

TABLE 7 – Red Blood Cell Morphology Summary Report by Session

Grading	Severity of Condition
1+	Rare – slight
2+	Moderately
3+	Marked

DRAFT

TABLE 7 - Red Blood Cell Morphology Summary Report By Session

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

		Males	
		Dosing Phase	
Group #	Category, Observations	Day of Phase:	16
		Session #:	1
		Scheduled:	5
Control		Number Examined:	5
		Number Normal:	5
2		Number Examined:	5
		Number Normal:	5
3		Number Examined:	5
		Number Normal:	5
4		Number Examined:	5
		Number Normal:	5

TABLE 7 - Red Blood Cell Morphology Summary Report By Session

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

		Females	
		Dosing Phase	
Group #	Category, Observations	Day of Phase:	16
		Session #:	1
		Scheduled:	S
		Number Examined:	4
Control		Number Normal:	4
2		Number Examined:	5
		Number Normal:	5
3		Number Examined:	5
		Number Normal:	5
4		Number Examined:	5
		Number Normal:	5

TABLE 8 – Coagulation Summary Report

Group	Sex	Dose Level
1*	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1*	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

* Control

Statistical evaluation performed using Prisma version 6.3.2

Bartlett's Test for homogeneity: used to determine homogeneity of the data.
Homogeneous data: ANOVA with Dunnett's test was used to determine statistical significance.

Non-homogeneous data: ANOVA with Kruskal Wallis test was used to determine statistical significance followed by Mann-Whitney U-Test to determine which groups were statistically different.

Statistically significant differences were determined from control group

*D = Dunnett Exact Homogeneous Test Significant: 0.05 level

+D = Dunnett Exact Homogeneous Test Significant: 0.01 level

#D = Dunnett Exact Homogeneous Test Significant: 0.001 level

*U = Mann Whitney U Test Significant: 0.05 level

+U = Mann Whitney U Test Significant: 0.01 level

#U = Mann Whitney U Test Significant: 0.001 level

TABLE 8 – Coagulation Summary Report

Abbr	Description	Units
PT	Prothrombin Time	Sec
APTT	Activated Partial Thromboplastin Time	Sec

DRAFT

TABLE 8 - Coagulation Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males

Coagulation (Stago)

Group #		PT (sec)	APTT (sec)
Control	(n)	5	5
	Means	16.10	9.90
	Sdevs	0.592	1.151
2	(n)	5	5
	Means	16.22	8.54
	Sdevs	0.311	0.451
3	(n)	4	4
	Means	16.70	8.75
	Sdevs	0.316	0.656
4	(n)	4	4
	Means	16.05	8.90
	Sdevs	0.387	1.268

TABLE 8 - Coagulation Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females

Coagulation (Stago)

Group #		PT (sec)	APTT (sec)
Control	(n)	4	4
	Means	16.13	8.60
	Sdevs	1.031	0.849
2	(n)	4	4
	Means	16.88	8.25
	Sdevs	1.468	1.328
3	(n)	4	4
	Means	16.40	8.38
	Sdevs	0.529	0.818
4	(n)	5	5
	Means	16.12	7.68
	Sdevs	0.795	0.415

TABLE 9 – Serum Chemistry Summary Report

Group	Sex	Dose Level
1*	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1*	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

* Control

Statistical evaluation performed using Pristima version 6.3.2

Bartlett's Test for homogeneity: used to determine homogeneity of the data.
Homogeneous data: ANOVA with Dunnett's test was used to determine statistical significance.

Non-homogeneous data: ANOVA with Kruskal Wallis test was used to determine statistical significance followed by Mann-Whitney U-Test to determine which groups were statistically different.

Statistically significant differences were determined from control group

*D = Dunnett Exact Homogeneous Test Significant: 0.05 level

+D = Dunnett Exact Homogeneous Test Significant: 0.01 level

#D = Dunnett Exact Homogeneous Test Significant: 0.001 level

*U = Mann Whitney U Test Significant: 0.05 level

+U = Mann Whitney U Test Significant: 0.01 level

#U = Mann Whitney U Test Significant: 0.001 level

TABLE 9 – Serum Chemistry Summary Report

Abbr	Description	Units
AST	Aspartate Aminotransferase	U/L
ALT	Alanine Aminotransferase	U/L
GLU	Glucose	mg/dL
BUN	Blood Urea Nitrogen	mg/dL
CREAT	Creatinine	mg/dL
PHOS	Inorganic Phosphorous	mg/dL
TRIG	Triglycerides	mg/dL
CHOL	Cholesterol	mg/dL
TP	Total Protein	g/dL
NA	Sodium	mEq/L
K	Potassium	mEq/L
CL	Chloride	mEq/L
TBILI	Total Bilirubin	mg/dL
ALP	Alkaline Phosphatase	U/L
CA	Calcium	mg/dL
GLOB	Globulin	g/dL
ALB	Albumin	g/dL
A/G	Albumin Globulin Ratio	None

TABLE 9 - Serum Chemistry Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

		Males								
		Serum Chemistry								
Group #		AST (U/L)	ALT (U/L)	GLU (mg/dL)	BUN (mg/dl)	CREAT (mg/dL)	PHOS (mg/dL)	TRIG (mg/dL)	CHOL (mg/dL)	TP (g/dL)
Control	(n)	5	5	5	5	5	5	5	5	5
	Means	81.8	35.0	108.2	19.0	0.410	9.50	39.6	99.6	6.18
	Sdevs	11.43	2.45	10.99	4.24	0.0616	0.464	18.23	32.97	0.249
2	(n)	5	5	5	5	5	5	5	5	5
	Means	84.2	37.0	111.6	17.0	0.386	9.80	46.6	102.2	6.18
	Sdevs	8.87	4.30	14.96	0.71	0.0134	0.447	21.13	8.35	0.164
3	(n)	5	5	5	5	5	5	5	5	5
	Means	80.6	38.0	104.0	17.2	0.378	9.98	28.2	91.2	6.34
	Sdevs	5.18	10.86	11.18	2.39	0.0377	0.545	6.53	10.31	0.288
4	(n)	5	5	5	5	5	5	5	5	5
	Means	87.6	38.4	110.2	19.2	0.394	9.72	31.4	87.2	6.44
	Sdevs	13.43	6.23	7.19	3.27	0.0230	0.622	5.18	7.29	0.134

TABLE 9 - Serum Chemistry Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

		Males								
		Serum Chemistry								
Group #		Na (mEq/L)	K (mEq/L)	Cl (mEq/L)	TBILI (mg/dL)	ALP (U/L)	CA (mg/dL)	GLOB (g/dL)	ALB (g/dL)	A/G
Control	(n)	5	5	5	5	5	5	5	5	5
	Means	144.6	6.12	101.6	0.090	120.2	10.410	2.86	3.32	1.16
	Sdevs	1.14	0.646	1.52	0.0100	17.75	0.1775	0.219	0.148	0.114
2	(n)	5	5	5	5	5	5	5	5	5
	Means	144.4	6.28	101.8	0.096	133.0	10.260	2.84	3.34	1.18
	Sdevs	0.55	0.148	0.84	0.0114	10.44	0.2164	0.089	0.114	0.045
3	(n)	5	5	5	5	5	5	5	5	5
	Means	145.2	6.74	101.8	0.098	125.0	10.312	3.04	3.30	1.10
	Sdevs	0.84	0.508	1.30	0.0311	13.51	0.1639	0.152	0.141	0.000
4	(n)	5	5	5	5	5	5	5	5	5
	Means	145.6	6.58	103.0	0.092	137.6	10.228	3.12*D	3.32	1.08
	Sdevs	0.55	0.606	1.41	0.0045	28.19	0.2607	0.130	0.084	0.045

*D = Dunnett LSD Test Significant at the 0.05 level

TABLE 9 - Serum Chemistry Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

		Females								
		Serum Chemistry								
Group #		AST (U/L)	ALT (U/L)	GLU (mg/dL)	BUN (mg/dl)	CREAT (mg/dL)	PHOS (mg/dL)	TRIG (mg/dL)	CHOL (mg/dL)	TP (g/dL)
Control	(n)	5	5	5	5	5	5	5	5	5
	Means	83.8	28.4	105.0	16.8	0.442	9.26	34.4	85.2	6.48
	Sdevs	27.73	7.16	8.86	1.92	0.0192	0.740	6.66	10.11	0.476
2	(n)	5	5	5	5	5	5	5	5	5
	Means	85.8	36.8	95.6	16.8	0.448	8.90	32.4	94.2	6.46
	Sdevs	8.87	6.26	7.20	2.59	0.0303	0.436	4.83	10.18	0.371
3	(n)	5	5	5	5	5	5	5	5	5
	Means	86.4	33.0	103.2	16.6	0.460	9.10	27.4	86.4	6.52
	Sdevs	10.99	3.39	13.74	1.67	0.0255	0.663	4.16	7.54	0.286
4	(n)	5	5	5	5	5	5	5	5	5
	Means	92.0	35.2	101.0	19.0	0.456	8.88	30.0	88.0	6.74
	Sdevs	7.25	6.61	7.04	1.00	0.0451	0.179	6.16	3.81	0.288

TABLE 9 - Serum Chemistry Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

		Females								
		Serum Chemistry								
Group #		Na (mEq/L)	K (mEq/L)	Cl (mEq/L)	TBILI (mg/dL)	ALP (U/L)	CA (mg/dL)	GLOB (g/dL)	ALB (g/dL)	A/G
Control	(n)	5	5	5	5	5	5	5	5	5
	Means	144.2	6.50	102.8	0.120	96.4	10.672	2.86	3.62	1.28
	Sdevs	0.84	0.374	1.10	0.0200	18.17	0.2572	0.321	0.179	0.084
2	(n)	5	5	5	5	5	5	5	5	5
	Means	143.8	6.26	101.6	0.130	101.4	10.576	2.90	3.56	1.24
	Sdevs	1.64	0.439	1.82	0.0122	33.22	0.3609	0.235	0.152	0.055
3	(n)	5	5	5	5	5	5	5	5	5
	Means	144.4	6.58	102.4	0.124	102.2	10.506	2.96	3.56	1.22
	Sdevs	1.14	0.327	1.52	0.0261	20.83	0.4535	0.182	0.114	0.045
4	(n)	5	5	5	5	5	5	5	5	5
	Means	145.4	6.42	103.2	0.138	94.4	10.354	3.14	3.60	1.18
	Sdevs	1.52	0.432	0.84	0.0259	12.42	0.2080	0.195	0.122	0.045

TABLE 10 – Quantitative Urinalysis Summary Report

Group	Sex	Dose Level
1*	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1*	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

* Control

Statistical evaluation performed using Pristima version 6.3.2

Bartlett's Test for homogeneity: used to determine homogeneity of the data.
Homogeneous data: ANOVA with Dunnett's test was used to determine statistical significance.

Non-homogeneous data: ANOVA with Kruskal Wallis test was used to determine statistical significance followed by Mann-Whitney U-Test to determine which groups were statistically different.

Statistically significant differences were determined from control group

*D = Dunnett Exact Homogeneous Test Significant: 0.05 level

+D = Dunnett Exact Homogeneous Test Significant: 0.01 level

#D = Dunnett Exact Homogeneous Test Significant: 0.001 level

*U = Mann Whitney U Test Significant: 0.05 level

+U = Mann Whitney U Test Significant: 0.01 level

#U = Mann Whitney U Test Significant: 0.001 level

TABLE 10 – Quantitative Urinalysis Summary Report

Abbr	Description
SPGR	Specific Gravity
pH	pH

DRAFT

TABLE 10 - Quantitative Urinalysis Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males

Urine - chemistry

Group #		pH	SPGR
Control	(n)	5	5
	Means	6.40	1.0250
	Sdevs	0.548	0.00570
2	(n)	5	5
	Means	6.20	1.0264
	Sdevs	0.447	0.00627
3	(n)	5	5
	Means	6.40	1.0226
	Sdevs	0.548	0.00702
4	(n)	5	5
	Means	6.00	1.0274
	Sdevs	0.000	0.00832

TABLE 10 - Quantitative Urinalysis Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

		Females	
		Urine - chemistry	
Group #		pH	SPGR
Control	(n)	5	5
	Means	6.60	1.0180
	Sdevs	0.548	0.00394
2	(n)	5	5
	Means	6.20	1.0234
	Sdevs	0.447	0.00808
3	(n)	5	5
	Means	6.60	1.0156
	Sdevs	0.548	0.00577
4	(n)	5	5
	Means	6.20	1.0196
	Sdevs	0.837	0.01501

TABLE 11 – Qualitative Urinalysis Report with Individual Values

Group	Sex	Dose Level
1*	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1*	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

* Control

TABLE 11 – Qualitative Urinalysis Report with Individual Values

STANDARD KEY FOR URINALYSIS		
<u>BLOOD (BLD)</u> Neg (Negative) / 1+ / 2+ / 3+	<u>BILIRUBIN (BIL)</u> Neg (Negative) / 1+ / 2+ <u>ICTOTEST (UIC)</u> Neg (Negative) / Pos (Positive)	<u>UROBILINOGEN (URO)</u> Neg (Negative) / 1+ / 2+
<u>KETONE (UKET)</u> Neg (Negative) / 1+ / 2+ / 3+	<u>GLUCOSE (UGLU)</u> Neg (Negative) / 1+ / 2+ / 3+	<u>PROTEIN (UPRO)</u> Neg (Negative) / 1+ / 2+ / 3+
<u>NITRITE (NIT)</u> Positive or Negative <u>LEUKOCYTES (LEU)</u> Neg (Negative) / TR (Trace) / 1+ / 2+ / 3+	<u>COLOR (UCOL)</u> A = Amber LA = Light Amber B = Brown LB = Light Brown BL = Blue LR = Light Red C = Colorless O = Orange DA = Dark Amber PY = Light Yellow DB = Dark Brown R = Red DR = Dark Red S = Straw DY = Dark Yellow Y = Yellow G = Green	<u>CLARITY (UCLA)</u> C = Clear CL = Cloudy H = Hazy T = Turbid
<u>MICROSCOPIC FINDINGS</u> Urine – Red Blood Cells (URBC): Number of red blood cells observed per high power field Urine – White Blood Cells (UWBC): Number of white blood cells observed per high power field		<u>BACTERIA (BACT):</u> Observations per high power field <u>Observations:</u> None seen / TR (Trace) / 1+ / 2+ / 3+
<u>YEAST:</u> Observations per high power field Budding Yeast (BYST) Hyphae Yeast (HYST) <u>Observations:</u> None seen / TR (Trace) / 1+ / 2+ / 3+	<u>EPITHELIAL CELLS:</u> Number of cells observed per high power field Squamous Epithelial Cells (SQEP) Transitional Epithelial Cells (TREP) Renal Epithelial Cells (REEP) <u>FAT (FAT):</u> Number of cells observed per high power field	<u>MUCOUS (MUCS):</u> Observations per low power field <u>Observations:</u> None seen / TR (Trace) / 1+ / 2+ / 3+
<u>FECAL COUNT (RBCC):</u> Observations per high power field <u>Observations:</u> None seen / TR (Trace) / 1+ / 2+ / 3+ <u>SPERM (SPRM):</u> Observations per high power field <u>Observations:</u> None seen / TR (Trace) / 1+ / 2+ / 3+	<u>CASTS:</u> Number of casts per low power field Hyaline Cast (HYAL) Epithelial Cell Cast (EPIC) White Blood Cell Cast (WBCT) Red Blood Cell Cast (RBCT) Fine granular Cast (GRAN) Cellular Cast (CELL) Broad Cast (BROAD) Fatty Cast (FATC) Waxy Cast (WAXY)	<u>CRYSTALS:</u> Observations per high power field Triple Phosphate Crystal (TPO4) Calcium Oxalate Crystal (CAOX) Calcium Phosphate Crystal (CAPH) Calcium Carbonate Crystal (CACB) Uric Acid Crystal (URIC) Bilirubin Crystal (BILC) Cholesterol Crystal (CYST) Tyrosine Crystal (TYRO) Amorphous Crystal (AMOR) <u>Observations:</u> None seen / TR (Trace) / 1+ / 2+ / 3+
<u>OTHER NOTATIONS/OBSERVATIONS</u> [none] = Test not run		

TABLE 11 - Qualitative Urinalysis Report with Individual Values

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males

Urine - chemistry

Group #	Animal #	BLD	BIL	URO	UKET	UGLU	UPRO	NIT	LEU	UCOL
Control	2701	Neg	Neg	Neg	1+	Neg	2+	Negative	3+	Y
	2702	Neg	Neg	Neg	1+	Neg	3+	Negative	3+	Y
	2703	Neg	Neg	Neg	1+	Neg	Neg	Negative	TR	PY
	2704	Neg	Neg	Neg	1+	Neg	1+	Negative	1+	Y
	2705	Neg	Neg	Neg	1+	Neg	1+	Negative	1+	Y
2	2711	Neg	Neg	Neg	1+	Neg	2+	Negative	3+	Y
	2712	Neg	Neg	Neg	Neg	Neg	Neg	Negative	TR	Y
	2713	Neg	Neg	Neg	1+	Neg	1+	Negative	1+	Y
	2714	Neg	Neg	Neg	1+	Neg	Neg	Positive	TR	Y
	2715	Neg	Neg	Neg	1+	Neg	1+	Negative	1+	DY
3	2721	Neg	Neg	Neg	1+	Neg	Neg	Negative	TR	PY
	2722	Neg	Neg	Neg	1+	Neg	1+	Negative	1+	LA
	2723	Neg	Neg	Neg	1+	Neg	1+	Negative	2+	Y
	2724	Neg	Neg	Neg	1+	Neg	Neg	Negative	TR	Y
	2725	2+	Neg	Neg	Neg	Neg	Neg	Negative	TR	PY
4	2731	Neg	Neg	Neg	1+	Neg	2+	Negative	2+	Y
	2732	Neg	Neg	Neg	1+	Neg	2+	Negative	2+	PY

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males

Urine - chemistry

Group #	Animal #	BLD	BIL	URO	UKET	UGLU	UPRO	NIT	LEU	UCOL
4	2733	Neg	Neg	Neg	Neg	Neg	1+	Negative	1+	Y
	2734	Neg	Neg	Neg	1+	Neg	Neg	Negative	TR	Y
	2735	Neg	Neg	Neg	1+	Neg	2+	Negative	1+	DY

TABLE 11 - Qualitative Urinalysis Report with Individual Values

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StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males

Urine - chemistry

Group #	Animal #	UCLA
Control	2701	Hazy
	2702	Cloudy
	2703	Hazy
	2704	Hazy
	2705	Hazy
2	2711	Hazy
	2712	Hazy
	2713	Cloudy
	2714	Hazy
	2715	Cloudy
3	2721	Hazy
	2722	Cloudy
	2723	Hazy
	2724	Hazy
	2725	Hazy
4	2731	Cloudy
	2732	Hazy

TABLE 11 - Qualitative Urinalysis Report with Individual Values

Calvert Labs Inc.
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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males

Urine - chemistry

Group #	Animal #	UCLA
4	2733	Cloudy
	2734	Hazy
	2735	Cloudy

TABLE 11 - Qualitative Urinalysis Report with Individual Values

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

Study Title: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males

Group #	Animal #	URBC	UWBC	BACT	Urine - microscopic		SQEP	TREP	REEP	MUCS
					BYST	HYST				
Control	2701	NONE	<1	NONE	NONE	NONE	NONE	NONE	NONE	TR
	2702	NONE	1	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2703	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	TR
	2704	NONE	<1	NONE	NONE	NONE	<1	NONE	NONE	NONE
	2705	NONE	<1	NONE	NONE	NONE	<1	NONE	NONE	TR
2	2711	NONE	<1	NONE	NONE	NONE	<1	NONE	NONE	TR
	2712	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	TR
	2713	NONE	<1	NONE	NONE	NONE	NONE	NONE	NONE	TR
	2714	NONE	<1	TR	NONE	NONE	<1	NONE	NONE	TR
	2715	NONE	<1	NONE	NONE	NONE	NONE	NONE	NONE	NONE
3	2721	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	TR
	2722	NONE	<1	NONE	NONE	NONE	<1	NONE	NONE	NONE
	2723	NONE	1	NONE	NONE	NONE	NONE	NONE	NONE	TR
	2724	NONE	<1	NONE	NONE	NONE	NONE	NONE	NONE	TR
	2725	<1	NONE	NONE	NONE	NONE	NONE	NONE	NONE	TR
4	2731	NONE	<1	NONE	NONE	NONE	<1	NONE	NONE	TR
	2732	NONE	<1	NONE	NONE	NONE	NONE	NONE	NONE	TR

TABLE 11 - Qualitative Urinalysis Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males										
Group #	Animal #	URBC	UWBC	BACT	Urine - microscopic		SQEP	TREP	REEP	MUCS
					BYST	HYST				
4	2733	NONE	<1	NONE	NONE	NONE	<1	NONE	NONE	TR
	2734	NONE	<1	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2735	NONE	<1	NONE	NONE	NONE	1	NONE	NONE	NONE

TABLE 11 - Qualitative Urinalysis Report with Individual Values

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males										
Group #	Animal #	RBCC	SPRM	HYAL	Urine - microscopic EPIC	WBCT	RBCT	GRAN	CELL	BROAD
Control	2701	NONE	TR	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2702	NONE	1+	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2703	NONE	TR	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2704	NONE	TR	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2705	NONE	TR	NONE	NONE	NONE	NONE	NONE	NONE	NONE
2	2711	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2712	NONE	TR	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2713	NONE	2+	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2714	NONE	1+	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2715	NONE	2+	NONE	NONE	NONE	NONE	NONE	NONE	NONE
3	2721	NONE	1+	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2722	NONE	TR	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2723	NONE	TR	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2724	NONE	1+	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2725	NONE	TR	NONE	NONE	NONE	NONE	NONE	NONE	NONE
4	2731	NONE	TR	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2732	NONE	TR	NONE	NONE	NONE	NONE	NONE	NONE	NONE

TABLE 11 - Qualitative Urinalysis Report with Individual Values

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males										
Group #	Animal #	RBCC	SPRM	HYAL	Urine - microscopic EPIC	WBCT	RBCT	GRAN	CELL	BROAD
4	2733	NONE	2+	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2734	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2735	NONE	2+	NONE	NONE	NONE	NONE	NONE	NONE	NONE

TABLE 11 - Qualitative Urinalysis Report with Individual Values

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

Study Title: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males

Group #	Animal #	FATC	WAXY	TPO4	Urine - microscopic CAOX	CAPH	CACB	URIC	BILC	CYST
Control	2701	NONE	NONE	TR	NONE	NONE	NONE	NONE	NONE	NONE
	2702	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2703	NONE	NONE	TR	NONE	NONE	NONE	NONE	NONE	NONE
	2704	NONE	NONE	TR	NONE	NONE	NONE	NONE	NONE	NONE
	2705	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
2	2711	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2712	NONE	NONE	TR	NONE	NONE	NONE	NONE	NONE	NONE
	2713	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2714	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2715	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
3	2721	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2722	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2723	NONE	NONE	TR	NONE	NONE	NONE	NONE	NONE	NONE
	2724	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2725	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
4	2731	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2732	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE

TABLE 11 - Qualitative Urinalysis Report with Individual Values

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Study: 0437RL35.002

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males

Group #	Animal #	FATC	WAXY	TPO4	Urine - microscopic		CACB	URIC	BILC	CYST
					CAOX	CAPH				
4	2733	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2734	NONE	NONE	TR	NONE	NONE	NONE	NONE	NONE	NONE
	2735	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE

TABLE 11 - Qualitative Urinalysis Report with Individual Values

Calvert Labs Inc.
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StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males			
Urine - microscopic			
Group #	Animal #	TYRO	AMOR
Control	2701	NONE	NONE
	2702	NONE	NONE
	2703	NONE	NONE
	2704	NONE	NONE
	2705	NONE	NONE
2	2711	NONE	NONE
	2712	NONE	NONE
	2713	NONE	NONE
	2714	NONE	NONE
	2715	NONE	NONE
3	2721	NONE	NONE
	2722	NONE	NONE
	2723	NONE	NONE
	2724	NONE	NONE
	2725	NONE	NONE
4	2731	NONE	NONE
	2732	NONE	NONE

TABLE 11 - Qualitative Urinalysis Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males

Urine - microscopic

Group #	Animal #	TYRO	AMOR
4	2733	NONE	NONE
	2734	NONE	NONE
	2735	NONE	NONE

TABLE 11 - Qualitative Urinalysis Report with Individual Values

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females										
Urine - chemistry										
Group #	Animal #	BLD	BIL	URO	UKET	UGLU	UPRO	NIT	LEU	UCOL
Control	2706	Neg	Neg	Neg	Neg	Neg	Neg	Negative	Neg	Y
	2707	Neg	Neg	Neg	Neg	Neg	Neg	Negative	Neg	PY
	2708	Neg	Neg	Neg	Neg	Neg	Neg	Negative	Neg	PY
	2709	Neg	Neg	Neg	Neg	Neg	1+	Negative	1+	Y
	2710	Neg	Neg	Neg	Neg	Neg	Neg	Positive	Neg	Y
2	2716	Neg	Neg	Neg	Neg	Neg	Neg	Negative	Neg	Y
	2717	Neg	Neg	Neg	Neg	Neg	Neg	Negative	Neg	PY
	2718	Neg	Neg	Neg	Neg	Neg	Neg	Negative	Neg	Y
	2719	Neg	Neg	Neg	1+	Neg	Neg	Negative	Neg	Y
	2720	Neg	Neg	Neg	Neg	Neg	Neg	Negative	Neg	Y
3	2726	2+	Neg	Neg	Neg	Neg	Neg	Positive	TR	PY
	2727	Neg	Neg	Neg	Neg	Neg	Neg	Negative	Neg	PY
	2728	Neg	Neg	Neg	Neg	Neg	1+	Positive	TR	Y
	2729	Neg	Neg	Neg	Neg	Neg	Neg	Negative	Neg	PY
	2730	Neg	Neg	Neg	Neg	Neg	Neg	Positive	Neg	Y
4	2736	Neg	Neg	Neg	1+	Neg	2+	Negative	Neg	A
	2737	Neg	Neg	Neg	Neg	Neg	Neg	Positive	Neg	PY

TABLE 11 - Qualitative Urinalysis Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females

Urine - chemistry

Group #	Animal #	BLD	BIL	URO	UKET	UGLU	UPRO	NIT	LEU	UCOL
4	2738	1+	Neg	Neg	Neg	Neg	Neg	Negative	Neg	PY
	2739	Neg	Neg	Neg	Neg	Neg	Neg	Negative	TR	PY
	2740	Neg	Neg	Neg	Neg	Neg	Neg	Positive	Neg	PY

TABLE 11 - Qualitative Urinalysis Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females

Urine - chemistry

Group #	Animal #	UCLA
Control	2706	Hazy
	2707	Hazy
	2708	Hazy
	2709	Hazy
	2710	Hazy
2	2716	Cloudy
	2717	Hazy
	2718	Hazy
	2719	Hazy
	2720	Hazy
3	2726	Hazy
	2727	Clear
	2728	Hazy
	2729	Hazy
	2730	Hazy
4	2736	Cloudy
	2737	Hazy

TABLE 11 - Qualitative Urinalysis Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females

Urine - chemistry

Group #	Animal #	UCLA
4	2738	Hazy
	2739	Clear
	2740	Hazy

TABLE 11 - Qualitative Urinalysis Report with Individual Values

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

Study Title: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females

Group #	Animal #	URBC	UWBC	BACT	Urine - microscopic		SQEP	TREP	REEP	MUCS
					BYST	HYST				
Control	2706	NONE	1	NONE	NONE	NONE	<1	NONE	NONE	TR
	2707	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	TR
	2708	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	TR
	2709	NONE	<1	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2710	NONE	<1	NONE	NONE	NONE	NONE	NONE	NONE	TR
2	2716	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	TR
	2717	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	TR
	2718	NONE	NONE	NONE	NONE	NONE	<1	NONE	NONE	TR
	2719	NONE	NONE	NONE	NONE	NONE	<1	NONE	NONE	TR
	2720	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	TR
3	2726	1	NONE	NONE	NONE	NONE	NONE	NONE	NONE	TR
	2727	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2728	NONE	<1	NONE	NONE	NONE	NONE	NONE	NONE	TR
	2729	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	TR
	2730	NONE	NONE	TR	NONE	NONE	NONE	NONE	NONE	TR
4	2736	NONE	NONE	NONE	NONE	NONE	<1	NONE	NONE	TR
	2737	NONE	<1	TR	NONE	NONE	NONE	NONE	NONE	TR

TABLE 11 - Qualitative Urinalysis Report with Individual Values

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females

Group #	Animal #	URBC	UWBC	BACT	Urine - microscopic		SQEP	TREP	REEP	MUCS
					BYST	HYST				
4	2738	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	TR
	2739	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2740	NONE	<1	TR	NONE	NONE	<1	NONE	NONE	TR

TABLE 11 - Qualitative Urinalysis Report with Individual Values

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

Study Title: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females										
Group #	Animal #	RBCC	SPRM	HYAL	Urine - microscopic EPIC	WBCT	RBCT	GRAN	CELL	BROAD
Control	2706	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2707	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2708	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2709	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2710	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
2	2716	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2717	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2718	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2719	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2720	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
3	2726	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2727	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2728	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2729	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2730	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
4	2736	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2737	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE

TABLE 11 - Qualitative Urinalysis Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females										
Group #	Animal #	RBCC	SPRM	HYAL	Urine - microscopic		RBCT	GRAN	CELL	BROAD
					EPIC	WBCT				
4	2738	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2739	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2740	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE

TABLE 11 - Qualitative Urinalysis Report with Individual Values

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females										
Group #	Animal #	FATC	WAXY	TPO4	Urine - microscopic		CACB	URIC	BILC	CYST
					CAOX	CAPH				
Control	2706	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2707	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2708	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2709	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2710	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
2	2716	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2717	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2718	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2719	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2720	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
3	2726	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2727	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2728	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2729	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2730	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
4	2736	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2737	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE

TABLE 11 - Qualitative Urinalysis Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Group #	Animal #	FATC	WAXY	TPO4	Urine - microscopic		CACB	URIC	BILC	CYST
					CAOX	CAPH				
4	2738	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2739	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE	NONE
	2740	NONE	NONE	TR	NONE	NONE	NONE	NONE	NONE	NONE

TABLE 11 - Qualitative Urinalysis Report with Individual Values

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females

Urine - microscopic

Group #	Animal #	TYRO	AMOR
Control	2706	NONE	NONE
	2707	NONE	NONE
	2708	NONE	NONE
	2709	NONE	NONE
	2710	NONE	NONE
2	2716	NONE	NONE
	2717	NONE	NONE
	2718	NONE	NONE
	2719	NONE	NONE
	2720	NONE	NONE
3	2726	NONE	NONE
	2727	NONE	NONE
	2728	NONE	NONE
	2729	NONE	NONE
	2730	NONE	NONE
4	2736	NONE	NONE
	2737	NONE	NONE

TABLE 11 - Qualitative Urinalysis Report with Individual Values

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females

Urine - microscopic

Group #	Animal #	TYRO	AMOR
4	2738	NONE	NONE
	2739	NONE	NONE
	2740	NONE	NONE

TABLE 12 – Summary Report of Gross Observations

Group	Sex	Dose Level
1*	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1*	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

* Control

TABLE 12 - Summary Report of Gross Observations

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

		Males				
		Dosage Group:	Control	2	3	4
		Number of Animals:	5	5	5	5
		Number Examined:	5	5	5	5
		Number Unremarkable:	5	4	5	5
Adrenal glands	Number Examined:	5	5	5	5	5
	Number Unremarkable:	5	5	5	5	5
Aorta	Number Examined:	5	5	5	5	5
	Number Unremarkable:	5	5	5	5	5
Brain	Number Examined:	5	5	5	5	5
	Number Unremarkable:	5	5	5	5	5
Epididymides	Number Examined:	5	5	5	5	5
	Number Unremarkable:	5	5	5	5	5

TABLE 12 - Summary Report of Gross Observations

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley	Subchronic/Intravenous Dosing				
	Males				
	Dosage Group:	Control	2	3	4
	Number of Animals:	5	5	5	5
	Number Examined:	5	5	5	5
	Number Unremarkable:	5	4	5	5
Esophagus	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Eye w/ optic nerve	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Femur w/ articular surface	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Heart	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5

TABLE 12 - Summary Report of Gross Observations

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley	Subchronic/Intravenous Dosing				
	Males				
	Dosage Group:	Control	2	3	4
	Number of Animals:	5	5	5	5
	Number Examined:	5	5	5	5
	Number Unremarkable:	5	4	5	5
Cecum	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Colon	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Rectum	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Duodenum	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5

TABLE 12 - Summary Report of Gross Observations

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley	Subchronic/Intravenous Dosing				
	Males				
	Dosage Group:	Control	2	3	4
	Number of Animals:	5	5	5	5
	Number Examined:	5	5	5	5
	Number Unremarkable:	5	4	5	5
Jejunum	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Ileum	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Kidneys	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Lacrimal gland(s)	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5

TABLE 12 - Summary Report of Gross Observations

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley	Subchronic/Intravenous Dosing				
	Males				
	Dosage Group:	Control	2	3	4
	Number of Animals:	5	5	5	5
	Number Examined:	5	5	5	5
	Number Unremarkable:	5	4	5	5
Larynx	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Liver	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Lung w/ mainstem bronchus	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Lymph node, mandibular	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5

TABLE 12 - Summary Report of Gross Observations

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley	Subchronic/Intravenous Dosing				
	Males				
	Dosage Group:	Control	2	3	4
	Number of Animals:	5	5	5	5
	Number Examined:	5	5	5	5
	Number Unremarkable:	5	4	5	5
Lymph node, mesenteric	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Mammary gland	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Nerve - sciatic	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Pancreas	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5

TABLE 12 - Summary Report of Gross Observations

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

	Males				
	Dosage Group:	Control	2	3	4
	Number of Animals:	5	5	5	5
	Number Examined:	5	5	5	5
	Number Unremarkable:	5	4	5	5
Pituitary gland	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Prostate gland	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Salivary glands	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Seminal vesicles	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5

TABLE 12 - Summary Report of Gross Observations

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley	Subchronic/Intravenous Dosing				
	Males				
	Dosage Group:	Control	2	3	4
	Number of Animals:	5	5	5	5
	Number Examined:	5	5	5	5
	Number Unremarkable:	5	4	5	5
Skeletal muscle (thigh)	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Skin	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Spinal cord, cervical	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Spinal cord, lumbar	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5

TABLE 12 - Summary Report of Gross Observations

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

	Males				
	Dosage Group:	Control	2	3	4
	Number of Animals:	5	5	5	5
	Number Examined:	5	5	5	5
	Number Unremarkable:	5	4	5	5
Spinal cord, midthoracic	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Sternum w/ bone marrow	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Stomach	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Testes	Number Examined:	5	5	5	5
	Number Unremarkable:	5	4	5	5
Decreased size, present, bilaterally, small		0	1	0	0

TABLE 12 - Summary Report of Gross Observations

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

Study Title: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

		Males				
		Dosage Group:	Control	2	3	4
		Number of Animals:	5	5	5	5
		Number Examined:	5	5	5	5
		Number Unremarkable:	5	4	5	5
Thymus		Number Examined:	5	5	5	5
		Number Unremarkable:	5	5	5	5
Thyroids/parathyroids		Number Examined:	5	5	5	5
		Number Unremarkable:	5	5	5	5
Tongue		Number Examined:	5	5	5	5
		Number Unremarkable:	5	5	5	5
Trachea		Number Examined:	5	5	5	5
		Number Unremarkable:	5	5	5	5

TABLE 12 - Summary Report of Gross Observations

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley	Subchronic/Intravenous Dosing				
	Males				
	Dosage Group:	Control	2	3	4
	Number of Animals:	5	5	5	5
	Number Examined:	5	5	5	5
	Number Unremarkable:	5	4	5	5
Urinary bladder	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Injection site(s)	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Abnormalities (gross)	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5

TABLE 12 - Summary Report of Gross Observations

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley	Subchronic/Intravenous Dosing				
	Females				
	Dosage Group:	Control	2	3	4
	Number of Animals:	5	5	5	5
	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Adrenal glands	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Aorta	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Brain	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Cervix	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5

TABLE 12 - Summary Report of Gross Observations

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Females					
Dosage Group:		Control	2	3	4
Number of Animals:		5	5	5	5
Number Examined:		5	5	5	5
Number Unremarkable:		5	5	5	5
Esophagus	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Eye w/ optic nerve	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Femur w/ articular surface	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Heart	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5

TABLE 12 - Summary Report of Gross Observations

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley	Subchronic/Intravenous Dosing				
	Females				
	Dosage Group:	Control	2	3	4
	Number of Animals:	5	5	5	5
	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Cecum	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Colon	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Rectum	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Duodenum	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5

TABLE 12 - Summary Report of Gross Observations

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

Study Title: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

		Females				
		Dosage Group:	Control	2	3	4
		Number of Animals:	5	5	5	5
		Number Examined:	5	5	5	5
		Number Unremarkable:	5	5	5	5
Jejunum		Number Examined:	5	5	5	5
		Number Unremarkable:	5	5	5	5
Ileum		Number Examined:	5	5	5	5
		Number Unremarkable:	5	5	5	5
Kidneys		Number Examined:	5	5	5	5
		Number Unremarkable:	5	5	5	5
Lacrimal gland(s)		Number Examined:	5	5	5	5
		Number Unremarkable:	5	5	5	5

TABLE 12 - Summary Report of Gross Observations

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley	Subchronic/Intravenous Dosing				
	Females				
	Dosage Group:	Control	2	3	4
	Number of Animals:	5	5	5	5
	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Larynx	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Liver	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Lung w/ mainstem bronchus	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Lymph node, mandibular	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5

TABLE 12 - Summary Report of Gross Observations

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Females					
Dosage Group:		Control	2	3	4
Number of Animals:		5	5	5	5
Number Examined:		5	5	5	5
Number Unremarkable:		5	5	5	5
Lymph node, mesenteric	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Mammary gland	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Ovaries	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Nerve - sciatic	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5

TABLE 12 - Summary Report of Gross Observations

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley	Subchronic/Intravenous Dosing				
	Females				
	Dosage Group:	Control	2	3	4
	Number of Animals:	5	5	5	5
	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Pancreas	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Pituitary gland	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Salivary glands	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Skeletal muscle (thigh)	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5

TABLE 12 - Summary Report of Gross Observations

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley	Subchronic/Intravenous Dosing				
	Females				
	Dosage Group:	Control	2	3	4
	Number of Animals:	5	5	5	5
	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Skin	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Spinal cord, cervical	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Spinal cord, lumbar	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Spinal cord, midthoracic	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5

TABLE 12 - Summary Report of Gross Observations

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

Study Title: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

	Females				
	Dosage Group:	Control	2	3	4
	Number of Animals:	5	5	5	5
	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Sternum w/ bone marrow	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Stomach	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Thymus	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Thyroids/parathyroids	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5

TABLE 12 - Summary Report of Gross Observations

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

	Females				
	Dosage Group:	Control	2	3	4
	Number of Animals:	5	5	5	5
	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Tongue	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Trachea	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Urinary bladder	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Uterus	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5

TABLE 12 - Summary Report of Gross Observations

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley	Subchronic/Intravenous Dosing				
	Females				
	Dosage Group:	Control	2	3	4
	Number of Animals:	5	5	5	5
	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Vagina	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Injection site(s)	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5
Abnormalities (gross)	Number Examined:	5	5	5	5
	Number Unremarkable:	5	5	5	5

TABLE 13 – Organ Weight Summary Report

Group	Sex	Dose Level
1*	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1*	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

* Control

Statistical evaluation performed using Pristima version 6.3.2

Bartlett's Test for homogeneity: used to determine homogeneity of the data.
Homogeneous data: ANOVA with Dunnett's test was used to determine statistical significance.

Non-homogeneous data: ANOVA with Kruskal Wallis test was used to determine statistical significance followed by Mann-Whitney U-Test to determine which groups were statistically different.

Statistically significant differences were determined from control group

*D = Dunnett Exact Homogeneous Test Significant: 0.05 level

+D = Dunnett Exact Homogeneous Test Significant: 0.01 level

#D = Dunnett Exact Homogeneous Test Significant: 0.001 level

*U = Mann Whitney U Test Significant: 0.05 level

+U = Mann Whitney U Test Significant: 0.01 level

#U = Mann Whitney U Test Significant: 0.001 level

TABLE 13 - Organ Weight Summary Report

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley		Subchronic/Intravenous Dosing					
		Males					
Group #		TBW (g)	(AD) Adrenal glands (g)	(BR) Brain (g)	(HT) Heart (g)	(KI) Kidneys (g)	(LI) Liver (g)
Control	(n)	5	5	5	5	5	5
	Means	303.4	0.0534	1.834	1.256	2.082	8.906
	Adj Means	-	-	-	-	-	-
	Sdevs	11.24	0.00555	0.1232	0.1588	0.1066	0.9715
2	(n)	5	5	5	5	5	5
	Means	309.2	0.0532	1.740	1.212	2.108	9.074
	Adj Means	-	-	-	-	-	-
	Sdevs	11.26	0.00642	0.0624	0.0327	0.0295	0.9097
3	(n)	5	5	5	5	5	5
	Means	309.6	0.0516	1.876	1.226	2.134	8.876
	Adj Means	-	-	-	-	-	-
	Sdevs	12.99	0.01045	0.1119	0.0862	0.1422	1.3279
4	(n)	5	5	5	5	5	5
	Means	307.8	0.0560	1.770	1.166	2.124	9.042
	Adj Means	-	-	-	-	-	-
	Sdevs	12.91	0.00809	0.0158	0.0805	0.0727	0.3925

TABLE 13 - Organ Weight Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

		Males			
Group #		TBW (g)	(SP) Spleen (g)	(TE) Testes (g)	(TDPT) Thyroids/parathyroids
Control	(n)	5	5	5	5
	Means	303.4	0.698	3.5594	0.0182
	Adj Means	-	-	-	-
	Sdevs	11.24	0.0657	0.29364	0.00228
2	(n)	5	5	5	5
	Means	309.2	0.670	3.0354	0.0184
	Adj Means	-	-	-	-
	Sdevs	11.26	0.0755	1.30826	0.00114
3	(n)	5	5	5	5
	Means	309.6	0.686	3.5682	0.0182
	Adj Means	-	-	-	-
	Sdevs	12.99	0.0391	0.26678	0.00335
4	(n)	5	5	5	5
	Means	307.8	0.694	3.8050	0.0176
	Adj Means	-	-	-	-
	Sdevs	12.91	0.0619	0.28171	0.00152

TABLE 13 - Organ Weight Summary Report

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley		Subchronic/Intravenous Dosing					
		Females					
Group #		TBW (g)	(AD) Adrenal glands (g)	(BR) Brain (g)	(HT) Heart (g)	(KI) Kidneys (g)	(LI) Liver (g)
Control	(n)	5	5	5	5	5	5
	Means	220.6	0.0718	1.716	0.928	1.518	6.396
	Adj Means	-	-	-	-	-	-
	Sdevs	8.44	0.00466	0.0691	0.0646	0.0766	0.5243
2	(n)	5	5	5	5	5	5
	Means	223.8	0.0634	1.800	0.868	1.424	5.796
	Adj Means	-	-	-	-	-	-
	Sdevs	9.42	0.00416	0.0745	0.0540	0.0658	0.2602
3	(n)	5	5	5	5	5	5
	Means	228.0	0.0798	1.708	0.984	1.506	6.286
	Adj Means	-	-	-	-	-	-
	Sdevs	9.82	0.01163	0.0390	0.1457	0.0796	0.7711
4	(n)	5	5	5	5	5	5
	Means	217.4	0.0640	1.692	0.882	1.450	5.782
	Adj Means	-	-	-	-	-	-
	Sdevs	3.21	0.00381	0.1158	0.0295	0.0667	0.2487

TABLE 13 - Organ Weight Summary Report

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

		Females			
Group #		TBW (g)	(OV) Ovaries (g)	(SP) Spleen (g)	(TDPT) Thyroids/parathyroids
Control	(n)	5	5	5	5
	Means	220.6	0.1526	0.630	0.0180
	Adj Means	-	-	-	-
	Sdevs	8.44	0.02704	0.1609	0.00255
2	(n)	5	5	5	5
	Means	223.8	0.1412	0.610	0.0180
	Adj Means	-	-	-	-
	Sdevs	9.42	0.02469	0.0587	0.00274
3	(n)	5	5	5	5
	Means	228.0	0.1634	0.654	0.0192
	Adj Means	-	-	-	-
	Sdevs	9.82	0.04898	0.0462	0.00277
4	(n)	5	5	5	5
	Means	217.4	0.1704	0.554	0.0184
	Adj Means	-	-	-	-
	Sdevs	3.21	0.02974	0.0467	0.00297

TABLE 14 – Organ to Terminal Body Weight Ratio Summary Report

Group	Sex	Dose Level
1*	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1*	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

* Control

Statistical evaluation performed using Pristima version 6.3.2

Bartlett's Test for homogeneity: used to determine homogeneity of the data.

Homogeneous data: ANOVA with Dunnett's test was used to determine statistical significance.

Non-homogeneous data: ANOVA with Kruskal Wallis test was used to determine statistical significance followed by Mann-Whitney U-Test to determine which groups were statistically different.

Statistically significant differences were determined from control group

*D = Dunnett Exact Homogeneous Test Significant: 0.05 level

+D = Dunnett Exact Homogeneous Test Significant: 0.01 level

#D = Dunnett Exact Homogeneous Test Significant: 0.001 level

*U = Mann Whitney U Test Significant: 0.05 level

+U = Mann Whitney U Test Significant: 0.01 level

#U = Mann Whitney U Test Significant: 0.001 level

TABLE 14 - Organ to Terminal Body Weight Ratio Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley		Subchronic/Intravenous Dosing					
Group #		TBW (g)	(AD) Adrenal glands	Males	(HT) Heart	(KI) Kidneys	(LI) Liver
				(BR) Brain			
Control	(n)	5	5	5	5	5	5
	Means	303.4	0.0176	0.6048	0.4136	0.6870	2.9338
	Sdevs	11.24	0.00190	0.03957	0.04641	0.04334	0.28043
2	(n)	5	5	5	5	5	5
	Means	309.2	0.0172	0.5633	0.3924	0.6823	2.9293
	Sdevs	11.26	0.00190	0.02747	0.01826	0.02027	0.19706
3	(n)	5	5	5	5	5	5
	Means	309.6	0.0167	0.6059	0.3960	0.6902	2.8574
	Sdevs	12.99	0.00362	0.02323	0.02238	0.05230	0.31480
4	(n)	5	5	5	5	5	5
	Means	307.8	0.0182	0.5757	0.3788	0.6913	2.9378
	Sdevs	12.91	0.00250	0.02078	0.02062	0.04208	0.04786

TABLE 14 - Organ to Terminal Body Weight Ratio Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

		Males			
Group #		TBW (g)	(SP) Spleen	(TE) Testes	(TDPT) Thyroids/parathyroids
Control	(n)	5	5	5	5
	Means	303.4	0.2301	1.1735	0.0060
	Sdevs	11.24	0.02084	0.08998	0.00074
2	(n)	5	5	5	5
	Means	309.2	0.2164	0.9828	0.0060
	Sdevs	11.26	0.01868	0.42067	0.00053
3	(n)	5	5	5	5
	Means	309.6	0.2217	1.1522	0.0059
	Sdevs	12.99	0.01290	0.06578	0.00093
4	(n)	5	5	5	5
	Means	307.8	0.2252	1.2349	0.0057
	Sdevs	12.91	0.01208	0.04080	0.00051

TABLE 14 - Organ to Terminal Body Weight Ratio Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley			Subchronic/Intravenous Dosing				
			Females				
Group #		TBW (g)	(AD) Adrenal glands	(BR) Brain	(HT) Heart	(KI) Kidneys	(LI) Liver
Control	(n)	5	5	5	5	5	5
	Means	220.6	0.0326	0.7780	0.4208	0.6884	2.8983
	Sdevs	8.44	0.00319	0.01606	0.02699	0.02990	0.19044
2	(n)	5	5	5	5	5	5
	Means	223.8	0.0284	0.8053	0.3886	0.6368	2.5900*D
	Sdevs	9.42	0.00239	0.04488	0.03384	0.03216	0.05423
3	(n)	5	5	5	5	5	5
	Means	228.0	0.0349	0.7503	0.4302	0.6604	2.7505
	Sdevs	9.82	0.00425	0.03756	0.04684	0.01312	0.23154
4	(n)	5	5	5	5	5	5
	Means	217.4	0.0294	0.7787	0.4057	0.6670	2.6594
	Sdevs	3.21	0.00179	0.05872	0.01314	0.03022	0.09909

*D = Dunnett LSD Test Significant at the 0.05 level

TABLE 14 - Organ to Terminal Body Weight Ratio Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley		Subchronic/Intravenous Dosing			
		Females			
Group #		TBW (g)	(OV) Ovaries	(SP) Spleen	(TDPT) Thyroids/parathyroids
Control	(n)	5	5	5	5
	Means	220.6	0.0690	0.2843	0.0082
	Sdevs	8.44	0.01094	0.06669	0.00110
2	(n)	5	5	5	5
	Means	223.8	0.0633	0.2725	0.0080
	Sdevs	9.42	0.01207	0.02265	0.00112
3	(n)	5	5	5	5
	Means	228.0	0.0713	0.2869	0.0084
	Sdevs	9.82	0.01959	0.01713	0.00137
4	(n)	5	5	5	5
	Means	217.4	0.0784	0.2547	0.0085
	Sdevs	3.21	0.01353	0.01860	0.00134

TABLE 15 –Organ To Brain Weight Ratio Summary Report

Group	Sex	Dose Level
1*	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1*	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

* Control

Statistical evaluation performed using Pristima version 6.3.2

Bartlett's Test for homogeneity: used to determine homogeneity of the data.

Homogeneous data: ANOVA with Dunnett's test was used to determine statistical significance.

Non-homogeneous data: ANOVA with Kruskal Wallis test was used to determine statistical significance followed by Mann-Whitney U-Test to determine which groups were statistically different.

Statistically significant differences were determined from control group

*D = Dunnett Exact Homogeneous Test Significant: 0.05 level

+D = Dunnett Exact Homogeneous Test Significant: 0.01 level

#D = Dunnett Exact Homogeneous Test Significant: 0.001 level

*U = Mann Whitney U Test Significant: 0.05 level

+U = Mann Whitney U Test Significant: 0.01 level

#U = Mann Whitney U Test Significant: 0.001 level

TABLE 15 - Organ to Brain Weight Ratio Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley			Subchronic/Intravenous Dosing				
			Males				
Group #		TBW (g)	(AD) Adrenal glands	(HT) Heart	(KI) Kidneys	(LI) Liver	(SP) Spleen
Control	(n)	5	5	5	5	5	5
	Means	303.4	2.9359	68.5267	113.9722	486.1787	38.2009
	Sdevs	11.24	0.48436	7.53659	10.02132	47.76023	4.32895
2	(n)	5	5	5	5	5	5
	Means	309.2	3.0524	69.7486	121.2523	522.0631	38.4831
	Sdevs	11.26	0.29940	3.68479	3.93231	56.56936	3.86869
3	(n)	5	5	5	5	5	5
	Means	309.6	2.7603	65.4816	114.0582	471.5236	36.5844
	Sdevs	12.99	0.61584	5.33760	9.91195	46.70756	1.12340
4	(n)	5	5	5	5	5	5
	Means	307.8	3.1639	65.8587	120.0216	510.8105	39.1923
	Sdevs	12.91	0.45495	4.20950	4.67230	20.64802	3.25209

TABLE 15 - Organ to Brain Weight Ratio Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

		Males		
		TBW (g)	(TE) Testes	(TDPT) Thyroids/parathyroids
Control	(n)	5	5	5
	Means	303.4	194.7275	0.9929
	Sdevs	11.24	19.47588	0.10839
2	(n)	5	5	5
	Means	309.2	173.0852	1.0583
	Sdevs	11.26	73.53579	0.07206
3	(n)	5	5	5
	Means	309.6	190.6134	0.9657
	Sdevs	12.99	16.68571	0.12390
4	(n)	5	5	5
	Means	307.8	214.9133	0.9942
	Sdevs	12.91	14.83667	0.08266

TABLE 15 - Organ to Brain Weight Ratio Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley			Subchronic/Intravenous Dosing				
			Females				
Group #		TBW (g)	(AD) Adrenal glands	(HT) Heart	(KI) Kidneys	(LI) Liver	(OV) Ovaries
Control	(n)	5	5	5	5	5	5
	Means	220.6	4.1946	54.1079	88.5368	372.5121	8.8621
	Sdevs	8.44	0.40546	3.56971	4.82152	22.63448	1.30159
2	(n)	5	5	5	5	5	5
	Means	223.8	3.5255	48.2184	79.3009	322.5760	7.8513
	Sdevs	9.42	0.24875	2.10732	6.32403	22.41524	1.38915
3	(n)	5	5	5	5	5	5
	Means	228.0	4.6778	57.6196	88.1953	368.0566	9.5561
	Sdevs	9.82	0.71152	8.33330	4.68172	43.87076	2.82592
4	(n)	5	5	5	5	5	5
	Means	217.4	3.7931	52.4052	86.2455	343.8391	10.1851
	Sdevs	3.21	0.28736	5.24069	10.03861	39.39968	2.39520

TABLE 15 - Organ to Brain Weight Ratio Summary Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

		Females		
		TBW (g)	(SP) Spleen	(TDPT) Thyroids/parathyroids
Group #				
Control	(n)	5	5	5
	Means	220.6	36.4700	1.0470
	Sdevs	8.44	8.02329	0.12115
2	(n)	5	5	5
	Means	223.8	34.0050	1.0026
	Sdevs	9.42	4.44000	0.17201
3	(n)	5	5	5
	Means	228.0	38.2800	1.1258
	Sdevs	9.82	2.36678	0.17481
4	(n)	5	5	5
	Means	217.4	32.9794	1.0880
	Sdevs	3.21	4.86547	0.15783

XII. Appendices

DRAFT

Appendix I—Individual Animal Data

DRAFT

Individual Animal Report of Clinical Signs Each Day Seen

Group	Sex	Dose Level
1	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

Individual Animal Report of Clinical Signs Each Day Seen

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group	Category, Observation	Days Seen
2701	M	1	Appears Normal	Dosing Phase: 1, 8, 15
2702	M	1	Appears Normal	Dosing Phase: 1, 8, 15
2703	M	1	Appears Normal	Dosing Phase: 1, 8, 15
2704	M	1	Appears Normal	Dosing Phase: 1, 8, 15
2705	M	1	Appears Normal	Dosing Phase: 1, 8, 15
2711	M	2	Appears Normal	Dosing Phase: 1, 8, 15
2712	M	2	Appears Normal	Dosing Phase: 1, 8, 15
2713	M	2	Appears Normal	Dosing Phase: 1, 8, 15
2714	M	2	Appears Normal	Dosing Phase: 1, 8, 15
2715	M	2	Appears Normal	Dosing Phase: 1, 8, 15
2721	M	3	Appears Normal	Dosing Phase: 1, 8, 15
2722	M	3	Appears Normal	Dosing Phase: 1, 8, 15
2723	M	3	Appears Normal	Dosing Phase: 1, 8, 15
2724	M	3	Appears Normal	Dosing Phase: 1, 8, 15
2725	M	3	Appears Normal	Dosing Phase: 1, 8, 15
2731	M	4	Appears Normal	Dosing Phase: 1, 8, 15
2732	M	4	Appears Normal	Dosing Phase: 1, 8, 15
2733	M	4	Appears Normal	Dosing Phase: 1, 8, 15
2734	M	4	Appears Normal	Dosing Phase: 1, 8, 15
2735	M	4	Appears Normal	Dosing Phase: 1, 8, 15
2706	F	1	Appears Normal	Dosing Phase: 1, 8, 15
2707	F	1	Appears Normal	Dosing Phase: 1, 8, 15
2708	F	1	Appears Normal	Dosing Phase: 1, 8, 15
2709	F	1	Appears Normal	Dosing Phase: 1, 8, 15
2710	F	1	Appears Normal	Dosing Phase: 1, 8, 15
2716	F	2	Appears Normal	Dosing Phase: 1, 8, 15
2717	F	2	Appears Normal	Dosing Phase: 1, 8, 15

Individual Animal Report of Clinical Signs Each Day Seen

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group	Category, Observation	Days Seen
2718	F	2	Appears Normal	Dosing Phase: 1, 8, 15
2719	F	2	Appears Normal	Dosing Phase: 1, 8, 15
2720	F	2	Appears Normal	Dosing Phase: 1, 8, 15
2726	F	3	Appears Normal	Dosing Phase: 1, 8, 15
2727	F	3	Appears Normal	Dosing Phase: 1, 8, 15
2728	F	3	Appears Normal	Dosing Phase: 1, 8, 15
2729	F	3	Appears Normal	Dosing Phase: 1, 8, 15
2730	F	3	Appears Normal	Dosing Phase: 1, 8, 15
2736	F	4	Appears Normal	Dosing Phase: 1, 8, 15
2737	F	4	Appears Normal	Dosing Phase: 1, 8, 15
2738	F	4	Appears Normal	Dosing Phase: 1, 8, 15
2739	F	4	Appears Normal	Dosing Phase: 1, 8, 15
2740	F	4	Appears Normal	Dosing Phase: 1, 8, 15

Individual Animal Report of Clinical Signs Each Day Seen

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group	Category, Observation	Days Seen
2701	M	1	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2702	M	1	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2703	M	1	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2704	M	1	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2705	M	1	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2711	M	2	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2712	M	2	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2713	M	2	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2714	M	2	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2715	M	2	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2721	M	3	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2722	M	3	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2723	M	3	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2724	M	3	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2725	M	3	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2731	M	4	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2732	M	4	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2733	M	4	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2734	M	4	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2735	M	4	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2706	F	1	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2707	F	1	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2708	F	1	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2709	F	1	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2710	F	1	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2716	F	2	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2717	F	2	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16

Individual Animal Report of Clinical Signs Each Day Seen

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group	Category, Observation	Days Seen
2718	F	2	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2719	F	2	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2720	F	2	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2726	F	3	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2727	F	3	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2728	F	3	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2729	F	3	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2730	F	3	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2736	F	4	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2737	F	4	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2738	F	4	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2739	F	4	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16
2740	F	4	Appears Normal	Dosing Phase: 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 16

Group Survival and Mortality Report

Group	Sex	Dose Level
1	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

Group Survival and Mortality Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley					Subchronic/Intravenous Dosing				
Animal #	Sex	Group	User #	Date/Time Entered	Phase Name	Day of Phase	Date/Time of Death	Approx	
2701	M	1	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y	
Death Status: Terminal Sacrifice					Death Type: Scheduled				
2702	M	1	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y	
Death Status: Terminal Sacrifice					Death Type: Scheduled				
2703	M	1	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y	
Death Status: Terminal Sacrifice					Death Type: Scheduled				
2704	M	1	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y	
Death Status: Terminal Sacrifice					Death Type: Scheduled				
2705	M	1	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y	
Death Status: Terminal Sacrifice					Death Type: Scheduled				
2711	M	2	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y	
Death Status: Terminal Sacrifice					Death Type: Scheduled				
2712	M	2	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y	
Death Status: Terminal Sacrifice					Death Type: Scheduled				
2713	M	2	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y	
Death Status: Terminal Sacrifice					Death Type: Scheduled				
2714	M	2	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y	
Death Status: Terminal Sacrifice					Death Type: Scheduled				
2715	M	2	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y	
Death Status: Terminal Sacrifice					Death Type: Scheduled				
2721	M	3	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y	
Death Status: Terminal Sacrifice					Death Type: Scheduled				

User # 11 = Michelle Zuerblis

Group Survival and Mortality Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley					Subchronic/Intravenous Dosing			
Animal #	Sex	Group	User #	Date/Time Entered	Phase Name	Day of Phase	Date/Time of Death	Approx
2722	M	3	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice					Death Type: Scheduled			
2723	M	3	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice					Death Type: Scheduled			
2724	M	3	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice					Death Type: Scheduled			
2725	M	3	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice					Death Type: Scheduled			
2731	M	4	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice					Death Type: Scheduled			
2732	M	4	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice					Death Type: Scheduled			
2733	M	4	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice					Death Type: Scheduled			
2734	M	4	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice					Death Type: Scheduled			
2735	M	4	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice					Death Type: Scheduled			
2706	F	1	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice					Death Type: Scheduled			
2707	F	1	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice					Death Type: Scheduled			

User # 11 = Michelle Zuerblis

Group Survival and Mortality Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley						Subchronic/Intravenous Dosing		
Animal #	Sex	Group	User #	Date/Time Entered	Phase Name	Day of Phase	Date/Time of Death	Approx
2708	F	1	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice						Death Type: Scheduled		
2709	F	1	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice						Death Type: Scheduled		
2710	F	1	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice						Death Type: Scheduled		
2716	F	2	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice						Death Type: Scheduled		
2717	F	2	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice						Death Type: Scheduled		
2718	F	2	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice						Death Type: Scheduled		
2719	F	2	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice						Death Type: Scheduled		
2720	F	2	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice						Death Type: Scheduled		
2726	F	3	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice						Death Type: Scheduled		
2727	F	3	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice						Death Type: Scheduled		
2728	F	3	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice						Death Type: Scheduled		

User # 11 = Michelle Zuerblis

Group Survival and Mortality Report

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley					Subchronic/Intravenous Dosing			
Animal #	Sex	Group	User #	Date/Time Entered	Phase Name	Day of Phase	Date/Time of Death	Approx
2729	F	3	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice					Death Type: Scheduled			
2730	F	3	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice					Death Type: Scheduled			
2736	F	4	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice					Death Type: Scheduled			
2737	F	4	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice					Death Type: Scheduled			
2738	F	4	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice					Death Type: Scheduled			
2739	F	4	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice					Death Type: Scheduled			
2740	F	4	11	22 Jul 2014 15:56:53	Dosing Phase	16	22 Jul 2014 15:56:42	Y
Death Status: Terminal Sacrifice					Death Type: Scheduled			

User # 11 = Michelle Zuerblis

Body Temperature Summary Report with Individual Values

Group	Sex	Dose Level
1	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Pretest (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal #	BWTP (°C)
1	2701	36.5
	2702	35.4
	2703	36.6
	2704	36.5
	2705	36.3
	(n)	5
	Means	36.26
2	SDevs	0.493
	2711	35.4
	2712	37.2
	2713	37.2
	2714	36.0
	2715	36.3
	(n)	5
3	Means	36.42
	SDevs	0.782
	2721	36.5
	2722	36.5

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Pretest (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal BWTP #	(°C)
3	2723	37.3
	2724	37.1
	2725	35.1
	(n)	5
	Means	36.50
	SDevs	0.860
4	2731	36.7
	2732	36.4
	2733	37.2
	2734	37.2
	2735	36.7
	(n)	5
	Means	36.84
	SDevs	0.351

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Pretest (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal #	BWTP (°C)
1	2706	37.8
	2707	37.1
	2708	36.1
	2709	36.6
	2710	36.8
	(n)	5
	Means	36.88
2	SDevs	0.630
	2716	37.9
	2717	38.0
	2718	36.8
	2719	37.6
	2720	37.6
	(n)	5
3	Means	37.58
	SDevs	0.471
	2726	38.4
	2727	37.8

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Pretest (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal BWTP #	(°C)
3	2728	37.4
	2729	38.3
	2730	37.6
	(n)	5
	Means	37.90
	SDevs	0.436
4	2736	37.4
	2737	37.6
	2738	38.0
	2739	37.4
	2740	37.9
	(n)	5
	Means	37.66
	SDevs	0.279

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 1 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal #	BWTP (°C)
1	2701	35.2
	2702	36.1
	2703	36.7
	2704	36.4
	2705	36.4
	(n)	5
	Means	36.16
	SDevs	0.577
2	2711	36.6
	2712	36.5
	2713	35.5
	2714	35.5
	2715	36.8
	(n)	5
	Means	36.18
	SDevs	0.630
3	2721	37.2
	2722	36.7

Body Temperature Summary Report with Individual Values

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 1 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal #	BWTP (°C)
3	2723	36.6
	2724	36.8
	2725	36.9
	(n)	5
	Means	36.84
	SDevs	0.230
4	2731	36.5
	2732	36.7
	2733	36.8
	2734	37.0
	2735	36.4
	(n)	5
	Means	36.68
	SDevs	0.239

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 1 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal #	BWTP (°C)
1	2706	36.8
	2707	36.8
	2708	36.9
	2709	37.3
	2710	37.0
	(n)	5
	Means	36.96
2	SDevs	0.207
	2716	37.8
	2717	37.9
	2718	37.1
	2719	37.9
	2720	37.1
	(n)	5
3	Means	37.56
	SDevs	0.422
	2726	37.6
	2727	38.0

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 1 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal #	BWTP (°C)
3	2728	37.7
	2729	38.0
	2730	37.1
	(n)	5
	Means	37.68
	SDevs	0.370
4	2736	37.4
	2737	38.2
	2738	37.6
	2739	38.0
	2740	37.7
	(n)	5
	Means	37.79
	SDevs	0.311

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 2 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal #	BWTP (°C)
1	2701	36.6
	2702	36.4
	2703	36.4
	2704	36.2
	2705	36.5
	(n)	5
	Means	36.42
	SDevs	0.148
2	2711	37.0
	2712	37.7
	2713	36.9
	2714	36.4
	2715	37.5
	(n)	5
	Means	37.10
	SDevs	0.515
3	2721	36.8
	2722	36.6

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 2 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal #	BWTP (°C)
3	2723	36.9
	2724	35.8
	2725	36.8
	(n)	5
	Means	36.58
	SDevs	0.449
4	2731	36.9
	2732	37.8
	2733	36.8
	2734	37.3
	2735	37.1
	(n)	5
	Means	37.19
	SDevs	0.408

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

Study Title: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 2 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal #	BWTP (°C)
1	2706	37.0
	2707	38.0
	2708	37.8
	2709	38.3
	2710	37.1
	(n)	5
	Means	37.64
2		SDevs 0.568
	2716	38.0
	2717	38.3
	2718	37.8
	2719	38.4
	2720	38.0
	(n)	5
3		Means 38.10
		SDevs 0.245
	2726	38.8
	2727	38.2

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 2 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal BWTP #	(°C)
3	2728	38.1
	2729	38.5
	2730	38.0
	(n)	5
	Means	38.32
	SDevs	0.327
4	2736	37.1
	2737	38.6
	2738	37.8
	2739	37.6
	2740	37.8
	(n)	5
	Means	37.78
	SDevs	0.540

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 3 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal #	BWTP (°C)
1	2701	34.7
	2702	34.5
	2703	36.6
	2704	36.3
	2705	36.4
	(n)	5
	Means	35.70
	SDevs	1.012
2	2711	36.6
	2712	36.5
	2713	36.6
	2714	36.2
	2715	36.5
	(n)	5
	Means	36.48
	SDevs	0.164
3	2721	37.0
	2722	36.2

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 3 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal #	BWTP (°C)
3	2723	37.3
	2724	36.9
	2725	36.7
	(n)	5
	Means	36.82
	SDevs	0.409
4	2731	37.4
	2732	36.6
	2733	36.7
	2734	37.2
	2735	36.2
	(n)	5
	Means	36.82
	SDevs	0.482

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 3 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal BWTP #	(°C)
1	2706	36.9
	2707	37.2
	2708	37.1
	2709	37.1
	2710	36.7
	(n)	5
	Means	37.00
2	SDevs	0.200
	2716	37.6
	2717	37.8
	2718	37.5
	2719	38.0
	2720	37.7
	(n)	5
3	Means	37.72
	SDevs	0.192
	2726	38.5
	2727	38.2

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 3 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal #	BWTP (°C)
3	2728	37.8
	2729	38.2
	2730	38.5
	(n)	5
	Means	38.24
	SDevs	0.288
4	2736	37.2
	2737	38.4
	2738	37.4
	2739	37.1
	2740	38.3
	(n)	5
	Means	37.68
	SDevs	0.622

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 4 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal #	BWTP (°C)
1	2701	35.8
	2702	36.2
	2703	36.8
	2704	36.7
	2705	36.6
	(n)	5
	Means	36.42
2	SDevs	0.415
	2711	36.5
	2712	37.3
	2713	37.1
	2714	36.2
	2715	37.4
	(n)	5
3	Means	36.90
	SDevs	0.524
	2721	37.2
	2722	36.9

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 4 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal BWTP #	(°C)
3	2723	37.2
	2724	37.1
	2725	37.2
	(n)	5
	Means	37.12
	SDevs	0.130
4	2731	36.7
	2732	37.9
	2733	36.4
	2734	38.0
	2735	36.8
	(n)	5
	Means	37.16
	SDevs	0.737

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 4 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal #	BWTP (°C)
1	2706	37.2
	2707	37.7
	2708	38.4
	2709	38.1
	2710	38.3
	(n)	5
	Means	37.94
	SDevs	0.493
2	2716	38.3
	2717	38.4
	2718	38.2
	2719	38.2
	2720	38.2
	(n)	5
	Means	38.26
	SDevs	0.089
3	2726	38.6
	2727	38.6

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 4 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal #	BWTP (°C)
3	2728	38.4
	2729	38.6
	2730	38.6
	(n)	5
	Means	38.56
	SDevs	0.089
4	2736	37.4
	2737	38.5
	2738	38.5
	2739	37.9
	2740	38.5
	(n)	5
	Means	38.16
	SDevs	0.498

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 5 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal #	BWTP (°C)
1	2701	36.5
	2702	36.7
	2703	36.4
	2704	36.4
	2705	36.4
	(n)	5
	Means	36.48
	SDevs	0.130
2	2711	36.8
	2712	37.0
	2713	37.1
	2714	36.4
	2715	36.8
	(n)	5
	Means	36.82
	SDevs	0.268
3	2721	36.6
	2722	36.3

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 5 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal #	BWTP (°C)
3	2723	36.8
	2724	35.9
	2725	36.8
	(n)	5
	Means	36.48
	SDevs	0.383
4	2731	36.6
	2732	37.8
	2733	36.5
	2734	37.2
	2735	36.6
	(n)	5
	Means	36.94
	SDevs	0.555

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 5 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal #	BWTP (°C)
1	2706	36.8
	2707	37.5
	2708	37.5
	2709	38.1
	2710	37.5
	(n)	5
	Means	37.48
	SDevs	0.460
2	2716	38.3
	2717	38.4
	2718	38.0
	2719	37.9
	2720	38.4
	(n)	5
	Means	38.20
	SDevs	0.235
3	2726	38.4
	2727	38.5

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 5 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal BWTP #	(°C)
3	2728	38.0
	2729	38.2
	2730	38.2
	(n)	5
	Means	38.26
	SDevs	0.195
4	2736	37.0
	2737	38.4
	2738	38.2
	2739	37.9
	2740	38.5
	(n)	5
	Means	38.00
	SDevs	0.604

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 6 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal #	BWTP (°C)
1	2701	36.4
	2702	36.3
	2703	36.4
	2704	36.6
	2705	36.5
	(n)	5
	Means	36.44
	SDevs	0.114
2	2711	36.8
	2712	37.8
	2713	37.0
	2714	36.6
	2715	37.3
	(n)	5
	Means	37.10
	SDevs	0.469
3	2721	37.6
	2722	37.0

Body Temperature Summary Report with Individual Values

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 6 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal BWTP #	(°C)
3	2723	37.7
	2724	37.4
	2725	37.6
	(n)	5
	Means	37.46
	SDevs	0.279
4	2731	37.5
	2732	37.9
	2733	36.5
	2734	38.0
	2735	37.7
	(n)	5
	Means	37.52
	SDevs	0.602

Body Temperature Summary Report with Individual Values

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 6 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal #	BWTP (°C)
1	2706	37.3
	2707	38.1
	2708	37.4
	2709	38.3
	2710	37.8
	(n)	5
	Means	37.78
2	SDevs	0.432
	2716	38.0
	2717	38.7
	2718	38.2
	2719	38.5
	2720	38.2
	(n)	5
3	Means	38.32
	SDevs	0.277
	2726	38.5
	2727	38.6

Body Temperature Summary Report with Individual Values

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 6 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal #	BWTP (°C)
3	2728	38.0
	2729	38.4
	2730	38.4
	(n)	5
	Means	38.38
	SDevs	0.228
4	2736	37.7
	2737	38.7
	2738	38.6
	2739	38.3
	2740	38.5
	(n)	5
	Means	38.36
	SDevs	0.397

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 7 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal #	BWTP (°C)
1	2701	36.4
	2702	36.3
	2703	36.7
	2704	36.6
	2705	36.7
	(n)	5
	Means	36.54
	SDevs	0.182
2	2711	37.4
	2712	37.4
	2713	37.1
	2714	36.9
	2715	37.3
	(n)	5
	Means	37.22
	SDevs	0.217
3	2721	37.2
	2722	36.8

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 7 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal #	BWTP (°C)
3	2723	37.8
	2724	37.6
	2725	37.4
	(n)	5
	Means	37.36
	SDevs	0.385
4	2731	36.9
	2732	37.8
	2733	36.8
	2734	38.0
	2735	37.7
	(n)	5
	Means	37.44
	SDevs	0.550

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 7 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal #	BWTP (°C)
1	2706	37.8
	2707	37.9
	2708	38.0
	2709	37.8
	2710	37.3
	(n)	5
	Means	37.76
2	SDevs	0.270
	2716	38.0
	2717	37.9
	2718	38.1
	2719	38.5
	2720	37.8
	(n)	5
3	Means	38.06
	SDevs	0.270
	2726	38.3
	2727	38.4

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 7 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal #	BWTP (°C)
3	2728	38.1
	2729	38.4
	2730	38.3
	(n)	5
	Means	38.30
	SDevs	0.122
4	2736	37.1
	2737	38.8
	2738	38.6
	2739	38.1
	2740	38.5
	(n)	5
	Means	38.22
	SDevs	0.676

Body Temperature Summary Report with Individual Values

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Pristima® Version 6.3.2 Build 20

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 8 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal #	BWTP (°C)
1	2701	36.4
	2702	36.1
	2703	36.7
	2704	36.8
	2705	36.6
	(n)	5
	Means	36.52
2	SDevs	0.277
	2711	37.1
	2712	36.9
	2713	36.0
	2714	36.5
	2715	36.5
	(n)	5
3	Means	36.60
	SDevs	0.424
	2721	37.0
	2722	36.5

Body Temperature Summary Report with Individual Values

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 8 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal #	BWTP (°C)
3	2723	36.4
	2724	36.1
	2725	36.4
	(n)	5
	Means	36.48
	SDevs	0.327
4	2731	36.5
	2732	37.5
	2733	36.4
	2734	37.0
	2735	36.3
	(n)	5
	Means	36.74
	SDevs	0.503

Body Temperature Summary Report with Individual Values

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Pristima® Version 6.3.2 Build 20

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 8 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal #	BWTP (°C)
1	2706	37.5
	2707	37.3
	2708	37.1
	2709	37.6
	2710	37.9
	(n)	5
	Means	37.48
2	SDevs	0.303
	2716	38.4
	2717	38.4
	2718	38.4
	2719	38.6
	2720	36.8
	(n)	5
3	Means	38.12
	SDevs	0.743
	2726	37.6
	2727	38.7

Body Temperature Summary Report with Individual Values

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 8 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal #	BWTP (°C)
3	2728	37.1
	2729	37.0
	2730	38.7
	(n)	5
	Means	37.82
	SDevs	0.835
4	2736	37.4
	2737	38.8
	2738	37.6
	2739	37.2
	2740	37.3
	(n)	5
	Means	37.66
	SDevs	0.654

Body Temperature Summary Report with Individual Values

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 9 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal #	BWTP (°C)
1	2701	36.3
	2702	36.3
	2703	36.5
	2704	36.4
	2705	36.7
	(n)	5
	Means	36.44
	SDevs	0.167
2	2711	36.8
	2712	37.7
	2713	36.9
	2714	37.5
	2715	36.5
	(n)	5
	Means	37.08
	SDevs	0.502
3	2721	36.3
	2722	37.0

Body Temperature Summary Report with Individual Values

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 9 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal #	BWTP (°C)
3	2723	37.7
	2724	37.8
	2725	36.3
	(n)	5
	Means	37.02
	SDevs	0.726
4	2731	36.5
	2732	37.9
	2733	36.6
	2734	37.8
	2735	37.2
	(n)	5
	Means	37.20
	SDevs	0.652

Body Temperature Summary Report with Individual Values

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 9 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal #	BWTP (°C)
1	2706	37.8
	2707	37.8
	2708	37.9
	2709	38.1
	2710	37.9
	(n)	5
	Means	37.90
2	SDevs	0.122
	2716	38.3
	2717	38.4
	2718	38.4
	2719	38.3
	2720	37.9
	(n)	5
3	Means	38.26
	SDevs	0.207
	2726	38.7
	2727	38.2

Body Temperature Summary Report with Individual Values

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 9 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal #	BWTP (°C)
3	2728	38.5
	2729	38.6
	2730	38.6
	(n)	5
	Means	38.52
	SDevs	0.192
4	2736	36.8
	2737	38.6
	2738	38.6
	2739	37.8
	2740	38.4
	(n)	5
	Means	38.04
	SDevs	0.767

Body Temperature Summary Report with Individual Values

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 10 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal #	BWTP (°C)
1	2701	36.7
	2702	36.5
	2703	36.8
	2704	36.9
	2705	36.5
	(n)	5
	Means	36.68
	SDevs	0.179
2	2711	36.9
	2712	37.9
	2713	37.0
	2714	37.0
	2715	36.8
	(n)	5
	Means	37.12
	SDevs	0.444
3	2721	37.0
	2722	36.5

Body Temperature Summary Report with Individual Values

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 10 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal BWTP #	(°C)
3	2723	37.9
	2724	37.9
	2725	36.3
	(n)	5
	Means	37.12
	SDevs	0.756
4	2731	37.0
	2732	37.9
	2733	36.3
	2734	37.9
	2735	37.9
	(n)	5
	Means	37.40
	SDevs	0.728

Body Temperature Summary Report with Individual Values

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 10 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal #	BWTP (°C)
1	2706	37.6
	2707	38.1
	2708	37.9
	2709	38.4
	2710	38.4
	(n)	5
	Means	38.08
2	SDevs	0.342
	2716	38.2
	2717	38.4
	2718	38.0
	2719	38.2
	2720	38.0
	(n)	5
3	Means	38.16
	SDevs	0.167
	2726	38.5
	2727	38.3

Body Temperature Summary Report with Individual Values

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 10 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal #	BWTP (°C)
3	2728	38.3
	2729	38.7
	2730	38.6
	(n)	5
	Means	38.48
	SDevs	0.179
4	2736	38.4
	2737	38.5
	2738	38.6
	2739	38.7
	2740	38.6
	(n)	5
	Means	38.56
	SDevs	0.114

Body Temperature Summary Report with Individual Values

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 11 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal #	BWTP (°C)
1	2701	36.5
	2702	36.6
	2703	36.5
	2704	36.3
	2705	36.6
	(n)	5
	Means	36.50
	SDevs	0.122
2	2711	37.2
	2712	38.0
	2713	37.0
	2714	36.9
	2715	37.5
	(n)	5
	Means	37.32
	SDevs	0.444
3	2721	37.2
	2722	36.3

Body Temperature Summary Report with Individual Values

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Pristima® Version 6.3.2 Build 20

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 11 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal #	BWTP (°C)
3	2723	37.9
	2724	37.7
	2725	36.6
	(n)	5
	Means	37.14
	SDevs	0.688
4	2731	36.7
	2732	38.1
	2733	36.5
	2734	38.4
	2735	37.0
	(n)	5
	Means	37.34
	SDevs	0.856

Body Temperature Summary Report with Individual Values

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 11 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal #	BWTP (°C)
1	2706	38.1
	2707	37.9
	2708	38.1
	2709	38.5
	2710	38.0
	(n)	5
	Means	38.12
	SDevs	0.228
2	2716	38.1
	2717	38.5
	2718	38.2
	2719	38.0
	2720	38.1
	(n)	5
	Means	38.18
	SDevs	0.192
3	2726	38.5
	2727	38.4

Body Temperature Summary Report with Individual Values

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 11 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal #	BWTP (°C)
3	2728	38.2
	2729	38.7
	2730	38.6
	(n)	5
	Means	38.48
	SDevs	0.192
4	2736	38.2
	2737	38.6
	2738	38.0
	2739	38.3
	2740	38.8
	(n)	5
	Means	38.38
	SDevs	0.319

Body Temperature Summary Report with Individual Values

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 12 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal #	BWTP (°C)
1	2701	33.8
	2702	33.0
	2703	34.3
	2704	34.0
	2705	34.2
	(n)	5
	Means	33.86
	SDevs	0.518
2	2711	34.0
	2712	34.5
	2713	35.1
	2714	35.0
	2715	35.6
	(n)	5
	Means	34.84
	SDevs	0.611
3	2721	33.8
	2722	35.1

Body Temperature Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 12 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal #	BWTP (°C)
3	2723	34.7
	2724	33.9
	2725	33.4
	(n)	5
	Means	34.18
	SDevs	0.698
4	2731	34.2
	2732	34.5
	2733	33.9
	2734	34.6
	2735	34.6
	(n)	5
	Means	34.36
	SDevs	0.305

Body Temperature Summary Report with Individual Values

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 12 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal #	BWTP (°C)
1	2706	36.5
	2707	34.8
	2708	35.3
	2709	34.6
	2710	36.1
	(n)	5
	Means	35.46
2	SDevs	0.820
	2716	35.8
	2717	35.5
	2718	36.2
	2719	35.9
	2720	36.4
	(n)	5
3	Means	35.96
	SDevs	0.351
	2726	36.4
	2727	35.1

Body Temperature Summary Report with Individual Values

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Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 12 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal #	BWTP (°C)
3	2728	35.7
	2729	36.0
	2730	36.6
	(n)	5
	Means	35.96
	SDevs	0.594
4	2736	35.8
	2737	36.1
	2738	36.1
	2739	35.7
	2740	35.5
	(n)	5
	Means	35.84
	SDevs	0.261

Body Temperature Summary Report with Individual Values

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Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 13 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal #	BWTP (°C)
1	2701	33.8
	2702	32.8
	2703	34.3
	2704	35.3
	2705	35.2
	(n)	5
	Means	34.28
2	SDevs	1.038
	2711	34.2
	2712	34.4
	2713	34.5
	2714	35.8
	2715	35.8
	(n)	5
3	Means	34.94
	SDevs	0.792
	2721	35.3
	2722	34.4

Body Temperature Summary Report with Individual Values

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Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 13 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal #	BWTP (°C)
3	2723	35.7
	2724	35.3
	2725	34.4
	(n)	5
	Means	35.02
	SDevs	0.589
4	2731	34.5
	2732	36.1
	2733	34.3
	2734	36.2
	2735	35.5
	(n)	5
	Means	35.32
	SDevs	0.884

Body Temperature Summary Report with Individual Values

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 13 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal #	BWTP (°C)
1	2706	35.4
	2707	35.9
	2708	35.4
	2709	35.8
	2710	35.4
	(n)	5
	Means	35.58
2	SDevs	0.249
	2716	36.3
	2717	37.5
	2718	37.1
	2719	37.5
	2720	36.2
	(n)	5
3	Means	36.92
	SDevs	0.634
	2726	37.6
	2727	37.2

Body Temperature Summary Report with Individual Values

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Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 13 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal #	BWTP (°C)
3	2728	37.4
	2729	37.8
	2730	38.2
	(n)	5
	Means	37.64
	SDevs	0.385
4	2736	37.8
	2737	38.5
	2738	38.2
	2739	37.8
	2740	37.8
	(n)	5
	Means	38.02
	SDevs	0.319

Body Temperature Summary Report with Individual Values

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 14 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal #	BWTP (°C)
1	2701	36.3
	2702	36.3
	2703	37.3
	2704	36.8
	2705	36.7
	(n)	5
	Means	36.68
	SDevs	0.415
2	2711	37.2
	2712	38.0
	2713	37.3
	2714	37.3
	2715	37.2
	(n)	5
	Means	37.40
	SDevs	0.339
3	2721	37.5
	2722	37.3

Body Temperature Summary Report with Individual Values

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 14 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal #	BWTP (°C)
3	2723	38.1
	2724	37.8
	2725	37.4
	(n)	5
	Means	37.62
	SDevs	0.327
4	2731	37.3
	2732	38.0
	2733	36.9
	2734	38.3
	2735	37.8
	(n)	5
	Means	37.66
	SDevs	0.559

Body Temperature Summary Report with Individual Values

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 14 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal #	BWTP (°C)
1	2706	37.7
	2707	38.0
	2708	38.3
	2709	38.2
	2710	37.7
	(n)	5
	Means	37.98
2	SDevs	0.277
	2716	38.4
	2717	38.3
	2718	38.3
	2719	38.5
	2720	38.3
	(n)	5
3	Means	38.36
	SDevs	0.089
	2726	38.5
	2727	38.1

Body Temperature Summary Report with Individual Values

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 14 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal #	BWTP (°C)
3	2728	38.4
	2729	38.7
	2730	38.4
	(n)	5
	Means	38.42
	SDevs	0.217
4	2736	38.4
	2737	38.8
	2738	38.7
	2739	38.6
	2740	38.8
	(n)	5
	Means	38.66
	SDevs	0.167

Body Temperature Summary Report with Individual Values

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Study: 0437RL35.002

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 15 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal #	BWTP (°C)
1	2701	35.2
	2702	34.7
	2703	34.7
	2704	35.2
	2705	35.6
	(n)	5
	Means	35.08
	SDevs	0.383
2	2711	35.4
	2712	36.0
	2713	36.2
	2714	35.5
	2715	35.4
	(n)	5
	Means	35.70
	SDevs	0.374
3	2721	35.4
	2722	34.3

Body Temperature Summary Report with Individual Values

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Study: 0437RL35.002

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 15 (Scheduled Animal Room - Session 1)

Males

Body Temperature

Group #	Animal #	BWTP (°C)
3	2723	35.5
	2724	35.4
	2725	33.8
	(n)	5
	Means	34.88
	SDevs	0.779
4	2731	34.6
	2732	34.7
	2733	34.7
	2734	34.9
	2735	34.8
	(n)	5
	Means	34.74
	SDevs	0.114

Body Temperature Summary Report with Individual Values

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 15 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal #	BWTP (°C)
1	2706	36.7
	2707	37.5
	2708	36.7
	2709	37.8
	2710	37.2
	(n)	5
	Means	37.18
2	SDevs	0.487
	2716	36.8
	2717	36.2
	2718	37.3
	2719	37.6
	2720	36.5
	(n)	5
3	Means	36.88
	SDevs	0.572
	2726	37.2
	2727	37.7

Body Temperature Summary Report with Individual Values

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 15 (Scheduled Animal Room - Session 1)

Females

Body Temperature

Group #	Animal #	BWTP (°C)
3	2728	37.1
	2729	37.0
	2730	37.4
	(n)	5
	Means	37.28
	SDevs	0.277
4	2736	36.0
	2737	37.9
	2738	37.7
	2739	37.0
	2740	36.5
	(n)	5
	Means	37.02
	SDevs	0.798

Body Weight Summary Report with Individual Values

Group	Sex	Dose Level
1	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

Body Weight Summary Report with Individual Values (g)

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Males				
Group #	Animal #	Dosing Phase Day: 1 Session 1	Day: 8 Session 1	Day: 15 Session 1
1	2701	285.	304.	325.
	2702	285.	312.	331.
	2703	297.	320.	334.
	2704	276.	293.	309.
	2705	285.	310.	333.
	(n)	5	5	5
	Means	285.6	307.8	326.4
	SDevs	7.47	10.06	10.33
2	2711	291.	315.	331.
	2712	284.	308.	336.
	2713	289.	310.	328.
	2714	288.	320.	346.
	2715	276.	295.	311.
	(n)	5	5	5
	Means	285.6	309.6	330.4
	SDevs	5.94	9.40	12.82
3	2721	284.	310.	334.
	2722	277.	303.	320.
	2723	289.	323.	340.
	2724	290.	326.	348.

Body Weight Summary Report with Individual Values (g)

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Males

Group #	Animal #	Dosing Phase		
		Day: 1 Session 1	Day: 8 Session 1	Day: 15 Session 1
3	2725	276.	293.	315.
	(n)	5	5	5
	Means	283.2	311.0	331.4
	SDevs	6.53	13.77	13.74
4	2731	277.	296.	308.
	2732	288.	312.	326.
	2733	291.	320.	340.
	2734	287.	312.	326.
	2735	297.	326.	344.
	(n)	5	5	5
	Means	288.0	313.2	328.8
	SDevs	7.28	11.28	14.18

Body Weight Summary Report with Individual Values (g)

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

		Females		
Group #	Animal #	Dosing Phase		
		Day: 1 Session 1	Day: 8 Session 1	Day: 15 Session 1
1	2706	215.	234.	252.
	2707	217.	226.	236.
	2708	212.	226.	241.
	2709	209.	218.	227.
	2710	220.	244.	250.
	(n)	5	5	5
	Means	214.6	229.6	241.2
	SDevs	4.28	9.84	10.28
2	2716	211.	227.	235.
	2717	215.	228.	242.
	2718	210.	225.	225.
	2719	219.	224.	244.
	2720	209.	224.	250.
	(n)	5	5	5
	Means	212.8	225.6	239.2
	SDevs	4.15	1.82	9.58
3	2726	213.	248.	269.
	2727	218.	238.	248.
	2728	214.	233.	243.
	2729	213.	230.	247.

Body Weight Summary Report with Individual Values (g)

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Females				
Group #	Animal #	Dosing Phase Day: 1 Session 1	Day: 8 Session 1	Day: 15 Session 1
3	2730	213.	221.	233.
	(n)	5	5	5
	Means	214.2	234.0	248.0
	SDevs	2.17	9.97	13.15
4	2736	211.	225.	235.
	2737	211.	225.	233.
	2738	202.	222.	232.
	2739	215.	228.	230.
	2740	218.	236.	244.
	(n)	5	5	5
	Means	211.4	227.2	234.8
	SDevs	6.02	5.36	5.45

Body Weight Gain Per Interval Summary Report with Individual Values

Group	Sex	Dose Level
1	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

Body Weight Gain Per Interval Summary Report with Individual Values (g)

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Males			
Group #	Animal #	Dosing Phase	
		Day: 8 Session 1	Day: 15 Session 1
1	2701	19.	21.
	2702	27.	19.
	2703	23.	14.
	2704	17.	16.
	2705	25.	23.
	(n)	5	5
	Means	22.2	18.6
	SDevs	4.15	3.65
2	2711	24.	16.
	2712	24.	28.
	2713	21.	18.
	2714	32.	26.
	2715	19.	16.
	(n)	5	5
	Means	24.0	20.8
	SDevs	4.95	5.76
3	2721	26.	24.
	2722	26.	17.
	2723	34.	17.

Body Weight Gain Per Interval Summary Report with Individual Values (g)

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Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Males

Group #	Animal #	Dosing Phase	
		Day: 8 Session 1	Day: 15 Session 1
3	2724	36.	22.
	2725	17.	22.
	(n)	5	5
	Means	27.8	20.4
	SDevs	7.56	3.21
4	2731	19.	12.
	2732	24.	14.
	2733	29.	20.
	2734	25.	14.
	2735	29.	18.
	(n)	5	5
	Means	25.2	15.6
	SDevs	4.15	3.29

Body Weight Gain Per Interval Summary Report with Individual Values (g)

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Females

Group #	Animal #	Dosing Phase	
		Day: 8 Session 1	Day: 15 Session 1
1	2706	19.	18.
	2707	9.	10.
	2708	14.	15.
	2709	9.	9.
	2710	24.	6.
	(n)	5	5
	Means	15.0	11.6
	SDevs	6.52	4.83
2	2716	16.	8.
	2717	13.	14.
	2718	15.	0.
	2719	5.	20.
	2720	15.	26.
	(n)	5	5
	Means	12.8	13.6
	SDevs	4.49	10.14
3	2726	35.	21.
	2727	20.	10.
	2728	19.	10.

Body Weight Gain Per Interval Summary Report with Individual Values (g)

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Females

Group #	Animal #	Dosing Phase	
		Day: 8 Session 1	Day: 15 Session 1
3	2729	17.	17.
	2730	8.	12.
	(n)	5	5
	Means	19.8	14.0
	SDevs	9.73	4.85
4	2736	14.	10.
	2737	14.	8.
	2738	20.	10.
	2739	13.	2.
	2740	18.	8.
	(n)	5	5
	Means	15.8	7.6
	SDevs	3.03	3.29

Food Consumption Summary Report with Individual Values

Group	Sex	Dose Level
1	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

Food Consumption Summary Report with Individual Values (g)

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StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Males

Group #	Animal #	Dosing Phase	
		Day: 8	Day: 15
1	2701	155.	159.
	2702	166.	171.
	2703	169.	155.
	2704	147.	146.
	2705	155.	158.
	Cages (N)	5	5
2	Means	158.4	157.8
	SDevs	8.99	8.98
	2711	178.	173.
	2712	155.	160.
	2713	161.	161.
	2714	162.	164.
3	2715	161.	148.
	Cages (N)	5	5
	Means	163.4	161.2
	SDevs	8.62	8.98
	2721	155.	164.
	2722	159.	157.

Food Consumption Summary Report with Individual Values (g)

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Males

Group #	Animal #	Dosing Phase	
		Day: 8	Day: 15
3	2723	170.	166.
	2724	175.	173.
	2725	146.	153.
	Cages (N)	5	5
	Means	161.0	162.6
	SDevs	11.64	7.83
4	2731	151.	156.
	2732	167.	153.
	2733	172.	164.
	2734	162.	155.
	2735	173.	170.
	Cages (N)	5	5
	Means	165.0	159.6
	SDevs	8.97	7.16

Food Consumption Summary Report with Individual Values (g)

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Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Females			
Group #	Animal #	Dosing Phase	
		Day: 8	Day: 15
1	2706	146.	151.
	2707	122.	130.
	2708	125.	134.
	2709	125.	124.
	2710	144.	152.
	Cages (N)	5	5
	Means	132.4	138.2
	SDevs	11.59	12.66
2	2716	126.	119.
	2717	126.	122.
	2718	121.	109.
	2719	114.	139.
	2720	125.	144.
	Cages (N)	5	5
	Means	122.4	126.6
	SDevs	5.13	14.54
3	2726	146.	161.
	2727	143.	137.

Food Consumption Summary Report with Individual Values (g)

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Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Females

Group #	Animal #	Dosing Phase	
		Day: 8	Day: 15
3	2728	137.	136.
	2729	120.	132.
	2730	126.	124.
	Cages (N)	5	5
	Means	134.4	138.0
	SDevs	11.10	13.84
4	2736	138.	126.
	2737	141.	149.
	2738	129.	128.
	2739	126.	126.
	2740	142.	137.
	Cages (N)	5	5
	Means	135.2	133.2
	SDevs	7.26	9.93

Hematology Summary Report with Individual Values

Group	Sex	Dose Level
1	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

Hematology Summary Report with Individual Values

Abbr	Description	Units
WBC	White Blood Cells	x10e3/ μ L
RBC	Red Blood Cells	x10e6/ μ L
HGB	Hemoglobin	g/dL
HCT	Hematocrit	%
#LYMPH	Lymphocytes	x10e3/ μ L
%LYMPH	Lymphocytes - %	%
MCV	Mean Corpuscular Volume	fL
MCHC	Mean Corpuscular Hemoglobin Concentration	g/dL
MCH	Mean Corpuscular Hemoglobin	pg
#MONO	Absolute Monocytes	x10e3/ μ L
%MONO	Monocytes - %	%
#NEUT	Absolute Neutrophil	x10e3/ μ L
%NEUT	Neutrophil - %	%
PLT	Platelets	x10e3/ μ L
#BASO	Absolute Basophil	x10e3/ μ L
%BASO	Basophil - %	%
#EOS	Absolute Eosinophil	x10e3/ μ L
%EOS	Eosinophil - %	%
%LUC	Large Unstained Cells - %	%
#LUC	Absolute Large Unstained Cells	x10e3/ μ L
%RETIC	Reticulocytes - %	%
#RETIC	Absolute Reticulocyte	x10e9/L

Hematology Summary Report with Individual Values

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Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males

Group #	Animal #	WBC (x10 ³ /μL)	RBC (x10 ⁶ /μL)	HGB (g/dL)	Complete Blood Count		MCH (pg)	MCHC (g/dL)	PLT (10 ³ /μL)	%NEUT (%)
					HCT (%)	MCV (fL)				
1	2701	13.18	9.24	17.1	56.6	61.3	18.5	30.2	949.	11.4
	2702	15.85	9.25	16.4	53.6	57.9	17.7	30.6	816.	13.4
	2703	7.72	8.51	14.9	48.5	57.0	17.5	30.7	834.	13.3
	2704	11.54	8.60	15.3	50.1	58.3	17.8	30.6	1004.	8.8
	2705	6.54	7.87	14.1	46.4	58.9	17.9	30.4	898.	12.7
	(n)	5	5	5	5	5	5	5	5	5
	Means	10.966	8.694	15.56	51.04	58.68	17.88	30.50	900.2	11.92
	SDevs	3.8474	0.5764	1.195	4.072	1.619	0.377	0.200	78.47	1.918
2	2711	10.28	8.50	15.7	51.5	60.7	18.5	30.5	1012.	11.9
	2712	11.40	8.19	15.0	49.5	60.4	18.3	30.2	886.	11.2
	2713	10.21	8.39	14.9	48.3	57.6	17.7	30.7	944.	13.1
	2714	12.70	8.22	14.4	48.5	59.0	17.6	29.8	969.	12.7
	2715	6.64	8.38	15.2	49.1	58.6	18.1	30.8	905.	13.0
	(n)	5	5	5	5	5	5	5	5	5
	Means	10.246	8.336	15.04	49.38	59.26	18.04	30.40	943.2	12.38
	SDevs	2.2561	0.1290	0.472	1.277	1.288	0.385	0.406	50.33	0.811
3	2721	7.68	8.55	15.4	50.9	59.5	18.0	30.3	1024.	16.9
	2722	11.25	8.44	15.1	49.0	58.0	17.9	30.9	925.	9.6

Hematology Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males

Group #	Animal #	WBC (x10 ³ /μL)	RBC (x10 ⁶ /μL)	HGB (g/dL)	Complete Blood Count		MCH (pg)	MCHC (g/dL)	PLT (10 ³ /μL)	%NEUT (%)
					HCT (%)	MCV (fL)				
3	2723	13.21	8.21	15.0	50.3	61.3	18.3	29.9	921.	14.9
	2724	14.11	8.25	14.9	49.0	59.5	18.0	30.3	873.	11.8
	2725	9.61	8.25	15.3	50.3	60.9	18.5	30.4	817.	13.1
	(n)	5	5	5	5	5	5	5	5	5
	Means	11.172	8.340	15.14	49.90	59.84	18.14	30.36	912.0	13.26
	SDevs	2.6185	0.1476	0.207	0.857	1.311	0.251	0.358	76.39	2.806
4	2731	11.46	8.39	15.1	49.4	58.9	18.0	30.6	905.	9.6
	2732	11.25	9.04	15.6	52.2	57.8	17.3	29.8	825.	13.5
	2733	7.52	8.32	15.2	49.2	59.2	18.3	30.9	925.	14.6
	2734	11.90	8.59	15.4	51.2	59.5	18.0	30.2	906.	12.2
	2735	9.88	8.14	14.8	48.3	59.4	18.2	30.7	1181.	13.8
	(n)	5	5	5	5	5	5	5	5	5
	Means	10.402	8.496	15.22	50.06	58.96	17.96	30.44	948.4	12.74
	SDevs	1.7791	0.3441	0.303	1.593	0.688	0.391	0.439	135.61	1.957

Hematology Summary Report with Individual Values

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males

Group #	Animal #	Complete Blood Count								
		%LYMP (%)	%MONO (%)	%EOS (%)	%BASO (%)	%LUC (%)	#NEUT (x10 ³ /μL)	#LYMPH (x10 ³ /μL)	#MONO (x10 ³ /μL)	#EOS (x10 ³ /μL)
1	2701	82.5	3.4	0.9	0.8	1.1	1.50	10.88	0.45	0.12
	2702	80.7	3.6	0.6	0.8	0.9	2.13	12.79	0.57	0.10
	2703	82.8	2.1	0.5	0.5	0.7	1.02	6.39	0.16	0.04
	2704	86.7	2.8	0.4	0.7	0.6	1.02	10.00	0.33	0.04
	2705	82.3	3.6	0.4	0.4	0.6	0.83	5.39	0.24	0.02
	(n)	5	5	5	5	5	5	5	5	5
	Means	83.00	3.10	0.56	0.64	0.78	1.300	9.090	0.350	0.064
2	2711	82.6	3.5	0.9	0.6	0.6	1.22	8.49	0.36	0.09
	2712	83.2	3.6	0.7	0.4	0.8	1.27	9.49	0.41	0.08
	2713	82.2	3.3	0.6	0.4	0.5	1.33	8.40	0.33	0.06
	2714	80.4	5.2	0.2	0.6	1.0	1.61	10.21	0.67	0.02
	2715	81.7	3.8	0.6	0.3	0.6	0.87	5.43	0.25	0.04
	(n)	5	5	5	5	5	5	5	5	5
	Means	82.02	3.88	0.60	0.46	0.70	1.260	8.404	0.404	0.058
3	2721	78.2	3.6	0.4	0.3	0.7	1.30	6.01	0.27	0.03
	2722	85.7	2.8	0.5	0.5	0.9	1.08	9.64	0.31	0.06

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Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males

Group #	Animal #	Complete Blood Count		%LYMP (%)	%MONO (%)	%EOS (%)	%BASO (%)	%LUC (%)	#NEUT (x10 ³ /μL)	#LYMPH (x10 ³ /μL)	#MONO (x10 ³ /μL)	#EOS (x10 ³ /μL)
		%LYMP (%)	%MONO (%)									
3	2723	77.7	5.5	0.6	0.5	0.8	1.97	10.26	0.73	0.08		
	2724	82.9	3.5	0.6	0.5	0.7	1.67	11.70	0.49	0.08		
	2725	81.2	3.8	0.6	0.4	0.9	1.26	7.80	0.36	0.06		
	(n)	5	5	5	5	5	5	5	5	5		
	Means	81.14	3.84	0.54	0.44	0.80	1.456	9.082	0.432	0.062		
	SDevs	3.331	1.001	0.089	0.089	0.100	0.3585	2.2155	0.1861	0.0205		
4	2731	83.0	5.5	0.5	0.5	0.8	1.11	9.52	0.63	0.05		
	2732	81.1	3.3	1.0	0.5	0.6	1.52	9.13	0.37	0.11		
	2733	79.1	4.8	0.6	0.3	0.7	1.10	5.94	0.36	0.04		
	2734	81.8	3.8	0.7	0.5	1.0	1.46	9.74	0.45	0.08		
	2735	80.6	4.1	0.7	0.4	0.4	1.36	7.96	0.41	0.07		
	(n)	5	5	5	5	5	5	5	5	5		
	Means	81.12	4.30	0.70	0.44	0.70	1.310	8.458	0.444	0.070		
	SDevs	1.445	0.863	0.187	0.089	0.224	0.1957	1.5661	0.1099	0.0274		

Hematology Summary Report with Individual Values

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Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males

Group #	Animal#	#BASO (x10 ³ /μL)	#LUC (x10 ³ /μL)	%RETIC (%)	Complete Blood Count
					#RETIC (x10 ⁹ /L)
1	2701	0.10	0.14	1.87	172.5
	2702	0.13	0.14	1.87	172.8
	2703	0.04	0.06	1.29	109.5
	2704	0.08	0.07	1.74	149.5
	2705	0.03	0.04	1.57	123.7
	(n)	5	5	5	5
	Means	0.076	0.090	1.668	145.60
	SDevs	0.0416	0.0469	0.2446	28.555
2	2711	0.06	0.06	1.77	150.4
	2712	0.05	0.09	2.26	185.4
	2713	0.04	0.05	1.70	142.8
	2714	0.07	0.12	1.93	158.7
	2715	0.02	0.04	1.96	163.9
	(n)	5	5	5	5
	Means	0.048	0.072	1.924	160.24
	SDevs	0.0192	0.0327	0.2169	16.200
3	2721	0.02	0.05	2.13	182.0
	2722	0.06	0.10	1.84	155.3

Hematology Summary Report with Individual Values

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Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males

Group #	Animal#	BASO (x10 ³ /μL)	#LUC (x10 ³ /μL)	%RETIC (%)	Complete Blood Count
					#RETIC (x10 ⁹ /L)
3	2723	0.07	0.10	1.89	155.4
	2724	0.08	0.10	1.40	115.9
	2725	0.04	0.08	1.88	154.7
	(n)	5	5	5	5
	Means	0.054	0.086	1.828	152.66
	SDevs	0.0241	0.0219	0.2651	23.616
4	2731	0.06	0.10	1.77	148.4
	2732	0.05	0.07	1.81	163.2
	2733	0.02	0.05	2.06	171.7
	2734	0.06	0.12	2.04	175.5
	2735	0.04	0.04	1.74	141.6
	(n)	5	5	5	5
	Means	0.046	0.076	1.884	160.08
	SDevs	0.0167	0.0336	0.1537	14.667

Hematology Summary Report with Individual Values

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Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females										
Group #	Animal #	WBC	RBC	HGB	Complete Blood Count		MCH	MCHC	PLT	%NEUT
		(x10 ³ /μL)	(x10 ⁶ /μL)	(g/dL)	HCT (%)	MCV (fL)	(pg)	(g/dL)	(10 ³ /μL)	(%)
1	2706	Clotted	Clotted	Clotted	Clotted	Clotted	Clotted	Clotted	Clotted	Clotted
	2707	6.00	8.63	15.9	52.5	60.8	18.5	30.4	942.	14.3
	2708	5.50	8.05	15.2	48.0	59.6	18.9	31.7	924.	19.2
	2709	6.82	8.08	14.6	46.2	57.1	18.1	31.7	771.	18.9
	2710	9.81	7.50	13.9	43.5	58.0	18.5	31.9	1001.	14.0
	(n)	4	4	4	4	4	4	4	4	4
	Means	7.033	8.065	14.90	47.55	58.88	18.50	31.43	909.5	16.60
	SDevs	1.9300	0.4615	0.852	3.783	1.648	0.327	0.690	98.02	2.834
2	2716	7.81	8.42	15.7	50.8	60.2	18.6	30.9	945.	12.3
	2717	9.90	8.86	15.9	51.5	58.1	17.9	30.9	1157.	13.6
	2718	9.15	8.61	15.9	50.2	58.4	18.5	31.6	729.	15.7
	2719	5.89	8.21	15.5	47.7	58.2	18.8	32.4	844.	12.9
	2720	10.59	7.43	13.9	43.3	58.2	18.6	32.0	652.	20.9
	(n)	5	5	5	5	5	5	5	5	5
	Means	8.668	8.306	15.38	48.70	58.62	18.48	31.56	865.4	15.08
	SDevs	1.8640	0.5452	0.844	3.341	0.890	0.342	0.666	197.46	3.497
3	2726	13.16	8.16	15.5	49.6	60.8	19.0	31.2	1056.	10.2
	2727	7.44	8.31	15.4	49.4	59.4	18.6	31.3	967.	15.5

Hematology Summary Report with Individual Values

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females										
Group #	Animal #	WBC (x10 ³ /μL)	RBC (x10 ⁶ /μL)	HGB (g/dL)	Complete Blood Count		MCH (pg)	MCHC (g/dL)	PLT (10 ³ /μL)	%NEUT (%)
					HCT (%)	MCV (fL)				
3	2728	10.63	8.32	15.5	49.0	58.9	18.6	31.6	787.	18.0
	2729	12.35	7.96	14.5	46.5	58.4	18.2	31.2	1128.	21.4
	2730	5.44	8.45	15.5	49.5	58.5	18.4	31.4	977.	16.8
	(n)	5	5	5	5	5	5	5	5	5
	Means	9.804	8.240	15.28	48.80	59.20	18.56	31.34	983.0	16.38
	SDevs	3.2810	0.1872	0.438	1.306	0.977	0.297	0.167	127.54	4.092
4	2736	8.37	8.20	15.6	49.9	60.8	19.0	31.3	1036.	8.3
	2737	14.81	8.25	15.1	47.6	57.6	18.3	31.8	749.	12.8
	2738	7.58	8.09	15.4	49.0	60.6	19.1	31.5	991.	12.6
	2739	9.33	7.92	15.1	47.3	59.7	19.0	31.9	1053.	22.2
	2740	9.50	7.94	14.6	47.8	60.3	18.4	30.6	1039.	18.0
	(n)	5	5	5	5	5	5	5	5	5
	Means	9.918	8.080	15.16	48.32	59.80	18.76	31.42	973.6	14.78
	SDevs	2.8423	0.1488	0.378	1.094	1.298	0.378	0.517	127.69	5.387

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Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females										
Group #	Animal #	Complete Blood Count								
		%LYMP (%)	%MONO (%)	%EOS (%)	%BASO (%)	%LUC (%)	#NEUT (x10 ³ /µL)	#LYMPH (x10 ³ /µL)	#MONO (x10 ³ /µL)	#EOS (x10 ³ /µL)
1	2706	Clotted	Clotted	Clotted	Clotted	Clotted	Clotted	Clotted	Clotted	Clotted
	2707	79.2	4.5	1.0	0.4	0.6	0.86	4.75	0.27	0.06
	2708	73.9	4.6	1.2	0.3	0.7	1.06	4.07	0.25	0.07
	2709	75.5	2.6	1.4	0.5	1.1	1.29	5.15	0.18	0.09
	2710	79.1	4.2	1.0	0.9	0.7	1.37	7.77	0.41	0.10
	(n)	4	4	4	4	4	4	4	4	4
	Means	76.93	3.98	1.15	0.53	0.78	1.145	5.435	0.278	0.080
	SDevs	2.651	0.932	0.191	0.263	0.222	0.2310	1.6192	0.0964	0.0183
2	2716	79.8	5.4	0.9	0.6	1.1	0.96	6.23	0.42	0.07
	2717	80.4	3.8	0.9	0.4	0.9	1.35	7.96	0.38	0.09
	2718	78.4	2.2	1.4	0.6	1.8	1.44	7.17	0.20	0.13
	2719	79.2	5.4	1.2	0.4	0.9	0.76	4.66	0.32	0.07
	2720	71.5	3.6	1.8	0.6	1.5	2.22	7.57	0.38	0.19
	(n)	5	5	5	5	5	5	5	5	5
	Means	77.86	4.08	1.24	0.52	1.24	1.346	6.718	0.340	0.110
	SDevs	3.632	1.354	0.378	0.110	0.397	0.5624	1.3178	0.0860	0.0510
3	2726	84.7	3.4	0.5	0.7	0.6	1.34	11.15	0.45	0.06
	2727	76.9	5.5	0.8	0.4	1.0	1.15	5.72	0.41	0.06

Hematology Summary Report with Individual Values

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females

Group #	Animal #	Complete Blood Count								
		%LYMP (%)	%MONO (%)	%EOS (%)	%BASO (%)	%LUC (%)	#NEUT (x10 ³ /μL)	#LYMPH (x10 ³ /μL)	#MONO (x10 ³ /μL)	#EOS (x10 ³ /μL)
3	2728	74.8	4.4	1.3	0.5	1.1	1.92	7.95	0.46	0.13
	2729	73.0	3.8	0.9	0.5	0.5	2.64	9.02	0.47	0.11
	2730	77.9	3.0	0.8	0.3	1.2	0.91	4.24	0.16	0.04
	(n)	5	5	5	5	5	5	5	5	5
	Means	77.46	4.02	0.86	0.48	0.88	1.592	7.616	0.390	0.080
	SDevs	4.469	0.976	0.288	0.148	0.311	0.6946	2.7186	0.1306	0.0381
4	2736	87.1	2.9	0.5	0.5	0.8	0.69	7.29	0.24	0.05
	2737	81.4	3.3	1.3	0.6	0.7	1.89	12.06	0.48	0.19
	2738	82.1	3.4	0.8	0.4	0.7	0.95	6.23	0.26	0.06
	2739	73.1	2.5	1.3	0.4	0.4	2.07	6.83	0.24	0.13
	2740	74.3	5.6	1.3	0.4	0.4	1.71	7.06	0.54	0.13
	(n)	5	5	5	5	5	5	5	5	5
	Means	79.60	3.54	1.04	0.46	0.60	1.462	7.894	0.352	0.112
	SDevs	5.833	1.205	0.371	0.089	0.187	0.6067	2.3620	0.1460	0.0576

Hematology Summary Report with Individual Values

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females

Complete Blood Count

Group #	Animal#	BASO # (x10 ³ /μL)	#LUC (x10 ³ /μL)	%RETIC (%)	#RETIC (x10 ⁹ /L)
1	2706	Clotted	Clotted	Clotted	Clotted
	2707	0.02	0.04	1.99	171.7
	2708	0.02	0.04	2.65	213.1
	2709	0.04	0.08	1.96	158.4
	2710	0.09	0.07	1.83	137.4
	(n)	4	4	4	4
	Means	0.043	0.058	2.108	170.15
	SDevs	0.0330	0.0206	0.3683	31.926
2	2716	0.04	0.08	2.22	187.1
	2717	0.04	0.09	1.46	129.5
	2718	0.05	0.17	1.41	121.8
	2719	0.02	0.05	2.34	191.7
	2720	0.07	0.16	2.37	176.1
	(n)	5	5	5	5
	Means	0.044	0.110	1.960	161.24
	SDevs	0.0182	0.0524	0.4829	33.092
3	2726	0.09	0.08	3.20	260.7
	2727	0.03	0.07	2.39	198.7

Hematology Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females

Group #	Animal#	BASO (x10 ³ /μL)	#LUC (x10 ³ /μL)	%RETIC (%)	Complete Blood Count
					#RETIC (x10 ⁹ /L)
3	2728	0.05	0.12	1.70	141.2
	2729	0.06	0.06	2.19	174.1
	2730	0.02	0.07	1.88	159.2
	(n)	5	5	5	5
	Means	0.050	0.080	2.272	186.78
	SDevs	0.0274	0.0235	0.5837	46.382
4	2736	0.04	0.06	2.25	184.6
	2737	0.09	0.11	1.40	115.9
	2738	0.03	0.06	1.99	160.6
	2739	0.03	0.04	1.65	130.5
	2740	0.04	0.04	2.07	164.3
	(n)	5	5	5	5
	Means	0.046	0.062	1.872	151.18
	SDevs	0.0251	0.0286	0.3421	27.614

Hematology Sample Condition

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

				Study Phase: Dosing Phase		Day of Phase: 16	Session: 1	Scheduled		
Study #	Cage #	Sex	Group	User #	Date/Time Data Entered		Data Entry Method		Date/Time Data Taken	
2701	1	M	1	306	22 Jul 2014 15:10:20		Key Real-time		22 Jul 2014 15:10:20	
	Condition				Platelet Clumps					
2705	5	M	1	306	22 Jul 2014 15:11:19		Key Real-time		22 Jul 2014 15:11:19	
	Condition				Platelet Clumps					
2712	12	M	2	306	22 Jul 2014 15:14:39		Key Real-time		22 Jul 2014 15:14:39	
	Condition				Platelet Clumps					
2721	21	M	3	306	22 Jul 2014 15:17:00		Key Real-time		22 Jul 2014 15:17:00	
	Condition				Platelet Clumps					
2722	22	M	3	306	22 Jul 2014 15:17:18		Key Real-time		22 Jul 2014 15:17:18	
	Condition				Platelet Clumps					
2724	24	M	3	306	22 Jul 2014 15:17:49		Key Real-time		22 Jul 2014 15:17:49	
	Condition				Platelet Clumps					
2725	25	M	3	306	22 Jul 2014 15:18:11		Key Real-time		22 Jul 2014 15:18:11	
	Condition				Platelet Clumps					
2731	31	M	4	306	22 Jul 2014 15:21:00		Key Real-time		22 Jul 2014 15:21:00	
	Condition				Platelet Clumps					
2709	9	F	1	306	22 Jul 2014 15:13:03		Key Real-time		22 Jul 2014 15:13:03	
	Condition				Platelet Clumps					
2718	18	F	2	306	22 Jul 2014 15:16:06		Key Real-time		22 Jul 2014 15:16:06	
	Condition				Platelet Clumps					

User # 306 = Amanda O'Brien

Hematology Sample Condition

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Study Phase: Dosing Phase Day of Phase: 16 Session: 1 Scheduled

Study #	Cage #	Sex	Group	User #	Date/Time Data Entered	Data Entry Method	Date/Time Data Taken
2720	20	F	2	306	22 Jul 2014 15:16:38	Key Real-time	22 Jul 2014 15:16:38
	Condition				Platelet Clumps		
2726	26	F	3	306	22 Jul 2014 15:18:31	Key Real-time	22 Jul 2014 15:18:31
	Condition				Platelet Clumps		
2728	28	F	3	306	22 Jul 2014 15:19:02	Key Real-time	22 Jul 2014 15:19:02
	Condition				Platelet Clumps		
2729	29	F	3	306	22 Jul 2014 15:19:38	Key Real-time	22 Jul 2014 15:19:38
	Condition				Platelet Clumps		
2737	37	F	4	306	22 Jul 2014 15:22:18	Key Real-time	22 Jul 2014 15:22:18
	Condition				Platelet Clumps		

User # 306 = Amanda O'Brien

Red Blood Cell Morphology

Group	Sex	Dose Level
1	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

Red Blood Cell Morphology Summary Report by Session

Grading Severity of Condition

- 1+ Rare – slight
- 2+ Moderately
- 3+ Marked

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Red Blood Cell Morphology

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

				Study Phase: Dosing Phase		Day of Phase: 16	Session: 1	Scheduled		
Study #	Cage #	Sex	Group	User #	Date/Time Data Entered		Data Entry Method		Date/Time Data Taken	
2701	1	M	1	306	22 Jul 2014 15:10:27		Key Real-time		22 Jul 2014 15:10:27	
			Normochromic & Normocytic							
2702	2	M	1	306	22 Jul 2014 15:10:39		Key Real-time		22 Jul 2014 15:10:39	
			Normochromic & Normocytic							
2703	3	M	1	306	22 Jul 2014 15:10:51		Key Real-time		22 Jul 2014 15:10:51	
			Normochromic & Normocytic							
2704	4	M	1	306	22 Jul 2014 15:11:03		Key Real-time		22 Jul 2014 15:11:03	
			Normochromic & Normocytic							
2705	5	M	1	306	22 Jul 2014 15:11:53		Key Real-time		22 Jul 2014 15:11:53	
			Normochromic & Normocytic							
2711	11	M	2	306	22 Jul 2014 15:13:38		Key Real-time		22 Jul 2014 15:13:38	
			Normochromic & Normocytic							
2712	12	M	2	306	22 Jul 2014 15:14:48		Key Real-time		22 Jul 2014 15:14:48	
			Normochromic & Normocytic							
2713	13	M	2	306	22 Jul 2014 15:14:58		Key Real-time		22 Jul 2014 15:14:58	
			Normochromic & Normocytic							
2714	14	M	2	306	22 Jul 2014 15:15:08		Key Real-time		22 Jul 2014 15:15:08	
			Normochromic & Normocytic							
2715	15	M	2	306	22 Jul 2014 15:15:18		Key Real-time		22 Jul 2014 15:15:18	
			Normochromic & Normocytic							

User # 306 = Amanda O'Brien

Red Blood Cell Morphology

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

				Study Phase: Dosing Phase		Day of Phase: 16	Session: 1	Scheduled		
Study #	Cage #	Sex	Group	User #	Date/Time Data Entered		Data Entry Method		Date/Time Data Taken	
2721	21	M	3	306	22 Jul 2014 15:17:06		Key Real-time		22 Jul 2014 15:17:06	
			Normochromic & Normocytic							
2722	22	M	3	306	22 Jul 2014 15:17:24		Key Real-time		22 Jul 2014 15:17:24	
			Normochromic & Normocytic							
2723	23	M	3	306	22 Jul 2014 15:17:36		Key Real-time		22 Jul 2014 15:17:36	
			Normochromic & Normocytic							
2724	24	M	3	306	22 Jul 2014 15:17:56		Key Real-time		22 Jul 2014 15:17:56	
			Normochromic & Normocytic							
2725	25	M	3	306	22 Jul 2014 15:18:18		Key Real-time		22 Jul 2014 15:18:18	
			Normochromic & Normocytic							
2731	31	M	4	306	22 Jul 2014 15:21:06		Key Real-time		22 Jul 2014 15:21:06	
			Normochromic & Normocytic							
2732	32	M	4	306	22 Jul 2014 15:21:18		Key Real-time		22 Jul 2014 15:21:18	
			Normochromic & Normocytic							
2733	33	M	4	306	22 Jul 2014 15:21:32		Key Real-time		22 Jul 2014 15:21:32	
			Normochromic & Normocytic							
2734	34	M	4	306	22 Jul 2014 15:21:43		Key Real-time		22 Jul 2014 15:21:43	
			Normochromic & Normocytic							
2735	35	M	4	306	22 Jul 2014 15:21:54		Key Real-time		22 Jul 2014 15:21:54	
			Normochromic & Normocytic							

User # 306 = Amanda O'Brien

Red Blood Cell Morphology

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

				Study Phase: Dosing Phase		Day of Phase: 16	Session: 1	Scheduled		
Study #	Cage #	Sex	Group	User #	Date/Time Data Entered		Data Entry Method		Date/Time Data Taken	
2707	7	F	1	306	22 Jul 2014 15:12:05		Key Real-time		22 Jul 2014 15:12:05	
			Normochromic & Normocytic							
2708	8	F	1	306	22 Jul 2014 15:12:40		Key Real-time		22 Jul 2014 15:12:40	
			Normochromic & Normocytic							
2709	9	F	1	306	22 Jul 2014 15:13:09		Key Real-time		22 Jul 2014 15:13:09	
			Normochromic & Normocytic							
2710	10	F	1	306	22 Jul 2014 15:13:27		Key Real-time		22 Jul 2014 15:13:27	
			Normochromic & Normocytic							
2716	16	F	2	306	22 Jul 2014 15:15:33		Key Real-time		22 Jul 2014 15:15:33	
			Normochromic & Normocytic							
2717	17	F	2	306	22 Jul 2014 15:15:44		Key Real-time		22 Jul 2014 15:15:44	
			Normochromic & Normocytic							
2718	18	F	2	306	22 Jul 2014 15:16:14		Key Real-time		22 Jul 2014 15:16:14	
			Normochromic & Normocytic							
2719	19	F	2	306	22 Jul 2014 15:16:24		Key Real-time		22 Jul 2014 15:16:24	
			Normochromic & Normocytic							
2720	20	F	2	306	22 Jul 2014 15:16:46		Key Real-time		22 Jul 2014 15:16:46	
			Normochromic & Normocytic							
2726	26	F	3	306	22 Jul 2014 15:18:37		Key Real-time		22 Jul 2014 15:18:37	
			Normochromic & Normocytic							

User # 306 = Amanda O'Brien

Red Blood Cell Morphology

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

				Study Phase: Dosing Phase		Day of Phase: 16	Session: 1	Scheduled		
Study #	Cage #	Sex	Group	User #	Date/Time Data Entered		Data Entry Method		Date/Time Data Taken	
2727	27	F	3	306	22 Jul 2014 15:18:48		Key Real-time		22 Jul 2014 15:18:48	
			Normochromic & Normocytic							
2728	28	F	3	306	22 Jul 2014 15:19:10		Key Real-time		22 Jul 2014 15:19:10	
			Normochromic & Normocytic							
2729	29	F	3	306	22 Jul 2014 15:19:45		Key Real-time		22 Jul 2014 15:19:45	
			Normochromic & Normocytic							
2730	30	F	3	306	22 Jul 2014 15:20:02		Key Real-time		22 Jul 2014 15:20:02	
			Normochromic & Normocytic							
2736	36	F	4	306	22 Jul 2014 15:22:06		Key Real-time		22 Jul 2014 15:22:06	
			Normochromic & Normocytic							
2737	37	F	4	306	22 Jul 2014 15:22:25		Key Real-time		22 Jul 2014 15:22:25	
			Normochromic & Normocytic							
2738	38	F	4	306	22 Jul 2014 15:22:35		Key Real-time		22 Jul 2014 15:22:35	
			Normochromic & Normocytic							
2739	39	F	4	306	22 Jul 2014 15:22:48		Key Real-time		22 Jul 2014 15:22:49	
			Normochromic & Normocytic							
2740	40	F	4	306	22 Jul 2014 15:23:01		Key Real-time		22 Jul 2014 15:23:01	
			Normochromic & Normocytic							

User # 306 = Amanda O'Brien

Coagulation Summary Report with Individual Values

Group	Sex	Dose Level
1	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

Coagulation Summary Report with Individual Values

Abbr	Description	Units
PT	Prothrombin Time	sec
APTT	Activated Partial Thromboplastin Time	sec

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Coagulation Summary Report with Individual Values

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males

Coagulation (Stago)

Group #	Animal #	PT (sec)	APTT (sec)
1	2701	15.2	8.4
	2702	16.0	10.3
	2703	16.4	10.7
	2704	16.8	11.1
	2705	16.1	9.0
	(n)	5	5
	Means	16.10	9.90
	SDevs	0.592	1.151
2	2711	15.9	9.1
	2712	16.3	8.5
	2713	16.7	7.9
	2714	16.0	8.8
	2715	16.2	8.4
	(n)	5	5
	Means	16.22	8.54
	SDevs	0.311	0.451
3	2721	16.3	9.3
	2722	16.6	8.0

NSO = No Sample Obtained

Coagulation Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males

Coagulation (Stago)

Group #	Animal #	PT (sec)	APTT (sec)
3	2723	16.9	9.3
	2724	Clotted	Clotted
	2725	17.0	8.4
	(n)	4	4
	Means	16.70	8.75
	SDevs	0.316	0.656
4	2731	Clotted	Clotted
	2732	16.2	10.0
	2733	16.4	9.9
	2734	15.5	7.4
	2735	16.1	8.3
	(n)	4	4
	Means	16.05	8.90
	SDevs	0.387	1.268

NSO = No Sample Obtained

Coagulation Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females

Coagulation (Stago)

Group #	Animal #	PT (sec)	APTT (sec)
1	2706	NSO	NSO
	2707	16.7	9.4
	2708	16.4	8.8
	2709	14.6	7.4
	2710	16.8	8.8
	(n)	4	4
	Means	16.13	8.60
	SDevs	1.031	0.849
2	2716	16.0	9.7
	2717	15.5	6.5
	2718	18.8	8.2
	2719	17.2	8.6
	2720	NSO	NSO
	(n)	4	4
	Means	16.88	8.25
	SDevs	1.468	1.328
3	2726	16.9	9.0
	2727	15.7	7.4

NSO = No Sample Obtained

Coagulation Summary Report with Individual Values

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females

Coagulation (Stago)

Group #	Animal #	PT (sec)	APTT (sec)
3	2728	Clotted	Clotted
	2729	16.7	8.0
	2730	16.3	9.1
	(n)	4	4
	Means	16.40	8.38
	SDevs	0.529	0.818
4	2736	15.1	7.3
	2737	15.9	7.7
	2738	17.3	8.3
	2739	16.0	7.3
	2740	16.3	7.8
	(n)	5	5
	Means	16.12	7.68
	SDevs	0.795	0.415

NSO = No Sample Obtained

Serum Chemistry Summary Report with Individual Values

Group	Sex	Dose Level
1	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

Serum Chemistry Summary Report with Individual Values

Abbr	Description	Units
AST	Aspartate Aminotransferase	U/L
ALT	Alanine Aminotransferase	U/L
GLU	Glucose	mg/dL
BUN	Blood Urea Nitrogen	mg/dL
CREAT	Creatinine	mg/dL
PHOS	Inorganic Phosphorous	mg/dL
TRIG	Triglycerides	mg/dL
CHOL	Cholesterol	mg/dL
TP	Total Protein	g/dL
NA	Sodium	mEq/L
K	Potassium	mEq/L
CL	Chloride	mEq/L
TBILI	Total Bilirubin	mg/dL
ALP	Alkaline Phosphatase	U/L
CA	Calcium	mg/dL
GLOB	Globulin	g/dL
ALB	Albumin	g/dL
A/G	Albumin Globulin Ratio	None

Serum Chemistry Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males

Group #	Animal #	Serum Chemistry								
		AST (U/L)	ALT (U/L)	GLU (mg/dL)	BUN (mg/dl)	CREAT (mg/dL)	PHOS (mg/dL)	TRIG (mg/dL)	CHOL (mg/dL)	TP (g/dL)
1	2701	99.	39.	123.	17.	0.43	9.8	29.	95.	6.6
	2702	80.	33.	116.	26.	0.51	10.1	70.	153.	6.2
	2703	68.	33.	100.	16.	0.37	9.0	33.	88.	6.0
	2704	85.	35.	105.	20.	0.37	9.1	24.	63.	6.1
	2705	77.	35.	97.	16.	0.37	9.5	42.	99.	6.0
	(n)	5	5	5	5	5	5	5	5	5
	Means	81.8	35.0	108.2	19.0	0.410	9.50	39.6	99.6	6.18
	SDevs	11.43	2.45	10.99	4.24	0.0616	0.464	18.23	32.97	0.249
2	2711	79.	30.	134.	17.	0.40	10.1	76.	100.	6.4
	2712	77.	37.	119.	18.	0.38	10.3	52.	110.	6.3
	2713	97.	40.	107.	16.	0.37	9.2	53.	99.	6.1
	2714	78.	37.	99.	17.	0.40	9.5	27.	111.	6.1
	2715	90.	41.	99.	17.	0.38	9.9	25.	91.	6.0
	(n)	5	5	5	5	5	5	5	5	5
	Means	84.2	37.0	111.6	17.0	0.386	9.80	46.6	102.2	6.18
	SDevs	8.87	4.30	14.96	0.71	0.0134	0.447	21.13	8.35	0.164
3	2721	76.	32.	121.	19.	0.38	10.7	25.	83.	6.7
	2722	89.	57.	97.	16.	0.34	10.2	37.	93.	6.1

Serum Chemistry Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males										
Group #	Animal #	Serum Chemistry								
		AST (U/L)	ALT (U/L)	GLU (mg/dL)	BUN (mg/dl)	CREAT (mg/dL)	PHOS (mg/dL)	TRIG (mg/dL)	CHOL (mg/dL)	TP (g/dL)
3	2723	82.	37.	93.	20.	0.44	9.2	32.	108.	6.6
	2724	78.	33.	109.	14.	0.36	9.9	27.	83.	6.1
	2725	78.	31.	100.	17.	0.37	9.9	20.	89.	6.2
	(n)	5	5	5	5	5	5	5	5	5
	Means	80.6	38.0	104.0	17.2	0.378	9.98	28.2	91.2	6.34
	SDevs	5.18	10.86	11.18	2.39	0.0377	0.545	6.53	10.31	0.288
4	2731	80.	42.	103.	18.	0.36	10.5	25.	90.	6.5
	2732	110.	47.	111.	25.	0.42	10.1	31.	76.	6.3
	2733	77.	35.	122.	18.	0.39	9.2	39.	86.	6.5
	2734	81.	31.	108.	18.	0.41	9.8	29.	88.	6.6
	2735	90.	37.	107.	17.	0.39	9.0	33.	96.	6.3
	(n)	5	5	5	5	5	5	5	5	5
	Means	87.6	38.4	110.2	19.2	0.394	9.72	31.4	87.2	6.44
	SDevs	13.43	6.23	7.19	3.27	0.0230	0.622	5.18	7.29	0.134

Serum Chemistry Summary Report with Individual Values

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males										
Serum Chemistry										
Group #	Animal #	Na (mEq/L)	K (mEq/L)	Cl (mEq/L)	TBILI (mg/dL)	ALP (U/L)	CA (mg/dL)	GLOB (g/dL)	ALB (g/dL)	A/G
1	2701	143.	6.8	100.	0.10	146.	10.61	3.1	3.5	1.1
	2702	144.	6.3	100.	0.09	119.	10.52	3.1	3.1	1.0
	2703	145.	5.6	103.	0.08	106.	10.14	2.7	3.3	1.2
	2704	145.	5.3	102.	0.10	102.	10.40	2.7	3.4	1.3
	2705	146.	6.6	103.	0.08	128.	10.38	2.7	3.3	1.2
	(n)	5	5	5	5	5	5	5	5	5
	Means	144.6	6.12	101.6	0.090	120.2	10.410	2.86	3.32	1.16
2	SDDevs	1.14	0.646	1.52	0.0100	17.75	0.1775	0.219	0.148	0.114
	2711	144.	6.1	101.	0.10	126.	10.41	2.9	3.5	1.2
	2712	144.	6.5	101.	0.11	129.	10.50	2.9	3.4	1.2
	2713	145.	6.2	102.	0.09	142.	10.26	2.9	3.2	1.1
	2714	144.	6.3	103.	0.08	146.	10.19	2.8	3.3	1.2
	2715	145.	6.3	102.	0.10	122.	9.94	2.7	3.3	1.2
	(n)	5	5	5	5	5	5	5	5	5
3	Means	144.4	6.28	101.8	0.096	133.0	10.260	2.84	3.34	1.18
	SDDevs	0.55	0.148	0.84	0.0114	10.44	0.2164	0.089	0.114	0.045
3	2721	144.	7.2	102.	0.10	113.	10.46	3.2	3.5	1.1
	2722	145.	6.9	101.	0.15	141.	10.33	2.9	3.2	1.1

Serum Chemistry Summary Report with Individual Values

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males										
Group #	Animal #	Serum Chemistry								
		Na (mEq/L)	K (mEq/L)	Cl (mEq/L)	TBILI (mg/dL)	ALP (U/L)	CA (mg/dL)	GLOB (g/dL)	ALB (g/dL)	A/G
3	2723	146.	6.2	101.	0.08	130.	10.26	3.2	3.4	1.1
	2724	146.	6.2	101.	0.07	132.	10.06	2.9	3.2	1.1
	2725	145.	7.2	104.	0.09	109.	10.45	3.0	3.2	1.1
	(n)	5	5	5	5	5	5	5	5	5
	Means	145.2	6.74	101.8	0.098	125.0	10.312	3.04	3.30	1.10
	SDevs	0.84	0.508	1.30	0.0311	13.51	0.1639	0.152	0.141	0.000
4	2731	145.	6.9	104.	0.10	107.	10.23	3.3	3.2	1.0
	2732	146.	7.0	102.	0.09	151.	10.18	3.0	3.3	1.1
	2733	146.	5.7	105.	0.09	135.	10.45	3.1	3.4	1.1
	2734	146.	7.1	102.	0.09	178.	10.46	3.2	3.4	1.1
	2735	145.	6.2	102.	0.09	117.	9.82	3.0	3.3	1.1
	(n)	5	5	5	5	5	5	5	5	5
	Means	145.6	6.58	103.0	0.092	137.6	10.228	3.12	3.32	1.08
	SDevs	0.55	0.606	1.41	0.0045	28.19	0.2607	0.130	0.084	0.045

Serum Chemistry Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females

Group #	Animal #	Serum Chemistry								
		AST (U/L)	ALT (U/L)	GLU (mg/dL)	BUN (mg/dl)	CREAT (mg/dL)	PHOS (mg/dL)	TRIG (mg/dL)	CHOL (mg/dL)	TP (g/dL)
1	2706	129.	38.	108.	15.	0.43	10.4	38.	98.	6.7
	2707	79.	32.	118.	16.	0.45	9.4	32.	83.	7.1
	2708	57.	23.	95.	16.	0.44	8.9	44.	91.	6.0
	2709	87.	29.	99.	20.	0.47	9.2	31.	83.	6.6
	2710	67.	20.	105.	17.	0.42	8.4	27.	71.	6.0
	(n)	5	5	5	5	5	5	5	5	5
	Means	83.8	28.4	105.0	16.8	0.442	9.26	34.4	85.2	6.48
	SDevs	27.73	7.16	8.86	1.92	0.0192	0.740	6.66	10.11	0.476
2	2716	85.	33.	105.	15.	0.43	8.8	33.	84.	6.7
	2717	91.	47.	95.	19.	0.49	9.6	34.	110.	7.0
	2718	98.	31.	85.	20.	0.46	9.0	36.	97.	6.2
	2719	77.	38.	98.	16.	0.45	8.6	24.	87.	6.2
	2720	78.	35.	95.	14.	0.41	8.5	35.	93.	6.2
	(n)	5	5	5	5	5	5	5	5	5
	Means	85.8	36.8	95.6	16.8	0.448	8.90	32.4	94.2	6.46
	SDevs	8.87	6.26	7.20	2.59	0.0303	0.436	4.83	10.18	0.371
3	2726	85.	38.	125.	18.	0.46	9.7	34.	95.	6.9
	2727	93.	34.	108.	16.	0.49	9.3	28.	93.	6.7

Serum Chemistry Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females										
Serum Chemistry										
Group #	Animal #	AST (U/L)	ALT (U/L)	GLU (mg/dL)	BUN (mg/dl)	CREAT (mg/dL)	PHOS (mg/dL)	TRIG (mg/dL)	CHOL (mg/dL)	TP (g/dL)
3	2728	98.	33.	98.	17.	0.46	8.3	25.	77.	6.3
	2729	69.	29.	92.	14.	0.42	9.7	23.	85.	6.2
	2730	87.	31.	93.	18.	0.47	8.5	27.	82.	6.5
	(n)	5	5	5	5	5	5	5	5	5
	Means	86.4	33.0	103.2	16.6	0.460	9.10	27.4	86.4	6.52
	SDevs	10.99	3.39	13.74	1.67	0.0255	0.663	4.16	7.54	0.286
4	2736	92.	40.	107.	18.	0.40	8.9	38.	92.	7.1
	2737	104.	40.	97.	18.	0.43	9.1	28.	85.	6.5
	2738	85.	32.	92.	20.	0.52	8.7	22.	83.	6.6
	2739	88.	25.	109.	19.	0.47	8.7	28.	90.	7.0
	2740	91.	39.	100.	20.	0.46	9.0	34.	90.	6.5
	(n)	5	5	5	5	5	5	5	5	5
	Means	92.0	35.2	101.0	19.0	0.456	8.88	30.0	88.0	6.74
	SDevs	7.25	6.61	7.04	1.00	0.0451	0.179	6.16	3.81	0.288

Serum Chemistry Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females										
Serum Chemistry										
Group #	Animal #	Na (mEq/L)	K (mEq/L)	Cl (mEq/L)	TBILI (mg/dL)	ALP (U/L)	CA (mg/dL)	GLOB (g/dL)	ALB (g/dL)	A/G
1	2706	143.	7.1	101.	0.13	104.	10.97	2.9	3.8	1.3
	2707	145.	6.2	103.	0.15	124.	10.74	3.3	3.8	1.2
	2708	145.	6.2	103.	0.11	79.	10.68	2.5	3.5	1.4
	2709	144.	6.6	104.	0.11	92.	10.71	3.0	3.6	1.2
	2710	144.	6.4	103.	0.10	83.	10.26	2.6	3.4	1.3
	(n)	5	5	5	5	5	5	5	5	5
	Means	144.2	6.50	102.8	0.120	96.4	10.672	2.86	3.62	1.28
	SDevs	0.84	0.374	1.10	0.0200	18.17	0.2572	0.321	0.179	0.084
2	2716	146.	5.6	102.	0.14	80.	10.57	3.1	3.6	1.2
	2717	145.	6.6	99.	0.14	94.	11.18	3.2	3.8	1.2
	2718	143.	6.7	104.	0.11	75.	10.33	2.8	3.4	1.2
	2719	143.	6.1	102.	0.13	158.	10.53	2.7	3.5	1.3
	2720	142.	6.3	101.	0.13	100.	10.27	2.7	3.5	1.3
	(n)	5	5	5	5	5	5	5	5	5
	Means	143.8	6.26	101.6	0.130	101.4	10.576	2.90	3.56	1.24
	SDevs	1.64	0.439	1.82	0.0122	33.22	0.3609	0.235	0.152	0.055
3	2726	144.	6.2	102.	0.12	114.	11.09	3.2	3.7	1.2
	2727	146.	6.3	101.	0.17	132.	10.30	3.1	3.6	1.2

Serum Chemistry Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females										
Serum Chemistry										
Group #	Animal #	Na (mEq/L)	K (mEq/L)	Cl (mEq/L)	TBILI (mg/dL)	ALP (U/L)	CA (mg/dL)	GLOB (g/dL)	ALB (g/dL)	A/G
3	2728	145.	6.6	104.	0.11	97.	10.15	2.8	3.5	1.3
	2729	143.	6.9	101.	0.11	81.	10.89	2.8	3.4	1.2
	2730	144.	6.9	104.	0.11	87.	10.10	2.9	3.6	1.2
	(n)	5	5	5	5	5	5	5	5	5
	Means	144.4	6.58	102.4	0.124	102.2	10.506	2.96	3.56	1.22
	SDevs	1.14	0.327	1.52	0.0261	20.83	0.4535	0.182	0.114	0.045
4	2736	146.	5.8	103.	0.17	105.	10.56	3.3	3.8	1.2
	2737	147.	6.3	102.	0.15	73.	10.30	3.0	3.5	1.2
	2738	143.	6.5	104.	0.14	98.	10.06	3.0	3.6	1.2
	2739	145.	7.0	103.	0.13	100.	10.30	3.4	3.6	1.1
	2740	146.	6.5	104.	0.10	96.	10.55	3.0	3.5	1.2
	(n)	5	5	5	5	5	5	5	5	5
	Means	145.4	6.42	103.2	0.138	94.4	10.354	3.14	3.60	1.18
	SDevs	1.52	0.432	0.84	0.0259	12.42	0.2080	0.195	0.122	0.045

Quantitative Urinalysis Summary Report with Individual Values

Group	Sex	Dose Level
1	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

0437RL35.002

Quantitative Urinalysis Summary Report with Individual Values

Abbr	Description
SPGR	Specific Gravity
pH	pH

DRAFT

Quantitative Urinalysis Summary Report with Individual Values

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males

Urine - chemistry

Group #	Animal #	pH	SPGR
1	2701	7.0	1.024
	2702	6.0	1.033
	2703	6.0	1.017
	2704	7.0	1.025
	2705	6.0	1.026
	(n)	5	5
	Means	6.40	1.0250
	SDevs	0.548	0.00570
2	2711	6.0	1.026
	2712	7.0	1.018
	2713	6.0	1.033
	2714	6.0	1.023
	2715	6.0	1.032
	(n)	5	5
	Means	6.20	1.0264
	SDevs	0.447	0.00627
3	2721	7.0	1.017
	2722	6.0	1.031

Quantitative Urinalysis Summary Report with Individual Values

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Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males

Urine - chemistry

Group #	Animal #	pH	SPGR
3	2723	6.0	1.027
	2724	6.0	1.024
	2725	7.0	1.014
	(n)	5	5
	Means	6.40	1.0226
	SDevs	0.548	0.00702
4	2731	6.0	1.025
	2732	6.0	1.024
	2733	6.0	1.021
	2734	6.0	1.025
	2735	6.0	1.042
	(n)	5	5
	Means	6.00	1.0274
	SDevs	0.000	0.00832

Quantitative Urinalysis Summary Report with Individual Values

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females

Urine - chemistry

Group #	Animal #	pH	SPGR
1	2706	7.0	1.018
	2707	7.0	1.012
	2708	6.0	1.018
	2709	6.0	1.023
	2710	7.0	1.019
	(n)	5	5
	Means	6.60	1.0180
	SDevs	0.548	0.00394
2	2716	6.0	1.018
	2717	7.0	1.017
	2718	6.0	1.027
	2719	6.0	1.036
	2720	6.0	1.019
	(n)	5	5
	Means	6.20	1.0234
	SDevs	0.447	0.00808
3	2726	7.0	1.015
	2727	7.0	1.010

Quantitative Urinalysis Summary Report with Individual Values

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females

Urine - chemistry

Group #	Animal #	pH	SPGR
3	2728	7.0	1.025
	2729	6.0	1.012
	2730	6.0	1.016
	(n)	5	5
	Means	6.60	1.0156
	SDevs	0.548	0.00577
4	2736	5.0	1.046
	2737	6.0	1.014
	2738	6.0	1.010
	2739	7.0	1.011
	2740	7.0	1.017
	(n)	5	5
	Means	6.20	1.0196
	SDevs	0.837	0.01501

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males

Urine - chemistry

Group #	Animal #	pH	SPGR
1	2701	7.0	1.024
	2702	6.0	1.033
	2703	6.0	1.017
	2704	7.0	1.025
	2705	6.0	1.026
	(n)	5	5
	Means	6.40	1.0250
	SDevs	0.548	0.00570
2	2711	6.0	1.026
	2712	7.0	1.018
	2713	6.0	1.033
	2714	6.0	1.023
	2715	6.0	1.032
	(n)	5	5
	Means	6.20	1.0264
	SDevs	0.447	0.00627
3	2721	7.0	1.017
	2722	6.0	1.031

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Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Males

Urine - chemistry

Group #	Animal #	pH	SPGR
3	2723	6.0	1.027
	2724	6.0	1.024
	2725	7.0	1.014
	(n)	5	5
	Means	6.40	1.0226
	SDevs	0.548	0.00702
4	2731	6.0	1.025
	2732	6.0	1.024
	2733	6.0	1.021
	2734	6.0	1.025
	2735	6.0	1.042
	(n)	5	5
	Means	6.00	1.0274
	SDevs	0.000	0.00832

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females

Urine - chemistry

Group #	Animal #	pH	SPGR
1	2706	7.0	1.018
	2707	7.0	1.012
	2708	6.0	1.018
	2709	6.0	1.023
	2710	7.0	1.019
	(n)	5	5
	Means	6.60	1.0180
	SDevs	0.548	0.00394
2	2716	6.0	1.018
	2717	7.0	1.017
	2718	6.0	1.027
	2719	6.0	1.036
	2720	6.0	1.019
	(n)	5	5
	Means	6.20	1.0234
	SDevs	0.447	0.00808
3	2726	7.0	1.015
	2727	7.0	1.010

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Dosing Phase Day 16 (Scheduled Animal Room - Session 1)

Females

Urine - chemistry

Group #	Animal #	pH	SPGR
3	2728	7.0	1.025
	2729	6.0	1.012
	2730	6.0	1.016
	(n)	5	5
	Means	6.60	1.0156
	SDevs	0.548	0.00577
4	2736	5.0	1.046
	2737	6.0	1.014
	2738	6.0	1.010
	2739	7.0	1.011
	2740	7.0	1.017
	(n)	5	5
	Means	6.20	1.0196
	SDevs	0.837	0.01501

0437RL35.002

Gross Observations Individual Animal Report

Group	Sex	Dose Level
1	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

Gross Observations Individual Animal Report

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal		
#	Sex	Group
2701	M	1
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands	Aorta	
Brain	Epididymides	
Esophagus	Eye w/ optic nerve	
Femur w/ articular surface	Heart	
Cecum	Colon	
Rectum	Duodenum	
Jejunum	Ileum	
Kidneys	Lacrimal gland(s)	
Larynx	Liver	
Lung w/ mainstem bronchus	Lymph node, mandibular	
Lymph node, mesenteric	Mammary gland	
Nerve - sciatic	Pancreas	
Pituitary gland	Prostate gland	
Salivary glands	Seminal vesicles	
Skeletal muscle (thigh)	Skin	
Spinal cord, cervical	Spinal cord, lumbar	
Spinal cord, midthoracic	Sternum w/ bone marrow	
Stomach	Testes	
Thymus	Thyroids/parathyroids	
Tongue	Trachea	
Urinary bladder	Injection site(s)	
Abnormalities (gross)		

Gross Observations Individual Animal Report

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group
2702	M	I
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands Brain Esophagus Femur w/ articular surface Cecum Rectum Jejunum Kidneys Larynx Lung w/ mainstem bronchus Lymph node, mesenteric Nerve - sciatic Pituitary gland Salivary glands Skeletal muscle (thigh) Spinal cord, cervical Spinal cord, midthoracic Stomach Thymus Tongue Urinary bladder Abnormalities (gross)		
Aorta Epididymides Eye w/ optic nerve Heart Colon Duodenum Ileum Lacrimal gland(s) Liver Lymph node, mandibular Mammary gland Pancreas Prostate gland Seminal vesicles Skin Spinal cord, lumbar Sternum w/ bone marrow Testes Thyroids/parathyroids Trachea Injection site(s)		

Gross Observations Individual Animal Report

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal		
#	Sex	Group
2703	M	1
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands	Aorta	
Brain	Epididymides	
Esophagus	Eye w/ optic nerve	
Femur w/ articular surface	Heart	
Cecum	Colon	
Rectum	Duodenum	
Jejunum	Ileum	
Kidneys	Lacrimal gland(s)	
Larynx	Liver	
Lung w/ mainstem bronchus	Lymph node, mandibular	
Lymph node, mesenteric	Mammary gland	
Nerve - sciatic	Pancreas	
Pituitary gland	Prostate gland	
Salivary glands	Seminal vesicles	
Skeletal muscle (thigh)	Skin	
Spinal cord, cervical	Spinal cord, lumbar	
Spinal cord, midthoracic	Sternum w/ bone marrow	
Stomach	Testes	
Thymus	Thyroids/parathyroids	
Tongue	Trachea	
Urinary bladder	Injection site(s)	
Abnormalities (gross)		

Gross Observations Individual Animal Report

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal		
#	Sex	Group
2704	M	1
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands	Aorta	
Brain	Epididymides	
Esophagus	Eye w/ optic nerve	
Femur w/ articular surface	Heart	
Cecum	Colon	
Rectum	Duodenum	
Jejunum	Ileum	
Kidneys	Lacrimal gland(s)	
Larynx	Liver	
Lung w/ mainstem bronchus	Lymph node, mandibular	
Lymph node, mesenteric	Mammary gland	
Nerve - sciatic	Pancreas	
Pituitary gland	Prostate gland	
Salivary glands	Seminal vesicles	
Skeletal muscle (thigh)	Skin	
Spinal cord, cervical	Spinal cord, lumbar	
Spinal cord, midthoracic	Sternum w/ bone marrow	
Stomach	Testes	
Thymus	Thyroids/parathyroids	
Tongue	Trachea	
Urinary bladder	Injection site(s)	
Abnormalities (gross)		

Gross Observations Individual Animal Report

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group
2705	M	1
Death Status: Scheduled, Terminal Sacrifice		
Phase/Death Day: Dosing Phase, Day 16		
Death Date and Time: 22 Jul 2014 15:56:42(approximate)		
The following Tissues are Within Normal Limits:		
Adrenal glands	Aorta	
Brain	Epididymides	
Esophagus	Eye w/ optic nerve	
Femur w/ articular surface	Heart	
Cecum	Colon	
Rectum	Duodenum	
Jejunum	Ileum	
Kidneys	Lacrimal gland(s)	
Larynx	Liver	
Lung w/ mainstem bronchus	Lymph node, mandibular	
Lymph node, mesenteric	Mammary gland	
Nerve - sciatic	Pancreas	
Pituitary gland	Prostate gland	
Salivary glands	Seminal vesicles	
Skeletal muscle (thigh)	Skin	
Spinal cord, cervical	Spinal cord, lumbar	
Spinal cord, midthoracic	Sternum w/ bone marrow	
Stomach	Testes	
Thymus	Thyroids/parathyroids	
Tongue	Trachea	
Urinary bladder	Injection site(s)	
Abnormalities (gross)		

Gross Observations Individual Animal Report

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group
2711	M	2
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) Testes (Required) Decreased size - present, bilaterally, small The following Tissues are Within Normal Limits:		
Adrenal glands Brain Esophagus Femur w/ articular surface Cecum Rectum Jejunum Kidneys Larynx Lung w/ mainstem bronchus Lymph node, mesenteric Nerve - sciatic Pituitary gland Salivary glands Skeletal muscle (thigh) Spinal cord, cervical Spinal cord, midthoracic Stomach Thyroids/parathyroids Trachea		
Aorta Epididymides Eye w/ optic nerve Heart Colon Duodenum Ileum Lacrimal gland(s) Liver Lymph node, mandibular Mammary gland Pancreas Prostate gland Seminal vesicles Skin Spinal cord, lumbar Sternum w/ bone marrow Thymus Tongue Urinary bladder		

Gross Observations Individual Animal Report

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal			
#	Sex	Group	
2711	M	2	
The following Tissues are Within Normal Limits:			
Injection site(s)			Abnormalities (gross)

Gross Observations Individual Animal Report

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group
2712	M	2
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands Brain Esophagus Femur w/ articular surface Cecum Rectum Jejunum Kidneys Larynx Lung w/ mainstem bronchus Lymph node, mesenteric Nerve - sciatic Pituitary gland Salivary glands Skeletal muscle (thigh) Spinal cord, cervical Spinal cord, midthoracic Stomach Thymus Tongue Urinary bladder Abnormalities (gross)		
Aorta Epididymides Eye w/ optic nerve Heart Colon Duodenum Ileum Lacrimal gland(s) Liver Lymph node, mandibular Mammary gland Pancreas Prostate gland Seminal vesicles Skin Spinal cord, lumbar Sternum w/ bone marrow Testes Thyroids/parathyroids Trachea Injection site(s)		

Gross Observations Individual Animal Report

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal		
#	Sex	Group
2713	M	2
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands	Aorta	
Brain	Epididymides	
Esophagus	Eye w/ optic nerve	
Femur w/ articular surface	Heart	
Cecum	Colon	
Rectum	Duodenum	
Jejunum	Ileum	
Kidneys	Lacrimal gland(s)	
Larynx	Liver	
Lung w/ mainstem bronchus	Lymph node, mandibular	
Lymph node, mesenteric	Mammary gland	
Nerve - sciatic	Pancreas	
Pituitary gland	Prostate gland	
Salivary glands	Seminal vesicles	
Skeletal muscle (thigh)	Skin	
Spinal cord, cervical	Spinal cord, lumbar	
Spinal cord, midthoracic	Sternum w/ bone marrow	
Stomach	Testes	
Thymus	Thyroids/parathyroids	
Tongue	Trachea	
Urinary bladder	Injection site(s)	
Abnormalities (gross)		

Gross Observations Individual Animal Report

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group
2714	M	2
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands Brain Esophagus Femur w/ articular surface Cecum Rectum Jejunum Kidneys Larynx Lung w/ mainstem bronchus Lymph node, mesenteric Nerve - sciatic Pituitary gland Salivary glands Skeletal muscle (thigh) Spinal cord, cervical Spinal cord, midthoracic Stomach Thymus Tongue Urinary bladder Abnormalities (gross)		
Aorta Epididymides Eye w/ optic nerve Heart Colon Duodenum Ileum Lacrimal gland(s) Liver Lymph node, mandibular Mammary gland Pancreas Prostate gland Seminal vesicles Skin Spinal cord, lumbar Sternum w/ bone marrow Testes Thyroids/parathyroids Trachea Injection site(s)		

Gross Observations Individual Animal Report

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group
2715	M	2
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands Brain Esophagus Femur w/ articular surface Cecum Rectum Jejunum Kidneys Larynx Lung w/ mainstem bronchus Lymph node, mesenteric Nerve - sciatic Pituitary gland Salivary glands Skeletal muscle (thigh) Spinal cord, cervical Spinal cord, midthoracic Stomach Thymus Tongue Urinary bladder Abnormalities (gross)		
Aorta Epididymides Eye w/ optic nerve Heart Colon Duodenum Ileum Lacrimal gland(s) Liver Lymph node, mandibular Mammary gland Pancreas Prostate gland Seminal vesicles Skin Spinal cord, lumbar Sternum w/ bone marrow Testes Thyroids/parathyroids Trachea Injection site(s)		

Gross Observations Individual Animal Report

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group
2721	M	3
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands Brain Esophagus Femur w/ articular surface Cecum Rectum Jejunum Kidneys Larynx Lung w/ mainstem bronchus Lymph node, mesenteric Nerve - sciatic Pituitary gland Salivary glands Skeletal muscle (thigh) Spinal cord, cervical Spinal cord, midthoracic Stomach Thymus Tongue Urinary bladder Abnormalities (gross)		
Aorta Epididymides Eye w/ optic nerve Heart Colon Duodenum Ileum Lacrimal gland(s) Liver Lymph node, mandibular Mammary gland Pancreas Prostate gland Seminal vesicles Skin Spinal cord, lumbar Sternum w/ bone marrow Testes Thyroids/parathyroids Trachea Injection site(s)		

Gross Observations Individual Animal Report

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StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group
2722	M	3
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands Brain Esophagus Femur w/ articular surface Cecum Rectum Jejunum Kidneys Larynx Lung w/ mainstem bronchus Lymph node, mesenteric Nerve - sciatic Pituitary gland Salivary glands Skeletal muscle (thigh) Spinal cord, cervical Spinal cord, midthoracic Stomach Thymus Tongue Urinary bladder Abnormalities (gross)		
Aorta Epididymides Eye w/ optic nerve Heart Colon Duodenum Ileum Lacrimal gland(s) Liver Lymph node, mandibular Mammary gland Pancreas Prostate gland Seminal vesicles Skin Spinal cord, lumbar Sternum w/ bone marrow Testes Thyroids/parathyroids Trachea Injection site(s)		

Gross Observations Individual Animal Report

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal			
#	Sex	Group	
2723	M	3	
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:			
Adrenal glands		Aorta	
Brain		Epididymides	
Esophagus		Eye w/ optic nerve	
Femur w/ articular surface		Heart	
Cecum		Colon	
Rectum		Duodenum	
Jejunum		Ileum	
Kidneys		Lacrimal gland(s)	
Larynx		Liver	
Lung w/ mainstem bronchus		Lymph node, mandibular	
Lymph node, mesenteric		Mammary gland	
Nerve - sciatic		Pancreas	
Pituitary gland		Prostate gland	
Salivary glands		Seminal vesicles	
Skeletal muscle (thigh)		Skin	
Spinal cord, cervical		Spinal cord, lumbar	
Spinal cord, midthoracic		Sternum w/ bone marrow	
Stomach		Testes	
Thymus		Thyroids/parathyroids	
Tongue		Trachea	
Urinary bladder		Injection site(s)	
Abnormalities (gross)			

Gross Observations Individual Animal Report

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal			
#	Sex	Group	
2724	M	3	
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:			
Adrenal glands		Aorta	
Brain		Epididymides	
Esophagus		Eye w/ optic nerve	
Femur w/ articular surface		Heart	
Cecum		Colon	
Rectum		Duodenum	
Jejunum		Ileum	
Kidneys		Lacrimal gland(s)	
Larynx		Liver	
Lung w/ mainstem bronchus		Lymph node, mandibular	
Lymph node, mesenteric		Mammary gland	
Nerve - sciatic		Pancreas	
Pituitary gland		Prostate gland	
Salivary glands		Seminal vesicles	
Skeletal muscle (thigh)		Skin	
Spinal cord, cervical		Spinal cord, lumbar	
Spinal cord, midthoracic		Sternum w/ bone marrow	
Stomach		Testes	
Thymus		Thyroids/parathyroids	
Tongue		Trachea	
Urinary bladder		Injection site(s)	
Abnormalities (gross)			

Gross Observations Individual Animal Report

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Study: 0437RL35.002

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group
2725	M	3
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands Brain Esophagus Femur w/ articular surface Cecum Rectum Jejunum Kidneys Larynx Lung w/ mainstem bronchus Lymph node, mesenteric Nerve - sciatic Pituitary gland Salivary glands Skeletal muscle (thigh) Spinal cord, cervical Spinal cord, midthoracic Stomach Thymus Tongue Urinary bladder Abnormalities (gross)		
Aorta Epididymides Eye w/ optic nerve Heart Colon Duodenum Ileum Lacrimal gland(s) Liver Lymph node, mandibular Mammary gland Pancreas Prostate gland Seminal vesicles Skin Spinal cord, lumbar Sternum w/ bone marrow Testes Thyroids/parathyroids Trachea Injection site(s)		

Gross Observations Individual Animal Report

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group
2731	M	4
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands Brain Esophagus Femur w/ articular surface Cecum Rectum Jejunum Kidneys Larynx Lung w/ mainstem bronchus Lymph node, mesenteric Nerve - sciatic Pituitary gland Salivary glands Skeletal muscle (thigh) Spinal cord, cervical Spinal cord, midthoracic Stomach Thymus Tongue Urinary bladder Abnormalities (gross)		
Aorta Epididymides Eye w/ optic nerve Heart Colon Duodenum Ileum Lacrimal gland(s) Liver Lymph node, mandibular Mammary gland Pancreas Prostate gland Seminal vesicles Skin Spinal cord, lumbar Sternum w/ bone marrow Testes Thyroids/parathyroids Trachea Injection site(s)		

Gross Observations Individual Animal Report

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group
2732	M	4
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands Brain Esophagus Femur w/ articular surface Cecum Rectum Jejunum Kidneys Larynx Lung w/ mainstem bronchus Lymph node, mesenteric Nerve - sciatic Pituitary gland Salivary glands Skeletal muscle (thigh) Spinal cord, cervical Spinal cord, midthoracic Stomach Thymus Tongue Urinary bladder Abnormalities (gross)		
Aorta Epididymides Eye w/ optic nerve Heart Colon Duodenum Ileum Lacrimal gland(s) Liver Lymph node, mandibular Mammary gland Pancreas Prostate gland Seminal vesicles Skin Spinal cord, lumbar Sternum w/ bone marrow Testes Thyroids/parathyroids Trachea Injection site(s)		

Gross Observations Individual Animal Report

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group
2733	M	4
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands Brain Esophagus Femur w/ articular surface Cecum Rectum Jejunum Kidneys Larynx Lung w/ mainstem bronchus Lymph node, mesenteric Nerve - sciatic Pituitary gland Salivary glands Skeletal muscle (thigh) Spinal cord, cervical Spinal cord, midthoracic Stomach Thymus Tongue Urinary bladder Abnormalities (gross)		
Aorta Epididymides Eye w/ optic nerve Heart Colon Duodenum Ileum Lacrimal gland(s) Liver Lymph node, mandibular Mammary gland Pancreas Prostate gland Seminal vesicles Skin Spinal cord, lumbar Sternum w/ bone marrow Testes Thyroids/parathyroids Trachea Injection site(s)		

Gross Observations Individual Animal Report

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group
2734	M	4
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands Brain Esophagus Femur w/ articular surface Cecum Rectum Jejunum Kidneys Larynx Lung w/ mainstem bronchus Lymph node, mesenteric Nerve - sciatic Pituitary gland Salivary glands Skeletal muscle (thigh) Spinal cord, cervical Spinal cord, midthoracic Stomach Thymus Tongue Urinary bladder Abnormalities (gross)		
Aorta Epididymides Eye w/ optic nerve Heart Colon Duodenum Ileum Lacrimal gland(s) Liver Lymph node, mandibular Mammary gland Pancreas Prostate gland Seminal vesicles Skin Spinal cord, lumbar Sternum w/ bone marrow Testes Thyroids/parathyroids Trachea Injection site(s)		

Gross Observations Individual Animal Report

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group
2735	M	4
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands Brain Esophagus Femur w/ articular surface Cecum Rectum Jejunum Kidneys Larynx Lung w/ mainstem bronchus Lymph node, mesenteric Nerve - sciatic Pituitary gland Salivary glands Skeletal muscle (thigh) Spinal cord, cervical Spinal cord, midthoracic Stomach Thymus Tongue Urinary bladder Abnormalities (gross)		
Aorta Epididymides Eye w/ optic nerve Heart Colon Duodenum Ileum Lacrimal gland(s) Liver Lymph node, mandibular Mammary gland Pancreas Prostate gland Seminal vesicles Skin Spinal cord, lumbar Sternum w/ bone marrow Testes Thyroids/parathyroids Trachea Injection site(s)		

Gross Observations Individual Animal Report

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group
2706	F	I
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands Brain Esophagus Femur w/ articular surface Cecum Rectum Jejunum Kidneys Larynx Lung w/ mainstem bronchus Lymph node, mesenteric Ovaries Pancreas Salivary glands Skin Spinal cord, lumbar Sternum w/ bone marrow Thymus Tongue Urinary bladder Vagina Abnormalities (gross)		
Aorta Cervix Eye w/ optic nerve Heart Colon Duodenum Ileum Lacrimal gland(s) Liver Lymph node, mandibular Mammary gland Nerve - sciatic Pituitary gland Skeletal muscle (thigh) Spinal cord, cervical Spinal cord, midthoracic Stomach Thyroids/parathyroids Trachea Uterus Injection site(s)		

Gross Observations Individual Animal Report

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Study: 0437RL35.002

Study Title: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group
2707	F	1
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands Brain Esophagus Femur w/ articular surface Cecum Rectum Jejunum Kidneys Larynx Lung w/ mainstem bronchus Lymph node, mesenteric Ovaries Pancreas Salivary glands Skin Spinal cord, lumbar Sternum w/ bone marrow Thymus Tongue Urinary bladder Vagina Abnormalities (gross)		
Aorta Cervix Eye w/ optic nerve Heart Colon Duodenum Ileum Lacrimal gland(s) Liver Lymph node, mandibular Mammary gland Nerve - sciatic Pituitary gland Skeletal muscle (thigh) Spinal cord, cervical Spinal cord, midthoracic Stomach Thyroids/parathyroids Trachea Uterus Injection site(s)		

Gross Observations Individual Animal Report

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group
2708	F	I
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands Brain Esophagus Femur w/ articular surface Cecum Rectum Jejunum Kidneys Larynx Lung w/ mainstem bronchus Lymph node, mesenteric Ovaries Pancreas Salivary glands Skin Spinal cord, lumbar Sternum w/ bone marrow Thymus Tongue Urinary bladder Vagina Abnormalities (gross)		
Aorta Cervix Eye w/ optic nerve Heart Colon Duodenum Ileum Lacrimal gland(s) Liver Lymph node, mandibular Mammary gland Nerve - sciatic Pituitary gland Skeletal muscle (thigh) Spinal cord, cervical Spinal cord, midthoracic Stomach Thyroids/parathyroids Trachea Uterus Injection site(s)		

Gross Observations Individual Animal Report

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group
2709	F	I
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands Brain Esophagus Femur w/ articular surface Cecum Rectum Jejunum Kidneys Larynx Lung w/ mainstem bronchus Lymph node, mesenteric Ovaries Pancreas Salivary glands Skin Spinal cord, lumbar Sternum w/ bone marrow Thymus Tongue Urinary bladder Vagina Abnormalities (gross)		
Aorta Cervix Eye w/ optic nerve Heart Colon Duodenum Ileum Lacrimal gland(s) Liver Lymph node, mandibular Mammary gland Nerve - sciatic Pituitary gland Skeletal muscle (thigh) Spinal cord, cervical Spinal cord, midthoracic Stomach Thyroids/parathyroids Trachea Uterus Injection site(s)		

Gross Observations Individual Animal Report

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal			
#	Sex	Group	
2710	F	1	
Death Status: Scheduled, Terminal Sacrifice			
Phase/Death Day: Dosing Phase, Day 16			
Death Date and Time: 22 Jul 2014 15:56:42(approximate)			
The following Tissues are Within Normal Limits:			
Adrenal glands			Aorta
Brain			Cervix
Esophagus			Eye w/ optic nerve
Femur w/ articular surface			Heart
Cecum			Colon
Rectum			Duodenum
Jejunum			Ileum
Kidneys			Lacrimal gland(s)
Larynx			Liver
Lung w/ mainstem bronchus			Lymph node, mandibular
Lymph node, mesenteric			Mammary gland
Ovaries			Nerve - sciatic
Pancreas			Pituitary gland
Salivary glands			Skeletal muscle (thigh)
Skin			Spinal cord, cervical
Spinal cord, lumbar			Spinal cord, midthoracic
Sternum w/ bone marrow			Stomach
Thymus			Thyroids/parathyroids
Tongue			Trachea
Urinary bladder			Uterus
Vagina			Injection site(s)
Abnormalities (gross)			

Gross Observations Individual Animal Report

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal			
#	Sex	Group	
2716	F	2	
Death Status: Scheduled, Terminal Sacrifice			
Phase/Death Day: Dosing Phase, Day 16			
Death Date and Time: 22 Jul 2014 15:56:42(approximate)			
The following Tissues are Within Normal Limits:			
Adrenal glands			Aorta
Brain			Cervix
Esophagus			Eye w/ optic nerve
Femur w/ articular surface			Heart
Cecum			Colon
Rectum			Duodenum
Jejunum			Ileum
Kidneys			Lacrimal gland(s)
Larynx			Liver
Lung w/ mainstem bronchus			Lymph node, mandibular
Lymph node, mesenteric			Mammary gland
Ovaries			Nerve - sciatic
Pancreas			Pituitary gland
Salivary glands			Skeletal muscle (thigh)
Skin			Spinal cord, cervical
Spinal cord, lumbar			Spinal cord, midthoracic
Sternum w/ bone marrow			Stomach
Thymus			Thyroids/parathyroids
Tongue			Trachea
Urinary bladder			Uterus
Vagina			Injection site(s)
Abnormalities (gross)			

Gross Observations Individual Animal Report

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group
2717	F	2
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands Brain Esophagus Femur w/ articular surface Cecum Rectum Jejunum Kidneys Larynx Lung w/ mainstem bronchus Lymph node, mesenteric Ovaries Pancreas Salivary glands Skin Spinal cord, lumbar Sternum w/ bone marrow Thymus Tongue Urinary bladder Vagina Abnormalities (gross)		
Aorta Cervix Eye w/ optic nerve Heart Colon Duodenum Ileum Lacrimal gland(s) Liver Lymph node, mandibular Mammary gland Nerve - sciatic Pituitary gland Skeletal muscle (thigh) Spinal cord, cervical Spinal cord, midthoracic Stomach Thyroids/parathyroids Trachea Uterus Injection site(s)		

Gross Observations Individual Animal Report

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group
2718	F	2
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands Brain Esophagus Femur w/ articular surface Cecum Rectum Jejunum Kidneys Larynx Lung w/ mainstem bronchus Lymph node, mesenteric Ovaries Pancreas Salivary glands Skin Spinal cord, lumbar Sternum w/ bone marrow Thymus Tongue Urinary bladder Vagina Abnormalities (gross)		
Aorta Cervix Eye w/ optic nerve Heart Colon Duodenum Ileum Lacrimal gland(s) Liver Lymph node, mandibular Mammary gland Nerve - sciatic Pituitary gland Skeletal muscle (thigh) Spinal cord, cervical Spinal cord, midthoracic Stomach Thyroids/parathyroids Trachea Uterus Injection site(s)		

Gross Observations Individual Animal Report

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal		
#	Sex	Group
2719	F	2
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands	Aorta	
Brain	Cervix	
Esophagus	Eye w/ optic nerve	
Femur w/ articular surface	Heart	
Cecum	Colon	
Rectum	Duodenum	
Jejunum	Ileum	
Kidneys	Lacrimal gland(s)	
Larynx	Liver	
Lung w/ mainstem bronchus	Lymph node, mandibular	
Lymph node, mesenteric	Mammary gland	
Ovaries	Nerve - sciatic	
Pancreas	Pituitary gland	
Salivary glands	Skeletal muscle (thigh)	
Skin	Spinal cord, cervical	
Spinal cord, lumbar	Spinal cord, midthoracic	
Sternum w/ bone marrow	Stomach	
Thymus	Thyroids/parathyroids	
Tongue	Trachea	
Urinary bladder	Uterus	
Vagina	Injection site(s)	
Abnormalities (gross)		

Gross Observations Individual Animal Report

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group
2720	F	2
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands Brain Esophagus Femur w/ articular surface Cecum Rectum Jejunum Kidneys Larynx Lung w/ mainstem bronchus Lymph node, mesenteric Ovaries Pancreas Salivary glands Skin Spinal cord, lumbar Sternum w/ bone marrow Thymus Tongue Urinary bladder Vagina Abnormalities (gross)		
Aorta Cervix Eye w/ optic nerve Heart Colon Duodenum Ileum Lacrimal gland(s) Liver Lymph node, mandibular Mammary gland Nerve - sciatic Pituitary gland Skeletal muscle (thigh) Spinal cord, cervical Spinal cord, midthoracic Stomach Thyroids/parathyroids Trachea Uterus Injection site(s)		

Gross Observations Individual Animal Report

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group
2726	F	3
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands Brain Esophagus Femur w/ articular surface Cecum Rectum Jejunum Kidneys Larynx Lung w/ mainstem bronchus Lymph node, mesenteric Ovaries Pancreas Salivary glands Skin Spinal cord, lumbar Sternum w/ bone marrow Thymus Tongue Urinary bladder Vagina Abnormalities (gross)		
Aorta Cervix Eye w/ optic nerve Heart Colon Duodenum Ileum Lacrimal gland(s) Liver Lymph node, mandibular Mammary gland Nerve - sciatic Pituitary gland Skeletal muscle (thigh) Spinal cord, cervical Spinal cord, midthoracic Stomach Thyroids/parathyroids Trachea Uterus Injection site(s)		

Gross Observations Individual Animal Report

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group
2727	F	3
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands Brain Esophagus Femur w/ articular surface Cecum Rectum Jejunum Kidneys Larynx Lung w/ mainstem bronchus Lymph node, mesenteric Ovaries Pancreas Salivary glands Skin Spinal cord, lumbar Sternum w/ bone marrow Thymus Tongue Urinary bladder Vagina Abnormalities (gross)		
Aorta Cervix Eye w/ optic nerve Heart Colon Duodenum Ileum Lacrimal gland(s) Liver Lymph node, mandibular Mammary gland Nerve - sciatic Pituitary gland Skeletal muscle (thigh) Spinal cord, cervical Spinal cord, midthoracic Stomach Thyroids/parathyroids Trachea Uterus Injection site(s)		

Gross Observations Individual Animal Report

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal		
#	Sex	Group
2728	F	3
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands	Aorta	
Brain	Cervix	
Esophagus	Eye w/ optic nerve	
Femur w/ articular surface	Heart	
Cecum	Colon	
Rectum	Duodenum	
Jejunum	Ileum	
Kidneys	Lacrimal gland(s)	
Larynx	Liver	
Lung w/ mainstem bronchus	Lymph node, mandibular	
Lymph node, mesenteric	Mammary gland	
Ovaries	Nerve - sciatic	
Pancreas	Pituitary gland	
Salivary glands	Skeletal muscle (thigh)	
Skin	Spinal cord, cervical	
Spinal cord, lumbar	Spinal cord, midthoracic	
Sternum w/ bone marrow	Stomach	
Thymus	Thyroids/parathyroids	
Tongue	Trachea	
Urinary bladder	Uterus	
Vagina	Injection site(s)	
Abnormalities (gross)		

Gross Observations Individual Animal Report

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group
2729	F	3
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands Brain Esophagus Femur w/ articular surface Cecum Rectum Jejunum Kidneys Larynx Lung w/ mainstem bronchus Lymph node, mesenteric Ovaries Pancreas Salivary glands Skin Spinal cord, lumbar Sternum w/ bone marrow Thymus Tongue Urinary bladder Vagina Abnormalities (gross)		
Aorta Cervix Eye w/ optic nerve Heart Colon Duodenum Ileum Lacrimal gland(s) Liver Lymph node, mandibular Mammary gland Nerve - sciatic Pituitary gland Skeletal muscle (thigh) Spinal cord, cervical Spinal cord, midthoracic Stomach Thyroids/parathyroids Trachea Uterus Injection site(s)		

Gross Observations Individual Animal Report

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal		
#	Sex	Group
2730	F	3
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands	Aorta	
Brain	Cervix	
Esophagus	Eye w/ optic nerve	
Femur w/ articular surface	Heart	
Cecum	Colon	
Rectum	Duodenum	
Jejunum	Ileum	
Kidneys	Lacrimal gland(s)	
Larynx	Liver	
Lung w/ mainstem bronchus	Lymph node, mandibular	
Lymph node, mesenteric	Mammary gland	
Ovaries	Nerve - sciatic	
Pancreas	Pituitary gland	
Salivary glands	Skeletal muscle (thigh)	
Skin	Spinal cord, cervical	
Spinal cord, lumbar	Spinal cord, midthoracic	
Sternum w/ bone marrow	Stomach	
Thymus	Thyroids/parathyroids	
Tongue	Trachea	
Urinary bladder	Uterus	
Vagina	Injection site(s)	
Abnormalities (gross)		

Gross Observations Individual Animal Report

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group
2736	F	4
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands Brain Esophagus Femur w/ articular surface Cecum Rectum Jejunum Kidneys Larynx Lung w/ mainstem bronchus Lymph node, mesenteric Ovaries Pancreas Salivary glands Skin Spinal cord, lumbar Sternum w/ bone marrow Thymus Tongue Urinary bladder Vagina Abnormalities (gross)		
Aorta Cervix Eye w/ optic nerve Heart Colon Duodenum Ileum Lacrimal gland(s) Liver Lymph node, mandibular Mammary gland Nerve - sciatic Pituitary gland Skeletal muscle (thigh) Spinal cord, cervical Spinal cord, midthoracic Stomach Thyroids/parathyroids Trachea Uterus Injection site(s)		

Gross Observations Individual Animal Report

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal				
#	Sex	Group		
2737	F	4		
Death Status: Scheduled, Terminal Sacrifice				
Phase/Death Day: Dosing Phase, Day 16				
Death Date and Time: 22 Jul 2014 15:56:42(approximate)				
The following Tissues are Within Normal Limits:				
Adrenal glands			Aorta	
Brain			Cervix	
Esophagus			Eye w/ optic nerve	
Femur w/ articular surface			Heart	
Cecum			Colon	
Rectum			Duodenum	
Jejunum			Ileum	
Kidneys			Lacrimal gland(s)	
Larynx			Liver	
Lung w/ mainstem bronchus			Lymph node, mandibular	
Lymph node, mesenteric			Mammary gland	
Ovaries			Nerve - sciatic	
Pancreas			Pituitary gland	
Salivary glands			Skeletal muscle (thigh)	
Skin			Spinal cord, cervical	
Spinal cord, lumbar			Spinal cord, midthoracic	
Sternum w/ bone marrow			Stomach	
Thymus			Thyroids/parathyroids	
Tongue			Trachea	
Urinary bladder			Uterus	
Vagina			Injection site(s)	
Abnormalities (gross)				

Gross Observations Individual Animal Report

Calvert Labs Inc.
www.calvertlabs.com

Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal				
#	Sex	Group		
2738	F	4		
Death Status: Scheduled, Terminal Sacrifice				
Phase/Death Day: Dosing Phase, Day 16				
Death Date and Time: 22 Jul 2014 15:56:42(approximate)				
The following Tissues are Within Normal Limits:				
Adrenal glands			Aorta	
Brain			Cervix	
Esophagus			Eye w/ optic nerve	
Femur w/ articular surface			Heart	
Cecum			Colon	
Rectum			Duodenum	
Jejunum			Ileum	
Kidneys			Lacrimal gland(s)	
Larynx			Liver	
Lung w/ mainstem bronchus			Lymph node, mandibular	
Lymph node, mesenteric			Mammary gland	
Ovaries			Nerve - sciatic	
Pancreas			Pituitary gland	
Salivary glands			Skeletal muscle (thigh)	
Skin			Spinal cord, cervical	
Spinal cord, lumbar			Spinal cord, midthoracic	
Sternum w/ bone marrow			Stomach	
Thymus			Thyroids/parathyroids	
Tongue			Trachea	
Urinary bladder			Uterus	
Vagina			Injection site(s)	
Abnormalities (gross)				

Gross Observations Individual Animal Report

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group
2739	F	4
Death Status: Scheduled, Terminal Sacrifice		
Phase/Death Day: Dosing Phase, Day 16		
Death Date and Time: 22 Jul 2014 15:56:42(approximate)		
The following Tissues are Within Normal Limits:		
Adrenal glands		Aorta
Brain		Cervix
Esophagus		Eye w/ optic nerve
Femur w/ articular surface		Heart
Cecum		Colon
Rectum		Duodenum
Jejunum		Ileum
Kidneys		Lacrimal gland(s)
Larynx		Liver
Lung w/ mainstem bronchus		Lymph node, mandibular
Lymph node, mesenteric		Mammary gland
Ovaries		Nerve - sciatic
Pancreas		Pituitary gland
Salivary glands		Skeletal muscle (thigh)
Skin		Spinal cord, cervical
Spinal cord, lumbar		Spinal cord, midthoracic
Sternum w/ bone marrow		Stomach
Thymus		Thyroids/parathyroids
Tongue		Trachea
Urinary bladder		Uterus
Vagina		Injection site(s)
Abnormalities (gross)		

Gross Observations Individual Animal Report

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group
2740	F	4
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) The following Tissues are Within Normal Limits:		
Adrenal glands Brain Esophagus Femur w/ articular surface Cecum Rectum Jejunum Kidneys Larynx Lung w/ mainstem bronchus Lymph node, mesenteric Ovaries Pancreas Salivary glands Skin Spinal cord, lumbar Sternum w/ bone marrow Thymus Tongue Urinary bladder Vagina Abnormalities (gross)		
Aorta Cervix Eye w/ optic nerve Heart Colon Duodenum Ileum Lacrimal gland(s) Liver Lymph node, mandibular Mammary gland Nerve - sciatic Pituitary gland Skeletal muscle (thigh) Spinal cord, cervical Spinal cord, midthoracic Stomach Thyroids/parathyroids Trachea Uterus Injection site(s)		

0437RL35.002

Organ Weight Summary Report with Individual Values

Group	Sex	Dose Level
1	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

Organ Weight Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Males							
Group #	Animal #	TBW (g)	(AD) Adrenal glands (g)	(BR) Brain (g)	(HT) Heart (g)	(KI) Kidneys (g)	(LI) Liver (g)
1	2701	301.	0.045	1.93	1.15	1.98	8.33
	2702	307.	0.052	1.92	1.52	2.25	10.54
	2703	312.	0.054	1.92	1.27	2.00	8.86
	2704	285.	0.056	1.72	1.12	2.09	8.04
	2705	312.	0.060	1.68	1.22	2.09	8.76
	(n)	5	5	5	5	5	5
	Means	303.4	0.0534	1.834	1.256	2.082	8.906
	AdjMean	-	-	-	-	-	-
2	SDevs	11.24	0.00555	0.1232	0.1588	0.1066	0.9715
	2711	312.	0.044	1.66	1.23	2.12	9.58
	2712	312.	0.053	1.75	1.16	2.11	9.38
	2713	307.	0.057	1.83	1.20	2.14	8.85
	2714	323.	0.061	1.75	1.24	2.11	9.95
	2715	292.	0.051	1.71	1.23	2.06	7.61

Organ Weight Summary Report with Individual Values

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley				Subchronic/Intravenous Dosing			
Group #	Animal #	TBW (g)	(AD) Adrenal glands (g)	Males	(HT) Heart (g)	(KI) Kidneys (g)	(LI) Liver (g)
				(BR) Brain (g)			
2	(n)	5	5	5	5	5	5
	Means	309.2	0.0532	1.740	1.212	2.108	9.074
	AdjMean	-	-	-	-	-	-
	SDevs	11.26	0.00642	0.0624	0.0327	0.0295	0.9097
3	2721	310.	0.034	1.83	1.20	2.25	9.07
	2722	301.	0.062	1.74	1.29	2.15	7.60
	2723	319.	0.053	1.92	1.20	1.91	9.29
	2724	325.	0.054	2.04	1.33	2.26	10.79
	2725	293.	0.055	1.85	1.11	2.10	7.63
	(n)	5	5	5	5	5	5
	Means	309.6	0.0516	1.876	1.226	2.134	8.876
	AdjMean	-	-	-	-	-	-
	SDevs	12.99	0.01045	0.1119	0.0862	0.1422	1.3279
4	2731	290.	0.052	1.75	1.04	2.13	8.63
	2732	302.	0.063	1.76	1.17	2.15	8.84
	2733	317.	0.063	1.79	1.16	2.00	9.11

Organ Weight Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Group #	Animal #	TBW (g)	(AD) Adrenal glands (g)	Males	(HT) Heart (g)	(KI) Kidneys (g)	(LI) Liver (g)
				(BR) Brain (g)			
4	2734	307.	0.044	1.78	1.26	2.19	8.96
	2735	323.	0.058	1.77	1.20	2.15	9.67
	(n)	5	5	5	5	5	5
	Means	307.8	0.0560	1.770	1.166	2.124	9.042
	AdjMean	-	-	-	-	-	-
	SDevs	12.91	0.00809	0.0158	0.0805	0.0727	0.3925

Organ Weight Summary Report with Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

		Males			
Group #	Animal #	TBW (g)	(SP) Spleen (g)	(TE) Testes (g)	(TDPT) Thyroids/parathyroids
1	2701	301.	0.62	3.482	0.021
	2702	307.	0.80	3.230	0.020
	2703	312.	0.68	4.033	0.016
	2704	285.	0.68	3.482	0.016
	2705	312.	0.71	3.570	0.018
	(n)	5	5	5	5
	Means	303.4	0.698	3.5594	0.0182
	AdjMean	-	-	-	-
	SDevs	11.24	0.0657	0.29364	0.00228
2	2711	312.	0.60	0.731	0.017
	2712	312.	0.70	3.540	0.018
	2713	307.	0.67	3.475	0.018
	2714	323.	0.78	4.001	0.019
	2715	292.	0.60	3.430	0.020

Organ Weight Summary Report with Individual Values

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Group #	Animal #	TBW (g)	(SP) Spleen (g)	Males	
				(TE) Testes (g)	(TDPT) Thyroids/parathyroids
2	(n)	5	5	5	5
	Means	309.2	0.670	3.0354	0.0184
	AdjMean	-	-	-	-
	SDevs	11.26	0.0755	1.30826	0.00114
3	2721	310.	0.65	3.855	0.016
	2722	301.	0.64	3.566	0.016
	2723	319.	0.70	3.527	0.017
	2724	325.	0.73	3.739	0.024
	2725	293.	0.71	3.154	0.018
	(n)	5	5	5	5
	Means	309.6	0.686	3.5682	0.0182
	AdjMean	-	-	-	-
	SDevs	12.99	0.0391	0.26678	0.00335
4	2731	290.	0.61	3.452	0.018
	2732	302.	0.65	3.651	0.015
	2733	317.	0.72	3.948	0.018

Organ Weight Summary Report with Individual Values

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Group #	Animal #	TBW (g)	(SP) Spleen (g)	Males		(TDPT) Thyroids/parathyroids
				(TE) Testes (g)		
4	2734	307.	0.73	3.784		0.019
	2735	323.	0.76	4.190		0.018
	(n)	5	5	5		5
	Means	307.8	0.694	3.8050		0.0176
	AdjMean	-	-	-		-
	SDevs	12.91	0.0619	0.28171		0.00152

Organ Weight Summary Report with Individual Values

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley				Subchronic/Intravenous Dosing			
Group #	Animal #	TBW (g)	(AD) Adrenal glands (g)	Females	(HT) Heart (g)	(KI) Kidneys (g)	(LI) Liver (g)
				(BR) Brain (g)			
1	2706	226.	0.071	1.73	0.89	1.65	6.98
	2707	220.	0.066	1.69	0.88	1.47	5.76
	2708	221.	0.071	1.78	0.91	1.48	6.77
	2709	207.	0.079	1.61	0.92	1.47	5.95
	2710	229.	0.072	1.77	1.04	1.52	6.52
	(n)	5	5	5	5	5	5
	Means	220.6	0.0718	1.716	0.928	1.518	6.396
	AdjMean	-	-	-	-	-	-
2	SDevs	8.44	0.00466	0.0691	0.0646	0.0766	0.5243
	2716	218.	0.057	1.76	0.81	1.45	5.63
	2717	222.	0.067	1.80	0.87	1.48	5.88
	2718	213.	0.067	1.85	0.95	1.32	5.43
	2719	229.	0.064	1.70	0.83	1.47	6.05
	2720	237.	0.062	1.89	0.88	1.40	5.99

Organ Weight Summary Report with Individual Values

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley				Subchronic/Intravenous Dosing			
				Females			
Group #	Animal #	TBW (g)	(AD) Adrenal glands (g)	(BR) Brain (g)	(HT) Heart (g)	(KI) Kidneys (g)	(LI) Liver (g)
2	(n)	5	5	5	5	5	5
	Means	223.8	0.0634	1.800	0.868	1.424	5.796
	AdjMean	-	-	-	-	-	-
	SDevs	9.42	0.00416	0.0745	0.0540	0.0658	0.2602
3	2726	242.	0.088	1.75	1.19	1.62	7.38
	2727	230.	0.089	1.68	1.04	1.55	6.71
	2728	224.	0.086	1.69	0.98	1.44	6.10
	2729	229.	0.074	1.67	0.91	1.49	5.82
	2730	215.	0.062	1.75	0.80	1.43	5.42
	(n)	5	5	5	5	5	5
	Means	228.0	0.0798	1.708	0.984	1.506	6.286
	AdjMean	-	-	-	-	-	-
	SDevs	9.82	0.01163	0.0390	0.1457	0.0796	0.7711
4	2736	217.	0.068	1.81	0.89	1.36	5.66
	2737	219.	0.065	1.77	0.84	1.41	5.61
	2738	215.	0.058	1.70	0.87	1.46	5.58

Organ Weight Summary Report with Individual Values

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

				Females			
Group #	Animal #	TBW (g)	(AD) Adrenal glands (g)	(BR) Brain (g)	(HT) Heart (g)	(KI) Kidneys (g)	(LI) Liver (g)
4	2739	214.	0.066	1.67	0.89	1.49	5.89
	2740	222.	0.063	1.51	0.92	1.53	6.17
	(n)	5	5	5	5	5	5
	Means	217.4	0.0640	1.692	0.882	1.450	5.782
	AdjMean	-	-	-	-	-	-
	SDevs	3.21	0.00381	0.1158	0.0295	0.0667	0.2487

Organ Weight Summary Report with Individual Values

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Females					
Group #	Animal #	TBW (g)	(OV) Ovaries (g)	(SP) Spleen (g)	(TDPT) Thyroids/parathyroids
1	2706	226.	0.172	0.62	0.019
	2707	220.	0.123	0.51	0.016
	2708	221.	0.183	0.82	0.022
	2709	207.	0.126	0.44	0.016
	2710	229.	0.159	0.76	0.017
	(n)	5	5	5	5
	Means	220.6	0.1526	0.630	0.0180
	AdjMean	-	-	-	-
	SDevs	8.44	0.02704	0.1609	0.00255
2	2716	218.	0.105	0.59	0.014
	2717	222.	0.163	0.56	0.020
	2718	213.	0.162	0.58	0.017
	2719	229.	0.148	0.71	0.021
	2720	237.	0.128	0.61	0.018

Organ Weight Summary Report with Individual Values

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StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Group #	Animal #	TBW (g)	(OV) Ovaries (g)	Females	
				(SP) Spleen (g)	(TDPT) Thyroids/parathyroids
2	(n)	5	5	5	5
	Means	223.8	0.1412	0.610	0.0180
	AdjMean	-	-	-	-
	SDevs	9.42	0.02469	0.0587	0.00274
3	2726	242.	0.215	0.71	0.017
	2727	230.	0.204	0.69	0.019
	2728	224.	0.168	0.62	0.024
	2729	229.	0.098	0.60	0.018
	2730	215.	0.132	0.65	0.018
	(n)	5	5	5	5
	Means	228.0	0.1634	0.654	0.0192
	AdjMean	-	-	-	-
	SDevs	9.82	0.04898	0.0462	0.00277
4	2736	217.	0.149	0.52	0.018
	2737	219.	0.156	0.56	0.023
	2738	215.	0.142	0.57	0.015

Organ Weight Summary Report with Individual Values

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Group #	Animal #	TBW (g)	(OV) Ovaries (g)	Females	
				(SP) Spleen (g)	(TDPT) Thyroids/parathyroids
4	2739	214.	0.204	0.50	0.019
	2740	222.	0.201	0.62	0.017
	(n)	5	5	5	5
	Means	217.4	0.1704	0.554	0.0184
	AdjMean	-	-	-	-
	SDevs	3.21	0.02974	0.0467	0.00297

0437RL35.002

Organ Weight Summary Report with Terminal Weight Ratio Individual Values

Group	Sex	Dose Level
1	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

Organ Weight Summary Report with Terminal Weight Ratio Individual Values

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Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley				Subchronic/Intravenous Dosing			
Group #	Animal #	TBW (g)	(AD) Adrenal glands	Males	(HT) Heart	(KI) Kidneys	(LI) Liver
				(BR) Brain			
1	2701	301.	0.015	0.641	0.382	0.658	2.767
	2702	307.	0.017	0.625	0.495	0.733	3.433
	2703	312.	0.017	0.615	0.407	0.641	2.840
	2704	285.	0.020	0.604	0.393	0.733	2.821
	2705	312.	0.019	0.538	0.391	0.670	2.808
	(n)	5	5	5	5	5	5
	Means	303.4	0.0176	0.6048	0.4136	0.6870	2.9338
	SDevs	11.24	0.00190	0.03957	0.04641	0.04334	0.28043
2	2711	312.	0.014	0.532	0.394	0.679	3.071
	2712	312.	0.017	0.561	0.372	0.676	3.006
	2713	307.	0.019	0.596	0.391	0.697	2.883
	2714	323.	0.019	0.542	0.384	0.653	3.080
	2715	292.	0.017	0.586	0.421	0.705	2.606

Organ Weight Summary Report with Terminal Weight Ratio Individual Values

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley			Subchronic/Intravenous Dosing				
Group #	Animal #	TBW (g)	Males				
			(AD) Adrenal glands	(BR) Brain	(HT) Heart	(KI) Kidneys	(LI) Liver
2	(n)	5	5	5	5	5	5
	Means	309.2	0.0172	0.5633	0.3924	0.6823	2.9293
	SDevs	11.26	0.00190	0.02747	0.01826	0.02027	0.19706
3	2721	310.	0.011	0.590	0.387	0.726	2.926
	2722	301.	0.021	0.578	0.429	0.714	2.525
	2723	319.	0.017	0.602	0.376	0.599	2.912
	2724	325.	0.017	0.628	0.409	0.695	3.320
	2725	293.	0.019	0.631	0.379	0.717	2.604
	(n)	5	5	5	5	5	5
	Means	309.6	0.0167	0.6059	0.3960	0.6902	2.8574
	SDevs	12.99	0.00362	0.02323	0.02238	0.05230	0.31480
4	2731	290.	0.018	0.603	0.359	0.734	2.976
	2732	302.	0.021	0.583	0.387	0.712	2.927
	2733	317.	0.020	0.565	0.366	0.631	2.874
	2734	307.	0.014	0.580	0.410	0.713	2.919

Organ Weight Summary Report with Terminal Weight Ratio Individual Values

Calvert Labs Inc.
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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley				Subchronic/Intravenous Dosing			
Group #	Animal #	TBW (g)	(AD) Adrenal glands	Males	(HT) Heart	(KI) Kidneys	(LI) Liver
				(BR) Brain			
4	2735	323.	0.018	0.548	0.372	0.666	2.994
	(n)	5	5	5	5	5	5
	Means	307.8	0.0182	0.5757	0.3788	0.6913	2.9378
	SDevs	12.91	0.00250	0.02078	0.02062	0.04208	0.04786

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Group #	Animal #	TBW (g)	(SP) Spleen	Males	
				(TE) Testes	(TDPT) Thyroids/parathyroids
1	2701	301.	0.206	1.157	0.007
	2702	307.	0.261	1.052	0.007
	2703	312.	0.218	1.293	0.005
	2704	285.	0.239	1.222	0.006
	2705	312.	0.228	1.144	0.006
	(n)	5	5	5	5
	Means	303.4	0.2301	1.1735	0.0060
	SDevs	11.24	0.02084	0.08998	0.00074
2	2711	312.	0.192	0.234	0.005
	2712	312.	0.224	1.135	0.006
	2713	307.	0.218	1.132	0.006
	2714	323.	0.241	1.239	0.006
	2715	292.	0.205	1.175	0.007

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Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Group #	Animal #	TBW (g)	(SP) Spleen	Males	
				(TE) Testes	(TDPT) Thyroids/parathyroids
2	(n)	5	5	5	5
	Means	309.2	0.2164	0.9828	0.0060
	SDevs	11.26	0.01868	0.42067	0.00053
3	2721	310.	0.210	1.244	0.005
	2722	301.	0.213	1.185	0.005
	2723	319.	0.219	1.106	0.005
	2724	325.	0.225	1.150	0.007
	2725	293.	0.242	1.076	0.006
	(n)	5	5	5	5
	Means	309.6	0.2217	1.1522	0.0059
	SDevs	12.99	0.01290	0.06578	0.00093
4	2731	290.	0.210	1.190	0.006
	2732	302.	0.215	1.209	0.005
	2733	317.	0.227	1.245	0.006
	2734	307.	0.238	1.233	0.006

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Group #	Animal #	TBW (g)	(SP) Spleen	Males	
				(TE) Testes	(TDPT) Thyroids/parathyroids
4	2735	323.	0.235	1.297	0.006
	(n)	5	5	5	5
	Means	307.8	0.2252	1.2349	0.0057
	SDevs	12.91	0.01208	0.04080	0.00051

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Test Article: LT3114

Rat/Sprague-Dawley				Subchronic/Intravenous Dosing			
Group #	Animal #	TBW (g)	(AD) Adrenal glands	Females	(HT) Heart	(KI) Kidneys	(LI) Liver
				(BR) Brain			
1	2706	226.	0.031	0.765	0.394	0.730	3.088
	2707	220.	0.030	0.768	0.400	0.668	2.618
	2708	221.	0.032	0.805	0.412	0.670	3.063
	2709	207.	0.038	0.778	0.444	0.710	2.874
	2710	229.	0.031	0.773	0.454	0.664	2.847
	(n)	5	5	5	5	5	5
	Means	220.6	0.0326	0.7780	0.4208	0.6884	2.8983
	SDevs	8.44	0.00319	0.01606	0.02699	0.02990	0.19044
2	2716	218.	0.026	0.807	0.372	0.665	2.583
	2717	222.	0.030	0.811	0.392	0.667	2.649
	2718	213.	0.031	0.869	0.446	0.620	2.549
	2719	229.	0.028	0.742	0.362	0.642	2.642
	2720	237.	0.026	0.797	0.371	0.591	2.527

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Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Group #	Animal #	TBW (g)	(AD) Adrenal glands	Females	(HT) Heart	(KI) Kidneys	(LI) Liver
				(BR) Brain			
2	(n)	5	5	5	5	5	5
	Means	223.8	0.0284	0.8053	0.3886	0.6368	2.5900
	SDevs	9.42	0.00239	0.04488	0.03384	0.03216	0.05423
3	2726	242.	0.036	0.723	0.492	0.669	3.050
	2727	230.	0.039	0.730	0.452	0.674	2.917
	2728	224.	0.038	0.754	0.437	0.643	2.723
	2729	229.	0.032	0.729	0.397	0.651	2.541
	2730	215.	0.029	0.814	0.372	0.665	2.521
	(n)	5	5	5	5	5	5
	Means	228.0	0.0349	0.7503	0.4302	0.6604	2.7505
	SDevs	9.82	0.00425	0.03756	0.04684	0.01312	0.23154
	2736	217.	0.031	0.834	0.410	0.627	2.608
	2737	219.	0.030	0.808	0.384	0.644	2.562
4	2738	215.	0.027	0.791	0.405	0.679	2.595
	2739	214.	0.031	0.780	0.416	0.696	2.752

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

		Females					
Group #	Animal #	TBW (g)	(AD) Adrenal glands	(BR) Brain	(HT) Heart	(KI) Kidneys	(LI) Liver
4	2740	222.	0.028	0.680	0.414	0.689	2.779
	(n)	5	5	5	5	5	5
	Means	217.4	0.0294	0.7787	0.4057	0.6670	2.6594
	SDevs	3.21	0.00179	0.05872	0.01314	0.03022	0.09909

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Group #	Animal #	TBW (g)	(OV) Ovaries	Females	(TDPT) Thyroids/parathyroids
				(SP) Spleen	
1	2706	226.	0.076	0.274	0.008
	2707	220.	0.056	0.232	0.007
	2708	221.	0.083	0.371	0.010
	2709	207.	0.061	0.213	0.008
	2710	229.	0.069	0.332	0.007
	(n)	5	5	5	5
	Means	220.6	0.0690	0.2843	0.0082
	SDevs	8.44	0.01094	0.06669	0.00110
2	2716	218.	0.048	0.271	0.006
	2717	222.	0.073	0.252	0.009
	2718	213.	0.076	0.272	0.008
	2719	229.	0.065	0.310	0.009
	2720	237.	0.054	0.257	0.008

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Group #	Animal #	TBW (g)	(OV) Ovaries	Females	(TDPT) Thyroids/parathyroids
				(SP) Spleen	
2	(n)	5	5	5	5
	Means	223.8	0.0633	0.2725	0.0080
	SDevs	9.42	0.01207	0.02265	0.00112
3	2726	242.	0.089	0.293	0.007
	2727	230.	0.089	0.300	0.008
	2728	224.	0.075	0.277	0.011
	2729	229.	0.043	0.262	0.008
	2730	215.	0.061	0.302	0.008
	(n)	5	5	5	5
	Means	228.0	0.0713	0.2869	0.0084
	SDevs	9.82	0.01959	0.01713	0.00137
4	2736	217.	0.069	0.240	0.008
	2737	219.	0.071	0.256	0.011
	2738	215.	0.066	0.265	0.007
	2739	214.	0.095	0.234	0.009

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Group #	Animal #	TBW (g)	(OV) Ovaries	Females	(TDPT) Thyroids/parathyroids
				(SP) Spleen	
4	2740	222.	0.091	0.279	0.008
	(n)	5	5	5	5
	Means	217.4	0.0784	0.2547	0.0085
	SDevs	3.21	0.01353	0.01860	0.00134

0437RL35.002

Organ Weight Summary Report with Brain Weight Ratio Individual Values

Group	Sex	Dose Level
1	Male	0 mg/kg/dose
2	Male	10 mg/kg/dose
3	Male	50 mg/kg/dose
4	Male	150 mg/kg/dose
1	Female	0 mg/kg/dose
2	Female	10 mg/kg/dose
3	Female	50 mg/kg/dose
4	Female	150 mg/kg/dose

Organ Weight Summary Report with Brain Weight Ratio Individual Values

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Test Article: LT3114

Rat/Sprague-Dawley				Subchronic/Intravenous Dosing			
				Males			
Group #	Animal #	TBW (g)	(AD) Adrenal glands	(HT) Heart	(KI) Kidneys	(LI) Liver	(SP) Spleen
1	2701	301.	2.332	59.585	102.591	431.606	32.124
	2702	307.	2.708	79.167	117.188	548.958	41.667
	2703	312.	2.813	66.146	104.167	461.458	35.417
	2704	285.	3.256	65.116	121.512	467.442	39.535
	2705	312.	3.571	72.619	124.405	521.429	42.262
	(n)	5	5	5	5	5	5
	Means	303.4	2.9359	68.5267	113.9722	486.1787	38.2009
	SDevs	11.24	0.48436	7.53659	10.02132	47.76023	4.32895
2	2711	312.	2.651	74.096	127.711	577.108	36.145
	2712	312.	3.029	66.286	120.571	536.000	40.000
	2713	307.	3.115	65.574	116.940	483.607	36.612
	2714	323.	3.486	70.857	120.571	568.571	44.571
	2715	292.	2.982	71.930	120.468	445.029	35.088

Organ Weight Summary Report with Brain Weight Ratio Individual Values

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Study: 0437RL35.002

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Test Article: LT3114

Rat/Sprague-Dawley			Subchronic/Intravenous Dosing				
Males							
Group #	Animal #	TBW (g)	(AD) Adrenal glands	(HT) Heart	(KI) Kidneys	(LI) Liver	(SP) Spleen
2	(n)	5	5	5	5	5	5
	Means	309.2	3.0524	69.7486	121.2523	522.0631	38.4831
	SDevs	11.26	0.29940	3.68479	3.93231	56.56936	3.86869
3	2721	310.	1.858	65.574	122.951	495.628	35.519
	2722	301.	3.563	74.138	123.563	436.782	36.782
	2723	319.	2.760	62.500	99.479	483.854	36.458
	2724	325.	2.647	65.196	110.784	528.922	35.784
	2725	293.	2.973	60.000	113.514	412.432	38.378
	(n)	5	5	5	5	5	5
	Means	309.6	2.7603	65.4816	114.0582	471.5236	36.5844
	SDevs	12.99	0.61584	5.33760	9.91195	46.70756	1.12340
4	2731	290.	2.971	59.429	121.714	493.143	34.857
	2732	302.	3.580	66.477	122.159	502.273	36.932
	2733	317.	3.520	64.804	111.732	508.939	40.223
	2734	307.	2.472	70.787	123.034	503.371	41.011

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Test Article: LT3114

Rat/Sprague-Dawley				Subchronic/Intravenous Dosing			
				Males			
Group #	Animal #	TBW (g)	(AD) Adrenal glands	(HT) Heart	(KI) Kidneys	(LI) Liver	(SP) Spleen
4	2735	323.	3.277	67.797	121.469	546.328	42.938
	(n)	5	5	5	5	5	5
	Means	307.8	3.1639	65.8587	120.0216	510.8105	39.1923
	SDevs	12.91	0.45495	4.20950	4.67230	20.64802	3.25209

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Group #	Animal #	TBW (g)	(TE) Testes	Males	
				(TDPT)	Thyroids/parathyroids
1	2701	301.	180.415	1.088	
	2702	307.	168.229	1.042	
	2703	312.	210.052	0.833	
	2704	285.	202.442	0.930	
	2705	312.	212.500	1.071	
	(n)	5	5	5	
	Means	303.4	194.7275	0.9929	
	SDevs	11.24	19.47588	0.10839	
2	2711	312.	44.036	1.024	
	2712	312.	202.286	1.029	
	2713	307.	189.891	0.984	
	2714	323.	228.629	1.086	
	2715	292.	200.585	1.170	

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Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Group #	Animal #	TBW (g)	(TE) Testes	Males
				(TDPT) Thyroids/parathyroids
2	(n)	5	5	5
	Means	309.2	173.0852	1.0583
	SDevs	11.26	73.53579	0.07206
3	2721	310.	210.656	0.874
	2722	301.	204.943	0.920
	2723	319.	183.698	0.885
	2724	325.	183.284	1.176
	2725	293.	170.486	0.973
	(n)	5	5	5
	Means	309.6	190.6134	0.9657
	SDevs	12.99	16.68571	0.12390
4	2731	290.	197.257	1.029
	2732	302.	207.443	0.852
	2733	317.	220.559	1.006
	2734	307.	212.584	1.067

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Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Group #	Animal #	TBW (g)	(TE) Testes	Males	
				(TDPT)	Thyroids/parathyroids
4	2735	323.	236.723	1.017	
	(n)	5	5	5	
	Means	307.8	214.9133	0.9942	
	SDevs	12.91	14.83667	0.08266	

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Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Group #	Animal #	TBW (g)	(AD) Adrenal glands	Females	(KI) Kidneys	(LI) Liver	(OV) Ovaries
				(HT) Heart			
1	2706	226.	4.104	51.445	95.376	403.468	9.942
	2707	220.	3.905	52.071	86.982	340.828	7.278
	2708	221.	3.989	51.124	83.146	380.337	10.281
	2709	207.	4.907	57.143	91.304	369.565	7.826
	2710	229.	4.068	58.757	85.876	368.362	8.983
	(n)	5	5	5	5	5	5
	Means	220.6	4.1946	54.1079	88.5368	372.5121	8.8621
	SDevs	8.44	0.40546	3.56971	4.82152	22.63448	1.30159
2	2716	218.	3.239	46.023	82.386	319.886	5.966
	2717	222.	3.722	48.333	82.222	326.667	9.056
	2718	213.	3.622	51.351	71.351	293.514	8.757
	2719	229.	3.765	48.824	86.471	355.882	8.706
	2720	237.	3.280	46.561	74.074	316.931	6.772

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Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Group #	Animal #	TBW (g)	(AD) Adrenal glands	Females	(KI) Kidneys	(LI) Liver	(OV) Ovaries
				(HT) Heart			
2	(n)	5	5	5	5	5	5
	Means	223.8	3.5255	48.2184	79.3009	322.5760	7.8513
	SDevs	9.42	0.24875	2.10732	6.32403	22.41524	1.38915
3	2726	242.	5.029	68.000	92.571	421.714	12.286
	2727	230.	5.298	61.905	92.262	399.405	12.143
	2728	224.	5.089	57.988	85.207	360.947	9.941
	2729	229.	4.431	54.491	89.222	348.503	5.868
	2730	215.	3.543	45.714	81.714	309.714	7.543
	(n)	5	5	5	5	5	5
	Means	228.0	4.6778	57.6196	88.1953	368.0566	9.5561
	SDevs	9.82	0.71152	8.33330	4.68172	43.87076	2.82592
4	2736	217.	3.757	49.171	75.138	312.707	8.232
	2737	219.	3.672	47.458	79.661	316.949	8.814
	2738	215.	3.412	51.176	85.882	328.235	8.353
	2739	214.	3.952	53.293	89.222	352.695	12.216

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Test Article: LT3114

Rat/Sprague-Dawley				Subchronic/Intravenous Dosing			
Group #	Animal #	TBW (g)	(AD) Adrenal glands	Females	(KI) Kidneys	(LI) Liver	(OV) Ovaries
				(HT) Heart			
4	2740	222.	4.172	60.927	101.325	408.609	13.311
	(n)	5	5	5	5	5	5
	Means	217.4	3.7931	52.4052	86.2455	343.8391	10.1851
	SDevs	3.21	0.28736	5.24069	10.03861	39.39968	2.39520

Organ Weight Summary Report with Brain Weight Ratio Individual Values

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Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

StudyTitle: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Group #	Animal #	TBW (g)	(SP) Spleen	Females	
				(TDPT)	Thyroids/parathyroids
1	2706	226.	35.838	1.098	
	2707	220.	30.178	0.947	
	2708	221.	46.067	1.236	
	2709	207.	27.329	0.994	
	2710	229.	42.938	0.960	
	(n)	5	5	5	
	Means	220.6	36.4700	1.0470	
	SDevs	8.44	8.02329	0.12115	
2	2716	218.	33.523	0.795	
	2717	222.	31.111	1.111	
	2718	213.	31.351	0.919	
	2719	229.	41.765	1.235	
	2720	237.	32.275	0.952	

Organ Weight Summary Report with Brain Weight Ratio Individual Values

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

		Females		
		(TDPT)		
		Thyroids/parathyroids		
Group #	Animal #	TBW (g)	(SP) Spleen	
2	(n)	5	5	5
	Means	223.8	34.0050	1.0026
	SDevs	9.42	4.44000	0.17201
3	2726	242.	40.571	0.971
	2727	230.	41.071	1.131
	2728	224.	36.686	1.420
	2729	229.	35.928	1.078
	2730	215.	37.143	1.029
	(n)	5	5	5
	Means	228.0	38.2800	1.1258
	SDevs	9.82	2.36678	0.17481
4	2736	217.	28.729	0.994
	2737	219.	31.638	1.299
	2738	215.	33.529	0.882
	2739	214.	29.940	1.138

Organ Weight Summary Report with Brain Weight Ratio Individual Values

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Group #	Animal #	TBW (g)	(SP) Spleen	Females (TDPT) Thyroids/parathyroids
4	2740	222.	41.060	1.126
	(n)	5	5	5
	Means	217.4	32.9794	1.0880
	SDevs	3.21	4.86547	0.15783

Appendix II—Histopathology Report

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ANATOMIC PATHOLOGY REPORT

**Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague
Dawley Rats with Toxicokinetics (non GLP)**

Calvert Laboratories, Inc. (Calvert) Study Number 0437RL35.002

Prepared by

Brett Saladino

Calvert Laboratories, Inc.

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Testing Facility

Calvert Laboratories, Inc.

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130 Discovery Drive

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15 December 2014

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1.0 Executive Summary

Purpose: The purpose of this study was to evaluate the toxicity of LT3114 when administered intravenously to Sprague Dawley rats. Animals received three doses, one week apart, and toxicokinetics were assessed following each dose.

Methods: In accordance with the protocol, in the toxicology portion of the study, 40 Sprague Dawley rats (20 male and 20 female) were assigned to four treatment groups. Animals were dosed by slow intravenous injection on Days 1, 8, and 15 with LT3114 at 0, 10, 50, or 150 mg/kg/dose. On Day 16, all surviving animals were euthanized and subjected to a complete gross necropsy. Protocol specified tissues were collected and forwarded to Histo-Scientific Research Laboratories, Inc. (HSRL). All tissues for animals in Groups 1 and 4 were processed, embedded in paraffin, sectioned, and stained with hematoxylin and eosin (H&E). The resulting slides were forwarded to Brett Saladino, DVM, Diplomate ACVP of Calvert Laboratories for evaluation by light microscopy.

Conclusion: All animals survived to scheduled sacrifice on Day 16. There were no test article-related macroscopic observations or microscopic findings. All macroscopic and microscopic findings were considered incidental background findings because they were also noted in control animals or are known background findings in animals of this species, strain and age.

2.0 Introduction

2.1 Protocol

This report represents the histopathology results of a repeat dose toxicity study of LT3114 when administered to Sprague Dawley rats via slow intravenous injection once weekly, Calvert Study Number 0437RL35.002, sponsored by Lpath Incorporated. All in-life procedures and tissue harvests were performed at Calvert Laboratories, Inc. under the direction of Jay Pennell, M.S., Study Director. Histology processing was performed at HSRL under the direction of Craig Zook. The microscopic evaluation was completed and reported by Brett H. Saladino, DVM, Diplomate ACVP of Calvert Laboratories.

2.2 Objective

The purpose of this study was to evaluate the toxicity of LT3114 when administered intravenously to Sprague Dawley rats. Animals received three doses, one week apart, and toxicokinetics were assessed following each dose.

3.0 Methods

3.1 Compliance Statement

Calvert study number 0437RL35.002 for Lpath Incorporated. was exploratory in nature and non-regulated. The portions that were conducted at Calvert Laboratories were conducted in accordance with the appropriate Standard Operating Procedures (SOP) of Calvert, although quality assurance inspections were neither required nor performed.

3.2 Study Design

As directed by protocol, 40 Hsd:SD rats (20 male and 20 female) from Harlan Laboratories, Inc. were enrolled in the toxicology portion of this study and dosed via intravenous injection in accordance with Table 1 below:

Table 1
Group Assignments and Dose Levels: Toxicology Groups

Group	Dose Level (mg/kg/dose)*	Concentration (mg/ml)	Dose Volume (ml/kg)	Number of Animals**	
				Male	Female
1. Control (vehicle)	0	0	5	5	5
2. Low-dose	10	2	5	5	5
3. Mid-dose	50	10	5	5	5
4. High dose	150	30	5	5	5

* Dosing occurs on Day 1, Day 8, and Day 15.

** Toxicology animals were euthanized and necropsied on Day 16.

3.3 Necropsy

On Day 16, all surviving animals were euthanized by CO₂ asphyxiation following terminal blood collection and subjected to a complete gross necropsy which included examination of the external body surface, all orifices, and the cranial, thoracic and abdominal cavities and their contents. Organs (when present) were weighed before fixation, with paired organs weighed together unless gross abnormalities were present,

in which case they were weighed separately: adrenals, brain, heart, kidneys, liver, testes, ovaries, and spleen. Thyroids/parathyroids were weighed after fixation at HSRL. The following tissues from all animals were preserved in 10% neutral buffered formalin (except for the eyes, which were preserved in Davidson's fixative and the testes which were preserved in Modified Davidson's fixative):

Aorta	Lung with mainstem bronchus	Adrenals
Heart	Sternum with bone marrow	Pituitary
Salivary gland(s)	Thymus	Thyroid/Parathyroid
Tongue	Spleen	Skin
Esophagus	Lymph node – mandibular	Mammary gland
Stomach	Lymph node - mesenteric	Skeletal muscle (thigh)
Duodenum	Kidneys	Femur with articular surface
Jejunum	Urinary bladder	Eye with optic nerve
Ileum	Ovaries	Sciatic nerve
Cecum	Uterus	Brain
Colon	Cervix	Spinal cord – cervical
Rectum	Vagina	Spinal cord – midthoracic
Pancreas	Testes	Spinal cord - lumbar
Liver	Epididymides	Lacrimal glands
Trachea	Prostate	Gross findings
Larynx	Seminal vesicles	Injection site(s)

3.4 Histological Processing

Collected tissues were shipped to HSRL where protocol-specified tissues (heart, liver, lung with mainstem bronchus, spleen, kidneys, ovaries, testes, adrenals, and brain) from all animals in Groups 1 and 4 were processed, embedded in paraffin, sectioned, and stained with hematoxylin and eosin (H&E). The resulting slides were forwarded to Brett Saladino, DVM, Diplomate ACVP of Calvert Laboratories for evaluation by light microscopy.

3.5 Histopathologic Evaluation

Anatomic pathology findings were entered directly into the Xybion data collection and reporting system (Pristima Version 6.3.2 Build 20) at Calvert Laboratories. Microscopic findings are presented in Appendix A and B of this report. For non-neoplastic findings, severity was graded on a five point scale where 1=minimal, 2=mild, 3=moderate, 4=marked, and 5=severe; select findings were simply marked "present".

4.0 Results

4.1 Animal Mortality

There were no unscheduled deaths on study; all animals survived to scheduled terminal sacrifice on Day 16.

4.2 Macroscopic observations

No test article-related macroscopic observations were noted at the terminal sacrifice on Day 16. One male in Group 2 (Animal No. 2711) had bilaterally small testes that correlated microscopically with diffuse, bilateral atrophied/hypoplastic seminiferous tubules, which was likely an incidental congenital anomaly.

4.3 Microscopic findings

No test article-related microscopic observations were noted at the terminal sacrifice on Day 16. All microscopic findings were considered incidental background findings of no toxicologic significance because they were also noted in control animals or are known background findings in animals of this species, strain and age. The high incidence of findings in the kidneys of both Group 1 and Group 4 animals represent early changes associated with the condition known as chronic progressive nephropathy of rats.

5.0 Conclusion

The purpose of this study was to evaluate the toxicity of LT3114 when administered intravenously to Sprague Dawley rats. Animals received three doses, one week apart, and toxicokinetics were assessed following each dose.

All animals survived to scheduled sacrifice on Day 16. There were no test article-related macroscopic observations or microscopic findings. All macroscopic and microscopic findings were considered incidental background findings because they were also noted in control animals or are known background findings in animals of this species, strain and age.

6.0 Signature:



Brett H. Saladino, DVM, Diplomate ACVP
Principal Investigator for Pathology
Calvert Laboratories, Inc.

15 Dec 2014

Date

DRAFT

APPENDIX A

Expanded Summary Report of Histopathology Evaluation

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Study: 0437RL35.002

Study Title: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Males			
	Dosage Group:	Control	4
	Number of Animals:	5	5
	Number Examined:	5	5
	Number Unremarkable:	0	0
Adrenal glands	Number examined:	5	5
	Number unremarkable:	4	5
One medulla	Present	1	0
	Total Finding Incidence	1	0
Brain	Number examined:	5	5
	Number unremarkable:	5	5
Death comment	Number examined:	5	5
	Number unremarkable:	0	0
Scheduled terminal sacrifice	Present	5	5
	Total Finding Incidence	5	5
Heart	Number examined:	5	5
	Number unremarkable:	2	2
Infiltrate, mononuclear cell	1>	2	2
	Total Finding Incidence	2	2
Inflammation, mixed cell, subepicardial	2>	1	1

APPENDIX A

Expanded Summary Report of Histopathology Evaluation

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Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Males			
	Dosage Group:	Control	4
	Number of Animals:	5	5
	Number Examined:	5	5
	Number Unremarkable:	0	0
Heart	Number examined:	5	5
	Number unremarkable:	2	2
Inflammation, mixed cell, subepicardial	Total Finding Incidence	1	1
Kidneys	Number examined:	5	5
	Number unremarkable:	0	0
Basophilic tubules	1>	4	5
	2>	1	0
	Total Finding Incidence	5	5
Dilation, tubules	1>	2	0
	Total Finding Incidence	2	0
Fibrosis, interstitial	1>	1	0
	Total Finding Incidence	1	0
Infiltrate, mononuclear cell	1>	3	1
	Total Finding Incidence	3	1
Mineralization	1>	3	1
	Total Finding Incidence	3	1
Protein, tubular	1>	3	4
	2>	1	0
	Total Finding Incidence	4	4
Thickening, basement membrane	1>	3	2
	Total Finding Incidence	3	2
Vacuolation, tubules	1>	1	2

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Expanded Summary Report of Histopathology Evaluation

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Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Males			
	Dosage Group:	Control	4
	Number of Animals:	5	5
	Number Examined:	5	5
	Number Unremarkable:	0	0
Kidneys	Number examined:	5	5
	Number unremarkable:	0	0
Vacuolation, tubules	Total Finding Incidence	1	2
Liver	Number examined:	5	5
	Number unremarkable:	0	0
Hematopoiesis, extramedullary	1>	0	1
	Total Finding Incidence	0	1
Infiltrate, mononuclear cell	1>	5	5
	Total Finding Incidence	5	5
Necrosis	1>	1	0
	Total Finding Incidence	1	0
Lung w/ mainstem bronchus	Number examined:	5	5
	Number unremarkable:	5	5
Spleen	Number examined:	5	5
	Number unremarkable:	5	5

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Expanded Summary Report of Histopathology Evaluation

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Males			
	Dosage Group:	Control	4
	Number of Animals:	5	5
	Number Examined:	5	5
	Number Unremarkable:	0	0
Spleen	Number examined:	5	5
	Number unremarkable:	5	5
Testes	Number examined:	5	5
	Number unremarkable:	5	5

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Expanded Summary Report of Histopathology Evaluation

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Females			
Dosage Group:		Control	4
Number of Animals:		5	5
Number Examined:		5	5
Number Unremarkable:		0	0
Adrenal glands			
Number examined:		5	5
Number unremarkable:		4	5
One medulla			
Present		1	0
Total Finding Incidence		1	0
Brain			
Number examined:		5	5
Number unremarkable:		5	5
Death comment			
Number examined:		5	5
Number unremarkable:		0	0
Scheduled terminal sacrifice			
Present		5	5
Total Finding Incidence		5	5
Heart			
Number examined:		5	5
Number unremarkable:		5	5

APPENDIX A

Expanded Summary Report of Histopathology Evaluation

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Females			
Dosage Group:		Control	4
Number of Animals:		5	5
Number Examined:		5	5
Number Unremarkable:		0	0
Kidneys	Number examined:	5	5
	Number unremarkable:	1	3
Basophilic tubules	1>	4	2
	Total Finding Incidence	4	2
Mineralization	1>	3	1
	Total Finding Incidence	3	1
Protein, tubular	1>	2	1
	Total Finding Incidence	2	1
Thickening, basement membrane	1>	0	1
	Total Finding Incidence	0	1
Liver	Number examined:	5	5
	Number unremarkable:	0	2
Infiltrate, mononuclear cell	1>	5	3
	Total Finding Incidence	5	3
Lung w/ mainstem bronchus	Number examined:	5	5
	Number unremarkable:	4	5
Metaplasia, osseous	1>	1	0
	Total Finding Incidence	1	0

APPENDIX A

Expanded Summary Report of Histopathology Evaluation

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Females			
	Dosage Group:	Control	4
	Number of Animals:	5	5
	Number Examined:	5	5
	Number Unremarkable:	0	0
Ovaries	Number examined:	5	5
	Number unremarkable:	5	5
Spleen	Number examined:	5	5
	Number unremarkable:	5	5

APPENDIX B

Individual Data Listing of Histopathology Evaluation

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal			
#	Sex	Group	
2701	M	1	
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) Death comment (Required) Scheduled terminal sacrifice - Present Kidneys (Required) Basophilic tubules - minimal, multifocal Protein, tubular - minimal, multifocal Liver (Required) Infiltrate, mononuclear cell - minimal, multifocal The following Tissues are No Abnormalities Detected: Brain Testes Heart Lung w/ mainstem bronchus Spleen Adrenal glands			
2702	M	1	
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) Death comment (Required) Scheduled terminal sacrifice - Present Heart (Required) Infiltrate, mononuclear cell - minimal, base, focal Kidneys (Required) Basophilic tubules - mild, multifocal Dilation, tubules - minimal, multifocal Infiltrate, mononuclear cell - minimal, multifocal Mineralization - minimal, multifocal			

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Individual Data Listing of Histopathology Evaluation

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Test Article: LT3114

Rat/Sprague-Dawley			Subchronic/Intravenous Dosing		
Animal #	Sex	Group			
2702	M	1			
	Kidneys (Required)				
	Protein, tubular - mild, multifocal				
	Thickening, basement membrane - minimal, multifocal				
	Vacuolation, tubules - minimal, multifocal				
	Liver (Required)				
	Infiltrate, mononuclear cell - minimal, multifocal				
	The following Tissues are No Abnormalities Detected:				
	Brain	Spleen	Testes		
	Lung w/ mainstem bronchus	Adrenal glands			
2703	M	1			
	Death Status: Scheduled, Terminal Sacrifice				
	Phase/Death Day: Dosing Phase, Day 16				
	Death Date and Time: 22 Jul 2014 15:56:42(approximate)				
	Death comment (Required)				
	Scheduled terminal sacrifice - Present				
	Kidneys (Required)				
	Basophilic tubules - minimal, multifocal				
	Liver (Required)				
	Infiltrate, mononuclear cell - minimal, multifocal				
	The following Tissues are No Abnormalities Detected:				
	Brain	Heart	Spleen		
	Testes	Lung w/ mainstem bronchus	Adrenal glands		
2704	M	1			
	Death Status: Scheduled, Terminal Sacrifice				
	Phase/Death Day: Dosing Phase, Day 16				
	Death Date and Time: 22 Jul 2014 15:56:42(approximate)				

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Individual Data Listing of Histopathology Evaluation

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group			
2704	M	1	Death comment (Required)		
			Scheduled terminal sacrifice - Present		
			Heart (Required)		
			Infiltrate, mononuclear cell - minimal, multifocal		
			Kidneys (Required)		
			Basophilic tubules - minimal, multifocal		
			Infiltrate, mononuclear cell - minimal, multifocal		
			Mineralization - minimal, focal		
			Protein, tubular - minimal, multifocal		
			Thickening, basement membrane - minimal, focal		
			Liver (Required)		
			Infiltrate, mononuclear cell - minimal, multifocal		
			Necrosis - minimal, focal		
			The following Tissues are No Abnormalities Detected:		
			Brain	Spleen	Testes
			Lung w/ mainstem bronchus	Adrenal glands	
2705	M	1	Death Status: Scheduled, Terminal Sacrifice		
			Phase/Death Day: Dosing Phase, Day 16		
			Death Date and Time: 22 Jul 2014 15:56:42(approximate)		
			Adrenal glands (Required)		
			One medulla - Present		
			Death comment (Required)		
			Scheduled terminal sacrifice - Present		
			Heart (Required)		
			Inflammation, mixed cell, subepicardial - mild, right ventricle, focal		

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Individual Data Listing of Histopathology Evaluation

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group			
2705	M	1	Kidneys (Required)		
			Basophilic tubules - minimal, multifocal		
			Dilation, tubules - minimal, multifocal		
			Fibrosis, interstitial - minimal, focal		
			Infiltrate, mononuclear cell - minimal, multifocal		
			Mineralization - minimal, focal		
			Protein, tubular - minimal, multifocal		
			Thickening, basement membrane - minimal, focal		
			Liver (Required)		
			Infiltrate, mononuclear cell - minimal, multifocal		
			The following Tissues are No Abnormalities Detected:		
			Brain	Spleen	Testes
			Lung w/ mainstem bronchus		
2711	M	2	Death Status: Scheduled, Terminal Sacrifice		
			Phase/Death Day: Dosing Phase, Day 16		
			Death Date and Time: 22 Jul 2014 15:56:42(approximate)		
			Testes (Non required)		
			Atrophy/hypoplasia, tubules - marked, bilateral, diffuse		
2731	M	4	Death Status: Scheduled, Terminal Sacrifice		
			Phase/Death Day: Dosing Phase, Day 16		
			Death Date and Time: 22 Jul 2014 15:56:42(approximate)		
			Death comment (Required)		
			Scheduled terminal sacrifice - Present		

APPENDIX B

Individual Data Listing of Histopathology Evaluation

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Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group			
2731	M	4			
			Kidneys (Required)		
			Basophilic tubules - minimal, multifocal		
			Protein, tubular - minimal, multifocal		
			Vacuolation, tubules - minimal, multifocal		
			Liver (Required)		
			Infiltrate, mononuclear cell - minimal, multifocal		
			The following Tissues are No Abnormalities Detected:		
			Brain	Heart	Spleen
			Testes	Lung w/ mainstem bronchus	Adrenal glands
2732	M	4			
			Death Status: Scheduled, Terminal Sacrifice		
			Phase/Death Day: Dosing Phase, Day 16		
			Death Date and Time: 22 Jul 2014 15:56:42(approximate)		
			Death comment (Required)		
			Scheduled terminal sacrifice - Present		
			Heart (Required)		
			Infiltrate, mononuclear cell - minimal, focal		
			Kidneys (Required)		
			Basophilic tubules - minimal, multifocal		
			Mineralization - minimal, focal		
			Protein, tubular - minimal, multifocal		
			Liver (Required)		
			Infiltrate, mononuclear cell - minimal, multifocal		
			The following Tissues are No Abnormalities Detected:		
			Brain	Spleen	Testes
			Lung w/ mainstem bronchus	Adrenal glands	

APPENDIX B

Individual Data Listing of Histopathology Evaluation

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Printed: 07 Aug 2014 11:33:52

Pristima® Version 6.3.2 Build 20

Calvert Labs Inc.
www.calvertlabs.com

Study: 0437RL35.002

Study Title: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group			
2733	M	4	Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) Death comment (Required) Scheduled terminal sacrifice - Present Kidneys (Required) Basophilic tubules - minimal, multifocal Protein, tubular - minimal, multifocal Thickening, basement membrane - minimal, multifocal Vacuolation, tubules - minimal, focal Liver (Required) Hematopoiesis, extramedullary - minimal, perivascular, focal Infiltrate, mononuclear cell - minimal, multifocal The following Tissues are No Abnormalities Detected: Brain Testes Heart Lung w/ mainstem bronchus	Spleen Adrenal glands	
2734	M	4	Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) Death comment (Required) Scheduled terminal sacrifice - Present Heart (Required) Infiltrate, mononuclear cell - minimal, focal		

APPENDIX B

Individual Data Listing of Histopathology Evaluation

Calvert Labs Inc.
www.calvertlabs.com

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Printed: 07 Aug 2014 11:33:52
Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

Study Title: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley			Subchronic/Intravenous Dosing		
Animal #	Sex	Group			
2734	M	4			
		Kidneys (Required)			
		Basophilic tubules - minimal, multifocal			
		Infiltrate, mononuclear cell - minimal, multifocal			
		Liver (Required)			
		Infiltrate, mononuclear cell - minimal, multifocal			
		The following Tissues are No Abnormalities Detected:			
		Brain	Spleen	Testes	
		Lung w/ mainstem bronchus	Adrenal glands		
2735	M	4			
		Death Status: Scheduled, Terminal Sacrifice			
		Phase/Death Day: Dosing Phase, Day 16			
		Death Date and Time: 22 Jul 2014 15:56:42(approximate)			
		Death comment (Required)			
		Scheduled terminal sacrifice - Present			
		Heart (Required)			
		Inflammation, mixed cell, subepicardial - mild, right ventricle, focal			
		Kidneys (Required)			
		Basophilic tubules - minimal, multifocal			
		Protein, tubular - minimal, multifocal			
		Thickening, basement membrane - minimal, focal			
		Liver (Required)			
		Infiltrate, mononuclear cell - minimal, multifocal			
		The following Tissues are No Abnormalities Detected:			
		Brain	Spleen	Testes	
		Lung w/ mainstem bronchus	Adrenal glands		

APPENDIX B

Individual Data Listing of Histopathology Evaluation

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Pristima® Version 6.3.2 Build 20

Calvert Labs Inc.
www.calvertlabs.com

Study: 0437RL35.002

Study Title: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal			
#	Sex	Group	
2706	F	1	
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) Adrenal glands (Required) One medulla - Present Death comment (Required) Scheduled terminal sacrifice - Present Kidneys (Required) Basophilic tubules - minimal, focal Liver (Required) Infiltrate, mononuclear cell - minimal, multifocal The following Tissues are No Abnormalities Detected: Brain Ovaries Heart Lung w/ mainstem bronchus Spleen			
2707	F	1	
Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) Death comment (Required) Scheduled terminal sacrifice - Present Kidneys (Required) Basophilic tubules - minimal, multifocal Mineralization - minimal, multifocal Protein, tubular - minimal, multifocal Liver (Required) Infiltrate, mononuclear cell - minimal, multifocal			

APPENDIX B

Individual Data Listing of Histopathology Evaluation

Calvert Labs Inc.
www.calvertlabs.com

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Printed: 07 Aug 2014 11:33:52
Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

Study Title: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group			
2707	F	1	The following Tissues are No Abnormalities Detected:		
			Brain	Heart	Spleen
			Ovaries	Lung w/ mainstem bronchus	Adrenal glands
2708	F	1	Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) Death comment (Required) Scheduled terminal sacrifice - Present Liver (Required) Infiltrate, mononuclear cell - minimal, multifocal Lung w/ mainstem bronchus (Required) Metaplasia, osseous - minimal, focal The following Tissues are No Abnormalities Detected:		
			Brain	Heart	Spleen
			Ovaries	Kidneys	Adrenal glands
2709	F	1	Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) Death comment (Required) Scheduled terminal sacrifice - Present Kidneys (Required) Basophilic tubules - minimal, multifocal Mineralization - minimal, multifocal Protein, tubular - minimal, multifocal		

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Individual Data Listing of Histopathology Evaluation

Calvert Labs Inc.
www.calvertlabs.com

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Printed: 07 Aug 2014 11:33:52
Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

Study Title: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group			
2709	F	1	Liver (Required) Infiltrate, mononuclear cell - minimal, multifocal The following Tissues are No Abnormalities Detected: Brain Heart Ovaries Lung w/ mainstem bronchus	Spleen Adrenal glands	
2710	F	1	Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) Death comment (Required) Scheduled terminal sacrifice - Present Kidneys (Required) Basophilic tubules - minimal, multifocal Mineralization - minimal, multifocal Liver (Required) Infiltrate, mononuclear cell - minimal, multifocal The following Tissues are No Abnormalities Detected: Brain Heart Ovaries Lung w/ mainstem bronchus	Spleen Adrenal glands	
2736	F	4	Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) Death comment (Required) Scheduled terminal sacrifice - Present		

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Individual Data Listing of Histopathology Evaluation

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Printed: 07 Aug 2014 11:33:52

Pristima® Version 6.3.2 Build 20

Calvert Labs Inc.
www.calvertlabs.com

Study: 0437RL35.002

Study Title: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley			Subchronic/Intravenous Dosing		
Animal #	Sex	Group			
2736	F	4	The following Tissues are No Abnormalities Detected:		
			Brain	Heart	Liver
			Spleen	Ovaries	Kidneys
			Lung w/ mainstem bronchus	Adrenal glands	
2737	F	4	Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) Death comment (Required) Scheduled terminal sacrifice - Present Liver (Required) Infiltrate, mononuclear cell - minimal, multifocal The following Tissues are No Abnormalities Detected:		
			Brain	Heart	Spleen
			Ovaries	Kidneys	Lung w/ mainstem bronchus
			Adrenal glands		
2738	F	4	Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) Death comment (Required) Scheduled terminal sacrifice - Present The following Tissues are No Abnormalities Detected:		
			Brain	Heart	Liver
			Spleen	Ovaries	Kidneys
			Lung w/ mainstem bronchus	Adrenal glands	

APPENDIX B

Individual Data Listing of Histopathology Evaluation

Calvert Labs Inc.
www.calvertlabs.com

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Printed: 07 Aug 2014 11:33:52
Pristima® Version 6.3.2 Build 20

Study: 0437RL35.002

Study Title: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group									
2739	F	4	Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) Death comment (Required) Scheduled terminal sacrifice - Present Kidneys (Required) Basophilic tubules - minimal, multifocal Mineralization - minimal, multifocal Protein, tubular - minimal, multifocal Liver (Required) Infiltrate, mononuclear cell - minimal, multifocal The following Tissues are No Abnormalities Detected: <table><tr><td>Brain</td><td>Heart</td><td>Spleen</td></tr><tr><td>Ovaries</td><td>Lung w/ mainstem bronchus</td><td>Adrenal glands</td></tr></table>	Brain	Heart	Spleen	Ovaries	Lung w/ mainstem bronchus	Adrenal glands		
Brain	Heart	Spleen									
Ovaries	Lung w/ mainstem bronchus	Adrenal glands									
2740	F	4	Death Status: Scheduled, Terminal Sacrifice Phase/Death Day: Dosing Phase, Day 16 Death Date and Time: 22 Jul 2014 15:56:42(approximate) Death comment (Required) Scheduled terminal sacrifice - Present Kidneys (Required) Basophilic tubules - minimal, multifocal Thickening, basement membrane - minimal, focal Liver (Required) Infiltrate, mononuclear cell - minimal, multifocal								

APPENDIX B

Individual Data Listing of Histopathology Evaluation

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Printed: 07 Aug 2014 11:33:52

Pristima® Version 6.3.2 Build 20

Calvert Labs Inc.
www.calvertlabs.com

Study: 0437RL35.002

Study Title: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Test Article: LT3114

Rat/Sprague-Dawley

Subchronic/Intravenous Dosing

Animal #	Sex	Group
The following Tissues are No Abnormalities Detected:		
	Brain	Heart
	Ovaries	Lung w/ mainstem bronchus
		Spleen
		Adrenal glands

DRAFT

Appendix III—Test Article Information

DRAFT

Lpath Inc.		Certificate of Analysis	
TITLE	Lpathomab-DS, 30 mg/mL, for use in GLP-conforming testing		REVISION No.
			1.0.0
			PAGE

Name		Part Number		Lot Number
Lpathomab-DS, 30 mg/mL		TBD		GA121459
Formulation Buffer				Storage Temperature
24.3 mM Sodium Citrate, 51.4 mM Sodium Phosphate, 300 mM D-Trehalose, 200 ppm Polysorbate-80, pH 5.0				2 to 8°C
Production Date		Test Date	Retest Date	Intended Use
JUN 2014		JUL 2014	DEC 2014	GLP-conforming Toxicity Testing
Manufacturer		Manufacturer Part Number		Manufacturer Lot Number
Gallus-MO, St. Louis, MO		TBD		Gallus-MO 103948
Attribute	Test		Specification	Results
Identity	Appearance		Particulate free, clear, colorless to pale yellow, slightly opalescent solution	Particulate free, clear, slightly yellow solution
	pH		5.0 ± 0.2	5.1
	CGE (Non-reduced)		Major band at 150 kDa	Conforms
	CGE (Reduced)		Two pronounced bands; One at ~50 kDa; One at ~ 25 kDa	Conforms
	iCE	Comparable to RS		Conforms
		Minor Peak at pI ~8.3 Major Peak at pI ~8.4 Minor Peak at pI ~8.6 Faint Peak at pI ~8.7		15% 60% 13% 4%
Potency	API Concentration by OD at 280 nm		30 ± 3 mg/mL	32.1 mg/mL
	LPA Binding		100% ±40% of RS	89%

Lpath Inc.

Certificate of Analysis

TITLE

Lpathomab-DS, 30 mg/mL, for use in GLP-conforming testing

REVISION No.

1.0.0

PAGE

2 of 2

Attribute	Test	Specifications		Results
Purity	CGE non-reduced	Monomer + Non-glycosylated Monomer	≥ 90%	98.8%
		Non-glycosylated Monomer	Report %	2.6%
		Other impurities	No single impurity > 5%	conforms
	CGE, reduced	HC + LC + Non-Glycosylated HC	≥ 90%	99.0%
		HC	Report %	66.6%
		Non-Glycosylated HC	Report %	1.0%
		LC	Report %	31.4%
		Other Impurities	No single impurity > 5%	0.4% (high molecular weight)
	SE-HPLC	Purity [%]	≥ 90%	99.5%
		Aggregates [%]	≤ 5%	0.5%
		Other Impurities [%]	No single impurity > 5%	None detected

Attribute	Test	Specification	Results
Safety	Residual Protein-A	≤ 10 ppm	<1 ppm
	Residual DNA	≤ 5 pg/mg	<0.05 pg/mg
	CHO HCP	≤ 100 ppm	15 ppm
	Endotoxin, USP <85>	Report	<0.06 EU/mL ¹
	Bioburden; USP <61> (Membrane Filtration)	≤ 5 CFU/10 mL	None detected

JUDGMENT

Based on a comprehensive review of the lot specific manufacturing and testing records, this API is approved for use in GLP-conforming pre-clinical animal toxicity tests.

Caution! Not for use in humans!

Quality Assurance:



Acting Head of QA - Frieder K. Hofmann, PhD

Date: 22AUG2014

¹ At 0.06 EU/mL and a tolerance of 5 EU/kg animal weight, the maximum volume that may be dosed is 83 mL/kg or 2,675 mg Lpathomab per kg of animal.

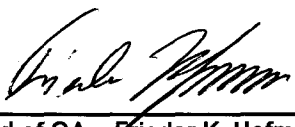
Confidential

Lpath Certificate of Analysis

Stability Assessment of Lpathomab-DS, lot 103948, stored at 5 ± 3 °C			Formulation Age							
Assay	Specification	Time 0		1 Month		2 Months		3 Months		
Appearance	Particulate-free, clear, colorless to slightly yellowish and slightly opalescent solution.	Conforms		Conforms		Conforms		Conforms		
pH	4.8 to 5.2	5.2		5.1		5.1		5.1		
ICE	Comparable to Reference Standard	Conforms		Conforms		Conforms		Conforms		
	Report pI and relative peak area of Major Isoform and total relative peak area of acidic and basic variants	pI	Area	pI	Area	pI	Area	pI	Area	
		Acidic	23%	Acidic	22%	Acidic	23%	Acidic	23%	
		8.2	53%	8.2	53%	8.2	53%	8.2	52%	
		Basic	24%	Basic	25%	Basic	24%	Basic	25%	
CGE-NR	IgG + IgG-NG ≥ 90%	98.4%		97.9%		98.8%		97.6%		
	IgG-NG	2.1%		1.8%		1.5%		2.2%		
	2H-1L	0.3%		0.4%		0.4%		0.4%		
	2H	0%		0%		0%		0%		
	HL	0%		0%		0%		0%		
	HC	0%		0%		0%		0%		
	LC	0.4%		0.4%		0.5%		0.6%		
CGE-R	HC + HC-NG + LC ≥ 90%	98.1%		95.5%		99.1%		98.1%		
	HC	66%		63%		67%		66%		
	HC-NG	1.1%		1.5%		1.1%		1.1%		
	LC	31%		31%		31%		31%		
qSE-HPLC	API concentration: 29 to 35 mg/mL	33.2		33.5%		34.3		34.5		
	Purity: ≥ 90%	99.2		98.6%		98.3%		98.2		
	Aggregates: ≥ 5%	0.8%		1.4%		1.7%		1.8		
	No single impurity > %	0%		0%		0%		0%		
LPA-binding ELISA	60% to 140% of reference standard activity	125%		106%		92%		136%		
Particulate	≤ 6,000 Particles ≥ 10 µm per vial	51						51		
	≤ 600 Particles ≥ 25 µm per vial	2						0		
Judgment										
Lot 103948 of Lpathomab-DS was placed in a validated 5 ± 3°C stability chamber at Lpath's qualified contract laboratory, Eurofins Lancaster Laboratories in Lancaster, PA, and tested monthly for three months with the stability-indicating assays appearance, pH, isoelectric focusing by capillary electrophoresis (ICE), non-reduced and reduced capillary gel electrophoresis (CGE-NR and CGE-R), quantitative size-exclusion high performance liquid chromatography (qSE-HPLC), LPA-binding ELISA and subvisible particulate matter. The obtained results and underlying raw data were reviewed and verified by Lpath's Quality Department. The results only varied within each test-specific accuracy tolerance and indicate that the lot exhibited excellent product stability under the storage conditions.										
Department	Title	Name		Signature				Date		
Quality Assurance	Head of Product Quality	Frieder K Hofmann, PhD						22DEC2014		

Lpath Inc.		Certificate of Compliance	
TITLE	24.3 mM Citrate, 51.4 mM Sodium Phosphate Dibasic, 300 mM D-Trehalose, pH 5.0 Buffer		
PAGE			1 of 1

Manufacturer				Lot Number	
Gallus Biopharmaceuticals, St. Louis, MO				104011-001	
Production Date	Expiry Dates¹				Intended Use
JUN 2014	Storage: 15 to 30°C; SEP 2014	Storage: 2 to 8°C; MAR 2015	Storage: -25 to -15°C; JUN 2016	Storage: -90°C to -60°C; JUN 2019	GLP-conforming toxicity testing and Lpathomab sample diluent
Attribute	Specification				Results
Appearance	Particulate free, clear, colorless				Conforms
pH	4.9 to 5.1				5.1
Sterility	Confirmed integrity of sterilizing membrane filter				Conforms

JUDGMENT	
Based on a comprehensive review of the executed production batch record, this buffer is approved for use in GLP-conforming toxicity testing and as sample diluent for Lpathomab. Caution – Not for use in humans	
Quality Assurance:  Acting Head of QA - Frieder K. Hofmann, PhD	Date: 18JUL2014

¹ Buffer not stored < -60°C or used for toxicity testing, requires aseptic handling.

Lpath Inc.	Certificate of Compliance
-------------------	----------------------------------

TITLE	10% (w/v) Polysorbate 80 in WFI
--------------	--

PAGE 1 of 1

Manufacturer	Lot Number
Gallus Biopharmaceuticals, St. Louis, MO	104012-001

Production Date	Expiry Dates	Intended Use
JUN 2014	DEC 2014, if stored at 15 to 30°C	GLP-conforming toxicity testing; Ingredient of Lpathomab Formulation Buffer (24.3 mM Citrate, 51.4 mM Sodium Phosphate Dibasic, 300 mM D-Trehalose, pH 5.0) at a concentration of 200 ppm

Attribute	Specification	Results
Appearance	Particulate free, clear, colorless	Conforms
Sterility	Confirmed integrity of sterilizing membrane filter	Conforms

JUDGMENT

Based on a review of the executed production batch record,
this buffer is approved as ingredient of Lpathomab formulation buffer in GLP-conforming toxicity testing

Caution – Not for use in humans

Quality Assurance:



Head of Product Quality- Frieder K. Hofmann, PhD

Date: 08JAN2015

**Appendix IV—Dose Formulation and Plasma Sample Analysis
Report**

DRAFT

STUDY PHASE REPORT

STUDY TITLE: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

REPORT TITLE: Determination of LT3114 in Dose Formulation Samples and Rat Plasma Toxicokinetic Samples

BRT Reference No.: BRT 20140723

Author: Victor J. Johnson, PhD (BRT)

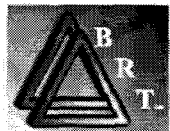
**Dose Formulation and
TK Analysis
Laboratory:** Burleson Research Technologies, Inc. (BRT)
120 First Flight Lane
Morrisville, NC 27560

Testing Facility: Calvert Laboratories, Inc.
Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447

Study Number: 0437RL35.002

Sponsor: Lpath Incorporated
4025 Sorrento Valley Blvd.
San Diego, CA, 92121

Report Date: 05 December 2014

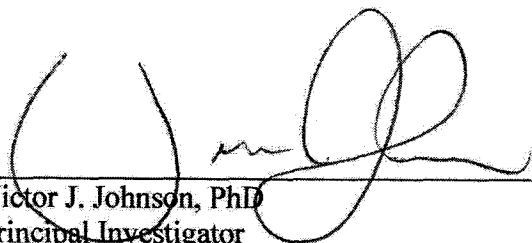


COMPLIANCE – NON-GLP STUDY

Although this study was performed as indicated in the study protocol and applicable BRT standard operating procedures (SOPs), it was investigational in nature and was not expected to conform to good laboratory practice (GLP) standards.

DRAFT

APPROVALS



Victor J. Johnson, PhD
Principal Investigator
Burleson Research Technologies, Inc.

05 Dec 14
Date

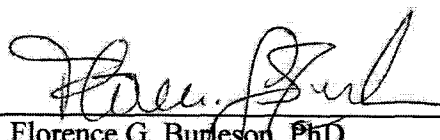


Gary R. Burleson, PhD
President (Management)
Burleson Research Technologies, Inc.

05 Dec 2014
Date

DRAFT

Report Review:



Florence G. Burleson, PhD
Director of Laboratory Operations
Burleson Research Technologies, Inc.

05 Dec 2014
Date

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STUDY INFORMATION

Determination of LT3114 in Dose Formulation Samples and Rat Plasma Toxicokinetic Samples

Study Title: Repeat Intravenous Dose Range Finding Toxicity
Study of LT3114 in Sprague Dawley Rats with
Toxicokinetics (non-GLP)

Testing Facility: Calvert Laboratories, Inc.

Testing Facility Study No: 0437RL35.002

Study Director: Jay Pennell, M.S.

Sponsor: Lpath Incorporated
4025 Sorrento Valley Blvd.
San Diego, CA, 92121

**Test Site (Dose Formulation
Analysis and Plasma TK
Analysis):** Burleson Research Technologies, Inc. (BRT)
120 First Flight Lane
Morrisville, NC 27560

Test Site Reference No.: BRT 20140723

Principal Investigator: Victor J. Johnson, PhD (BRT)

Analysts: Kristen Fisher (BRT)
Kim Barfield (BRT)

Study Initiation Date: 05 Mar 14

Sample Receipt Date: Dose Formulation Samples: 23 Jul 14
Toxicokinetic Plasma Samples: 12 Aug 14

Experimental Start Date: 18 Aug 14

Experimental Completion Date: 09 Sep 14

Study Phase Completion Date: 05 Dec 14

SUMMARY

The purpose of this study phase was to determine the concentration of LT3114 in dose formulation samples and rat plasma Toxicokinetic (TK) samples generated by Calvert Study No. 0437RL35.002: “Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)”.

Dose formulation samples were provided for three dose concentrations, 2 mg/mL, 10 mg/mL, and 30 mg/mL. Dose formulation samples were collected from each dosing solution and the vehicle from each of three independent preparations during the study. Analyses demonstrated the absence of LT3114 in the vehicle dosing solutions. The LT3114 concentration ranges were 2.1 – 2.3 mg/mL, 9.2 – 10.1 mg/mL, and 27.3 – 35.2 mg/mL for the 2, 10, and 30 mg/mL dosing solutions, respectively. The LT3114 concentrations determined for all dose formulation samples met acceptance criteria for accuracy and precision.

LT3114 was not detectable in any of the pre-dose rat plasma samples. LT3114 was detectable in all of the post-dose in rat plasma TK samples and ranged from 56.1 to 355.0 µg/mL, 235.7 to 1365.4 µg/mL, and 686.8 to 3882.8 µg/mL in rats treated with 10, 50, or 150 mg/kg, respectively.

DRAFT

INTRODUCTION

The objective of this study phase was to determine the concentration of LT3114 in dose formulation samples and rat plasma toxicokinetic (TK) samples in support of the study entitled "Repeat Intravenous Dose Range Finding Toxicity Study of L T3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)".

STUDY DESIGN

Dose Formulation Samples

Dose formulations (2 and 10 mg/mL) were prepared by Calvert on 07 Jul 14, 14 Jul 14, and 21 Jul 14 according to the study protocol using LT3114 in a citrate buffer formulation supplied by the Sponsor. Duplicate 1 mL samples were collected on 07 Jul 14, 14 Jul 14, and 21 Jul 14 from the middle of each dosing solution and were stored at 2-8°C until single aliquots from each sample were shipped to BRT and received on 23 Jul 14 (Table 1). Dose formulation samples were logged into BRT inventory and stored at 2-8°C until analyzed for LT3114 concentration. Samples were returned to 2-8°C following analysis and will be stored at BRT until authorization to discard is provided by the Study Director. Final disposition of the samples will be documented in the study records and provided to the Study Director.

Table 1: Dose formulation samples received at BRT on 23 Jul 14.

Date of Collection	Date of Receipt at BRT	Aliquots	Description
07 Jul 14	23 Jul 14	1	Vehicle
07 Jul 14	23 Jul 14	1	LT3114 Low-Dose: 2 mg/mL
07 Jul 14	23 Jul 14	1	LT3114 Mid-Dose: 10 mg/mL
07 Jul 14	23 Jul 14	1	LT3114 High-Dose: 30 mg/mL
14 Jul 14	23 Jul 14	1	Vehicle
14 Jul 14	23 Jul 14	1	LT3114 Low-Dose: 2 mg/mL
14 Jul 14	23 Jul 14	1	LT3114 Mid-Dose: 10 mg/mL
14 Jul 14	23 Jul 14	1	LT3114 High-Dose: 30 mg/mL
21 Jul 14	23 Jul 14	1	Vehicle
21 Jul 14	23 Jul 14	1	LT3114 Low-Dose: 2 mg/mL
21 Jul 14	23 Jul 14	1	LT3114 Mid-Dose: 10 mg/mL
21 Jul 14	23 Jul 14	1	LT3114 High-Dose: 30 mg/mL

Rat Plasma Toxicokinetic Samples

For the assessment of LT3114 toxicokinetics, male and female Sprague Dawley rats were administered LT3114 Citrate Drug Substance at 10, 50, or 150 mg/kg. Route of

administration was intravenous injection as a manual slow bolus injection over a period of approximately 1 minute via the lateral tail vein. Doses were administered on Days 1, 8, and 15. On Day 1, whole blood samples (Table 2) were obtained by retroorbital puncture from 3 male and 3 female rats at pre-dose, 30 minutes, and 6 hours post-dose. Subsequent blood samples were collected on Day 2 (~24 hours post Day 1 dose), Day 3 (~48 hours post Day 1 dose), Day 5 (~96 hours post Day 1 dose), Day 8 (~168 hours post Day 1 dose and prior to Day 8 dose), and Day 15 (pre Day 15 dose). Blood was collected into K₂EDTA plasma tubes and plasma was isolated by centrifugation at 3000 PRM for 10 minutes. Plasma samples (144) were stored at $\leq -70^{\circ}\text{C}$ until shipped to BRT on dry ice and received on 12 Aug 14. Plasma samples were logged into inventory at BRT and stored at $\leq -70^{\circ}\text{C}$ until analyzed for LT3114 concentration. Samples were returned to $\leq -70^{\circ}\text{C}$ following analysis and will be stored at BRT until authorization to discard is provided by the Study Director. Final disposition of the samples will be documented in the study records and provided to the Study Director.

Table 2: Rat plasma TK samples received at BRT on 12 Aug 14.

Animal #	Sex	Time Point	Group #	Dose Level (mg/kg)	Dose Concentration (mg/mL)	Dose Volume (mL/kg)
2741	M	Day 1 Pre	5	10	2	5
2743	M	Day 1 Pre	5	10	2	5
2745	M	Day 1 Pre	5	10	2	5
2747	F	Day 1 Pre	5	10	2	5
2749	F	Day 1 Pre	5	10	2	5
2751	F	Day 1 Pre	5	10	2	5
2753	M	Day 1 Pre	6	50	10	5
2755	M	Day 1 Pre	6	50	10	5
2757	M	Day 1 Pre	6	50	10	5
2759	F	Day 1 Pre	6	50	10	5
2761	F	Day 1 Pre	6	50	10	5
2763	F	Day 1 Pre	6	50	10	5
2765	M	Day 1 Pre	7	150	30	5
2767	M	Day 1 Pre	7	150	30	5
2769	M	Day 1 Pre	7	150	30	5
2771	F	Day 1 Pre	7	150	30	5
2773	F	Day 1 Pre	7	150	30	5
2775	F	Day 1 Pre	7	150	30	5
2742	M	30minPD	5	10	2	5
2744	M	30minPD	5	10	2	5
2746	M	30minPD	5	10	2	5
2748	F	30minPD	5	10	2	5
2750	F	30minPD	5	10	2	5
2752	F	30minPD	5	10	2	5
2754	M	30minPD	6	50	10	5
2756	M	30minPD	6	50	10	5
2758	M	30minPD	6	50	10	5

Animal #	Sex	Time Point	Group #	Dose Level (mg/kg)	Dose Concentration (mg/mL)	Dose Volume (mL/kg)
2760	F	30minPD	6	50	10	5
2762	F	30minPD	6	50	10	5
2764	F	30minPD	6	50	10	5
2766	M	30minPD	7	150	30	5
2768	M	30minPD	7	150	30	5
2770	M	30minPD	7	150	30	5
2772	F	30minPD	7	150	30	5
2774	F	30minPD	7	150	30	5
2776	F	30minPD	7	150	30	5
2741	M	6hr PD	5	10	2	5
2743	M	6hr PD	5	10	2	5
2745	M	6hr PD	5	10	2	5
2747	F	6hr PD	5	10	2	5
2749	F	6hr PD	5	10	2	5
2751	F	6hr PD	5	10	2	5
2753	M	6hr PD	6	50	10	5
2755	M	6hr PD	6	50	10	5
2757	M	6hr PD	6	50	10	5
2759	F	6hr PD	6	50	10	5
2761	F	6hr PD	6	50	10	5
2763	F	6hr PD	6	50	10	5
2765	M	6hr PD	7	150	30	5
2767	M	6hr PD	7	150	30	5
2769	M	6hr PD	7	150	30	5
2771	F	6hr PD	7	150	30	5
2773	F	6hr PD	7	150	30	5
2775	F	6hr PD	7	150	30	5
2742	M	24h PD	5	10	2	5
2744	M	24h PD	5	10	2	5
2746	M	24h PD	5	10	2	5
2748	F	24h PD	5	10	2	5
2750	F	24h PD	5	10	2	5
2752	F	24h PD	5	10	2	5
2754	M	24h PD	6	50	10	5
2756	M	24h PD	6	50	10	5
2758	M	24h PD	6	50	10	5
2760	F	24h PD	6	50	10	5
2762	F	24h PD	6	50	10	5
2764	F	24h PD	6	50	10	5
2766	M	24h PD	7	150	30	5
2768	M	24h PD	7	150	30	5
2770	M	24h PD	7	150	30	5
2772	F	24h PD	7	150	30	5
2774	F	24h PD	7	150	30	5
2776	F	24h PD	7	150	30	5
2741	M	48hr PD	5	10	2	5
2743	M	48hr PD	5	10	2	5
2745	M	48hr PD	5	10	2	5
2747	F	48hr PD	5	10	2	5

Animal #	Sex	Time Point	Group #	Dose Level (mg/kg)	Dose Concentration (mg/mL)	Dose Volume (mL/kg)
2749	F	48hr PD	5	10	2	5
2751	F	48hr PD	5	10	2	5
2753	M	48hr PD	6	50	10	5
2755	M	48hr PD	6	50	10	5
2757	M	48hr PD	6	50	10	5
2759	F	48hr PD	6	50	10	5
2761	F	48hr PD	6	50	10	5
2763	F	48hr PD	6	50	10	5
2765	M	48hr PD	7	150	30	5
2767	M	48hr PD	7	150	30	5
2769	M	48hr PD	7	150	30	5
2771	F	48hr PD	7	150	30	5
2773	F	48hr PD	7	150	30	5
2775	F	48hr PD	7	150	30	5
2742	M	96h PD	5	10	2	5
2744	M	96h PD	5	10	2	5
2746	M	96h PD	5	10	2	5
2748	F	96h PD	5	10	2	5
2750	F	96h PD	5	10	2	5
2752	F	96h PD	5	10	2	5
2754	M	96h PD	6	50	10	5
2756	M	96h PD	6	50	10	5
2758	M	96h PD	6	50	10	5
2760	F	96h PD	6	50	10	5
2762	F	96h PD	6	50	10	5
2764	F	96h PD	6	50	10	5
2766	M	96h PD	7	150	30	5
2768	M	96h PD	7	150	30	5
2770	M	96h PD	7	150	30	5
2772	F	96h PD	7	150	30	5
2774	F	96h PD	7	150	30	5
2776	F	96h PD	7	150	30	5
2741	M	168hr PD	5	10	2	5
2743	M	168hr PD	5	10	2	5
2745	M	168hr PD	5	10	2	5
2747	F	168hr PD	5	10	2	5
2749	F	168hr PD	5	10	2	5
2751	F	168hr PD	5	10	2	5
2753	M	168hr PD	6	50	10	5
2755	M	168hr PD	6	50	10	5
2757	M	168hr PD	6	50	10	5
2759	F	168hr PD	6	50	10	5
2761	F	168hr PD	6	50	10	5
2763	F	168hr PD	6	50	10	5
2765	M	168hr PD	7	150	30	5
2767	M	168hr PD	7	150	30	5
2769	M	168hr PD	7	150	30	5
2771	F	168hr PD	7	150	30	5
2773	F	168hr PD	7	150	30	5

Animal #	Sex	Time Point	Group #	Dose Level (mg/kg)	Dose Concentration (mg/mL)	Dose Volume (mL/kg)
2775	F	168hr PD	7	150	30	5
2742	M	Day15 Pre	5	10	2	5
2744	M	Day15 Pre	5	10	2	5
2746	M	Day15 Pre	5	10	2	5
2748	F	Day15 Pre	5	10	2	5
2750	F	Day15 Pre	5	10	2	5
2752	F	Day15 Pre	5	10	2	5
2754	M	Day15 Pre	6	50	10	5
2756	M	Day15 Pre	6	50	10	5
2758	M	Day15 Pre	6	50	10	5
2760	F	Day15 Pre	6	50	10	5
2762	F	Day15 Pre	6	50	10	5
2764	F	Day15 Pre	6	50	10	5
2766	M	Day15 Pre	7	150	30	5
2768	M	Day15 Pre	7	150	30	5

METHODOLOGY

Dose Formulation and Plasma TK Samples

Dose formulation samples (Vehicle plus 3 dose levels with 1 aliquot each from 2 preparation/collection dates; 12 total samples) were received frozen on dry ice on 23 Jul 14. Upon receipt at BRT, all samples were stored at $\leq -70^{\circ}\text{C}$. The samples were analyzed for LT3114 concentrations at BRT.

Rat plasma TK samples (8 time points * 6 animals/group/time point * 3 treatment groups = 144 total samples) were received frozen on dry ice on 12 Aug 14. Upon receipt at BRT, all samples were stored at $\leq -70^{\circ}\text{C}$. The samples were analyzed for LT3114 concentrations at BRT.

LT3114 ELISA Method

Dose formulation samples and plasma TK samples were analyzed for LT3114 concentration using an ELISA method that was transferred to BRT from the Sponsor (BRT 20140420). The ELISA method is provided in Appendix 1.

LT3114 is a human monoclonal IgG antibody specific for lysophosphatidic acid (LPA). The ELISA method for quantification of LT3114 utilized LPA conjugated to BSA as the capture reagent. High protein binding microtiter plates were coated overnight with LPA-BSA. Standards, quality control (QC) samples, dilipidized bovine serum (DBS) assay buffer, dose formulation samples, and plasma TK samples were added to the coated plates and incubated to allow for capture of the LT3114. A horse radish peroxidase (HRP)-conjugated secondary antibody specific for human IgG was added and 3,3',5,5'-

tetramethylbenzidine (TMB) used as the substrate. Quantification of substrate absorbance was performed using a microplate spectrophotometer using a 450 nm wavelength. Concentrations were extrapolated from the calibration curve and adjusted for dilution factor, if required.

Data Analysis

Data analysis was performed using SoftMax®Pro (Molecular Devices) software. Calibration curves were generated by plotting the mean OD₄₅₀ values (y-axis) versus concentration (x-axis) and fitting the curve using the 4-parameter logistic function. Computer printouts included OD₄₅₀ values (raw data), the calibration curve, the standard deviation, the coefficient of variation between replicates, and the interpolated values of the unknown test samples. Concentrations of LT3114 in test samples were obtained by multiplying the mean interpolated value (result), from duplicate wells, by the sample dilution factor. Results were reported to one decimal place.

Acceptance Criteria

Calibration Curve

- Determined by the total number of points (with no more than 2 points masked, and no 2 points from the same standard masked).
- Correlation Coefficient ≥ 0.9
- Precision (%CV) between replicates $\leq 20\%$
- Accuracy of each standard within 20% of nominal from 3.9-500 ng/mL.

Sample Acceptance

- Precision (%CV) between replicates $\leq 20\%$
- Accuracy within 20% of nominal (Dose formulation samples only)
- Samples with absorbance values outside the upper limit of quantification (ULOQ at 500 ng/mL) were retested at a higher dilution.
- Samples with absorbance values outside the lower limit of quantification (LLOQ at 5 ng/mL) were retested at a lower dilution, unless already tested at the minimum dilution (1:10).

RESULTS AND CONCLUSIONS

Dose Formulation Sample Analysis

LT3114 concentrations for the triplicate samples of each dose formulation are provided in Table 3. Analysis confirmed the absence of LT3114 in the vehicle preparations. The concentration of LT3114 in the 2 mg/mL dose formulations was 2.2, 2.3, and 2.1 mg/mL for the dose formulation samples collected on 07 Jul 14, 14, Jul 14, and 21, Jul 14, respectively. The concentration of LT3114 in the 10 mg/mL dose formulations was 9.2, 10.1, and 10.0 mg/mL for the dose formulation samples collected on 07 Jul 14, 14, Jul 14, and 21, Jul 14, respectively. The concentration of LT3114 in the 30 mg/mL dose formulations was 35.2, 33.1, and 27.3 mg/mL for the dose formulation samples collected on 07 Jul 14, 14, Jul 14, and 21, Jul 14, respectively. All dose formulation concentration met acceptance criteria for accuracy and precision.

Rat plasma TK samples were analyzed for LT3114 concentrations (Table 4) All plasma samples provided results with acceptable precision. LT3114 was not detectable in any of the pre-dose rat plasma samples. LT3114 was detectable in all of the post-dose in rat plasma TK samples and ranged from 56.1 to 355.0 µg/mL, 235.7 to 1365.4 µg/mL, and 686.8 to 3882.8 µg/mL in rats treated with 10, 50, or 150 mg/kg, respectively.

DOCUMENTATION AND RETENTION OF DATA

Description of Circumstances Affecting Data Quality or Integrity

BRT is not aware of any circumstances surrounding this study phase that would adversely affect data quality or integrity.

Maintenance of Study Records

All raw data, protocol, amendments, study records, and the final study phase report will be archived at the Test Site (BRT) for a period of at least 1 year from the date that the final report is issued unless alternative arrangements are made with the Sponsor. After that time, or as per contract agreement, the Sponsor will be notified and the data and report will be returned to the Sponsor, unless alternative arrangements are made. All raw data not specific to this study (e.g., instrument logs, CVs, etc.) will be archived at BRT.

Deviations from the Study Protocol

There were no deviations from the Study Protocol during the study phase.

TABLES

Table 1: Dose formulation samples received at BRT on 23 Jul 14.	7
Table 2: Rat plasma TK samples received at BRT on 12 Aug 14.	8
Table 3: LT3114 concentrations in dose formulation samples.	15
Table 4: LT3114 concentrations in rat plasma TK samples.	16

DRAFT

Table 3: LT3114 concentrations in dose formulation samples.

Dose Formulation	Samples	Sampling Dates	LT3114 (mg/mL)	% of Target
Vehicle	Middle (Set 1)	07 July 14	BLQ	N/A
LT3114 Low-Dose: 2mg/ml	Middle (Set 1)		2.2	110
LT3114 Mid-Dose: 10mg/ml	Middle (Set 1)		9.2	92
LT3114 High-Dose: 30mg/ml	Middle (Set 1)		35.2	117
Vehicle	Middle (Set 1)	14 July 14	BLQ	N/A
LT3114 Low-Dose: 2mg/ml	Middle (Set 1)		2.3	115
LT3114 Mid-Dose: 10mg/ml	Middle (Set 1)		10.1	101
LT3114 High-Dose: 30mg/ml	Middle (Set 1)		33.1	110
Vehicle	Middle (Set 1)	21 July 14	BLQ	N/A
LT3114 Low-Dose: 2mg/ml	Middle (Set 1)		2.1	105
LT3114 Mid-Dose: 10mg/ml	Middle (Set 1)		10.0	100
LT3114 High-Dose: 30mg/ml	Middle (Set 1)		27.3	91

BLQ – Below Lower Limit of Quantification (3.9 ng/mL); N/A – Not applicable.

DRAFT

Table 4: LT3114 concentrations in rat plasma TK samples.

Collection Time Point	Dose (mg/kg)	Group/Sex	Animal Number	LT3114 (µg/mL)
Day 1 Predose (07Jul14)	10	5 Male	2741	BLQ
		5 Male	2743	BLQ
		5 Male	2745	BLQ
		5 Female	2747	BLQ
		5 Female	2749	BLQ
		5 Female	2751	BLQ
	50	6 Male	2753	BLQ
		6 Male	2755	BLQ
		6 Male	2757	BLQ
		6 Female	2759	BLQ
		6 Female	2761	BLQ
		6 Female	2763	BLQ
	150	7 Male	2765	BLQ
		7 Male	2767	BLQ
		7 Male	2769	BLQ
		7 Female	2771	BLQ
		7 Female	2773	BLQ
		7 Female	2775	BLQ
Day 1 30 min post-Day 1 dose (07Jul14)	10	5 Male	2742	300.0
		5 Male	2744	243.6
		5 Male	2746	280.4
		5 Female	2748	248.4
		5 Female	2750	239.7
		5 Female	2752	240.6
	50	6 Male	2754	953.1
		6 Male	2756	1246.9
		6 Male	2758	1255.1
		6 Female	2760	941.7
		6 Female	2762	977.4
		6 Female	2764	922.3
	150	7 Male	2766	2180.9
		7 Male	2768	2991.8
		7 Male	2770	3074.2
		7 Female	2772	3882.8
		7 Female	2774	3146.0
		7 Female	2776	3075.1

BLQ – Below Lower Limit of Quantification (3.9 ng/mL)

Collection Time Point	Dose (mg/kg)	Group/Sex	Animal Number	LT3114 (µg/mL)
Day 1 6 Hours Post-Day 1 dose (07Jul14)	10	5 Male	2741	213.8
		5 Male	2743	341.5
		5 Male	2745	320.5
		5 Female	2747	355.0
		5 Female	2749	324.9
		5 Female	2751	319.4
	50	6 Male	2753	1365.4
		6 Male	2755	869.2
		6 Male	2757	922.8
		6 Female	2759	1126.6
		6 Female	2761	989.5
		6 Female	2763	1111.3
	150	7 Male	2765	3015.4
		7 Male	2767	3549.1
		7 Male	2769	3125.5
		7 Female	2771	2839.8
		7 Female	2773	2597.9
		7 Female	2775	2283.0
Day 2 24 Hours Post-Day 1 dose (08Jul14)	10	5 Male	2742	108.9
		5 Male	2744	123.4
		5 Male	2746	126.7
		5 Female	2748	143.8
		5 Female	2750	117.9
		5 Female	2752	136.5
	50	6 Male	2754	457.0
		6 Male	2756	542.0
		6 Male	2758	559.4
		6 Female	2760	493.8
		6 Female	2762	455.2
		6 Female	2764	421.5
	150	7 Male	2766	1270.0
		7 Male	2768	1263.7
		7 Male	2770	1297.2
		7 Female	2772	1580.3
		7 Female	2774	1347.0
		7 Female	2776	1435.7

Collection Time Point	Dose (mg/kg)	Group/Sex	Animal Number	LT3114 (µg/mL)
Day 3 48 Hours Post-Day 1 dose (09Jul14)	10	5 Male	2741	90.1
		5 Male	2743	126.5
		5 Male	2745	109.3
		5 Female	2747	120.8
		5 Female	2749	121.7
		5 Female	2751	142.4
	50	6 Male	2753	508.4
		6 Male	2755	378.7
		6 Male	2757	385.1
		6 Female	2759	509.1
		6 Female	2761	467.9
		6 Female	2763	456.1
	150	7 Male	2765	1702.9
		7 Male	2767	1405.7
		7 Male	2769	1664.2
		7 Female	2771	1251.0
		7 Female	2773	1421.0
		7 Female	2775	1165.0
Day 5 96 Hours Post-Day 1 dose (11Jul14)	10	5 Male	2742	75.5
		5 Male	2744	63.1
		5 Male	2746	63.8
		5 Female	2748	73.1
		5 Female	2750	71.8
		5 Female	2752	71.1
	50	6 Male	2754	320.5
		6 Male	2756	422.0
		6 Male	2758	420.1
		6 Female	2760	346.6
		6 Female	2762	326.5
		6 Female	2764	305.7
	150	7 Male	2766	964.8
		7 Male	2768	878.4
		7 Male	2770	1018.0
		7 Female	2772	850.0
		7 Female	2774	746.2
		7 Female	2776	894.2

Collection Time Point	Dose (mg/kg)	Group/Sex	Animal Number	LT3114 (µg/mL)
Day 8 168 Hours Post-Day 1 dose; pre-Day 8 dose (14Jul14)	10	5 Male	2741	75.5
		5 Male	2743	75.3
		5 Male	2745	88.8
		5 Female	2747	82.6
		5 Female	2749	90.5
		5 Female	2751	91.1
	50	6 Male	2753	264.0
		6 Male	2755	235.7
		6 Male	2757	257.0
		6 Female	2759	257.7
		6 Female	2761	273.5
		6 Female	2763	271.1
	150	7 Male	2765	874.0
		7 Male	2767	686.8
		7 Male	2769	776.1
		7 Female	2771	801.9
		7 Female	2773	744.1
		7 Female	2775	988.3
Day 15 Pre-dose (21Jul14)	10	5 Male	2742	99.9
		5 Male	2744	103.0
		5 Male	2746	56.1
		5 Female	2748	136.0
		5 Female	2750	130.8
		5 Female	2752	137.2
	50	6 Male	2754	543.2
		6 Male	2756	580.0
		6 Male	2758	549.0
		6 Female	2760	757.3
		6 Female	2762	605.6
		6 Female	2764	541.3
	150	7 Male	2766	1408.8
		7 Male	2768	1366.5
		7 Male	2770	1319.7
		7 Female	2772	1222.7
		7 Female	2774	1332.6
		7 Female	2776	1317.0

APPENDIX 1: ELISA METHOD FOR QUANTIFICATION OF LT3114.

Materials and Equipment

- LPA-BSA Conjugated Provided by Lpath
- Human Reference Anti-LPA IgG (Provided by Lpath; LT3114)
- Rat Plasma
- 0.05 M Carbonate Buffer, pH 9.6 (Sigma C3041, or equivalent)
- Phosphate Buffered Saline (PBS, 1X) (Gibco 10010, or equivalent)
- Polyoxyethylene sorbitan monolaurate (Tween-20) (Sigma-Aldrich – Cat # P-1379 or equivalent)
- Wash solution (PBS, 0.1% Tween-20)
- Blocking Solution (PBS, 0.1% Tween-20, 1% BSA)
- Delipidized Bovine serum (DBS); assay diluent (Biocell cat 1231-00 or equivalent)
- Albumin: Bovine Serum, Fraction V, Fatty Acid-Free (Cat # 126575, CALBIOCHEM phone # 858-450-5558)
- Peroxidase Conjugated anti-human IgG (Jackson 709-035-149)
- TMB substrate (3,3',5,5'-Tetramethylbenzidine) liquids substrate system for ELISA (Sigma T0440, or equivalent)
- 1M H₂SO₄ (Mallinkrodt H381, or equivalent)
- Microtiter plates for ELISA: 96 well flat bottom, high binding polystyrene (Greiner 655001, or equivalent)
- Cluster tubes (VWR 29442-610, or equivalent)
- Plate Sealers
- Disposable polypropylene containers: 15 mL, 50 mL, 100 mL
- Weighing Boats
- Pipette tips for pipettors
- Sterile 0.2 µm filter units (VWR 87006-066, or equivalent)
- Disposable serological pipettes: 5 mL, 10 mL, 25 mL, 50 mL, 100 mL
- Micropipettors
- Microtiter plate washer
- Bench top vortex mixer
- ELISA microtiter plate reader

Solutions and Storage

1. LPA-Conjugated-BSA (provided by the Sponsor)
 - 1.1 LPA-Conjugated-BSA coating material should be kept frozen until use.
 - 1.2 After dilution, LPA-Conjugated-BSA coating material should be stored at 2-8° C for up to 4 months
2. Peroxidase conjugated-anti-human IgG secondary antibody
 - 2.1 Reconstitute following vendor instructions, Dilute in glycerol (50%)

- 2.2 Store at -20° C for up to 1 year
3. Carbonate Buffer (0.05 M Carbonate Buffer pH 9.6)
 - 3.1 Add 1 tablet per 100 mL of deionized water, mix thoroughly
 - 3.2 Store at 2-8°C for up to 3 months
4. Blocking Solution 1xPBS, 1% BSA 1, 0.1% Tween 20
 - 4.1 Dissolve 10 g BSA and 1 mL Tween-20 in 999 mL PBS (1X)
 - 4.2 Mix thoroughly and sterile filter
 - 4.3 Store at 2-8°C for up to 1 month

ELISA Method

1. Coating Plates
 - 1.1 Retrieve the appropriate frozen aliquot of LPA-Conjugated-BSA that was stored at $\leq -70^{\circ}\text{C}$. Thaw the aliquot at room temperature.
 - 1.2 Dilute LPA-Conjugated-BSA coating material to 1 $\mu\text{g/ml}$ in 0.05 M Carbonate Buffer.
 - 1.3 Coat a 96-well high-binding microtiter plate(s) with 100 μL /well coating solution.
 - 1.4 Cover the plate(s) with plate sealers and incubate plate overnight at 2-8°C.
2. Bring all materials and plate(s) to room temperature before starting assay (except TMB).
3. Blocking Plates
 - 3.1 Wash coated plate(s) 4 times with wash solution.
 - 3.2 Dispense 150 μL /well of blocking solution to each well and cover the plate(s) with plate sealers.
 - 3.3 Incubate plate(s) at 37°C for 1 hour $\pm$ 5 minutes.
4. Prepare Samples/Reference Controls/Blanks while blocking plate(s)
 - 4.1 Retrieve samples from storage and allow them to thaw at room temperature. (Note: Minimize freeze thaw cycles)
 - 4.2 Prepare unknown samples by diluting in DBS to appropriate dilutions.
 - 4.3 Prepare Human Reference Anti-LPA IgG 12 point standard curve by diluting in DBS to achieve desired range (0 – 4,000 ng/ml)
 - 4.4 Wash plate(s) 4 times (300 μL /well) with wash solution.
 - 4.5 Add samples (controls, calibrators and unknowns) at 100 μL /well.

- 5 Add samples to appropriate wells
 - 5.1 Dispense 100 µl of appropriate sample to designated wells in duplicate.
Add 100 µl of DBS to assay blank wells.
 - 5.2 Cover the plate(s) with plate sealers and incubate plate at 37°C for 1 hour ± 5 minutes.
- 6 Add HRP-goat-anti-human secondary antibody solution.
 - 6.1 Wash plate(s) 4 times (300µl/well) with wash solution.
 - 6.2 Create the working dilution of 1:40,000 of HRP-goat-anti-human secondary antibody
 - 6.2.1 Dilute HRP conjugated anti-human IgG with blocking solution to 1:20 from the stock (50% glycerol), creating a 1:40 dilution.
 - 6.2.2 Using 1:40 dilution, further dilute to a 1:1,000 final dilution in blocking solution.
 - 6.3 Dispense 100 µl of diluted HRP-goat-anti-human secondary antibody to each well.
 - 6.4 Cover the plate(s) with plate sealers and incubate plate at 37°C for 1 hour ± 5 minutes.
- 7 Substrate Addition
 - 7.1 Wash coated plate(s) 4 times with wash solution.
 - 7.2 Dispense 100 µl **COLD** TMB solution to each well.
 - 7.3 Cover the plate(s) with plate sealers and incubate for 5-30 minutes protected from light.
- 8 Stop Solution Addition
 - 8.1 Dispense 100 µl 1.0 M H₂SO₄ solution to each well.
 - 8.2 Measure absorbance at 450 nm on microplate reader

Appendix V—Toxicokinetic Report

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Toxicokinetic Final Report

TITLE: Repeat Intravenous Dose Range Finding Toxicity Study of
LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

**Calvert
Study No.:** 0437RL35.002

**Testing
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II. List of Abbreviations

°C	degree Celsius
µg	Microgram
µg/ml	Microgram per milliliter
µg*hr/ml	Microgram x hour per milliliter
%	Percent
%RSD	Percent relative standard deviation
AUC _{0-last}	Area under the curve from time zero to the time of the last quantifiable concentration
AUC _{0-inf}	Area under the curve from time zero to infinity
BLQ	Below the limit of quantitation
C ₀	Extrapolated concentration at time zero
C _{last}	Last quantifiable concentration
Cl	Apparent plasma clearance
C _{max}	Maximum observed concentration; for intravenous bolus dosing the C _{max} is considered to occur at the first sampling time after dose administration
Conc	Concentration
F	Female
g	Gram
Grp	Group
hr	Hour

IV	Intravenous
K ₂ EDTA	Dipotassium ethylenediaminetetraacetic acid
kg	Kilogram
M	Male
mg	Milligram
mg/kg	Milligram per kilogram
mg/kg/dose	Milligram per kilogram per dose
mg/ml	Milligram per milliliter
min	Minute
ml	Milliliter
NA	Not applicable
ND	Not determined
No.	Number
rpm	Revolutions per minute
SD	Standard deviation
SOP	Standard operating procedure
T _{last}	Time of last quantifiable concentration
T _{max}	Time when the maximum concentration was observed; for intravenous bolus dosing the C _{max} is considered to occur at the first sampling time after dose administration
T _{1/2}	Apparent elimination phase plasma half-life
V _c	Apparent volume of distribution

V_{ss}	Apparent volume of distribution at steady-state
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Vol	Volume
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III. Regulatory Compliance and Signature

Calvert Study No.: 0437RL35.002

Study Title: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

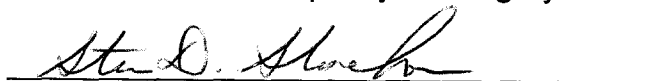
A. Regulatory Compliance

1. Calvert Laboratories-Conducted Study Elements

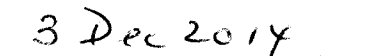
This was a non-regulated study. However, it was run according to the Standard Operating Procedures of Calvert and there was no formal involvement of the Quality Assurance Unit. Bioanalytical evaluation was conducted by Burleson Research Technologies.

B. Principal Investigator Signature

I, the undersigned, hereby declare that this toxicokinetic report is a true and accurate record of the toxicokinetic results obtained. No circumstances occurred during the toxicokinetic evaluation that were considered to have significantly affected the overall quality or integrity of the data obtained.



Steven D. Sloneker, B.S., Principal Investigator
Calvert Laboratories, Inc.



Date

C. Signatures of Other Responsible Personnel

Scientific Oversight and Report Review



Scientific Management
Calvert Laboratories, Inc.



Date

IV. Experimental Design

Introduction

A toxicokinetic assessment of plasma levels of LT3114 from plasma in male and female Sprague-Dawley rats was conducted when LT3114 was administered intravenously at three different dose levels on Days 1, 8, and 15. Animals received three doses, one week apart, and toxicokinetic analysis was assessed following each dose.

Dosing Details

For the toxicokinetic component of this repeat intravenous dose range finding toxicity study of LT3114, three toxicokinetic groups of three male and three female animals per group (Group 5, 6, and 7) were administered an intravenous injection at 5 ml/kg. Doses were administered on three separate occasions (Days 1, 8, and 15) at 10, 50, and 150 mg/kg in Groups 5, 6, and 7, respectively. Nominal dose concentrations were 2, 10, and 30 mg/ml. Burleson Research Technologies performed analysis of the dosing formulation samples.

Sample Collection

On Day 1, whole blood samples (~0.5 ml/sample) were collected from 3 animals/sex/group in Groups 5-7 at 0 (pre-dose), 0.5 and 6 hours post-dose, and on Days 2 (~24 hours post Day 1 dose), 3 (~48 hours post Day 1 dose), 5 (~96 hours post Day 1 dose), 8 (~168 hour post Day 1 dose and prior to the dose on Day 8), and 15 (prior to the dose on Day 15). Whole blood was collected by retro-orbital puncture into tubes containing K₂-EDTA from anesthetized animals by CO₂ inhalation. Blood samples were spun in a refrigerated centrifuge (~+4°C) at ~3000 rpm for ~10 minutes. Harvested plasma samples were stored frozen at ~ -70 °C until shipment to Burleson Research Technologies (Morrisville, NC).

Sample Analysis

Plasma analysis was conducted under the direction of Victor J. Johnson, Ph.D., Burleson Research Technologies, Inc., 120 First Flight Lane,

Morrisville, NC 27560. Plasma assay results were provided by BRT and were retained in the study file.

Data Analysis

Composite-averaged concentration-time data (per sex per dose level) were used and Office Excel 2010 (Microsoft Corp., Redmond, WA) was used for graphing. Plasma concentration assay results listed as BLQ (below the limit of quantitation) were reported as 0 µg/ml.

Non-compartmental toxicokinetic analysis, using WinNonlin Version 5.3, calculated the LT3114 toxicokinetic parameters: C_0 , C_{max} , T_{max} , C_{last} , T_{last} , $T_{1/2}$, AUC_{0-last} , AUC_{0-inf} , Cl , and V_{ss} . $T_{1/2}$ was calculated using the LT3114 concentrations from the 24, 48, 96 and 168 hour timepoints. V_c was calculated by $Dose / C_0$. In some cases, C_0 could not be determined by log-linear regression of the first two concentration-time measurements and was set to the first observed measurement.

Nominal dose concentrations, nominal dose levels, and non-truncated values were used in the calculation of all toxicokinetic parameters.

V. Records and Reports

Storage of Records

The original toxicokinetic report, toxicokinetic data analysis, and related study records will be archived at Calvert, 105 Edella Road, Suite 100, Clarks Summit, PA 18411. All analytical and bioanalytical data will be archived at Burleson Research Technologies, Morrisville, NC. After 2 years, the Sponsor will be contacted to determine final disposition of all study materials archived at Calvert.

VI. Results

In this study to evaluate the toxicity of LT3114 when administered intravenously to Sprague-Dawley rats, the plasma toxicokinetics of LT3114 were assessed following each dose level per sex.

Individual and mean $\pm$ SD (%RSD) animal plasma concentration-time data for LT3114 are summarized in Table 1. The predose plasma of all animals in Groups 5, 6, and 7 were devoid of LT3114 on Day 1. Toxicokinetic parameter results for LT3114 are summarized in Table 2. Composite-averaged animal plasma concentration-time profiles for LT3114 are shown in Figure 1.

Animals were exposed to LT3114 following an intravenous dose of LT3114 at 10, 50, and 150 mg/kg/dose. Exposure was LT3114 dose dependent and this exposure increased as the LT3114 dose increased; this exposure appeared to be dose proportional. There did not appear to be any gender-related differences with regard to exposure or elimination. $T_{1/2}$ ranged from 147 to 225 hours. Cl was ~ 0.2 to ~ 0.3 ml/hr/kg. V_{ss} ranged from 55 to 80 ml/kg and V_c ranged from 36 to 55 ml/kg.

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VII. Tables

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Table 1 – Individual and Mean $\pm$ SD (%RSD) Rat Plasma Concentrations of LT3114 Following an Intravenous Administration of LT3114 at Three Different Dose Levels

Group #	Dose Level (mg/kg/dose)	Rat #	Sex	Day	Time (hr)	LT3114 Conc. (μ g/ml)	Mean Conc. (μ g/ml)	SD	%RSD
5	10	2747	F	1	0	0	0	0.0	
5	10	2749	F	1	0	0			
5	10	2751	F	1	0	0			
5	10	2748	F	1	0.5	248.4	242.9	4.78	2.0
5	10	2750	F	1	0.5	239.7			
5	10	2752	F	1	0.5	240.6			
5	10	2747	F	1	6	355.0	333.1	19.16	5.8
5	10	2749	F	1	6	324.9			
5	10	2751	F	1	6	319.4			
5	10	2748	F	2	24	143.8	132.7	13.35	10.1
5	10	2750	F	2	24	117.9			
5	10	2752	F	2	24	136.5			
5	10	2747	F	3	48	120.8	128.3	12.22	9.5
5	10	2749	F	3	48	121.7			
5	10	2751	F	3	48	142.4			
5	10	2748	F	5	96	73.1	72.0	1.01	1.4
5	10	2750	F	5	96	71.8			
5	10	2752	F	5	96	71.1			
5	10	2747	F	8	168	82.6	88.1	4.74	5.4
5	10	2749	F	8	168	90.5			
5	10	2751	F	8	168	91.1			
5	10	2748	F	15	336	136.0	134.7	3.40	2.5
5	10	2750	F	15	336	130.8			
5	10	2752	F	15	336	137.2			
5	10	2741	M	1	0	0	0	0.0	
5	10	2743	M	1	0	0			
5	10	2745	M	1	0	0			
5	10	2742	M	1	0.5	300.0	274.7	28.63	10.4
5	10	2744	M	1	0.5	243.6			
5	10	2746	M	1	0.5	280.4			
5	10	2741	M	1	6	213.8	291.9	68.48	23.5
5	10	2743	M	1	6	341.5			
5	10	2745	M	1	6	320.5			
5	10	2742	M	2	24	108.9	119.7	9.47	7.9
5	10	2744	M	2	24	123.4			
5	10	2746	M	2	24	126.7			
5	10	2741	M	3	48	90.1	108.6	18.21	16.8
5	10	2743	M	3	48	126.5			
5	10	2745	M	3	48	109.3			

Table 1 (continued) – Individual and Mean $\pm$ SD (%RSD) Rat Plasma Concentrations of LT3114 Following an Intravenous Administration of LT3114 at Three Different Dose Levels

Group #	Dose Level (mg/kg/dose)	Rat #	Sex	Day	Time (hr)	LT3114 Conc. (μ g/ml)	Mean Conc. (μ g/ml)	SD	%RSD
5	10	2742	M	5	96	75.5	67.5	6.97	10.3
5	10	2744	M	5	96	63.1			
5	10	2746	M	5	96	63.8			
5	10	2741	M	8	168	75.5	79.9	7.74	9.7
5	10	2743	M	8	168	75.3			
5	10	2745	M	8	168	88.8			
5	10	2742	M	15	336	99.9	86.3	26.23	30.4
5	10	2744	M	15	336	103.0			
5	10	2746	M	15	336	56.1			
6	50	2759	F	1	0	0	0	0.0	
6	50	2761	F	1	0	0			
6	50	2763	F	1	0	0			
6	50	2760	F	1	0.5	941.7	947.1	27.95	3.0
6	50	2762	F	1	0.5	977.4			
6	50	2764	F	1	0.5	922.3			
6	50	2759	F	1	6	1126.6	1075.8	75.13	7.0
6	50	2761	F	1	6	989.5			
6	50	2763	F	1	6	1111.3			
6	50	2760	F	2	24	493.8	456.8	36.18	7.9
6	50	2762	F	2	24	455.2			
6	50	2764	F	2	24	421.5			
6	50	2759	F	3	48	509.1	477.7	27.83	5.8
6	50	2761	F	3	48	467.9			
6	50	2763	F	3	48	456.1			
6	50	2760	F	5	96	346.6	326.3	20.45	6.3
6	50	2762	F	5	96	326.5			
6	50	2764	F	5	96	305.7			
6	50	2759	F	8	168	257.7	267.4	8.51	3.2
6	50	2761	F	8	168	273.5			
6	50	2763	F	8	168	271.1			
6	50	2760	F	15	336	757.3	634.7	110.91	17.5
6	50	2762	F	15	336	605.6			
6	50	2764	F	15	336	541.3			
6	50	2753	M	1	0	0	0	0.0	
6	50	2755	M	1	0	0			
6	50	2757	M	1	0	0			
6	50	2754	M	1	0.5	953.1	1151.7	172.04	14.9
6	50	2756	M	1	0.5	1246.9			
6	50	2758	M	1	0.5	1255.1			

Table 1 (continued) – Individual and Mean $\pm$ SD (%RSD) Rat Plasma Concentrations of LT3114 Following an Intravenous Administration of LT3114 at Three Different Dose Levels

Group #	Dose Level (mg/kg/dose)	Rat #	Sex	Day	Time (hr)	LT3114 Conc. (μ g/ml)	Mean Conc. (μ g/ml)	SD	%RSD
6	50	2753	M	1	6	1365.4	1052.5	272.33	25.9
6	50	2755	M	1	6	869.2			
6	50	2757	M	1	6	922.8			
6	50	2754	M	2	24	457.0	519.5	54.79	10.5
6	50	2756	M	2	24	542.0			
6	50	2758	M	2	24	559.4			
6	50	2753	M	3	48	508.4	424.1	73.10	17.2
6	50	2755	M	3	48	378.7			
6	50	2757	M	3	48	385.1			
6	50	2754	M	5	96	320.5	387.5	58.06	15.0
6	50	2756	M	5	96	422.0			
6	50	2758	M	5	96	420.1			
6	50	2753	M	8	168	264.0	252.2	14.74	5.8
6	50	2755	M	8	168	235.7			
6	50	2757	M	8	168	257.0			
6	50	2754	M	15	336	543.2	557.4	19.79	3.5
6	50	2756	M	15	336	580.0			
6	50	2758	M	15	336	549.0			
7	150	2771	F	1	0	0	0	0.0	
7	150	2773	F	1	0	0			
7	150	2775	F	1	0	0			
7	150	2772	F	1	0.5	3882.8	3368.0	447.27	13.3
7	150	2774	F	1	0.5	3146.0			
7	150	2776	F	1	0.5	3075.1			
7	150	2771	F	1	6	2839.8	2573.6	279.20	10.8
7	150	2773	F	1	6	2597.9			
7	150	2775	F	1	6	2283.0			
7	150	2772	F	2	24	1580.3	1454.3	117.76	8.1
7	150	2774	F	2	24	1347.0			
7	150	2776	F	2	24	1435.7			
7	150	2771	F	3	48	1251.0	1279.0	130.28	10.2
7	150	2773	F	3	48	1421.0			
7	150	2775	F	3	48	1165.0			
7	150	2772	F	5	96	850.0	830.1	75.97	9.2
7	150	2774	F	5	96	746.2			
7	150	2776	F	5	96	894.2			
7	150	2771	F	8	168	801.9	844.8	127.62	15.1
7	150	2773	F	8	168	744.1			
7	150	2775	F	8	168	988.3			

Table 1 (continued) – Individual and Mean $\pm$ SD (%RSD) Rat Plasma Concentrations of LT3114 Following an Intravenous Administration of LT3114 at Three Different Dose Levels

Group #	Dose Level (mg/kg/dose)	Rat #	Sex	Day	Time (hr)	LT3114 Conc. (μ g/ml)	Mean Conc. (μ g/ml)	SD	%RSD
7	150	2772	F	15	336	1222.7	1290.8	59.46	4.6
7	150	2774	F	15	336	1332.6			
7	150	2776	F	15	336	1317.0			
7	150	2765	M	1	0	0	0	0.0	
7	150	2767	M	1	0	0			
7	150	2769	M	1	0	0			
7	150	2766	M	1	0.5	2180.9	2749.0	493.68	18.0
7	150	2768	M	1	0.5	2991.8			
7	150	2770	M	1	0.5	3074.2			
7	150	2765	M	1	6	3015.4	3230.0	281.78	8.7
7	150	2767	M	1	6	3549.1			
7	150	2769	M	1	6	3125.5			
7	150	2766	M	2	24	1270.0	1277.0	17.80	1.4
7	150	2768	M	2	24	1263.7			
7	150	2770	M	2	24	1297.2			
7	150	2765	M	3	48	1702.9	1590.9	161.58	10.2
7	150	2767	M	3	48	1405.7			
7	150	2769	M	3	48	1664.2			
7	150	2766	M	5	96	964.8	953.7	70.45	7.4
7	150	2768	M	5	96	878.4			
7	150	2770	M	5	96	1018.0			
7	150	2765	M	8	168	874.0	779.0	93.63	12.0
7	150	2767	M	8	168	686.8			
7	150	2769	M	8	168	776.1			
7	150	2766	M	15	336	1408.8	1365.0	44.57	3.3
7	150	2768	M	15	336	1366.5			
7	150	2770	M	15	336	1319.7			

Table 2 – Toxicokinetic Parameters of LT3114 in Rat Plasma Following Intravenous Administration at 10 mg/kg/dose (Group 5), 50 mg/kg/dose (Group 6), and 150 mg/kg/dose (Group 7) of LT3114

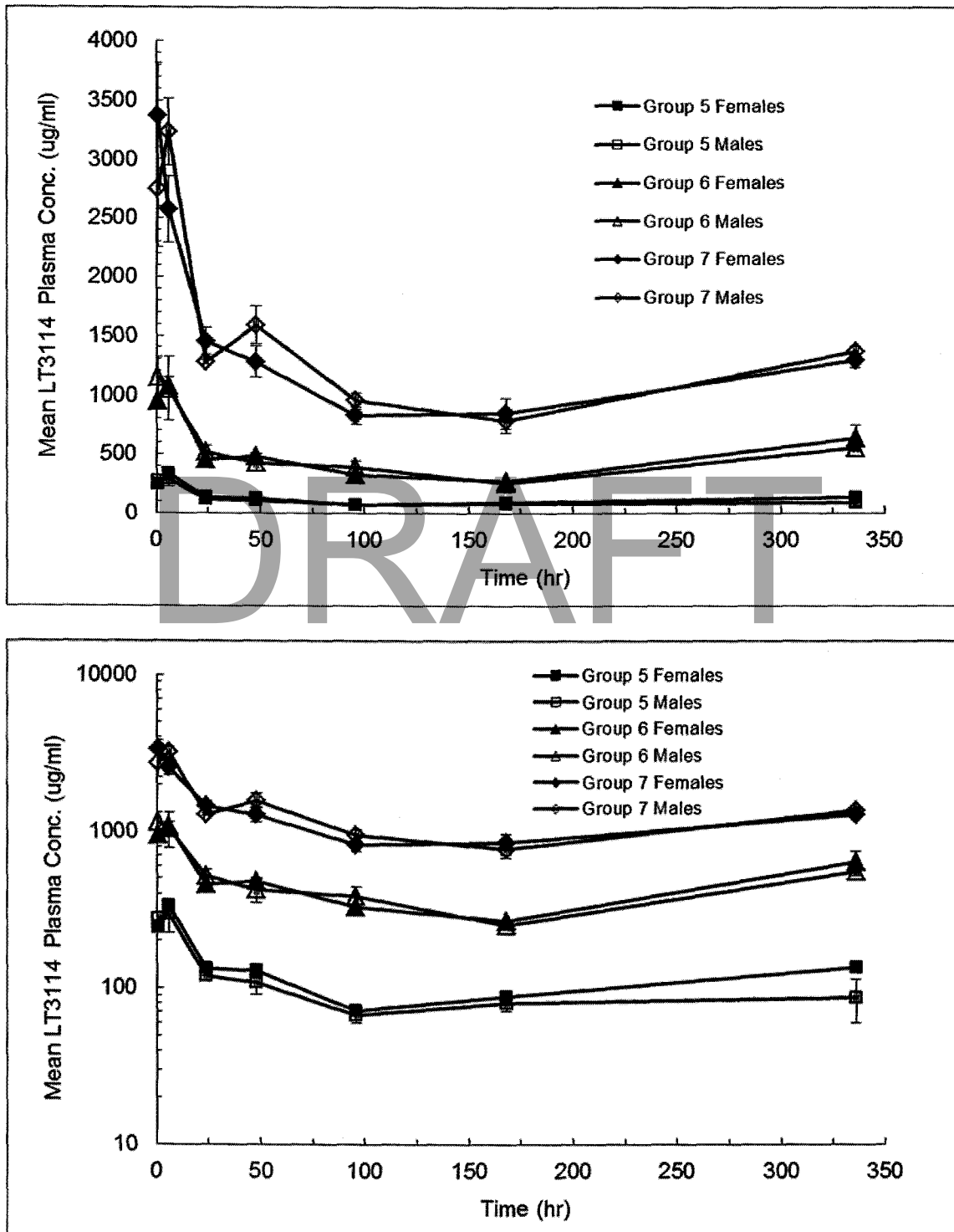
Group #	Sex	Dose (mg/kg/dose)	C ₀ (µg/ml)	C _{max} (µg/ml)	T _{max} (hr)	C _{last} (µg/ml)	T _{last} (hr)	AUC _{0-last} (µg*hr/ml)	AUC _{0-inf} (µg*hr/ml)	T _{1/2} (hr)	Cl (ml/hr/kg)	V _{ss} (ml/kg)	V _c (ml/kg)
5	F	10	243 ^a	333	6	135	336	38316	51392	207	0.195	55.3	41.2
5	M	10	275 ^a	292	6	86.3	336	31633	45259	225	0.221	66.9	36.4
6	F	50	947 ^a	1076	6	635	336	147490	178397	166	0.280	69.3	52.8
6	M	50	1161	1152	0.5	557	336	142625	167497	147	0.299	67.3	43.1
7	F	150	3451	3368	0.5	1291	336	377400	477588	176	0.314	79.2	43.5
7	M	150	2749 ^a	3230	6	1365	336	396338	480416	158	0.312	72.0	54.6

^a C₀ could not be determined by log-linear regression of the first two concentration-time measurements and was set to the first observed measurement.

VIII. Figure

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Figure 1 – Composite-Averaged Rat Plasma Concentrations (Mean $\pm$ SD) of LT3114 Plotted as a Function of Time Following Intravenous Administration at 10 mg/kg/dose (Group 5), 50 mg/kg/dose (Group 6), and 150 mg/kg/dose (Group 7) of LT3114 (Rectilinear and Semi-Logarithmic Plots)



The LT3114 plasma concentrations on Day 8 and Day 15 were trough concentrations which were harvested just before the Day 8 and Day 15 dose administrations.

Appendix VI—Historical Clinical Pathology Data

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CALVERT LABORATORIES, INC.
SPRAGUE DAWLEY RAT - CARDIAC PUNCTURE
HISTORICAL DATABASE
2013

HEMATOLOGY DATA – ADVIA 120

MALE SPRAGUE DAWLEY RAT > 8 WEEKS

HEMATOLOGY TEST	N	MEAN	S.D.	LOWER-UPPER LIMIT
LEUKOCYTES (WBC) $10^3/\text{mm}^3$	1640	9.57	2.975	3.63 – 15.53
ERYTHROCYTES (RBC) $10^6/\text{mm}^3$	1642	8.47	0.602	7.28 – 9.67
HEMOGLOBIN (HGB) g/dl	1640	15.9	0.91	14.0 – 17.7
HEMATOCRIT (HCT) %	1642	49.59	2.76	44.1 – 55.0
MEAN CORPUSCULAR VOLUME (MCV) fl	1638	58.5	3.04	52.7 – 64.6
MEAN CORPUSCULAR HEMOGLOBIN (MCH) pg	1646	18.8	0.97	16.8 – 20.7
MEAN CORPUSCULAR HEMOGLOBIN CONC. (MCHC) g/dl	1645	32.1	1.34	29.4 – 34.8
PLATELET (PLT) $10^3/\text{mm}^3$	1612	1016	209.4	597 – 1435
NEUTROPHILS% (NEUT%)	1621	16.2	5.91	4.2 – 28.0
LYMPHOCYTES% (LYMPH%)	1628	77.6	6.47	64.6 – 90.5
MONOCYTES% (MONO%)	1640	3.8	1.30	1.2 – 6.4
EOSINOPHILS% (EOS%)	1639	0.9	0.55	0.0 – 2.0
BASOPHILS% (BASO%)	1633	0.6	0.55	0.0 – 1.7
LUC% (LUC%)	1646	0.8	0.46	0.0 – 1.7
NEUTROPHILS (NEUT) $10^3/\text{mm}^3$	1621	1.53	0.755	0.02 – 3.04
LYMPHOCYTES (LYMPH) $10^3/\text{mm}^3$	1645	7.43	2.459	2.52 – 12.35
MONOCYTES (MONO) $10^3/\text{mm}^3$	1642	0.37	0.178	0.01 – 0.72
EOSINOPHILS (EOS) $10^3/\text{mm}^3$	1641	0.10	0.064	0.00 – 0.22
BASOPHILS (BASO) $10^3/\text{mm}^3$	1646	0.06	0.063	0.00 – 0.19
LUC (LUC) $10^3/\text{mm}^3$	1642	0.08	0.056	0.00 – 0.19
RETICULOCYTES (RETIC) $10^9/\text{L}$	1636	239.7	95.0	49.7 – 426.5
RETICULOCYTES% (%RETIC)	1634	2.89	1.343	0.2 – 5.57
RED CELL DISTRIBUTION WIDTH (RDW) %	25	12.1	0.62	10.9 – 13.3

CALVERT LABORATORIES, INC.
SPRAGUE DAWLEY RAT - CARDIAC PUNCTURE
HISTORICAL DATABASE
2013

HEMATOLOGY DATA – ADVIA 120

FEMALE SPRAGUE DAWLEY RAT > 8 WEEKS

HEMATOLOGY TEST	N	MEAN	S.D.	LOWER-UPPER LIMIT
LEUKOCYTES (WBC) $10^3/\text{mm}^3$	1585	7.01	2.58	1.85 – 12.17
ERYTHROCYTES (RBC) $10^6/\text{mm}^3$	1575	8.02	0.572	6.88 – 9.16
HEMOGLOBIN (HGB) g/dl	1572	15.3	1.05	13.2 – 17.4
HEMATOCRIT (HCT) %	1574	46.7	3.20	40.4 – 53.0
MEAN CORPUSCULAR VOLUME (MCV) fl	1582	58.2	2.52	53.2 – 63.3
MEAN CORPUSCULAR HEMOGLOBIN (MCH) pg	1587	19.1	0.80	17.5 – 20.7
MEAN CORPUSCULAR HEMOGLOBIN CONC. (MCHC) g/dl	1589	32.8	1.28	30.2 – 35.3
PLATELET (PLT) $10^3/\text{mm}^3$	1526	1044	212.6	620 – 1470
NEUTROPHILS% (NEUT%)	1556	14.1	5.31	3.5 – 24.7
LYMPHOCYTES% (LYMPH%)	1574	79.2	6.61	66.0 – 92.5
MONOCYTES% (MONO%)	1588	3.5	1.28	0.9 – 6.0
EOSINOPHILS% (EOS%)	1586	1.4	0.77	0.0 – 2.9
BASOPHILS% (BASO%)	1590	0.7	0.62	0.0 – 2.0
LUC% (LUC%)	1588	0.8	0.60	0.0 – 2.0
NEUTROPHILS (NEUT) $10^3/\text{mm}^3$	1579	1.01	0.590	0.0 – 2.19
LYMPHOCYTES (LYMPH) $10^3/\text{mm}^3$	1586	5.53	2.10	1.34 – 9.72
MONOCYTES (MONO) $10^3/\text{mm}^3$	1588	0.25	0.143	0.00 – 0.53
EOSINOPHILS (EOS) $10^3/\text{mm}^3$	1586	0.10	0.068	0.00 – 0.23
BASOPHILS (BASO) $10^3/\text{mm}^3$	1590	0.05	0.048	0.00 – 0.14
LUC (LUC) $10^3/\text{mm}^3$	1589	0.06	0.050	0.00 – 0.16
RETICULOCYTES (RETIC%)	1564	2.68	1.292	0.10 – 5.27
RETICULOCYTES (RETIC) $10^9/\text{L}$	1571	212.8	88.90	34.9 – 390.6
RED CELL DISTRIBUTION WIDTH (RDW) %	25	11.2	0.54	10.1 – 12.3

**CALVERT LABORATORIES, INC.
SPRAGUE DAWLEY RAT - CARDIAC PUNCTURE
HISTORICAL DATABASE
2013**

MALE COAGULATION DATA > 8 WEEKS
DIAGNOSTICA STAGO STA COMPACT

COAGULATION TEST	N	MEAN	S.D.	LOWER – UPPER LIMIT
PROTHROMBIN TIME (PT) sec.	1357	17.4	1.35	14.7 – 20.3
ACTIVATED PARTIAL THROMBOPLASTIN TIME (APTT) sec.	1357	10.7	2.02	6.6 – 14.7
FIBRINOGEN (FIB) mg/dl	5	291.2	19.7	251.8 – 330.6

FEMALE COAGULATION DATA > 8 WEEKS
DIAGNOSTICA STA COMPACT

COAGULATION TEST	N	MEAN	S.D.	LOWER – UPPER LIMIT
PROTHROMBIN TIME (PT) sec.	1269	17.6	1.88	13.8 – 21.4
ACTIVATED PARTIAL THROMBOPLASTIN TIME (APTT) sec.	1273	9.5	2.34	4.8 – 14.2

CALVERT LABORATORIES, INC.
SPRAGUE DAWLEY RAT - CARDIAC PUNCTURE
HISTORICAL DATABASE
2013

CLINICAL CHEMISTRY DATA – AU400 CHEMISTRY ANALYZER

MALE SPRAGUE DAWLEY RAT > 8 WEEKS

CLINICAL CHEMISTRY TEST	N	MEAN	S.D.	LOWER – UPPER LIMIT
GLUCOSE (GLU) mg/dl	1684	119	18.6	82 – 156
BLOOD UREA NITROGEN (BUN) mg/dl	1700	18	3.3	11 – 24
CREATININE (CREAT) mg/dl	1699	0.46	0.128	0.21 – 0.72
PHOSPHORUS (PHOS) mg/dl	1690	9.1	1.38	6.4 – 11.9
TOTAL PROTEIN (TP) g/dl	1690	6.2	0.41	5.4 – 7.1
ALBUMIN (ALB) g/dl	1689	3.4	0.17	3.0 – 3.7
TOTAL BILIRUBIN (TBIL) mg/dl	1679	0.12	0.024	0.07 – 0.17
ALKALINE PHOSPHATASE (ALP) U/L	1676	135	41.2	53 – 218
ASPARTATE AMINO TRANSFERASE (AST) U/L	1679	93	29.2	35 – 151
ALANINE AMINO TRANSFERASE (ALT) U/L	1680	46	15.5	15 – 78
CHOLESTEROL (CHOL) mg/dl	1690	90	18.5	53 – 127
CALCIUM (CA) mg/dl	1690	10.88	0.442	9.99 – 11.77
SODIUM (NA) mEq/L	1679	146	2.8	141 – 152
POTASSIUM (K) mEq/L	1673	6.2	0.64	5.0 – 7.5
CHLORIDE (CL) mEq/L	1680	101	2.3	96 – 105
TRIGLYCERIDE (TRIG) mg/dl	1683	33	15.0	3 – 62
GLOBULIN (GLOB) g/dl - calc.	1679	2.9	0.35	2.2 – 3.6
A/G RATIO (A/G) calc.	1671	1.2	0.15	0.9 – 1.5
CREATINE KINASE (CK) U/L	129	176	84.8	6 - 346
LACTATE DEHYDROGENASE (LD) U/L	112	174	167.3	0 – 509
AMYLASE (AMYL) U/L	18	602	123.6	355 – 849
IRON (FE) µg/dl	51	99	29.1	41 – 157
HDL CHOLESTEROL (HDL) mg/dl	75	26	4.0	18 – 34
LDL CHOLESTEROL (LDL) mg/dl	75	12	2.8	6 – 18
LIPASE (LIP) U/L	15	27	11.9	3 – 51
AMMONIA (AMM) mmol/L	4	162.2	90.17	0.0 – 342.5
DIRECT BILIRUBIN (DBILI) mg/dl	19	0.02	0.042	0.0 – 0.1
UNCONJUGATED IRON BINDING CAPACITY(UIBC)ug/dl	15	351.9	43.65	264.6 – 439.2
TOTAL IRON BINDING CAPACITY(TIBC)ug/dl	15	464.4	38.6	387 – 542
BILE ACID (BA) mmol/L	30	8.64	6.5	0 – 21.6
INDIRECT BILIRUBIN (IBILI) mg/dl	15	0.08	0.04	0 – 0.16

CALVERT LABORATORIES, INC.
SPRAGUE DAWLEY RAT - CARDIAC PUNCTURE
HISTORICAL DATABASE
2013

CLINICAL CHEMISTRY DATA – AU400 CHEMISTRY ANALYZER

FEMALE SPRAGUE DAWLEY RAT > 8 WEEKS

CLINICAL CHEMISTRY TEST	N	MEAN	S.D.	LOWER - UPPER LIMIT
GLUCOSE (GLU) mg/dl	1617	111	15.4	80 - 142
BLOOD UREA NITROGEN (BUN) mg/dl	1651	19	3.5	12 - 26
CREATININE (CREAT) mg/dl	1637	0.52	0.134	0.25 - 0.79
PHOSPHORUS (PHOS) mg/dl	1628	8.5	1.54	5.5 - 11.6
TOTAL PROTEIN (TP) g/dl	1628	6.4	0.55	5.3 - 7.5
ALBUMIN (ALB) g/dl	1614	3.5	0.26	3.0 - 4.1
TOTAL BILIRUBIN (TBIL) mg/dl	1631	0.14	0.037	0.07 - 0.22
ALKALINE PHOSPHATASE (ALP) U/L	1626	96	32.9	30 - 161
ASPARTATE AMINO TRANSFERASE (AST) U/L	1613	90	30.6	29 - 151
ALANINE AMINO TRANSFERASE (ALT) U/L	1615	37	12.2	12 - 61
CHOLESTEROL (CHOL) mg/dl	1619	86	20.6	44 - 127
CALCIUM (CA) mg/dl	1614	10.93	0.462	10.00 - 11.85
SODIUM (NA) mEq/L	1611	144	2.4	139 - 149
POTASSIUM (K) mEq/L	1607	6.3	0.63	5.1 - 7.6
CHLORIDE (CL) mEq/L	1615	101	2.4	96 - 106
TRIGLYCERIDE (TRIG) mg/dl	1597	30	12.6	5 - 55
GLOBULIN (GLOB) g/dl - calc.	1631	2.8	0.40	2.0 - 3.6
A/G RATIO (A/G) calc.	1594	1.3	0.16	0.9 - 1.6
CREATINE KINASE (CK) U/L	115	199	122.9	0 - 444
LACTATE DEHYDROGENASE (LD) U/L	106	193	156.8	0 - 507
IRON (FE) µg/dl	42	272	120.3	31 - 513
HDL CHOLESTEROL (HDL) mg/dl	70	23	4.9	13 - 33
LDL CHOLESTEROL (LDL) mg/dl	69	5	1.2	3 - 7
AMYLASE (AMY) U/L	14	288	43.8	200 - 376
LIPASE (LIP) U/L	15	32	5.4	21 - 43
AMMONIA (AMM) mmol/L	4	153.3	46.08	61.1 - 245.5
DIRECT BILIRUBIN (DBILI) mg/dl	19	0.04	0.051	0.00 - 0.14
UNCONJUGATED IRON BINDING CAPACITY(UIBC)ug/dl	14	212.5	93.38	25.7 - 399.3
TOTAL IRON BINDING CAPACITY(TIBC)ug/dl	14	529.1	62.5	404 - 654
BILE ACID (BA) mmol/L	29	12.4	6.6	0 - 25.6
INDIRECT BILIRUBIN (IBILI) mg/dl	15	0.09	0.05	0 - 0.20

CALVERT LABORATORIES, INC.
SPRAGUE DAWLEY RAT - CARDIAC PUNCTURE
HISTORICAL DATABASE
2013

SPRAGUE DAWLEY RATS > 8 WEEKS

MALE URINALYSIS DATA

URINALYSIS TEST	N	MEAN	S.D.	LOWER – UPPER LIMIT
SPECIFIC GRAVITY (S.G.)	768	1.023	0.0114	1.000 – 1.046
pH	765	6.8	0.54	5.7 – 7.9

FEMALE URINALYSIS DATA

URINALYSIS TEST	N	MEAN	S.D.	LOWER – UPPER LIMIT
SPECIFIC GRAVITY (S.G.)	778	1.019	0.0089	1.001 – 1.036
pH	783	6.4	0.55	5.4 – 7.5

Appendix VII—Protocol, Amendments and Deviations

DRAFT



Study Protocol

Title: Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

Calvert Study No.: 0437RL35.002

Testing Facility: Calvert Laboratories, Inc.
Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447

Study Sponsor: Lpath Incorporated
4025 Sorrento Valley Blvd.
San Diego, CA, 92121

LABORATORY

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II. Introduction

A. Title

Repeat Intravenous Dose Range Finding Toxicity Study of LT3114 in Sprague Dawley Rats with Toxicokinetics (non-GLP)

B. Objective

The purpose of this study is to evaluate the toxicity of LT3114 when administered intravenously to Sprague Dawley rats. Animals will receive two doses, one week apart, and toxicokinetics will be assessed following each dose.

C. Regulatory Compliance

This is a non-regulated study. However, it will be run according to the Standard Operating Procedures (SOPs) of Calvert. There will be no formal involvement of the Quality Assurance Unit.

D. Testing Guidelines

ICH Guidance for Industry: M3(R2) Nonclinical Safety Studies for the Conduct of Human Clinical Trials and Marketing Authorization for Pharmaceuticals

E. Calvert Study Number

0437RL35.002

F. Testing Facility

Calvert Laboratories, Inc. (Calvert)
Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447

G. Sponsor

Lpath Incorporated
4025 Sorrento Valley Blvd.
San Diego, CA, 92121

H. Study Director

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L. Principal Investigator – Dosing Formulation and Toxicokinetic Sample Analysis

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M. Principal Investigator - Histopathology Slide Preparation

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O. Key Study Dates

Proposed Experimental Start Date:	Mar 2013
Proposed First Day of Dosing:	Mar 2013
Proposed Necropsy:	Apr 2013
Proposed Experimental Completion Date:	Apr 2013

III. Materials and Methods

A. Test Article

1. Test Article

Identification: LT3114

Lot/Batch No.: To be documented in the study data

Physical Description: To be documented in the study data

Storage Conditions: Stored refrigerated (2-8°C) and protected from light.

2. Dose Preparation

The Sponsor will provide the test article already formulated at 30 mg/ml in sterile containers. Diluent will also be provided in separate sterile containers. Diluent and test article need to come to room temperature before the formulation is prepared. Do not shake the formulation. Group 1 control animals will receive the diluent. When necessary depending on the dose level, doses will be prepared by dilution using the provided 30 mg/ml and diluent solutions under aseptic conditions. Before dosing, visual inspection of the formulation should be conducted against a black background for any irregularities.

3. Dose Formulation Samples

On Days 1, 8 and 15, duplicate 1-ml samples of each dosing solution, including the vehicle control, will be obtained to determine the concentration of the test article in vehicle. These samples will be stored at refrigerated temperatures (2-8°C).

4. Dose Formulation Analysis

Dose formulation samples will be returned to, and analytical evaluation will be done under the direction of:

Florence Burleson, Ph.D.
Executive VP - Director, Laboratory Operations
Burleson Research Technologies
120 First Flight Lane
Morrisville, NC 27560

Targeted

Number of Samples: 12 in duplicate (24 total collected)

Sample Storage

Conditions: Refrigerated temperatures (2-8°C)

Testing Requirements: Concentration and homogeneity

5. Accountability and Disposition

Unused test article will be returned to the Sponsor or designee at the completion of this study or, if necessary, retained for use on related future studies. The Sponsor will be notified in advance of shipping and a transmittal letter will accompany the shipment. The material will be packed in a suitable container to maintain the conditions specified by the Sponsor during transit plus an adequate margin of safety to account for any possible transit delays. If material is returned, the Sponsor will provide shipping instructions.

B. Test System (Animals and Animal Care)

1. Description

Species:	Rat
Stock:	Sprague-Dawley, Hsd:SD or appropriate per vendor
Total Number:	76 (38 male, 38 female)
Gender:	Male and female
Age Range:	8-10 weeks at start of dosing; records of dates of birth for animals used in this study will be retained in the Calvert archives.

Body Weight Range: 150 to 300 g at the outset (Day 1) of the study.

Animal Source: Harlan Laboratories, Inc.

Experimental History: Purpose-bred and experimentally naïve at the outset of the study.

Identification: Ear tag and cage card

2. *Rationale for Choice of Species and Number of Animals*

The Sprague Dawley rat was chosen for this study as it is a species that is used for non-clinical toxicity and pharmacokinetic evaluations and satisfies the regulatory requirement for non-clinical safety studies in rodents (1). Previous research has been conducted with the test article in rats. The test article, a human anti-LPA antibody (LT3114) and the murine parental molecule have both been evaluated for efficacy in a variety of rat and mouse animal models of disease. In models using Sprague Dawley rats efficacy has been examined in the STZ induced model of diabetic neuropathy, a chemotherapeutic-induced model of peripheral neuropathy and in models of acute inflammatory pain (carrageen or Freund's adjuvant induced) with doses (intravenous or subcutaneous) ranging from 10 to 40mg/Kg. In addition single and multiple doses pharmacokinetic studies have been conducted with LT 3114 in murine at dose levels ranging from 10 to 30mg/Kg.

The number of animals used in this study is needed to provide an assessment of the acute intravenous dose toxicity of the test article at three dose levels, account for variability among animals and allow for statistical evaluation of numerical data (1, 2).

3. *Husbandry*

Housing: Animals will be group housed by sex upon receipt and individually housed upon assignment to study in compliance with National Research Council "Guide for the Care and Use of Laboratory Animals". The room in which the animals will be

kept will be documented in the study records. No other species will be kept in the same room.

- Lighting:** 12 hours light/12 hours dark, except when room lights will be turned on during the dark cycle to accommodate blood sampling or other study procedures.
- Room Temperature:** 20 to 26°C
- Relative Humidity:** 30-70%
- Food:** All animals will have access to Harlan Teklad Rodent Diet (certified) or equivalent *ad libitum*, unless otherwise specified. The lot number(s) and specifications of each lot used are archived at Calvert. No contaminants are known to be present in the certified diet at levels that would be expected to interfere with the results of this study. Analysis of the diet was limited to that performed by the manufacturer, records of which will be maintained in the Calvert archives.
- Water:** Water will be available *ad libitum*, to each animal via an automatic watering device. The water is routinely analyzed for contaminants as per Calvert SOPs. No contaminants are known to be present in the water at levels that would be expected to interfere with the results of this study. Results of the water analysis will be maintained in the Calvert archives.
- Acclimation:** Study animals will be acclimated to their housing for a minimum of 5 days prior to their first day of dosing.

4. *Prestudy Health Screen and Selection Criteria*

All animals received for this study will be assessed as to their general health by a member of the technical staff or other authorized personnel. During the acclimation period, each animal will be observed at least once daily for any abnormalities or for the development of infectious disease. Only animals that are determined to be suitable for use will be assigned to this study. Any animals considered unacceptable for use in this study will be replaced with animals of similar age and weight from the same vendor.

5. *Assignment to Study Groups*

Animals will be assigned to study groups using Pristima version 6.3.2 (Xybio Corporation, 201 Littleton Road, Morris Plains, NJ 07950). Animals will be randomly assigned to groups using selection designed to achieve similar group mean body weights.

6. *Humane Care of Animals*

Treatment of animals will be in accordance with the study protocol and also in accordance with Calvert SOPs which adhere to the regulations outlined in the USDA Animal Welfare Act (9 CFR Parts 1, 2 and 3) and the conditions specified in the Guide for the Care and Use of Laboratory Animals (ILAR publication, NRC, 2011, The National Academies Press). The Calvert Institutional Animal Care and Use Committee (IACUC) will approve the study protocol prior to finalization to insure compliance with acceptable standard animal welfare and humane care.

No alternative test systems exist which have been adequately validated to permit replacement of the use of live animals in this study. Every effort has been made to obtain the maximum amount of information while reducing to a minimum the number of animals required for this study. The assessment of pain and distress in study animals and the use or non-use of pain alleviating medications will be in accordance with Standard Operating Procedure VET-19, Criteria for Assessing Pain and Distress in Laboratory Animals. The study will be terminated in part or whole for humane reasons if unnecessary pain occurs. To the best of our knowledge, this study is not unnecessary or duplicative.

C. Test Article Administration

1. Group Assignments and Dose Levels

Toxicology Groups

Group	Dose Level (mg/kg/dose)*	Concentration (mg/ml)	Dose Volume (ml/kg)	Number of Animals**	
				Male	Female
1. Control (vehicle)	0	0	5	5	5
2. Test Article Low-dose	10	2	5	5	5
3. Test Article Mid-dose	50	10	5	5	5
4. Test Article High-dose	150	30	5	5	5

*Dosing occurs on Day 1, Day 8, and Day 15

**Toxicology animals will be euthanized and necropsied on Day 16

Toxicokinetic Groups

Group	Dose Level (mg/kg dose)*	Concentration (mg/ml)	Dose Volume (ml/kg)	Number of Animals**	
				Male	Female
5. Test Article Low-dose	10	2	5	6	6
6. Test Article Mid-dose	50	10	5	6	6
7. Test Article High-dose	150	30	5	6	6

*Dosing occurs on Day 1, Day 8, and Day 15 - no Day 15 dose

**Toxicokinetic animals will be euthanized after the last toxicokinetic sample is taken

2. Dosing

Route: Slow intravenous injection (~1 minute)

Frequency: On Days 1, 8, and 15

Procedure: Doses will be administered on Days 1, 8, and 15 by intravenous injection via the lateral tail vein at a dose volume of 5 ml/kg. Each intravenous dose will be administered as a manual slow injection (~1 minute). Each animal will receive a mg/kg dose based on its most recent body weight.

3. Justification for Route, Dose Levels and Dosing Schedule

The intravenous route was chosen as it is the intended route of administration in humans.

Dose levels were selected by the Sponsor. The anticipated lowest pharmacologically active dose is 30 mg/kg. The high dose selected for this study represents a several-fold increase from the anticipated lowest pharmacologic dose. The study animals will be administered the test formulations in order of ascending dosage on every dosing day. Should any adverse signs be observed during administration of the low- or mid-dose levels, dosing will be aborted and the next highest dose(s) will not be administered.

The frequency of dosing was chosen by the Sponsor based on the test article half-life in rodents (~100 hr) and supports the safety of the product under its intended clinical use.

D. In-Life Observations and Measurements

1. Mortality/Morbidity

Frequency: Twice daily (a.m. and p.m.) on Days 1 to Day 15.
Once on Day 16 prior to euthanasia.

Each animal observed for evidence of death or impending death (as per Calvert SOP VET-14).

2. Clinical Observations

Frequency: On Days 1, 8, and 15 animals in Groups 1-4 will be observed prior to dose administration, immediately post-dose, 1-2 hours post-dose, and additionally if needed. Animals will be observed a minimum of once daily on non-dosing days.

3. Body Weight

Frequency: Prior to dose administration on Days 1, 8, and 15.

A fasted body weight will be recorded for animals in Groups 1-4 on Day 16 for calculation of organ to body weight ratios.

4. Food Consumption

Frequency: Full feeder weights and/or feeder weigh backs will be recorded weekly for Groups 1-4 for determination of food consumption.

5. Body Temperature

Frequency: Body temperature will be recorded once daily on Day -1 through Day 15 for animals in Groups 1-4. On dosing days, body temperatures will be recorded post-dose. Temperatures will be recorded at approximately the same time on each day of measurement.

E. Clinical Pathology Evaluation**1. Sample Collection**

Blood samples for evaluation of serum chemistry, hematology and coagulation parameters will be collected from all surviving animals in Groups 1-4 at terminal sacrifice on Day 16. Animals will be anesthetized by CO₂ inhalation prior to blood collection. Immediately following exsanguination by cardiocentesis for terminal blood collection, rats will be returned to the CO₂ chamber to ensure euthanasia. Urine will be collected overnight (approximately 12-18 hours). Animals will be fasted overnight (approximately 12-24 hours) prior to blood collection for clinical pathology evaluation.

2. Collection Procedures, Processing and Analysis**a) Hematology**

Method of Collection: Cardiocentesis

Anticoagulant: K2-EDTA

Parameters Analyzed:

Hematology Parameters	
Red Blood Cell Count (RBC) and	Platelet count (PLT)

Morphology	
White Blood Cell Count (WBC)*	Hematocrit (HCT)
Mean Corpuscular Hemoglobin (MCH)	Hemoglobin (HGB)
Mean Corpuscular Hemoglobin Concentration (MCHC)	Reticulocyte Count (Retic)
Mean Corpuscular Volume (MCV)	

*Total and differential white blood cell counts, including neutrophils, basophils, eosinophils, monocytes, lymphocytes and large unstained cells

b) Coagulation

Method of Collection: Cardiocentesis

Anticoagulant: Sodium Citrate

Parameters Analyzed:

Coagulation Parameters	
Activated Partial Thromboplastin Time (APTT)	Prothrombin Time (PT)

c) Serum Clinical Chemistry

Method of Collection: Cardiocentesis

Anticoagulant: None

Parameters Analyzed:

Clinical Chemistry Parameters	
Alanine Aminotransferase (ALT)	Globulin (calculated)(GLOB)
Albumin (ALB)	Glucose (GLU)
Albumin/Globulin ratio (calculated)(A/G)	Phosphorus (PHOS)
Alkaline Phosphatase (ALP)	Potassium (K)
Aspartate Aminotransferase (AST)	Sodium (NA)
Calcium (CA)	Total Bilirubin (T-BIL)
Chloride (CL)	Total Protein (TP)
Cholesterol (CHOL)	Triglycerides (TRIG)
Creatinine (CREAT)	Urea Nitrogen (BUN)

d) Urinalysis

Method of Collection: Collection in metabolism cages

Parameters Analyzed:

Urinalysis Parameters	
Specific gravity	Nitrite
pH	Appearance/Color
Protein	Bilirubin
Glucose	Blood
Ketone	Leukocytes
Urobilinogen	Microscopic examination of formed elements

F. Antibody Sampling

Whole blood samples (approximately 0.5 ml/sample) will be collected from all animals in Groups 1-4 via cardiocentesis at terminal sacrifice on Day 16 into K₂EDTA collection tubes. Animals will be anesthetized by CO₂ inhalation prior to blood collection. Blood samples will be spun in a centrifuge (approximately 3000 rpm for approximately 10 minutes) to separate the plasma. Plasma will then be collected and stored frozen at approximately -70°C. Plasma samples will be shipped to the Sponsor for use in assay development. There will be no assay results to report for this study.

G. Toxicokinetics (Group 5-7 only)**1. Blood Sampling**

On Day 1, whole blood samples (approximately 0.5 ml/sample) will be collected from 3 animals/sex/group at the following timepoints:

0 hour (pre-dose)	30 minutes post-dose
	6 hours post-dose

On the following days post the Day 1 dosing, whole blood samples (approximately 0.5 ml/sample) will be collected from 3 animals/sex/group animals at the following timepoints:

Day 2 (~24 hours post Day 1 dose)

Day 3 (~48 hours post Day 1 dose)

Day 5 (~96 hours post Day 1 dose)

Day 8 (~168 hours post Day 1 dose and prior to next dose)

Day 15 (pre dose)

The total volume collected, in consideration with other parameters that require blood collection, is not to exceed 1% of body weight for the animals during a two-week period. Animals will not be bled more than 4 times throughout the duration of the study. Animals will be anesthetized by CO₂ inhalation prior to blood collection. Whole blood (approximately 0.5 ml/sample) will be collected by retroorbital puncture into K₂EDTA plasma collection tubes. Blood samples will be spun in a centrifuge (approximately 3000 rpm for approximately 10 minutes) to separate the serum. Plasma will then be collected and stored frozen at approximately -70°C.

2. Bioanalytical Evaluation

Plasma samples will be returned to, and bioanalytical evaluation will be done under the direction of:

Florence Burleson, Ph.D.
Executive VP - Director, Laboratory Operations
Burleson Research Technologies
120 First Flight Lane
Morrisville, NC 27560

Targeted Number

of Samples: 144 (8 timepoints * 6 animals/group/timepoint * 3 groups)

Sample Storage

Conditions: approximately -70° C or lower

Testing Requirements: Testing for levels of test article

3. Toxicokinetic Evaluation

Data from the bioanalytical evaluation will be provided to, and toxicokinetic evaluation will be done under the direction of:

Steven Sloneker
Calvert Laboratories, Inc.
Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447

Parameters Examined: At least the following areas will be examined,

AUC_{0-inf} , AUC_{0-last} , C_{max} , Cl , C_{last} , C_0 , T_{last} , T_{max} ,
 $T_{1/2}$, V_{ss} , V_c

Note: If the terminal half-life cannot be reliably calculated, the dependent PK parameters of AUC_{0-inf} , Cl , V_{ss} , and V_c will not be reported.

Data Evaluation

Software: Win Nonlin Version 5.3

H. Terminal Procedures and Anatomic Pathology

1. Termination

a) Scheduled Sacrifice

All surviving animals will be euthanized by CO₂ asphyxiation following terminal blood collection on Day 16.

b) Unscheduled Sacrifice

Any animals sacrificed for humane reasons will be euthanized by CO₂ asphyxiation. A final body weight (non-fasted) will be recorded for such animals. Animals found dead will undergo a complete necropsy.

c) Final Body Weight

A fasted terminal body weight will be recorded prior to sacrifice on Day 16. This body weight will be used to calculate organ to body weight ratios.

2. Gross Necropsy

A complete gross necropsy will be performed by Calvert personnel on all animals that are sacrificed or found dead during the study. The necropsy will include examination of:

- the external body surface
- all orifices
- the cranial, thoracic and abdominal cavities and their contents.

All abnormalities will be described completely and recorded.

3. *Organ Weights*

At scheduled sacrifice on Day 16, the following organs (when present) will be weighed before fixation, after dissection of excess fat and other excess tissues. Organ weights will not be recorded for animals found dead or sacrificed moribund. Paired organs will be weighed together unless gross abnormalities are present, in which case they will be weighed separately. The thyroids/parathyroids will be weighed fixed at the histology laboratory.

Organs Weighed	
Adrenals	Testes
Brain	Ovaries
Heart	Spleen
Kidneys	Thyroids/parathyroids
Liver	

Organ to body weight ratios will be calculated (using the final body weight obtained prior to necropsy), as well as organ to brain weight ratios.

4. *Tissue Collection and Preservation*

For all animals necropsied, the tissues listed in the table below will be preserved in 10% neutral buffered formalin (except for the eyes, which will be preserved in Davidson's fixative and the testes that will be preserved in modified Davidson's fixative for optimum fixation).

Tissues Collected	
Cardiovascular	Urogenital
Aorta	Kidneys
Heart	Urinary Bladder
Digestive	Ovaries

Tissues Collected	
Salivary gland(s)	Uterus
Tongue	Cervix
Esophagus	Vagina
Stomach	Testes
Small Intestine	Epididymides
Duodenum	Prostate
Jejunum	Seminal Vesicles
Ileum	Endocrine
Large Intestine	Adrenals
Cecum	Pituitary
Colon	Thyroid/Parathyroid
Rectum	Skin/Musculoskeletal
Pancreas	Skin
Liver	Mammary Gland
Respiratory	Skeletal Muscle (thigh)
Trachea	Femur with articular surface
Larynx	Nervous/Special Sense
Lung with mainstem bronchus	Eye with optic nerve
Lymphoid/Hematopoietic	Sciatic Nerve
Sternum with bone marrow	Brain
Thymus	Spinal Cord – cervical
Spleen	Spinal Cord – midthoracic
Lymph Nodes	Spinal Cord – lumbar
Mandibular	Lacrimal Glands
Mesenteric	Other
	Unique Animal Identifier (not for evaluation)
	Gross Findings
	Injection site(s)

5. Histology

Preparation of slides will be performed under the direction of:

Craig Zook

Histo-Scientific Research Laboratories, Inc. (HSRL)
5930 Main St.
Mt. Jackson, VA 22842

Tissues for evaluation will be processed to paraffin blocks and prepared to slides. Slides will be stained with hematoxylin and eosin. Occasionally, other stains may be required by the Study Pathologist to aid in the diagnosis of lesions; these will be documented in the final report.

Slides will be prepared for all tissues for animals in Groups 1 and 4 for the following tissues: adrenals, brain, heart, kidneys, liver, lung, spleen, ovaries and testes. If test article-related lesions are noted, additional slides will be prepared on those tissues from Groups 2 and 3 at additional cost to the Sponsor, following the Sponsor's consent.

6. *Histopathology Evaluation*

Histopathology evaluation will be performed by:

Brett H. Saladino, DVM, Diplomate ACVP
Director of Pathology
Calvert Laboratories, Inc.
Scott Technology Park
130 Discovery Drive
Scott Township, PA 18447

Histopathological evaluations will be done on all tissues for animals in Groups 1 and 4 for the following tissues: adrenals, brain, heart, kidneys, liver, lung, spleen, ovaries and testes. If test article-related lesions are noted, additional histopathological evaluation will be performed on those tissues from Groups 2 and 3 at additional cost to the Sponsor, following the Sponsor's consent.

IV. Records and Reports

A. Data Collection and Analysis

In-Life data (clinical observations, body weights, feeder weights, dose administration and/or other related data) will be collected using Pristima version 6.3.2 or on paper when necessary. Hematology data will be collected using the

Advia 120 Hematology analyzer. Coagulation data will be collected using the Diagnostica STA Compact Coagulation Analyzer. Clinical chemistry and urine chemistry data will be collected using the AU400 Chemistry analyzer. Qualitative and quantitative urinalysis will be recorded using iRICELL 1500 UA Analyzer. Once collected on their respective instruments, hematology, coagulation, clinical chemistry and urinalysis data will be transferred to Pristima version 6.3.2 via a validated interface. In the event of hardware or software maintenance, testing or malfunction or network connection issues, data will be collected on paper or on the appropriate clinical pathology instrument and then entered into Pristima version 6.3.2 at a later date. Necropsy data will be collected on paper. Anatomic pathology findings will be directly entered into Pristima version 6.3.2. Any other data not collected on-line will be manually tabulated for inclusion in the report.

In-Life data, clinical pathology data, necropsy and anatomic pathology data will be tabulated and/or statistically evaluated using Pristima version 6.3.2. Pristima software was developed by Xybion Corporation (201 Littleton Road, Morris Plains, NJ 07950).

Statistical evaluation will be performed on In-Life, Clinical Pathology, and Organ Weight numerical data. The software will determine statistical significance by following a decision tree. First, the homogeneity of the data will be determined by Bartlett's Test. If the data is homogeneous, a one-way analysis of variance will be performed to assess statistical significance. If statistically significant differences between the means are found, Dunnett's test will be used to determine the degree of significance from the control means ($p < 0.05$, $p < 0.01$ and $p < 0.001$). If the data is non-homogeneous, the Kruskal-Wallis non-parametric analysis will be performed to assess statistical significance. If statistically significant differences between the means are found ($p < 0.05$, $p < 0.01$ and $p < 0.001$), the Mann-Whitney U-Test will be used to determine the degree of significance from the control means ($p < 0.05$, $p < 0.01$ and $p < 0.001$).

B. Storage of Records

Test article preparation and test article information, test article tracking, in-life data, necropsy data, clinical pathology data, protocol, protocol amendments (if applicable), slides, pathology evaluations (which may include blocks and wet tissues), and the original final report generated as a result of this study will be

archived at Calvert, 105 Edella Road, Suite 100, Clarks Summit, PA 18411. After 2 years, the Sponsor will be contacted to determine final disposition of all study materials.

Six months following the submission of the draft report, if there are no client comments generated by the Sponsor and/or Study Monitor, the Sponsor/Study Monitor will be notified and the report will be finalized and archived according to the terms stated in the protocol.

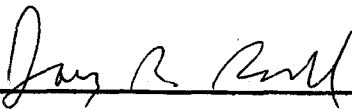
V. *Miscellaneous*

A. Confidentiality Statement

The information contained herein is for the personal use of the intended recipient(s).

B. References

1. Gad, Shayne C. (ed.). (1995). Safety Assessment for Pharmaceuticals. Van Nostrand Reinhold, New York, pp. 111-128.
2. Speid, L.H., Lumley, C.E., and Walker, S.R.. (1990). Harmonization of Guidelines for Toxicity Testing of Pharmaceuticals by 1992. Regulatory Toxicology and Pharmacology, 12: 179-211.

VI. Protocol Approval Signatures

Jay Pennell, M.S.
Study Director
Calvert Laboratories, Inc.

5 Mar 2014
Date



Scientific Management
Calvert Laboratories, Inc.

5 MAR 2014
Date



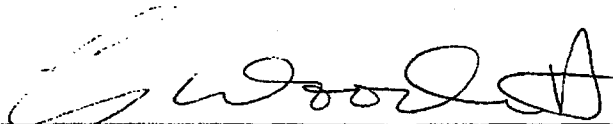
IACUC
Calvert Laboratories, Inc.

5 MARCH 2014
Date



Marlene Modi, Ph.D.
Study Monitor

04 March 2014
Date



Gary Woodnutt, PhD
Sponsor Representative

4 March 2014
Date



Customized. Responsive. Proven.

PROTOCOL AMENDMENT

STUDY NUMBER: 0437RL35.002
STUDY TITLE: Repeat Intravenous Dose Range Finding Toxicity
Study of LT3114 in Sprague Dawley Rats with
Toxicokinetics (non-GLP)
AMENDMENT NUMBER: 1
EFFECTIVE DATE: 7 Jul 2014
PROTOCOL SECTION CHANGE (Page 4):

II. Introduction

J. Sponsor Representative

~~Rosalia Matteo, PhD
Assistant Director
Lpath Inc
4025 Sorrento Valley Blvd
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Phone: (858) 678-0800 ext 112
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**Gary Woodnutt, PhD
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Phone: (858) 678-0800 ext 130
Email: gwoodnutt@lpath.com**

REASON FOR CHANGE:

The Sponsor Representative has been changed at the Sponsor's request.

PROTOCOL SECTION CHANGE (Page 6):

II. Introduction

L. Principal Investigator – Dosing Formulation and Toxicokinetic Sample Analysis

~~Florence Burleson, Ph.D.
Executive VP – Director, Laboratory Operations
Burleson Research Technologies
120 First Flight Lane
Morrisville, NC 27560
Phone: (919) 719-2500
Fax: (866) 971-2128
Email: fburleson@brt-labs.com~~

**Victor J. Johnson, Ph.D.
Burleson Research Technologies, Inc.
120 First Flight Lane
Morrisville, NC 27560
Phone: 919-719-2500
Email: vjohnson@brt-labs.com**

REASON FOR CHANGE:

At the request of the analytical laboratory, the PI has been changed to Victor J. Johnson, Ph.D.

PROTOCOL SECTION CHANGE (Page 6):

II. Introduction**O. Key Study Dates**

Proposed Experimental Start Date:	Mar 2013 1 July 2014
Proposed First Day of Dosing:	Mar 2013 7 July 2014
Proposed Necropsy:	Apr 2013 22 July 2014
Proposed Experimental Completion Date:	Apr 2013 22 July 2014

REASON FOR CHANGE:

To update the key study dates.

PROTOCOL SECTION CHANGE (Page 7):

III. Materials and Methods**A. Test Article*****2. Dose Preparation***

Before dose preparation, 2 ml of the 10% polysorbate 80 solution will be added to each liter of diluent formulation buffer under aseptic conditions and mixed gently to ensure distribution throughout the volume of buffer. The Sponsor will provide the polysorbate 80 solution and the diluent formulation. This diluent may then be used on multiple studies for placebo dosing as well as

for diluting test article to achieve the appropriate dosing formulations.

The Sponsor will provide the test article already formulated at ~~30~~ **32.1** mg/ml in sterile containers. Diluent will also be provided in separate sterile containers. Diluent and test article need to come to room temperature before the formulation is prepared. Do not shake the formulation. Group 1 control animals will receive the diluent. When necessary depending on the dose level, doses will be prepared by dilution using the provided ~~30~~ **32.1** mg/ml and diluent solutions under aseptic conditions. Before dosing, visual inspection of the formulation should be conducted against a black background for any irregularities.

REASON FOR CHANGE:

At the Sponsor's request, polysorbate 80 will be added to the diluent formulation buffer prior to use.

PROTOCOL SECTION CHANGE (Page 8):

III. Materials and Methods

A. Test Article

2. Dose Preparation

Dose formulation samples will be returned to, and analytical evaluation will be done under the direction of:

Victor J. Johnson, Ph.D.

~~Florence Burleson, Ph.D.~~

~~Executive VP — Director, Laboratory Operations~~

Burleson Research Technologies

120 First Flight Lane

Morrisville, NC 27560

REASON FOR CHANGE:

At the request of the analytical laboratory, the PI has been changed to Victor J. Johnson, Ph.D.

PROTOCOL SECTION CHANGE (Page 17):***III. Materials and Methods*****G. Toxicokinetics (Group 5-7 only)*****2. Bioanalytical Evaluation***

Plasma samples will be returned to, and bioanalytical evaluation will be done under the direction of:

Victor J. Johnson, Ph.D.

~~Florence Burleson, Ph.D.~~

~~Executive VP—Director, Laboratory Operations~~

Burleson Research Technologies

120 First Flight Lane

Morrisville, NC 27560

REASON FOR CHANGE:

At the request of the analytical laboratory, the PI has been changed to Victor J. Johnson, Ph.D.

SIGNATURES:



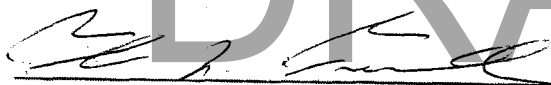
Jay Pennell, M.S.
Study Director
Calvert Laboratories, Inc.

7 Jul 2014
Date



Scientific Management
Calvert Laboratories, Inc.

7 JUL 2014
Date



IACUC
Calvert Laboratories, Inc.

3 JUL 2014
Date



Marlene Modi, Ph.D.
Study Monitor

July 2, 2014
Date



Gary Woodnutt, PhD
Sponsor Representative

June 30th 2014
Date

Calvert Study No. 0437RL35.002
Protocol Deviations

1. The 1-2 hours post-dose observations for animals #2723-#2725 (Group 3 Males), #2726-#2730 (Group 3 Females), and #2731 (Group 4 Male) were recorded at 48-58 minutes post-dose on Day 15 (21 Jul 2014) by technician JS. The observations were recorded slightly before the targeted time range. All animals appeared normal throughout the study. The deviation was minimal in magnitude and had no apparent impact on the clinical observation data obtained.

DRAFT